

LLMagogy or fast multi-stage pre-training of Large Language Models with pre-selected limited datasets. Version 2.

“knowing a few principles frees you from knowing many facts” R. Descartes

“the knowledge of certain principles easily compensates the lack of knowledge of certain facts” C. Helvétius

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Abstract

This paper experimentally tests the hypothesis regarding the benefits of multi-stage pretraining of large language models (LLMs), an approach inspired by human learning. Instead of traditionally training a fixed-size model on a huge, chaotic corpus, we propose gradually "growing" the model: starting with a small architecture and expanding its depth and breadth in successive stages, all the while training on increasingly comprehensive datasets.

Experiments on a controlled corpus of Python documentation show that the key idea—dividing the training material into stages of increasing complexity—is not supported: this approach is inferior to training on the full dataset at all stages. However, the strategy of gradually growing the model on a single dataset proves effective. It allows for a reduction in overall training time by 5–25% without loss of quality, and sometimes even improves model accuracy.

The mechanisms for controlling learning during expansion are also explored: warming up new weights demonstrates a moderate improvement in stability, while partially freezing old parameters yields no positive effect. The results indicate that the advantage of the stepwise approach is not due to the "pedagogical" structure of the data, but to the computational efficiency of training small models in the early stages. The findings suggest directions for future research, including the use of datasets with repetitive information and optimizing the number of growth stages.

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1. LLMagogy Version 1

The first version of LLMagogy proposed a multi-stage pretraining approach for large-scale language models (LLMs), inspired by human learning. Unlike the traditional approach—training a fixed-size model on trillions of tokens of chaotic internet content—LLMagogy offers:

- Gradual "growing up" of the model: starting with a small depth (n_{layer}) and width (n_{embd}), the model expands in stages, simulating the development of a child's brain.
- Using selected educational texts: In the early stages, the model is trained on small, structured corpora (e.g., school textbooks), which promotes comprehension rather than memorization.
- Preservation of knowledge between stages: through the "freezing" mechanism – the F coefficient, which slows down the updating of parameters obtained in previous stages.
- Dividing training into two phases: first, the formation of a basic model of the world ("knowledge"), then further training on the full corpus ("facts"), while the basic parameters remain almost unchanged.

The key hypothesis is that this strategy allows for the creation of an LLM that understands rather than simply memorizes, while requiring fewer resources in the early stages and potentially accelerating final learning.

Version 1 remained theoretical—it articulated the idea but lacked experimental verification. This paper (LLMagogyV2) aims to fill this gap. The full text of Version 1 can be found at <https://github.com/loftyara/LLMagogy> or <https://loftyara.github.io/llmagogy.html>

2. Goals and objectives

This paper aims to experimentally validate the concept proposed in LLMagogy (Version 1). While the first version remained theoretical, LLMagogyV2 implements it in practice using a modified GPT model and controlled experiments.

Main objectives of the project:

- Selection and compilation of datasets for hypothesis testing.
- Implement multi-stage expansion of the model (by depth n_{layer} and width n_{embd}) with preservation of parameters between stages.
- Compare methods of initializing new parameters (zeros vs. random values) and their impact on losses.
- To evaluate the role of the `weight_freeze` parameter (analogous to the freezing factor F from Version 1) in preserving "knowledge" between stages.
- To investigate the impact of the warmup strategy when expanding a model: is it beneficial for stabilizing learning or, on the contrary, hinders the adaptation of new weights?
- To test the hypothesis that training on cumulative, structured datasets (`train_cum_01.bin`, `train_cum_02.bin`, ...) promotes the formation of general "understanding" rather than memorization.

Unlike traditional pretraining, where a fixed-size model is trained on a large corpus, LLMagogyV2 focuses on controlled, step-by-step model maturation that mimics the teaching process.

3. Selection and composition of datasets

3.1. Dataset Preparation: From Ideal to Reality

The original concept for LLMagogy envisioned the use of pedagogically structured texts—for example, school textbooks, organized by grade and topic, as is done in human education. This is logical: if we want to simulate "understanding" rather than memorization, then the material should be structured, sequential, and gradually

increase in complexity. However, in practice, it turned out that ready-made datasets categorized by difficulty level do not exist. Moreover, even collections of school textbooks in open digital format are rare. Those that are available are often copyrighted, fragmented, or do not cover the entire learning cycle.

After considerable thought and research, a practical compromise was chosen: the official Python documentation, available in text format and logically organized into sections—from basic syntax to advanced topics (decorators, async, metaclasses, etc.)—was used as the training corpus. This structure allowed for a natural categorization of the material by increasing complexity, creating cumulative datasets. The Python corpus from w3schools.com, which was not part of the training material but covered similar topics, was used as the validation set. This allowed us to assess the extent to which the model generalizes knowledge rather than simply reproducing memorized documentation fragments.

The solution is, of course, not perfect:

- documentation is not a literary or interdisciplinary text,
- it does not cover the "world model" in the broad sense,
- and certainly does not replace the school curriculum.

But in terms of testing the hypothesis—that the structure and phasing of learning are more important than the volume of data—this corpus proved sufficient, controllable, and easily reproducible. Most importantly, it allows us to separate the influence of the learning method from the influence of chaotic, unstructured content.

In the future, given resources and access to licensed materials, this corpus could be replaced with actual textbooks. But for the first experimental validation of LLMagogy's idea, Python documentation became the workhorse.

3.2. Dataset for models training

The official Python programming language documentation, available in open-text format, was chosen as the basis for the training corpus. This documentation is logically structured into sections, covering topics from basic syntax to advanced concepts such as asynchronous programming, metaclasses, and low-level system interaction. This organization allowed for a natural implementation of the principle of increasing complexity inherent in the LLMagogy concept.

The training data is located in the `data/raw/train/` directory and is divided into five stages:

- Stage 1 (`data/raw/train/1/`) includes materials appropriate for the introductory level: an introduction to the language, basic data types, control structures, and functions. This stage includes the `tutorial/`, `using/`, `faq/` directories, and supporting files.
- Stage 2 (`data/raw/train/2/`) contains practical tutorials from the `howto/` directory, covering working with files, regular expressions, logging, functional programming, and other topics that require a basic understanding of the language.
- Stage 3 (`data/raw/train/3/`) is devoted to the formal specification of the language and includes the contents of the `reference/` directory, which describes in detail the syntax, semantics, exception system, and other fundamental aspects of Python.
- Stages 4 and 5 (`data/raw/train/4/` and `data/raw/train/5/`) contain material from the `library/` directory—the standard library reference. These materials have been manually divided according to pedagogical logic: Stage 4 includes modules typically studied at the intermediate level (working with the operating system, serialization, unit testing, command-line arguments, basic data structures, etc.), while Stage 5 contains advanced and system-specific topics: multithreading, multiprocessing, networking, GUI, C integration, and other sections requiring a mature understanding of the language and its ecosystem.

3.3. Dataset for models validation

The validation dataset was compiled independently of the training materials and is designed to objectively assess the model's ability to generalize, rather than simply reproduce memorized fragments. The core of the validation corpus is the Python documentation from w3schools.com—a source not used to prepare the training data. The texts from this resource cover basic and intermediate topics: syntax, data structures, functions, file handling, modules, and other aspects relevant to the early and middle stages of training. Additionally, the validation dataset was expanded with additional materials on the same topics to ensure more comprehensive coverage of key concepts and improve the reliability of the evaluation. All texts are preserved in their original form and placed in the `data/raw/val/` directory with arbitrary nesting of subdirectories, as with the training data. Importantly, none of the files in the validation dataset overlap with the training materials. This ensures that the reduction in validation loss reflects increased model understanding, rather than the memorization of specific token sequences.

4. Preparing for launch

4.1. Launch platform

All experiments were conducted on the author's PC with the following characteristics:

- OC Windows 11
- 32 GB DRAM
- Nvidia 5070 Ti graphics card with 16 GB of memory (CUDA support)
- Python version 3.12

4.2. Downloading the project

The LLMagogyV2 project is hosted in an open repository on GitHub. To get it to your computer, follow these steps:

- Open the command prompt (cmd)
- Navigate to the directory where you want to place the project (for example, `D:\Projects` with the command `cd D:\Projects`)
- Run the command `git clone https://github.com/loftyara/LLMagogyV2.git`
- Instead of the command line, you can use the GitHub Desktop program.
- Once complete, you'll see a new folder called LLMagogyV2. The entire project structure will be contained within it.

4.3. Preparing the virtual environment

To isolate project dependencies, it's recommended to use a Python virtual environment (venv). Navigate to the downloaded project directory and run the following command:

```
python -m venv venv
```

Activate the virtual environment:

```
venv\Scripts\activate
```

install dependencies:

```
pip install -r requirements.txt
```

Install pytorch with cuda support:

```
pip install torch --index-url https://download.pytorch.org/whl/cu128
```

5. Preparing datasets

Based on the raw text data located in the directories `data/raw/train/1-5/` and `data/raw/val/`, the process of preparing training and validation datasets was performed.

For this, the script `scripts/prepare_datasets.py` was used, which:

- recursively collects all `.txt` files from the specified directories (taking into account arbitrary nesting of subdirectories),
- forms cumulative text corpora for each stage of learning:
 - `train_cum_1.txt` — stage 1 data,
 - `train_cum_2.txt` — stages 1 + 2 data
 - ...
 - `train_cum_5.txt` — stages 1 + 2 + 3 + 4 + 5 data,
- Combines all validation texts into a single file `val.txt`.

The results are saved in the `data/processed/` directory. These text files are subsequently used for tokenization and the creation of binary datasets suitable for model training.

6. Traditional LLM Training

To establish a baseline performance, a series of experiments using traditional pretraining of the language model was conducted on the full `train_cum_5.txt` corpus. Unlike LLMagogy's multi-stage approach, this fixed-size model is trained on all available data at once, without gradual expansion or knowledge retention between stages.

The experiments are implemented in the `src/llmagogy_step0.py` program. The model is evaluated on the validation dataset `val.txt`, which does not overlap with the training data. The primary metric is the validation loss, which reflects the model's ability to generalize rather than learn.

The following variables are used:

No	Variable	Description
1	<code>n_layer</code>	Number of transformer blocks (model depth)
2	<code>n_embd</code>	Embedding dimensions (model width)
3	<code>n_head</code>	Number of attention heads in each block
4	<code>block_size</code>	Maximum context length (number of tokens/symbols in one training block)
5	<code>dropout</code>	Probability of neurons being switched off during training (regularization)
6	<code>batch_size</code>	The number of sequences processed in one training step
7	<code>max_iters</code>	Total number of training steps (iterations)
8	<code>eval_interval</code>	The interval (in steps) at which the current loss is displayed
9	<code>eval_iters</code>	The number of batches on which the loss is averaged during evaluation
10	<code>learning_rate</code>	Learning rate (gradient descent step)
11	<code>grad_clip</code>	Gradient norm clipping threshold (prevents "gradient explosion")
12	<code>final_loss_sampels</code>	The number of batches on which the final loss is calculated at the end of training

Table 1. Model variable parameters in the `llmagogy_step0.py` file

6.1. Selecting model parameters

The model parameters are selected taking into account three factors:

1. LLMagogy's hypotheses: the model should be small enough to avoid memorizing text verbatim, but large enough to capture structural patterns ("comprehension").

2. Hardware limitations: A video card with 16 GB of memory places an upper limit on `block_size` and `batch_size`.
3. Data structures: Python documentation contains long logical blocks (functions, classes), possibly requiring significant length of context.

From now on, we'll name models using the template `n_layer/n_embd/n_head/block_size/dropout/batch_size`. For example, model `1/32/1/32/0.00/64` is the minimum possible model with which we'll begin testing. `Batch_size` is set to the maximum possible size so that the model fits in the video card's memory, but not exceeding 64. We'll determine the optimal model parameters experimentally, starting with the minimum model.

The results of experiments with different parameter values are presented in the following sections.

6.2. Model training results. Step 1.1

We'll set the dropout parameter to 0, start with the minimum model of `1/32/1/32/0.00/64`, and test all models until overfitting occurs and the val loss begins to increase. The model training results are presented in Table 2.

№	Модель	model size M	n_layer	n_embd	n_head	block_size	dropout	batch_size	train iterations	train loss	val iterations	val loss
1	1/32/1/32/0.00/64	0.03	1	32	1	32	0.00	64	800 000	1.6392	970 000	1.9899
2	1/32/1/64/0.00/64	0.03	1	32	1	64	0.00	64	800 000	1.6274	820 000	1.9898
3	1/32/1/128/0.00/64	0.03	1	32	1	128	0.00	64	890 000	1.6290	840 000	1.9990
4	1/32/1/256/0.00/64	0.03	1	32	1	256	0.00	64	930 000	1.6695	980 000	2.0392
5	2/32/1/32/0.00/64	0.04	2	32	1	32	0.00	64	890 000	1.4792	790 000	1.8494
6	2/32/1/64/0.00/64	0.04	2	32	1	64	0.00	64	890 000	1.4491	700 000	1.8496
7	2/32/1/128/0.00/64	0.04	2	32	1	128	0.00	64	1 000 000	1.4176	780 000	1.8180
8	2/32/1/256/0.00/64	0.05	2	32	1	256	0.00	64	960 000	1.4366	850 000	1.8479
9	1/64/2/64/0.00/64	0.08	1	64	2	64	0.00	64	530 000	1.3790	530 000	1.8150
10	1/64/2/128/0.00/64	0.08	1	64	2	128	0.00	64	450 000	1.3693	330 000	1.7953
11	1/64/2/256/0.00/64	0.09	1	64	2	256	0.00	64	530 000	1.3586	590 000	1.8198
12	1/64/2/512/0.00/64	0.11	1	64	2	512	0.00	64	610 000	1.3688	490 000	1.8295
13	2/64/2/128/0.00/64	0.13	2	64	2	128	0.00	64	940 000	1.1380	800 000	1.6839
14	2/64/2/256/0.00/64	0.14	2	64	2	256	0.00	64	870 000	1.1594	810 000	1.6891
15	2/64/2/512/0.00/64	0.16	2	64	2	512	0.00	64	930 000	1.1585	940 000	1.7283
16	3/64/2/128/0.00/64	0.18	3	64	2	128	0.00	64	760 000	1.0889	670 000	1.6369
17	3/64/2/256/0.00/64	0.19	3	64	2	256	0.00	64	940 000	1.0563	860 000	1.6660
18	3/64/2/512/0.00/64	0.21	3	64	2	512	0.00	64	470 000	1.0597	480 000	1.7291
19	4/64/2/128/0.00/64	0.23	4	64	2	128	0.00	64	360 000	1.0589	240 000	1.6577
20	4/64/2/256/0.00/64	0.24	4	64	2	256	0.00	64	360 000	1.0490	30 000	1.7256
21	3/96/3/128/0.00/64	0.38	3	96	3	128	0.00	64	280 000	0.9955	120 000	1.6499
22	3/96/3/256/0.00/64	0.40	3	96	3	256	0.00	64	240 000	0.9670	100 000	1.6361
23	2/128/4/128/0.00/64	0.46	2	128	4	128	0.00	64	540 000	0.9595	450 000	1.6471
24	2/128/4/256/0.00/64	0.48	2	128	4	256	0.00	64	430 000	0.9496	380 000	1.6579
25	2/128/4/512/0.00/64	0.51	2	128	4	512	0.00	64	240 000	0.9979	180 000	1.6875
26	4/96/3/128/0.00/64	0.50	4	96	3	128	0.00	64	410 000	0.9288	470 000	1.6639
27	4/96/3/256/0.00/64	0.51	4	96	3	256	0.00	64	320 000	0.9099	50 000	1.6767
28	3/128/4/128/0.00/64	0.66	3	128	4	128	0.00	64	610 000	0.8894	120 000	1.6289
29	3/128/4/256/0.00/64	0.67	3	128	4	256	0.00	64	450 000	0.8399	20 000	1.6159

Table 2. Results of training models at Step 1.1.

6.3. Analysis of the results of training models in Step 1.1

Small models, as expected, are insufficient to achieve the minimum val loss. Moreover, models with 3 or 4 layers tend to overfit, requiring the dropout parameter.

6.4. Model training results. Step 1.2

Let's start with a dropout parameter of 0.05 and increase it in 0.05 increments if we see signs of overtraining. We'll select 3/128/4 as the starting models—the model we used in Step 1.1. We'll continue to gradually increase the model to 8-9 million parameters. Further increases may result in a decrease in loss, but the GPU memory will no longer be sufficient for training.

The obtained results of model training are presented in Table 3.

№	Модель	model size M	n_layer	n_embd	n_head	block_size	dropout	batch_size	train iterations	train loss	val iterations	val loss
1	3/128/4/128/0.05/64	0.66	3	128	4	128	0.05	64	910 000	0.8969	850 000	1.4479
2	3/128/4/256/0.05/64	0.67	3	128	4	256	0.05	64	630 000	0.8386	350 000	1.3999
3	3/128/4/128/0.10/64	0.66	3	128	4	128	0.10	64	740 000	0.9397	620 000	1.4592
4	3/128/4/256/0.10/64	0.67	3	128	4	256	0.10	64	760 000	0.8686	380 000	1.4080
5	4/128/4/256/0.05/64	0.87	4	128	4	256	0.05	64	400 000	0.8095	450 000	1.4447
6	4/128/4/512/0.05/64	0.91	4	128	4	512	0.05	64	270 000	0.7781	180 000	1.3395
7	3/192/6/512/0.05/32	1.50	3	192	6	512	0.05	32	350 000	0.7043	130 000	1.3392
8	3/192/6/1024/0.05/32	1.60	3	192	6	1024	0.05	32	120 000	0.7457	100 000	1.2775
9	4/192/6/1024/0.05/8	2.05	4	192	6	1024	0.05	8	250 000	0.6898	230 000	1.2453
10	4/192/6/1536/0.05/8	2.15	4	192	6	1536	0.05	8	260 000	0.6677	240 000	1.2115
11	10/128/4/1536/0.05/8	2.23	10	128	4	1536	0.05	8	80 000	0.6757	50 000	1.1731
12	5/192/6/1536/0.05/8	2.59	5	192	6	1536	0.05	8	220 000	0.6193	80 000	1.2045
13	5/192/6/2048/0.05/8	2.69	5	192	6	2048	0.05	8	160 000	0.6268	50 000	1.1879
14	12/128/4/2048/0.05/4	2.69	12	128	4	2048	0.05	4	90 000	0.6440	70 000	1.1260
15	12/128/4/2048/0.10/4	2.69	12	128	4	2048	0.10	4	140 000	0.6378	120 000	1.0961
16	8/160/5/2048/0.05/8	2.86	8	160	5	2048	0.05	8	90 000	0.6182	30 000	1.1527
17	10/160/5/2048/0.05/4	3.48	10	160	5	2048	0.05	4	100 000	0.5924	50 000	1.1397
18	10/160/5/2048/0.10/4	3.48	10	160	5	2048	0.10	4	90 000	0.6422	80 000	1.0957
19	4/256/8/1536/0.05/8	3.65	4	256	8	1536	0.05	8	170 000	0.5997	110 000	1.2714
20	4/256/8/1536/0.10/8	3.65	4	256	8	1536	0.10	8	190 000	0.6499	110 000	1.2346
21	4/256/8/2048/0.10/8	3.78	4	256	8	2048	0.10	8	170 000	0.6435	70 000	1.1927
22	8/192/6/2048/0.10/4	4.02	8	192	6	2048	0.10	4	70 000	0.6545	40 000	1.1061
23	12/160/5/2048/0.10/4	4.10	12	160	5	2048	0.10	4	80 000	0.6189	80 000	1.1228
24	5/256/8/2048/0.10/4	4.57	5	256	8	2048	0.10	4	170 000	0.6099	130 000	1.1457
25	10/192/6/2048/0.10/4	4.91	10	192	6	2048	0.10	4	110 000	0.5654	50 000	1.0973
26	6/256/8/2048/0.10/4	5.36	6	256	8	2048	0.10	4	170 000	0.5556	60 000	1.1235
27	8/224/7/2048/0.10/4	5.38	8	224	7	2048	0.10	4	130 000	0.5524	40 000	1.0994
28	12/192/6/2048/0.10/4	5.80	12	192	6	2048	0.10	4	50 000	0.6036	30 000	1.1191
29	9/224/7/2048/0.10/4	5.99	9	224	7	2048	0.10	4	70 000	0.5900	30 000	1.0856
30	7/256/8/2048/0.10/4	6.15	7	256	8	2048	0.10	4	170 000	0.5092	50 000	1.1179
31	10/224/7/2048/0.10/4	6.59	10	224	7	2048	0.10	4	160 000	0.4737	30 000	1.0975

32	8/256/8/2048/0.10/4	6.94	8	256	8	2048	0.10	4	120 000	0.5013	40 000	1.0878
33	8/256/8/2048/0.15/4	6.94	8	256	8	2048	0.15	4	80 000	0.5877	40 000	1.0789
34	5/320/10/2048/0.10/4	6.94	5	320	10	2048	0.10	4	60 000	0.5599	60 000	1.1507
35	5/320/10/2048/0.15/4	6.94	5	320	10	2048	0.15	4	170 000	0.5806	60 000	1.1446
36	9/256/8/2048/0.15/4	7.73	9	256	8	2048	0.15	4	100 000	0.5453	50 000	1.0855
37	6/320/10/2048/0.15/4	8.17	6	320	10	2048	0.15	4	110 000	0.5625	40 000	1.1026
38	6/320/10/2048/0.20/4	8.17	6	320	10	2048	0.20	4	140 000	0.5852	40 000	1.1196
39	10/256/8/2048/0.15/4	8.52	10	256	8	2048	0.15	4	100 000	0.5084	40 000	1.1019
40	8/288/9/2048/0.15/4	8.69	8	288	9	2048	0.15	4	70 000	0.5733	50 000	1.0852

Table 3. Results of training models at Step 1.2

6.5. Analysis of the results of training models in Step 1.2

Adding dropout improves the performance of even the models already tested in Step 1.1. During testing, it was found that the minimum achievable val loss is approximately 1.07 for the 8/256/8/2048/0.15/4 model. When the limit of model size growth was reached, variants with smaller embedding dimensions but a larger number of layers were tested and found that these models achieve similar accuracy even at significantly smaller sizes—12/128/4/2048/0.10/4.

For further testing, we will use three models: the minimal 12/128/4/2048/0.10/4 with a small width and large depth, the most accurate 8/256/8/2048/0.15/4 with a larger width and smaller depth, and an intermediate 10/192/6/2048/0.10/4. All these models were able to achieve a val loss of 1.07 – 1.10.

6.6. Model training results. Step 1.3

For each of the three selected models (12/128/4/2048/0.10/4, 8/256/8/2048/0.15/4, 10/192/6/2048/0.10/4), we will perform long-term training with a small step size. For each model, we will find checkpoints and determine the number of iterations, the training time, and the train loss and val loss results for the following cases:

- Reaching a plateau in val loss
- Minimum value of val loss
- Reaching a plateau in train loss or the minimum value of train loss before val loss increases

The obtained results of model training are presented in Table 4.

No	iterations	Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1		model size, M	2.69	4.91	6.94
2		n_layer	12	10	8
3		n_embd	128	192	256
4		n_head	4	6	8
5		block_size	2 048	2 048	2 048
6		dropout	0.10	0.10	0.15
7		batch_size	4	4	4
8	0	duration / val/train	5 1.0148	5 1.0378	5 1.0246
9		train loss	4.6763	4.5850	4.5445
10		val loss	4.7457	4.7585	4.6561
11	2 000	duration / val/train	297 1.1154	403 1.1176	408 1.1226
12		train loss / train diff %	2.4462 -47.69%	2.4113 -47.41%	2.3913 -47.38%
13		val loss / val diff %	2.7286 -42.50%	2.6949 -43.37%	2.6845 -42.34%
14	4 000	duration / val/train	590 1.2040	799 1.2713	814 1.2720
15		train loss / train diff %	1.9095 -21.94%	1.5336 -36.40%	1.3686 -42.77%
16		val loss / val diff %	2.2991 -15.74%	1.9496 -27.66%	1.7409 -35.15%
17	6 000	duration / val/train	880 1.2991	1 192 1.3369	1 215 1.3296
18		train loss / train diff %	1.2713 -33.42%	1.0791 -29.64%	1.0370 -24.23%
19		val loss / val diff %	1.6516 -28.16%	1.4426 -26.01%	1.3788 -20.80%
20	8 000	duration / val/train	1 170 1.3448	1 570 1.3612	1 618 1.3587
21		train loss / train diff %	1.0816 -14.92%	0.9568 -11.33%	0.9385 -9.50%
22		val loss / val diff %	1.4545 -11.93%	1.3024 -9.72%	1.2751 -7.52%
23	10 000	duration / val/train	1 460 1.3393	1 938 1.3748	2 033 1.3622
24		train loss / train diff %	1.0036 -7.21%	0.9056 -5.35%	0.8785 -6.39%
25		val loss / val diff %	1.3441 -7.59%	1.2450 -4.41%	1.1967 -6.15%

26	12 000	<i>duration / val/train</i>	1 750	1.3885	2 303	1.4016	2 436	1.4081
27		<i>train loss / train diff %</i>	0.9448	-5.86%	0.8621	-4.80%	0.8398	-4.41%
28		<i>val loss / val diff %</i>	1.3119	-2.40%	1.2083	-2.95%	1.1825	-1.19%
29	14 000	<i>duration / val/train</i>	2 040	1.4096	2 668	1.4630	2 845	1.4315
30		<i>train loss / train diff %</i>	0.9185	-2.78%	0.8294	-3.79%	0.8098	-3.57%
31		<i>val loss / val diff %</i>	1.2947	-1.31%	1.2134	0.42%	1.1592	-1.97%
32	16 000	<i>duration / val/train</i>	2 331	1.3991	3 033	1.4392	3 261	1.4477
33		<i>train loss / train diff %</i>	0.8840	-3.76%	0.8136	-1.90%	0.7920	-2.20%
34		<i>val loss / val diff %</i>	1.2368	-4.47%	1.1709	-3.50%	1.1466	-1.09%
35	18 000	<i>duration / val/train</i>	2 621	1.4195	3 397	1.4661	3 676	1.4897
36		<i>train loss / train diff %</i>	0.8665	-1.98%	0.7867	-3.31%	0.7685	-2.97%
37		<i>val loss / val diff %</i>	1.2300	-0.55%	1.1534	-1.49%	1.1448	-0.16%
38	20 000	<i>duration / val/train</i>	2 911	1.4446	3 762	1.4733	4 072	1.5007
39		<i>train loss / train diff %</i>	0.8471	-2.24%	0.7707	-2.03%	0.7570	-1.50%
40		<i>val loss / val diff %</i>	1.2237	-0.51%	1.1355	-1.55%	1.1360	-0.77%
41	22 000	<i>duration / val/train</i>	3 201	1.4417	4 138	1.5248	4 483	1.5284
42		<i>train loss / train diff %</i>	0.8318	-1.81%	0.7528	-2.32%	0.7413	-2.07%
43		<i>val loss / val diff %</i>	1.1992	-2.00%	1.1479	1.09%	1.1330	-0.26%
44	24 000	<i>duration / val/train</i>	3 491	1.4743	4 510	1.5060	4 876	1.5532
45		<i>train loss / train diff %</i>	0.8144	-2.09%	0.7453	-1.00%	0.7215	-2.67%
46		<i>val loss / val diff %</i>	1.2007	0.13%	1.1224	-2.22%	1.1206	-1.09%
47	26 000	<i>duration / val/train</i>	3 781	1.4702	4 887	1.5506	5 271	1.5404
48		<i>train loss / train diff %</i>	0.8039	-1.29%	0.7318	-1.81%	0.7250	0.49%
49		<i>val loss / val diff %</i>	1.1819	-1.57%	1.1347	1.10%	1.1168	-0.34%
50	28 000	<i>duration / val/train</i>	4 071	1.4609	5 258	1.5409	5 663	1.5581
51		<i>train loss / train diff %</i>	0.7870	-2.10%	0.7240	-1.07%	0.7137	-1.56%
52		<i>val loss / val diff %</i>	1.1497	-2.72%	1.1156	-1.68%	1.1120	-0.43%
53	30 000	<i>duration / val/train</i>	4 361	1.4881	5 642	1.5666	6 052	1.6109
54		<i>train loss / train diff %</i>	0.7884	0.18%	0.7143	-1.34%	0.6957	-2.52%
55		<i>val loss / val diff %</i>	1.1732	2.04%	1.1190	0.30%	1.1207	0.78%
56	32 000	<i>duration / val/train</i>	4 651	1.5002	6 010	1.5817	6 441	1.5717
57		<i>train loss / train diff %</i>	0.7775	-1.38%	0.7078	-0.91%	0.6914	-0.62%
58		<i>val loss / val diff %</i>	1.1664	-0.58%	1.1195	0.04%	1.0867	-3.03%
59	34 000	<i>duration / val/train</i>	4 941	1.5264	6 385	1.5946	6 830	1.6082
60		<i>train loss / train diff %</i>	0.7618	-2.02%	0.6966	-1.58%	0.6841	-1.06%
61		<i>val loss / val diff %</i>	1.1628	-0.31%	1.1108	-0.78%	1.1002	1.24%
62	36 000	<i>duration / val/train</i>	5 231	1.5095	6 767	1.6345	7 219	1.6398
63		<i>train loss / train diff %</i>	0.7580	-0.50%	0.6914	-0.75%	0.6672	-2.47%
64		<i>val loss / val diff %</i>	1.1442	-1.60%	1.1301	1.74%	1.0941	-0.55%
65	38 000	<i>duration / val/train</i>	5 521	1.5177	7 151	1.6060	7 608	1.6377
66		<i>train loss / train diff %</i>	0.7473	-1.41%	0.6842	-1.04%	0.6702	0.45%
67		<i>val loss / val diff %</i>	1.1342	-0.87%	1.0988	-2.77%	1.0976	0.32%
68	40 000	<i>duration / val/train</i>	5 812	1.5256	7 547	1.6360	7 997	1.6378
69		<i>train loss / train diff %</i>	0.7502	0.39%	0.6717	-1.83%	0.6604	-1.46%
70		<i>val loss / val diff %</i>	1.1445	0.91%	1.0989	0.01%	1.0816	-1.46%
71	42 000	<i>duration / val/train</i>	6 102	1.5505	7 936	1.6438	8 386	1.6391
72		<i>train loss / train diff %</i>	0.7426	-1.01%	0.6710	-0.10%	0.6587	-0.26%
73		<i>val loss / val diff %</i>	1.1514	0.60%	1.1030	0.37%	1.0797	-0.18%
74	44 000	<i>duration / val/train</i>	6 392	1.5540	8 340	1.6529	8 775	1.6933

75		<i>train loss / train diff %</i>	0.7374	-0.70%	0.6670	-0.60%	0.6489	-1.49%
76		<i>val loss / val diff %</i>	1.1459	-0.48%	1.1025	-0.05%	1.0988	1.77%
77	46 000	<i>duration / val/train</i>	6 682	1.5539	8 742	1.6863	9 164	1.6959
78		<i>train loss / train diff %</i>	0.7321	-0.72%	0.6615	-0.82%	0.6475	-0.22%
79		<i>val loss / val diff %</i>	1.1376	-0.72%	1.1155	1.18%	1.0981	-0.06%
80	48 000	<i>duration / val/train</i>	6 972	1.5242	9 139	1.6802	9 553	1.7063
81		<i>train loss / train diff %</i>	0.7326	0.07%	0.6523	-1.39%	0.6415	-0.93%
82		<i>val loss / val diff %</i>	1.1166	-1.85%	1.0960	-1.75%	1.0946	-0.32%
83	50 000	<i>duration / val/train</i>	7 262	1.5500	9 523	1.6917	9 942	1.7087
84		<i>train loss / train diff %</i>	0.7246	-1.09%	0.6513	-0.15%	0.6357	-0.90%
85		<i>val loss / val diff %</i>	1.1231	0.58%	1.1018	0.53%	1.0862	-0.77%
86	52 000	<i>duration / val/train</i>	7 552	1.5624	9 897	1.7167	10 331	1.7433
87		<i>train loss / train diff %</i>	0.7205	-0.57%	0.6480	-0.51%	0.6322	-0.55%
88		<i>val loss / val diff %</i>	1.1257	0.23%	1.1124	0.96%	1.1021	1.46%
89	54 000	<i>duration / val/train</i>	7 842	1.5673	10 278	1.6963	10 720	1.7482
90		<i>train loss / train diff %</i>	0.7160	-0.62%	0.6437	-0.66%	0.6207	-1.82%
91		<i>val loss / val diff %</i>	1.1222	-0.31%	1.0919	-1.84%	1.0851	-1.54%
92	56 000	<i>duration / val/train</i>	8 132	1.5921	10 658	1.7447	11 109	1.7322
93		<i>train loss / train diff %</i>	0.7125	-0.49%	0.6327	-1.71%	0.6195	-0.19%
94		<i>val loss / val diff %</i>	1.1344	1.09%	1.1039	1.10%	1.0731	-1.11%
95	58 000	<i>duration / val/train</i>	8 422	1.5948	11 025	1.7447	11 498	1.7681
96		<i>train loss / train diff %</i>	0.7076	-0.69%	0.6337	0.16%	0.6174	-0.34%
97		<i>val loss / val diff %</i>	1.1285	-0.52%	1.1056	0.15%	1.0916	1.72%
98	60 000	<i>duration / val/train</i>	8 712	1.5893	11 390	1.7733	11 886	1.7730
99		<i>train loss / train diff %</i>	0.7030	-0.65%	0.6299	-0.60%	0.6142	-0.52%
100		<i>val loss / val diff %</i>	1.1173	-0.99%	1.1170	1.03%	1.0890	-0.24%
101	62 000	<i>duration / val/train</i>	9 002	1.6282	11 766	1.7521	12 279	1.7839
102		<i>train loss / train diff %</i>	0.7020	-0.14%	0.6269	-0.48%	0.6179	0.60%
103		<i>val loss / val diff %</i>	1.1430	2.30%	1.0984	-1.67%	1.1023	1.22%
104	64 000	<i>duration / val/train</i>	9 292	1.6219	12 133	1.7716	12 669	1.8138
105		<i>train loss / train diff %</i>	0.6991	-0.41%	0.6229	-0.64%	0.6043	-2.20%
106		<i>val loss / val diff %</i>	1.1339	-0.80%	1.1035	0.46%	1.0961	-0.56%
107	66 000	<i>duration / val/train</i>	9 583	1.5970	12 498	1.7954	13 058	1.8289
108		<i>train loss / train diff %</i>	0.7007	0.23%	0.6194	-0.56%	0.6060	0.28%
109		<i>val loss / val diff %</i>	1.1190	-1.31%	1.1121	0.78%	1.1083	1.11%
110	68 000	<i>duration / val/train</i>	9 873	1.6310	12 863	1.7894	13 447	1.8198
111		<i>train loss / train diff %</i>	0.6937	-1.00%	0.6167	-0.44%	0.6037	-0.38%
112		<i>val loss / val diff %</i>	1.1314	1.11%	1.1035	-0.77%	1.0986	-0.88%
113	70 000	<i>duration / val/train</i>	10 163	1.6410	13 227	1.7727	13 836	1.8219
114		<i>train loss / train diff %</i>	0.6869	-0.98%	0.6165	-0.03%	0.6046	0.15%
115		<i>val loss / val diff %</i>	1.1272	-0.37%	1.0929	-0.96%	1.1015	0.26%
116	72 000	<i>duration / val/train</i>	10 453	1.6242	13 592	1.7964	14 226	1.8089
117		<i>train loss / train diff %</i>	0.6913	0.64%	0.6165	0.00%	0.5971	-1.24%
118		<i>val loss / val diff %</i>	1.1228	-0.39%	1.1075	1.34%	1.0801	-1.94%
119	74 000	<i>duration / val/train</i>	10 743	1.6192	13 957	1.8277	14 625	1.8294
120		<i>train loss / train diff %</i>	0.6877	-0.52%	0.6058	-1.74%	0.5950	-0.35%
121		<i>val loss / val diff %</i>	1.1135	-0.83%	1.1072	-0.03%	1.0885	0.78%
122	76 000	<i>duration / val/train</i>	11 033	1.6382	14 321	1.7896	15 034	1.8338
123		<i>train loss / train diff %</i>	0.6849	-0.41%	0.6078	0.33%	0.5938	-0.20%

124		val loss / val diff %	1.1220	0.76%	1.0877	-1.76%	1.0889	0.04%
125	78 000	duration / val/train	11 323	1.6192	14 686	1.8425	15 450	1.8682
126		train loss / train diff %	0.6820	-0.42%	0.6005	-1.20%	0.5873	-1.09%
127		val loss / val diff %	1.1043	-1.58%	1.1064	1.72%	1.0972	0.76%
128	80 000	duration / val/train	11 613	1.6530	15 050	1.8571	15 861	1.8966
129		train loss / train diff %	0.6818	-0.03%	0.6005	0.00%	0.5822	-0.87%
130		val loss / val diff %	1.1270	2.06%	1.1152	0.80%	1.1042	0.64%
131	82 000	duration / val/train	11 903	1.6480	15 415	1.8707	16 265	1.8779
132		train loss / train diff %	0.6758	-0.88%	0.5969	-0.60%	0.5864	0.72%
133		val loss / val diff %	1.1137	-1.18%	1.1166	0.13%	1.1012	-0.27%
134	84 000	duration / val/train	12 193	1.6507	15 780	1.8756	16 670	1.8977
135		train loss / train diff %	0.6767	0.13%	0.5971	0.03%	0.5766	-1.67%
136		val loss / val diff %	1.1170	0.30%	1.1199	0.30%	1.0942	-0.64%
137	86 000	duration / val/train	12 483	1.6441	16 144	1.8867	17 078	1.9037
138		train loss / train diff %	0.6782	0.22%	0.5949	-0.37%	0.5792	0.45%
139		val loss / val diff %	1.1150	-0.18%	1.1224	0.22%	1.1026	0.77%
140	88 000	duration / val/train	12 773	1.6727	16 509	1.9105	17 484	1.9256
141		train loss / train diff %	0.6707	-1.11%	0.5885	-1.08%	0.5792	0.00%
142		val loss / val diff %	1.1219	0.62%	1.1243	0.17%	1.1153	1.15%
143	90 000	duration / val/train	13 064	1.6611	16 873	1.8793	17 881	1.9060
144		train loss / train diff %	0.6715	0.12%	0.5898	0.22%	0.5721	-1.23%
145		val loss / val diff %	1.1154	-0.58%	1.1084	-1.41%	1.0904	-2.23%
146	92 000	duration / val/train	13 354	1.6737	17 238	1.8982	18 279	1.9020
147		train loss / train diff %	0.6693	-0.33%	0.5873	-0.42%	0.5723	0.03%
148		val loss / val diff %	1.1202	0.43%	1.1148	0.58%	1.0885	-0.17%
149	94 000	duration / val/train	13 644	1.6702	17 603	1.8805	18 692	1.9257
150		train loss / train diff %	0.6707	0.21%	0.5882	0.15%	0.5749	0.45%
151		val loss / val diff %	1.1202	0.00%	1.1061	-0.78%	1.1071	1.71%
152	96 000	duration / val/train	13 934	1.6527	17 967	1.9304	19 121	1.9290
153		train loss / train diff %	0.6692	-0.22%	0.5775	-1.82%	0.5692	-0.99%
154		val loss / val diff %	1.1060	-1.27%	1.1148	0.79%	1.0980	-0.82%
155	98 000	duration / val/train	14 232	1.6549	18 332	1.8947	19 532	1.9612
156		train loss / train diff %	0.6667	-0.37%	0.5851	1.32%	0.5666	-0.46%
157		val loss / val diff %	1.1033	-0.24%	1.1086	-0.56%	1.1112	1.20%
158	100 000	duration / val/train	14 526	1.6534	18 696	1.9252	19 924	1.9306
159		train loss / train diff %	0.6688	0.31%	0.5775	-1.30%	0.5662	-0.07%
160		val loss / val diff %	1.1058	0.23%	1.1118	0.29%	1.0931	-1.63%
161	102 000	duration / val/train	14 824	1.6864	19 061	1.9538	20 313	1.9802
162		train loss / train diff %	0.6581	-1.60%	0.5734	-0.71%	0.5592	-1.24%
163		val loss / val diff %	1.1098	0.36%	1.1203	0.76%	1.1073	1.30%
164	104 000	duration / val/train	15 124	1.6445	19 425	1.9270	20 703	1.9383
165		train loss / train diff %	0.6663	1.25%	0.5725	-0.16%	0.5626	0.61%
166		val loss / val diff %	1.0957	-1.27%	1.1032	-1.53%	1.0905	-1.52%
167	106 000	duration / val/train	15 423	1.6811	19 790	1.9793	21 092	1.9384
168		train loss / train diff %	0.6626	-0.56%	0.5700	-0.44%	0.5621	-0.09%
169		val loss / val diff %	1.1139	1.66%	1.1282	2.27%	1.0896	-0.08%
170	108 000	duration / val/train	15 729	1.6779	20 155	1.9372	21 481	2.0023
171		train loss / train diff %	0.6584	-0.63%	0.5681	-0.33%	0.5576	-0.80%
172		val loss / val diff %	1.1047	-0.83%	1.1005	-2.46%	1.1165	2.47%

173	110 000	<i>duration / val/train</i>	16 039	1.7104	20 519	1.9730	21 870	1.9959
174		<i>train loss / train diff %</i>	0.6481	-1.56%	0.5676	-0.09%	0.5552	-0.43%
175		<i>val loss / val diff %</i>	1.1085	0.34%	1.1199	1.76%	1.1081	-0.75%
176	112 000	<i>duration / val/train</i>	16 344	1.6719	20 884	1.9878	22 260	2.0155
177		<i>train loss / train diff %</i>	0.6531	0.77%	0.5720	0.78%	0.5545	-0.13%
178		<i>val loss / val diff %</i>	1.0919	-1.50%	1.1370	1.53%	1.1176	0.86%
179	114 000	<i>duration / val/train</i>	16 644	1.7001	21 248	1.9718	22 649	2.0277
180		<i>train loss / train diff %</i>	0.6498	-0.51%	0.5674	-0.80%	0.5519	-0.47%
181		<i>val loss / val diff %</i>	1.1047	1.17%	1.1188	-1.60%	1.1191	0.13%
182	116 000	<i>duration / val/train</i>	16 937	1.7488	21 613	1.9954	23 038	2.0356
183		<i>train loss / train diff %</i>	0.6454	-0.68%	0.5641	-0.58%	0.5476	-0.78%
184		<i>val loss / val diff %</i>	1.1287	2.17%	1.1256	0.61%	1.1147	-0.39%
185	118 000	<i>duration / val/train</i>	17 230	1.6938	21 978	1.9825	23 427	2.0099
186		<i>train loss / train diff %</i>	0.6489	0.54%	0.5655	0.25%	0.5477	0.02%
187		<i>val loss / val diff %</i>	1.0991	-2.62%	1.1211	-0.40%	1.1008	-1.25%
188	120 000	<i>duration / val/train</i>	17 521	1.6952	22 350	2.0077	23 816	2.0378
189		<i>train loss / train diff %</i>	0.6466	-0.35%	0.5620	-0.62%	0.5508	0.57%
190		<i>val loss / val diff %</i>	1.0961	-0.27%	1.1283	0.64%	1.1224	1.96%
191	122 000	<i>duration / val/train</i>	17 820	1.6938	22 726	2.0124	24 205	2.0682
192		<i>train loss / train diff %</i>	0.6531	1.01%	0.5575	-0.80%	0.5472	-0.65%
193		<i>val loss / val diff %</i>	1.1062	0.92%	1.1219	-0.57%	1.1317	0.83%
194	124 000	<i>duration / val/train</i>	18 122	1.6940	23 105	2.0027	24 632	2.0894
195		<i>train loss / train diff %</i>	0.6457	-1.13%	0.5632	1.02%	0.5470	-0.04%
196		<i>val loss / val diff %</i>	1.0938	-1.12%	1.1279	0.53%	1.1429	0.99%
197	126 000	<i>duration / val/train</i>	18 417	1.7144	23 476	1.9969	25 025	2.1031
198		<i>train loss / train diff %</i>	0.6375	-1.27%	0.5570	-1.10%	0.5459	-0.20%
199		<i>val loss / val diff %</i>	1.0929	-0.08%	1.1123	-1.38%	1.1481	0.45%
200	128 000	<i>duration / val/train</i>	18 722	1.6985	23 843	2.0394	25 420	2.0617
201		<i>train loss / train diff %</i>	0.6447	1.13%	0.5527	-0.77%	0.5447	-0.22%
202		<i>val loss / val diff %</i>	1.0950	0.19%	1.1272	1.34%	1.1230	-2.19%
203	130 000	<i>duration / val/train</i>	19 030	1.6994	24 208	2.0152	25 825	2.0583
204		<i>train loss / train diff %</i>	0.6448	0.02%	0.5540	0.24%	0.5418	-0.53%
205		<i>val loss / val diff %</i>	1.0958	0.07%	1.1164	-0.96%	1.1152	-0.69%
206	132 000	<i>duration / val/train</i>	19 336	1.7244	24 572	2.0154	26 218	2.1261
207		<i>train loss / train diff %</i>	0.6448	0.00%	0.5576	0.65%	0.5359	-1.09%
208		<i>val loss / val diff %</i>	1.1119	1.47%	1.1238	0.66%	1.1394	2.17%
209	134 000	<i>duration / val/train</i>	19 637	1.7314	24 946	2.0561	26 607	2.0803
210		<i>train loss / train diff %</i>	0.6438	-0.16%	0.5521	-0.99%	0.5353	-0.11%
211		<i>val loss / val diff %</i>	1.1147	0.25%	1.1352	1.01%	1.1136	-2.26%
212	136 000	<i>duration / val/train</i>	19 934	1.7197	25 332	2.0721	26 996	2.0769
213		<i>train loss / train diff %</i>	0.6417	-0.33%	0.5466	-1.00%	0.5387	0.64%
214		<i>val loss / val diff %</i>	1.1035	-1.00%	1.1326	-0.23%	1.1188	0.47%
215	138 000	<i>duration / val/train</i>	20 228	1.7228	25 710	2.0886	27 386	2.0627
216		<i>train loss / train diff %</i>	0.6388	-0.45%	0.5460	-0.11%	0.5378	-0.17%
217		<i>val loss / val diff %</i>	1.1005	-0.27%	1.1404	0.69%	1.1093	-0.85%
218	140 000	<i>duration / val/train</i>	20 522	1.7295	26 083	2.1252	27 775	2.1399
219		<i>train loss / train diff %</i>	0.6378	-0.16%	0.5438	-0.40%	0.5352	-0.48%
220		<i>val loss / val diff %</i>	1.1031	0.24%	1.1557	1.34%	1.1453	3.25%
221	142 000	<i>duration / val/train</i>	20 822	1.7336	26 451	2.0760	28 164	2.1061

222		<i>train loss / train diff %</i>	0.6367	-0.17%	0.5436	-0.04%	0.5305	-0.88%
223		<i>val loss / val diff %</i>	1.1038	0.06%	1.1285	-2.35%	1.1173	-2.44%
224	144 000	<i>duration / val/train</i>	21 125	1.6974	26 816	2.1049	28 553	2.1313
225		<i>train loss / train diff %</i>	0.6420	0.83%	0.5463	0.50%	0.5315	0.19%
226		<i>val loss / val diff %</i>	1.0897	-1.28%	1.1499	1.90%	1.1328	1.39%
227	146 000	<i>duration / val/train</i>	21 429	1.7342	27 180	2.0828	28 942	2.1555
228		<i>train loss / train diff %</i>	0.6344	-1.18%	0.5437	-0.48%	0.5305	-0.19%
229		<i>val loss / val diff %</i>	1.1002	0.96%	1.1324	-1.52%	1.1435	0.94%
230	148 000	<i>duration / val/train</i>	21 733	1.7343	27 545	2.1359	29 350	2.1462
231		<i>train loss / train diff %</i>	0.6360	0.25%	0.5403	-0.63%	0.5268	-0.70%
232		<i>val loss / val diff %</i>	1.1030	0.25%	1.1540	1.91%	1.1306	-1.13%
233	150 000	<i>duration / val/train</i>	22 045	1.7335	27 909	2.1442	29 749	2.1602
234		<i>train loss / train diff %</i>	0.6350	-0.16%	0.5352	-0.94%	0.5319	0.97%
235		<i>val loss / val diff %</i>	1.1008	-0.20%	1.1476	-0.55%	1.1490	1.63%
236	152 000	<i>duration / val/train</i>	22 343	1.7383	28 274	2.1129	30 157	2.1696
237		<i>train loss / train diff %</i>	0.6373	0.36%	0.5414	1.16%	0.5235	-1.58%
238		<i>val loss / val diff %</i>	1.1078	0.64%	1.1439	-0.32%	1.1358	-1.15%
239	154 000	<i>duration / val/train</i>	22 647	1.7224	28 639	2.1171	30 595	2.1508
240		<i>train loss / train diff %</i>	0.6351	-0.35%	0.5405	-0.17%	0.5273	0.73%
241		<i>val loss / val diff %</i>	1.0939	-1.25%	1.1443	0.03%	1.1341	-0.15%
242	156 000	<i>duration / val/train</i>	22 945	1.7542	29 003	2.0915	31 012	2.1166
243		<i>train loss / train diff %</i>	0.6307	-0.69%	0.5462	1.05%	0.5316	0.82%
244		<i>val loss / val diff %</i>	1.1064	1.14%	1.1424	-0.17%	1.1252	-0.78%
245	158 000	<i>duration / val/train</i>	23 254	1.7465	29 368	2.1230	31 421	2.1479
246		<i>train loss / train diff %</i>	0.6295	-0.19%	0.5397	-1.19%	0.5252	-1.20%
247		<i>val loss / val diff %</i>	1.0994	-0.63%	1.1458	0.30%	1.1281	0.26%
248	160 000	<i>duration / val/train</i>	23 559	1.7968	29 735	2.1562	31 822	2.1533
249		<i>train loss / train diff %</i>	0.6318	0.37%	0.5341	-1.04%	0.5239	-0.25%
250		<i>val loss / val diff %</i>	1.1352	3.26%	1.1516	0.51%	1.1281	0.00%
251	162 000	<i>duration / val/train</i>	23 871	1.7689	30 112	2.1478	32 222	2.1584
252		<i>train loss / train diff %</i>	0.6252	-1.04%	0.5365	0.45%	0.5241	0.04%
253		<i>val loss / val diff %</i>	1.1059	-2.58%	1.1523	0.06%	1.1312	0.27%
254	164 000	<i>duration / val/train</i>	24 174	1.7880	30 494	2.1190	32 635	2.2009
255		<i>train loss / train diff %</i>	0.6292	0.64%	0.5336	-0.54%	0.5206	-0.67%
256		<i>val loss / val diff %</i>	1.1250	1.73%	1.1307	-1.87%	1.1458	1.29%
257	166 000	<i>duration / val/train</i>	24 467	1.7708	30 877	2.1046	33 049	2.1578
258		<i>train loss / train diff %</i>	0.6257	-0.56%	0.5373	0.69%	0.5223	0.33%
259		<i>val loss / val diff %</i>	1.1080	-1.51%	1.1308	0.01%	1.1270	-1.64%
260	168 000	<i>duration / val/train</i>	24 759	1.7438	31 248	2.1157	33 457	2.1988
261		<i>train loss / train diff %</i>	0.6245	-0.19%	0.5409	0.67%	0.5172	-0.98%
262		<i>val loss / val diff %</i>	1.0890	-1.71%	1.1444	1.20%	1.1372	0.91%
263	170 000	<i>duration / val/train</i>	25 049	1.7704	31 645	2.1757	33 861	2.1821
264		<i>train loss / train diff %</i>	0.6286	0.66%	0.5360	-0.91%	0.5201	0.56%
265		<i>val loss / val diff %</i>	1.1129	2.19%	1.1662	1.90%	1.1349	-0.20%
266	172 000	<i>duration / val/train</i>	25 339	1.7475	32 029	2.1890	34 263	2.1998
267		<i>train loss / train diff %</i>	0.6249	-0.59%	0.5270	-1.68%	0.5145	-1.08%
268		<i>val loss / val diff %</i>	1.0920	-1.88%	1.1536	-1.08%	1.1318	-0.27%
269	174 000	<i>duration / val/train</i>	25 629	1.7505	32 429	2.1586	34 690	2.2179
270		<i>train loss / train diff %</i>	0.6288	0.62%	0.5359	1.69%	0.5141	-0.08%

271		<i>val loss / val diff %</i>	1.1007	0.80%	1.1568	0.28%	1.1402	0.74%
272	176 000	<i>duration / val/train</i>	25 919	1.7727	32 825	2.1555	35 094	2.2053
273		<i>train loss / train diff %</i>	0.6251	-0.59%	0.5338	-0.39%	0.5174	0.64%
274		<i>val loss / val diff %</i>	1.1081	0.67%	1.1506	-0.54%	1.1410	0.07%
275	178 000	<i>duration / val/train</i>	26 209	1.7778	33 218	2.1363	35 488	2.2145
276		<i>train loss / train diff %</i>	0.6239	-0.19%	0.5310	-0.52%	0.5161	-0.25%
277		<i>val loss / val diff %</i>	1.1092	0.10%	1.1344	-1.41%	1.1429	0.17%
278	180 000	<i>duration / val/train</i>	26 499	1.7889	33 606	2.1951	35 936	2.2233
279		<i>train loss / train diff %</i>	0.6249	0.16%	0.5224	-1.62%	0.5127	-0.66%
280		<i>val loss / val diff %</i>	1.1179	0.78%	1.1467	1.08%	1.1399	-0.26%
281	182 000	<i>duration / val/train</i>	26 790	1.7623	33 997	2.2136	36 364	2.2560
282		<i>train loss / train diff %</i>	0.6281	0.51%	0.5252	0.54%	0.5110	-0.33%
283		<i>val loss / val diff %</i>	1.1069	-0.98%	1.1626	1.39%	1.1528	1.13%
284	184 000	<i>duration / val/train</i>	27 080	1.7917	34 388	2.1785	36 774	2.2024
285		<i>train loss / train diff %</i>	0.6203	-1.24%	0.5305	1.01%	0.5157	0.92%
286		<i>val loss / val diff %</i>	1.1114	0.41%	1.1557	-0.59%	1.1358	-1.47%
287	186 000	<i>duration / val/train</i>	27 370	1.7917	34 782	2.2215	37 177	2.2845
288		<i>train loss / train diff %</i>	0.6206	0.05%	0.5238	-1.26%	0.5089	-1.32%
289		<i>val loss / val diff %</i>	1.1119	0.04%	1.1636	0.68%	1.1626	2.36%
290	188 000	<i>duration / val/train</i>	27 660	1.7815	35 171	2.1725	37 604	2.1704
291		<i>train loss / train diff %</i>	0.6184	-0.35%	0.5293	1.05%	0.5176	1.71%
292		<i>val loss / val diff %</i>	1.1017	-0.92%	1.1499	-1.18%	1.1234	-3.37%
293	190 000	<i>duration / val/train</i>	27 950	1.7994	35 560	2.2014	38 010	2.2247
294		<i>train loss / train diff %</i>	0.6217	0.53%	0.5258	-0.66%	0.5110	-1.28%
295		<i>val loss / val diff %</i>	1.1187	1.54%	1.1575	0.66%	1.1368	1.19%
296	192 000	<i>duration / val/train</i>	28 240	1.7757	35 949	2.2280	38 403	2.2470
297		<i>train loss / train diff %</i>	0.6183	-0.55%	0.5228	-0.57%	0.5069	-0.80%
298		<i>val loss / val diff %</i>	1.0979	-1.86%	1.1648	0.63%	1.1390	0.19%
299	194 000	<i>duration / val/train</i>	28 530	1.7874	36 346	2.2327	38 793	2.2757
300		<i>train loss / train diff %</i>	0.6177	-0.10%	0.5216	-0.23%	0.5099	0.59%
301		<i>val loss / val diff %</i>	1.1041	0.56%	1.1646	-0.02%	1.1604	1.88%
302	196 000	<i>duration / val/train</i>	28 820	1.7813	36 748	2.2284	39 193	2.2846
303		<i>train loss / train diff %</i>	0.6192	0.24%	0.5189	-0.52%	0.5035	-1.26%
304		<i>val loss / val diff %</i>	1.1030	-0.10%	1.1563	-0.71%	1.1503	-0.87%
305	198 000	<i>duration / val/train</i>	29 110	1.7897	37 132	2.2297	39 598	2.2476
306		<i>train loss / train diff %</i>	0.6171	-0.34%	0.5186	-0.06%	0.5072	0.73%
307		<i>val loss / val diff %</i>	1.1044	0.13%	1.1563	0.00%	1.1400	-0.90%
308	200 000	<i>duration / val/train</i>	29 401	1.7993	37 500	2.2207	40 003	2.2453
309		<i>train loss / train diff %</i>	0.6178	0.11%	0.5193	0.13%	0.5068	-0.08%
310		<i>val loss / val diff %</i>	1.1116	0.65%	1.1532	-0.27%	1.1379	-0.18%
311	202 000	<i>duration / val/train</i>	29 691	1.7933	37 865	2.2164	40 405	2.2578
312		<i>train loss / train diff %</i>	0.6184	0.10%	0.5198	0.10%	0.5062	-0.12%
313		<i>val loss / val diff %</i>	1.1090	-0.23%	1.1521	-0.10%	1.1429	0.44%
314	204 000	<i>duration / val/train</i>	29 989	1.7928	38 229	2.2635	40 809	2.3004
315		<i>train loss / train diff %</i>	0.6139	-0.73%	0.5218	0.38%	0.5023	-0.77%
316		<i>val loss / val diff %</i>	1.1006	-0.76%	1.1811	2.52%	1.1555	1.10%
317	206 000	<i>duration / val/train</i>	30 306	1.7795	38 594	2.2667	41 216	2.3021
318		<i>train loss / train diff %</i>	0.6151	0.20%	0.5166	-1.00%	0.5034	0.22%
319		<i>val loss / val diff %</i>	1.0946	-0.55%	1.1710	-0.86%	1.1589	0.29%

320	208 000	<i>duration / val/train</i>	30 610	1.7851	38 958	2.2767	41 621	2.3049
321		<i>train loss / train diff %</i>	0.6189	0.62%	0.5161	-0.10%	0.5028	-0.12%
322		<i>val loss / val diff %</i>	1.1048	0.93%	1.1750	0.34%	1.1589	0.00%
323	210 000	<i>duration / val/train</i>	30 909	1.7848	39 323	2.1701	42 027	2.2787
324		<i>train loss / train diff %</i>	0.6152	-0.60%	0.5226	1.26%	0.4999	-0.58%
325		<i>val loss / val diff %</i>	1.0980	-0.62%	1.1341	-3.48%	1.1391	-1.71%
326	212 000	<i>duration / val/train</i>	31 205	1.7755	39 687	2.2706	42 432	2.2938
327		<i>train loss / train diff %</i>	0.6147	-0.08%	0.5092	-2.56%	0.5044	0.90%
328		<i>val loss / val diff %</i>	1.0914	-0.60%	1.1562	1.95%	1.1570	1.57%
329	214 000	<i>duration / val/train</i>	31 512	1.7971	40 052	2.2688	42 838	2.2698
330		<i>train loss / train diff %</i>	0.6116	-0.50%	0.5126	0.67%	0.5018	-0.52%
331		<i>val loss / val diff %</i>	1.0991	0.71%	1.1630	0.59%	1.1390	-1.56%
332	216 000	<i>duration / val/train</i>	31 816	1.7861	40 416	2.2749	43 246	2.2647
333		<i>train loss / train diff %</i>	0.6102	-0.23%	0.5147	0.41%	0.4994	-0.48%
334		<i>val loss / val diff %</i>	1.0899	-0.84%	1.1709	0.68%	1.1310	-0.70%
335	218 000	<i>duration / val/train</i>	32 120	1.7710	40 781	2.2746	43 653	2.2757
336		<i>train loss / train diff %</i>	0.6144	0.69%	0.5110	-0.72%	0.5049	1.10%
337		<i>val loss / val diff %</i>	1.0881	-0.17%	1.1623	-0.73%	1.1490	1.59%
338	220 000	<i>duration / val/train</i>	32 415	1.7888	41 145	2.2776	44 059	2.2519
339		<i>train loss / train diff %</i>	0.6127	-0.28%	0.5122	0.23%	0.5014	-0.69%
340		<i>val loss / val diff %</i>	1.0960	0.73%	1.1666	0.37%	1.1291	-1.73%
341	222 000	<i>duration / val/train</i>	32 708	1.7818	41 510	2.2695		
342		<i>train loss / train diff %</i>	0.6142	0.24%	0.5113	-0.18%		
343		<i>val loss / val diff %</i>	1.0944	-0.15%	1.1604	-0.53%		
344	224 000	<i>duration / val/train</i>	32 998	1.7983	41 874	2.2982		
345		<i>train loss / train diff %</i>	0.6119	-0.37%	0.5128	0.29%		
346		<i>val loss / val diff %</i>	1.1004	0.55%	1.1785	1.56%		
347	226 000	<i>duration / val/train</i>	33 289	1.8064	42 239	2.2538		
348		<i>train loss / train diff %</i>	0.6090	-0.47%	0.5149	0.41%		
349		<i>val loss / val diff %</i>	1.1001	-0.03%	1.1605	-1.53%		
350	228 000	<i>duration / val/train</i>	33 579	1.7921	42 603	2.2568		
351		<i>train loss / train diff %</i>	0.6151	1.00%	0.5121	-0.54%		
352		<i>val loss / val diff %</i>	1.1023	0.20%	1.1557	-0.41%		
353	230 000	<i>duration / val/train</i>	33 869	1.8111	42 973	2.3057		
354		<i>train loss / train diff %</i>	0.6078	-1.19%	0.5106	-0.29%		
355		<i>val loss / val diff %</i>	1.1008	-0.14%	1.1773	1.87%		
356	232 000	<i>duration / val/train</i>	34 159	1.7614				
357		<i>train loss / train diff %</i>	0.6136	0.95%				
358		<i>val loss / val diff %</i>	1.0808	-1.82%				
359	234 000	<i>duration / val/train</i>	34 451	1.8235				
360		<i>train loss / train diff %</i>	0.6102	-0.55%				
361		<i>val loss / val diff %</i>	1.1127	2.95%				
362	236 000	<i>duration / val/train</i>	34 744	1.8126				
363		<i>train loss / train diff %</i>	0.6157	0.90%				
364		<i>val loss / val diff %</i>	1.1160	0.30%				
365	238 000	<i>duration / val/train</i>	35 044	1.7901				
366		<i>train loss / train diff %</i>	0.6132	-0.41%				
367		<i>val loss / val diff %</i>	1.0977	-1.64%				
368	240 000	<i>duration / val/train</i>	35 338	1.7860				

369		<i>train loss / train diff %</i>	0.6052	-1.30%		
370		<i>val loss / val diff %</i>	1.0809	-1.53%		
371	242 000	<i>duration / val/train</i>	35 635	1.7928		
372		<i>train loss / train diff %</i>	0.6081	0.48%		
373		<i>val loss / val diff %</i>	1.0902	0.86%		
374	244 000	<i>duration / val/train</i>	35 934	1.8033		
375		<i>train loss / train diff %</i>	0.6055	-0.43%		
376		<i>val loss / val diff %</i>	1.0919	0.16%		
377	246 000	<i>duration / val/train</i>	36 252	1.8397		
378		<i>train loss / train diff %</i>	0.6056	0.02%		
379		<i>val loss / val diff %</i>	1.1141	2.03%		
380	248 000	<i>duration / val/train</i>	36 563	1.8022		
381		<i>train loss / train diff %</i>	0.6052	-0.07%		
382		<i>val loss / val diff %</i>	1.0907	-2.10%		
383	250 000	<i>duration / val/train</i>	36 873	1.8188		
384		<i>train loss / train diff %</i>	0.6070	0.30%		
385		<i>val loss / val diff %</i>	1.1040	1.22%		
386	252 000	<i>duration / val/train</i>	37 172	1.8446		
387		<i>train loss / train diff %</i>	0.6035	-0.58%		
388		<i>val loss / val diff %</i>	1.1132	0.83%		
389	254 000	<i>duration / val/train</i>	37 469	1.8037		
390		<i>train loss / train diff %</i>	0.6047	0.20%		
391		<i>val loss / val diff %</i>	1.0907	-2.02%		
392	256 000	<i>duration / val/train</i>	37 779	1.8533		
393		<i>train loss / train diff %</i>	0.6004	-0.71%		
394		<i>val loss / val diff %</i>	1.1127	2.02%		
395	258 000	<i>duration / val/train</i>	38 087	1.8487		
396		<i>train loss / train diff %</i>	0.6007	0.05%		
397		<i>val loss / val diff %</i>	1.1105	-0.20%		
398	260 000	<i>duration / val/train</i>	38 395	1.8163		
399		<i>train loss / train diff %</i>	0.6098	1.51%		
400		<i>val loss / val diff %</i>	1.1076	-0.26%		
401	262 000	<i>duration / val/train</i>	38 714	1.8432		
402		<i>train loss / train diff %</i>	0.6054	-0.72%		
403		<i>val loss / val diff %</i>	1.1159	0.75%		
404	264 000	<i>duration / val/train</i>	39 019	1.8332		
405		<i>train loss / train diff %</i>	0.5978	-1.26%		
406		<i>val loss / val diff %</i>	1.0959	-1.79%		
407	266 000	<i>duration / val/train</i>	39 323	1.8647		
408		<i>train loss / train diff %</i>	0.5992	0.23%		
409		<i>val loss / val diff %</i>	1.1173	1.95%		
410	268 000	<i>duration / val/train</i>	39 616	1.8228		
411		<i>train loss / train diff %</i>	0.6009	0.28%		
412		<i>val loss / val diff %</i>	1.0953	-1.97%		
413	270 000	<i>duration / val/train</i>	39 923	1.8239		
414		<i>train loss / train diff %</i>	0.6009	0.00%		
415		<i>val loss / val diff %</i>	1.0960	0.06%		
416	272 000	<i>duration / val/train</i>	40 217	1.8092		
417		<i>train loss / train diff %</i>	0.6012	0.05%		

418		<i>val loss / val diff %</i>	1.0877	-0.76%		
419	274 000	<i>duration / val/train</i>	40 521	1.8379		
420		<i>train loss / train diff %</i>	0.5985	-0.45%		
421		<i>val loss / val diff %</i>	1.1000	1.13%		
422	276 000	<i>duration / val/train</i>	40 819	1.8077		
423		<i>train loss / train diff %</i>	0.6064	1.32%		
424		<i>val loss / val diff %</i>	1.0962	-0.35%		
425	278 000	<i>duration / val/train</i>	41 118	1.8107		
426		<i>train loss / train diff %</i>	0.6064	0.00%		
427		<i>val loss / val diff %</i>	1.0980	0.16%		
428	280 000	<i>duration / val/train</i>	41 416	1.8270		
429		<i>train loss / train diff %</i>	0.5984	-1.32%		
430		<i>val loss / val diff %</i>	1.0933	-0.43%		
431	282 000	<i>duration / val/train</i>	41 716	1.8306		
432		<i>train loss / train diff %</i>	0.6046	1.04%		
433		<i>val loss / val diff %</i>	1.1068	1.23%		
434	284 000	<i>duration / val/train</i>	42 019	1.8437		
435		<i>train loss / train diff %</i>	0.6025	-0.35%		
436		<i>val loss / val diff %</i>	1.1108	0.36%		
437	286 000	<i>duration / val/train</i>	42 320	1.8390		
438		<i>train loss / train diff %</i>	0.6013	-0.20%		
439		<i>val loss / val diff %</i>	1.1058	-0.45%		

Table 4. Results of training models at Step 1.3

6.7. Analysis of the results of training models in Step 1.3

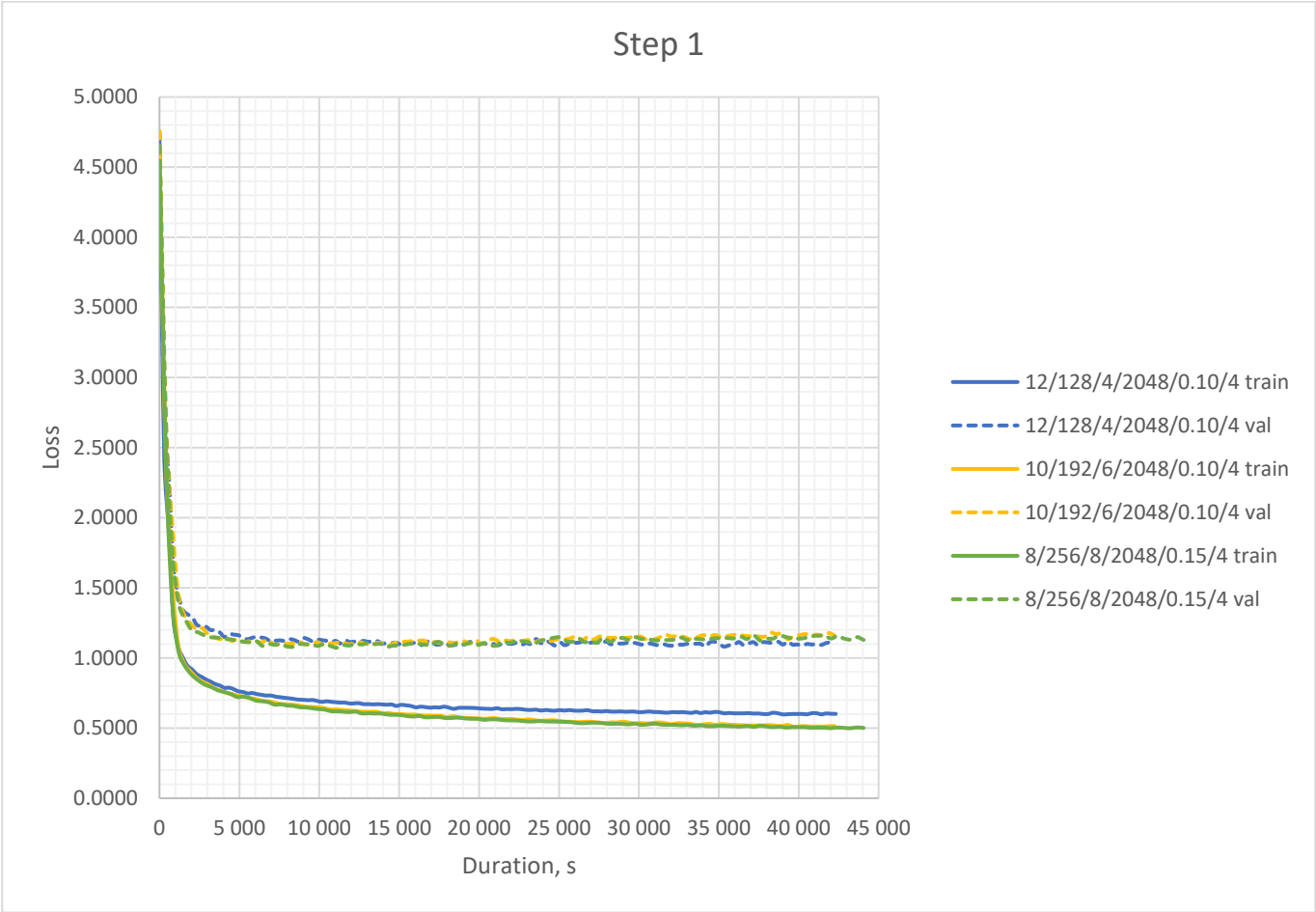
The selected results for each model are presented in Table 5. In the future, during the step-by-step training, we will try to achieve the same control points and their parameters, and by comparing the results, we will be able to conclude whether and how step-by-step training affects the results.

No	Checkpoints	Params	12/128/4/2048/ 0.10/4	10/192/6/2048/ 0.10/4	8/256/8/2048 /0.15/4
1		<i>model size, M</i>	2.69	4.91	6.94
2		<i>n_layer</i>	12	10	8
3		<i>n_embd</i>	128	192	256
4		<i>n_head</i>	4	6	8
5		<i>block_size</i>	2 048	2 048	2 048
6		<i>dropout</i>	0.10	0.10	0.15
7		<i>batch_size</i>	4	4	4
8	val loss plateau	<i>iterations</i>	96 000	38 000	32 000
9		<i>duration</i>	13 934	7 151	6 441
10		<i>train loss</i>	0.6692	0.6842	0.6914
11		<i>val loss</i>	1.1060	1.0988	1.0867
12	min val loss	<i>iterations</i>	232 000	76 000	56 000
13		<i>duration</i>	34 159	14 321	11 109
14		<i>train loss</i>	0.6136	0.6078	0.6195
15		<i>val loss</i>	1.0808	1.0877	1.0731
16	min train loss	<i>iterations</i>	252 000	180 000	196 000
17		<i>Duration</i>	37 172	33 606	39 193

18		<i>train loss</i>	0.6035	0.5224	0.5035
19		<i>val loss</i>	1.1132	1.1467	1.1503

Table 5. Model checkpoints at Step 1.3

Figure 1 shows graphs of the change in loss depending on the duration of model training in Step 1.



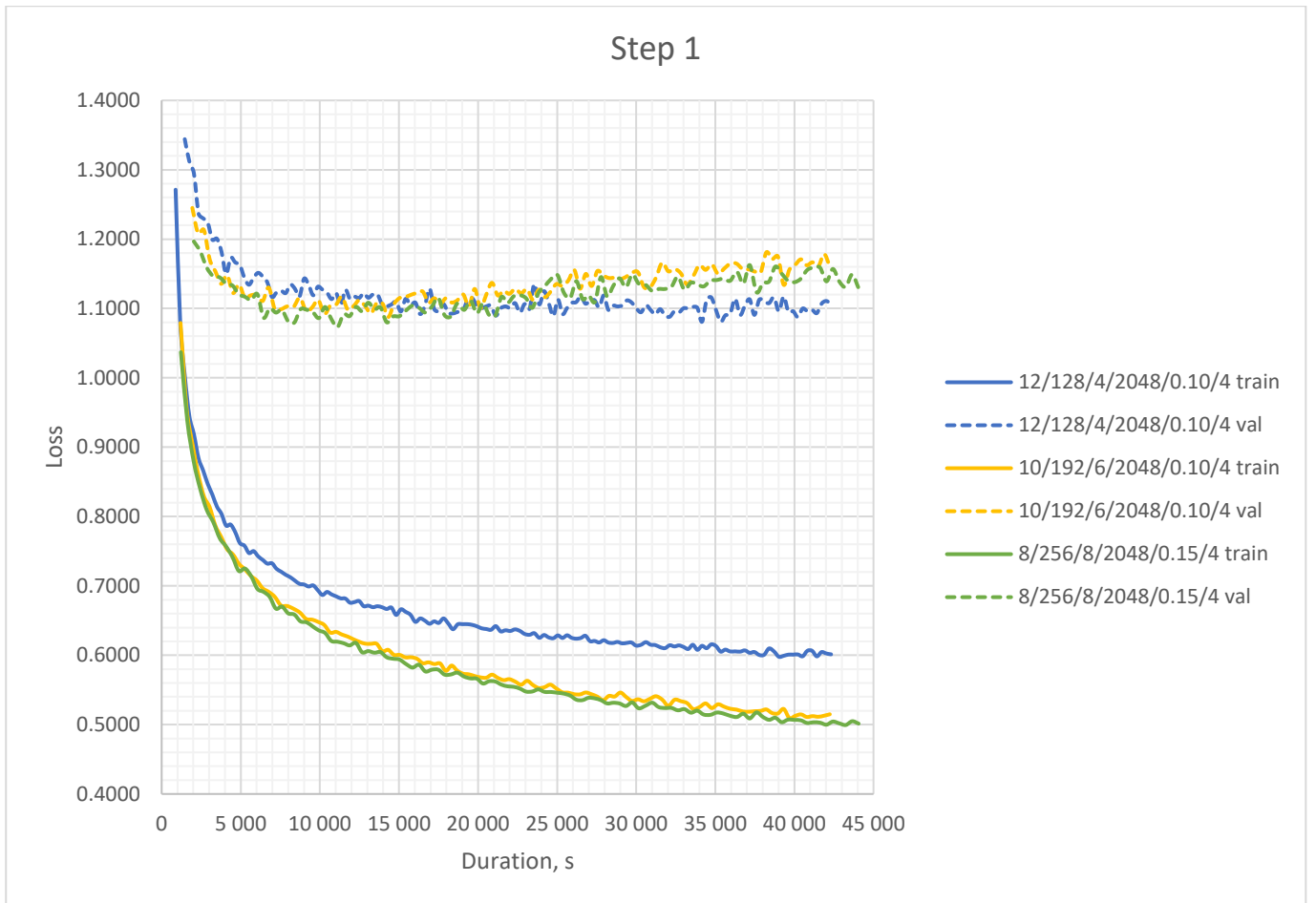


Figure 1. Graphs of loss depending on the duration of model training at Step 1.

7. LLM step-by-step training

For step-by-step model training, we'll start with a small model and train it. We'll then increase the model size and train it on the same dataset or a larger dataset, and so on. In the final step, we train the final-sized model on the same dataset as the traditional model from the previous section. We're trying to test the hypothesis that step-by-step model training improves accuracy and/or reduces the required number of computations and decreases the time needed to train the neural network. Based on the training results, we'll identify three key points for each of the three models and compare them with the results from the previous section.

The experiments are implemented in the programs `src/llmagogy_step2.py`, `src/llmagogy_step3.py`, and `src/llmagogy_step4.py`. In `llmagogy_step2.py`, the model is trained in stages on the full dataset. In `llmagogy_step2.py`, the model is trained on a gradually increasing dataset (files `train_cum_1.txt`, `train_cum_2.txt`, `train_cum_3.txt`, `train_cum_4.txt`, `train_cum_5.txt`). In `llmagogy_step4.py`, the model is again trained on the full dataset, but with a larger number of stages (more than in steps 2 and 3). The models are evaluated on the same validation dataset `val.txt` as the models from the previous section, which does not overlap with the training data. We will be able to compare the results not only with Step 1, but also with each other and see the dependence on splitting the dataset and increasing the number of model generations.

The model modification plan is contained in the scheduler variable (a list with 5 or more elements). Each element contains the same parameters as those listed in Table 1 and the name of the training data file. Since our training dataset is divided into 5 parts, we will go through 5 stages in steps 2 and 3. One model training stage corresponds to one entry in this array. Between stages, the model grows in size. It can grow in depth (number of layers), width (embedding size), or both simultaneously. Old parameters are kept unchanged, and new ones are filled with zeros (if `zero_new_weights == True`) or random values (if `zero_new_weights == False`).

We are testing the step-by-step training of three models: 12/128/4/2048/0.10/4, 10/192/6/2048/0.10/4, and 8/256/8/2048/0.15/4. Each model will have its own model modification plan. The optimal number of iterations for each stage (except the final one) is determined experimentally. The plateauing of the val loss is used as a sufficient criterion.

7.1. Step-by-step model training. Step 2.

Models grow from the minimum to the final size uniformly and simultaneously across all parameters. The model modification plan for Step 2 is shown in Table 6.

No	Stage/Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1	model size, M	2.69	4.91	6.94
2	n_{layer}	12	10	8
3	n_{embd}	128	192	256
4	n_{head}	4	6	8
5	block_size	2 048	2 048	2 048
6	dropout	0.10	0.10	0.15
7	batch_size	4	4	4
8	Start stage 1	2/32/1/128/0.00/64	2/64/2/128/0.00/64	2/64/2/128/0.00/64
9	Stage 2	4/64/2/256/0.05/64	4/96/3/256/0.05/64	3/96/3/256/0.05/64
10	Stage 3	6/64/2/512/0.05/64	6/128/4/512/0.05/64	4/128/4/512/0.05/64
11	Stage 4	9/96/3/1024/0.05/32	8/160/5/1536/0.10/8	6/192/6/1536/0.10/8
12	Final Stage 5	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4

Table 6. Step 2 model change plan.

The results of training the models in Step 2 are shown in Table 7.

No	iterations	Params	12/128/4/2048/0.10/ 4	10/192/6/2048/0.10/ 4	8/256/8/2048/0.15/4	12/128/4/2048/0.10/ 4	10/192/6/2048/0.10/ 4	8/256/8/2048/0.15/4
1		model size, M	2.69	4.91	6.94	2.69	4.91	6.94
2		n_layer	12	10	8	12	10	8
3		n_embd	128	192	256	128	192	256
4		n_head	4	6	8	4	6	8
5		block_size	2 048	2 048	2 048	2 048	2 048	2 048
6		dropout	0.10	0.10	0.15	0.10	0.10	0.15
7		batch_size	4	4	4	4	4	4
8		zero_new_weights	True	True	True	False	False	False
9	Stage 1	model	2/32/1/128/0.00/64/ 50k	2/64/2/128/0.00/64/ 80k	2/64/2/128/0.00/64/ 80k	2/32/1/128/0.00/64/ 50k	2/64/2/128/0.00/64/ 80k	2/64/2/128/0.00/64/ 80k
10		iterations	50 000	80 000	80 000	50 000	80 000	80 000
11		duration / val/train	278 1.2229	435 1.3522	430 1.3522	277 1.2229	407 1.3522	443 1.3522
12		train loss	1.6531	1.3102	1.3102	1.6531	1.3102	1.3102
13		val loss	2.0216	1.7717	1.7717	2.0216	1.7717	1.7717
14	Stage 2	model	4/64/2/256/0.05/64/ 50k	4/96/3/256/0.05/64/ 75k	3/96/3/256/0.05/64/ 125k	4/64/2/256/0.05/64/ 30k	4/96/3/256/0.05/64/ 85k	3/96/3/256/0.05/64/ 80k
15		iterations	50 000	75 000	125 000	30 000	85 000	80 000
16		duration / val/train	454 1.3572	1 102 1.4636	1 434 1.4316	287 1.3710	1 249 1.5175	921 1.4330
17		train loss	1.2112 -26.73%	0.9723 -25.79%	1.0301 -21.38%	1.2140 -26.56%	0.9518 -27.35%	1.0087 -23.01%
18		val loss	1.6439 -18.68%	1.4231 -19.68%	1.4747 -16.76%	1.6644 -17.67%	1.4444 -18.47%	1.4455 -18.41%
19	Stage 3	model	6/64/2/512/0.05/64/ 30k	6/128/4/512/0.05/64/ /20k	4/128/4/512/0.05/64/ /25k	6/64/2/512/0.05/64/ 40k	6/128/4/512/0.05/64/ /25k	4/128/4/512/0.05/64/ /45k
20		iterations	30 000	20 000	25 000	40 000	25 000	45 000
21		duration / val/train	1 325 1.3827	1 755 1.5964	1 547 1.4852	1 841 1.4109	2 198 1.7175	2 664 1.6439
22		train loss	1.0269 -15.22%	0.8109 -16.60%	0.8749 -15.07%	0.9893 -18.51%	0.7654 -19.58%	0.8024 -20.45%
23		val loss	1.4199 -13.63%	1.2945 -9.04%	1.2994 -11.89%	1.3958 -16.14%	1.3146 -8.99%	1.3191 -8.74%
24	Stage 4	model	9/96/3/1024/0.05/32/ /35k	8/160/5/1536/0.10/8/ /50k	6/192/6/1536/0.10/8/ /60k	9/96/3/1024/0.05/32/ /35k	8/160/5/1536/0.10/8/ /45k	6/192/6/1536/0.10/8/ /40k
25		iterations	35 000	50 000	60 000	35 000	45 000	40 000
26		duration / val/train	6 102 1.5204	7 230 1.6179	8 004 1.6069	6 290 1.5974	6 630 1.7247	5 217 1.7249
27		train loss	0.8051 -21.60%	0.6943 -14.38%	0.6991 -20.09%	0.7522 -23.97%	0.6568 -14.19%	0.6582 -17.97%
28		val loss	1.2241 -13.79%	1.1233 -13.23%	1.1234 -13.54%	1.2016 -13.91%	1.1328 -13.83%	1.1353 -13.93%
29		Total time	8 159	10 522	11 415	8 695	10 484	9 245

30	0	duration / val/train	0	1.1006	0	0.9245	0	1.1017	0	1.0783	0	1.0046	0	1.0872
31		train loss	3.8584		3.7819		3.5651		3.9528		3.7379		3.1878	
32		val loss	4.2464		3.4962		3.9275		4.2622		3.7550		3.4657	
33	2 000	duration / val/train	290	1.2689	364	1.4599	409	1.3747	291	1.3146	370	1.5118	390	1.4309
34		train loss / train diff %	1.0156	-73.68%	0.7691	-79.66%	0.8393	-76.46%	0.9555	-75.83%	0.7605	-79.65%	0.8107	-74.57%
35		val loss / val diff %	1.2887	-69.65%	1.1228	-67.89%	1.1538	-70.62%	1.2561	-70.53%	1.1497	-69.38%	1.1600	-66.53%
36	4 000	duration / val/train	580	1.3101	728	1.4812	819	1.4192	581	1.3608	742	1.5400	779	1.5284
37		train loss / train diff %	0.9428	-7.17%	0.7300	-5.08%	0.7943	-5.36%	0.8706	-8.89%	0.7058	-7.19%	0.7415	-8.54%
38		val loss / val diff %	1.2352	-4.15%	1.0813	-3.70%	1.1273	-2.30%	1.1847	-5.68%	1.0869	-5.46%	1.1333	-2.30%
39	6 000	duration / val/train	869	1.3180	1 092	1.5517	1 234	1.4591	871	1.3777	1 110	1.5894	1 168	1.5697
40		train loss / train diff %	0.9169	-2.75%	0.7125	-2.40%	0.7733	-2.64%	0.8467	-2.75%	0.6836	-3.15%	0.7220	-2.63%
41		val loss / val diff %	1.2085	-2.16%	1.1056	2.25%	1.1283	0.09%	1.1665	-1.54%	1.0865	-0.04%	1.1333	0.00%
42	8 000	duration / val/train	1 159	1.3446	1 456	1.5889	1 636	1.4661	1 162	1.4112	1 479	1.6583	1 573	1.5851
43		train loss / train diff %	0.8939	-2.51%	0.7015	-1.54%	0.7560	-2.24%	0.8261	-2.43%	0.6705	-1.92%	0.7018	-2.80%
44		val loss / val diff %	1.2019	-0.55%	1.1146	0.81%	1.1084	-1.76%	1.1658	-0.06%	1.1119	2.34%	1.1124	-1.84%
45	10 000	duration / val/train	1 449	1.3520	1 820	1.5707	2 041	1.5047	1 452	1.4316	1 846	1.6567	1 978	1.6148
46		train loss / train diff %	0.8718	-2.47%	0.6965	-0.71%	0.7403	-2.08%	0.8046	-2.60%	0.6595	-1.64%	0.6929	-1.27%
47		val loss / val diff %	1.1787	-1.93%	1.0940	-1.85%	1.1139	0.50%	1.1519	-1.19%	1.0926	-1.74%	1.1189	0.58%
48	12 000	duration / val/train	1 739	1.3458	2 184	1.6142	2 441	1.5190	1 749	1.4234	2 211	1.6869	2 378	1.6122
49		train loss / train diff %	0.8633	-0.97%	0.6779	-2.67%	0.7307	-1.30%	0.7950	-1.19%	0.6435	-2.43%	0.6860	-1.00%
50		val loss / val diff %	1.1618	-1.43%	1.0943	0.03%	1.1099	-0.36%	1.1316	-1.76%	1.0855	-0.65%	1.1060	-1.15%
51	14 000	duration / val/train	2 028	1.3756	2 548	1.6171	2 837	1.5310	2 048	1.4523	2 579	1.6747	2 787	1.6619
52		train loss / train diff %	0.8516	-1.36%	0.6741	-0.56%	0.7124	-2.50%	0.7812	-1.74%	0.6372	-0.98%	0.6660	-2.92%
53		val loss / val diff %	1.1715	0.83%	1.0901	-0.38%	1.0907	-1.73%	1.1345	0.26%	1.0671	-1.70%	1.1068	0.07%
54	16 000	duration / val/train	2 318	1.3845	2 912	1.6345	3 229	1.5411	2 349	1.4700	2 953	1.7153	3 197	1.6732
55		train loss / train diff %	0.8385	-1.54%	0.6749	0.12%	0.7075	-0.69%	0.7670	-1.82%	0.6357	-0.24%	0.6551	-1.64%
56		val loss / val diff %	1.1609	-0.90%	1.1031	1.19%	1.0903	-0.04%	1.1275	-0.62%	1.0904	2.18%	1.0961	-0.97%
57	18 000	duration / val/train	2 608	1.4182	3 276	1.6611	3 619	1.5541	2 660	1.5160	3 331	1.7317	3 607	1.7300
58		train loss / train diff %	0.8214	-2.04%	0.6613	-2.02%	0.6947	-1.81%	0.7520	-1.96%	0.6202	-2.44%	0.6397	-2.35%
59		val loss / val diff %	1.1649	0.34%	1.0985	-0.42%	1.0796	-0.98%	1.1400	1.11%	1.0740	-1.50%	1.1067	0.97%
60	20 000	duration / val/train	2 897	1.4116	3 640	1.6854	4 007	1.5733	2 973	1.5132	3 715	1.7720	4 012	1.7679
61		train loss / train diff %	0.8192	-0.27%	0.6573	-0.60%	0.6857	-1.30%	0.7504	-0.21%	0.6153	-0.79%	0.6282	-1.80%
62		val loss / val diff %	1.1564	-0.73%	1.1078	0.85%	1.0788	-0.07%	1.1355	-0.39%	1.0903	1.52%	1.1106	0.35%
63	22 000	duration / val/train	3 187	1.4194	4 004	1.6846	4 395	1.5948	3 279	1.5016	4 098	1.7693	4 416	1.8004

64		<i>train loss / train diff %</i>	0.8104	-1.07%	0.6537	-0.55%	0.6785	-1.05%	0.7393	-1.48%	0.6108	-0.73%	0.6188	-1.50%
65		<i>val loss / val diff %</i>	1.1503	-0.53%	1.1012	-0.60%	1.0821	0.31%	1.1101	-2.24%	1.0807	-0.88%	1.1141	0.32%
66	24 000	<i>duration / val/train</i>	3 477	1.4204	4 368	1.7224	4 784	1.5926	3 587	1.5295	4 467	1.8040	4 818	1.7980
67		<i>train loss / train diff %</i>	0.8013	-1.12%	0.6458	-1.21%	0.6762	-0.34%	0.7294	-1.34%	0.6024	-1.38%	0.6173	-0.24%
68		<i>val loss / val diff %</i>	1.1382	-1.05%	1.1123	1.01%	1.0769	-0.48%	1.1156	0.50%	1.0867	0.56%	1.1099	-0.38%
69	26 000	<i>duration / val/train</i>	3 767	1.4338	4 732	1.7178	5 172	1.6329	3 888	1.5413	4 836	1.8559	5 221	1.8347
70		<i>train loss / train diff %</i>	0.7969	-0.55%	0.6400	-0.90%	0.6616	-2.16%	0.7273	-0.29%	0.5955	-1.15%	0.6012	-2.61%
71		<i>val loss / val diff %</i>	1.1426	0.39%	1.0994	-1.16%	1.0803	0.32%	1.1210	0.48%	1.1052	1.70%	1.1030	-0.62%
72	28 000	<i>duration / val/train</i>	4 056	1.4258	5 096	1.7146	5 560	1.6388	4 188	1.5643	5 202	1.7987	5 629	1.8877
73		<i>train loss / train diff %</i>	0.7891	-0.98%	0.6405	0.08%	0.6580	-0.54%	0.7174	-1.36%	0.5970	0.25%	0.5903	-1.81%
74		<i>val loss / val diff %</i>	1.1251	-1.53%	1.0982	-0.11%	1.0783	-0.19%	1.1222	0.11%	1.0738	-2.84%	1.1143	1.02%
75	30 000	<i>duration / val/train</i>	4 346	1.4346	5 462	1.7669	5 950	1.6382	4 484	1.5571	5 567	1.8741	6 034	1.8504
76		<i>train loss / train diff %</i>	0.7856	-0.44%	0.6276	-2.01%	0.6506	-1.12%	0.7134	-0.56%	0.5820	-2.51%	0.5870	-0.56%
77		<i>val loss / val diff %</i>	1.1270	0.17%	1.1089	0.97%	1.0658	-1.16%	1.1108	-1.02%	1.0907	1.57%	1.0862	-2.52%
78	32 000	<i>duration / val/train</i>	4 636	1.4521	5 839	1.7514	6 347	1.6704	4782	1.5555	5 932	1.8856	6 432	1.9296
79		<i>train loss / train diff %</i>	0.7755	-1.29%	0.6282	0.10%	0.6454	-0.80%	0.7064	-0.98%	0.5811	-0.15%	0.5784	-1.47%
80		<i>val loss / val diff %</i>	1.1261	-0.08%	1.1002	-0.78%	1.0781	1.15%	1.0988	-1.08%	1.0957	0.46%	1.1161	2.75%
81	34 000	<i>duration / val/train</i>	4 925	1.4670	6 212	1.8086	6 742	1.6877	5 079	1.6150	6 297	1.9264	6 832	1.9418
82		<i>train loss / train diff %</i>	0.7709	-0.59%	0.6191	-1.45%	0.6420	-0.53%	0.7003	-0.86%	0.5690	-2.08%	0.5726	-1.00%
83		<i>val loss / val diff %</i>	1.1309	0.43%	1.1197	1.77%	1.0835	0.50%	1.1310	2.93%	1.0961	0.04%	1.1119	-0.38%
84	36 000	<i>duration / val/train</i>	5 215	1.4570	6 582	1.8010	7 145	1.6727	5 381	1.6114	6 661	1.9297	7 253	1.9607
85		<i>train loss / train diff %</i>	0.7692	-0.22%	0.6132	-0.95%	0.6397	-0.36%	0.6974	-0.41%	0.5633	-1.00%	0.5679	-0.82%
86		<i>val loss / val diff %</i>	1.1207	-0.90%	1.1044	-1.37%	1.0700	-1.25%	1.1238	-0.64%	1.0870	-0.83%	1.1135	0.14%
87	38 000	<i>duration / val/train</i>	5 505	1.4596	6 957	1.7869	7 545	1.6980	5 681	1.6053	7 033	1.9738	7 677	1.9804
88		<i>train loss / train diff %</i>	0.7626	-0.86%	0.6147	0.24%	0.6367	-0.47%	0.6882	-1.32%	0.5621	-0.21%	0.5650	-0.51%
89		<i>val loss / val diff %</i>	1.1131	-0.68%	1.0984	-0.54%	1.0811	1.04%	1.1048	-1.69%	1.1095	2.07%	1.1189	0.48%
90	40 000	<i>duration / val/train</i>	5 795	1.4886	7 334	1.8107	7 942	1.7087	5 982	1.6454	7 419	1.9928	8 088	2.0016
91		<i>train loss / train diff %</i>	0.7531	-1.25%	0.6075	-1.17%	0.6312	-0.86%	0.6830	-0.76%	0.5534	-1.55%	0.5575	-1.33%
92		<i>val loss / val diff %</i>	1.1211	0.72%	1.1000	0.15%	1.0785	-0.24%	1.1238	1.72%	1.1028	-0.60%	1.1159	-0.27%
93	42 000	<i>duration / val/train</i>	6 084	1.4781	7 719	1.8481	8 347	1.7183	6 275	1.6393	7 797	2.0129	8 497	2.0262
94		<i>train loss / train diff %</i>	0.7582	0.68%	0.6025	-0.82%	0.6297	-0.24%	0.6879	0.72%	0.5485	-0.89%	0.5541	-0.61%
95		<i>val loss / val diff %</i>	1.1207	-0.04%	1.1135	1.23%	1.0820	0.32%	1.1277	0.35%	1.1041	0.12%	1.1227	0.61%
96	44 000	<i>duration / val/train</i>	6 374	1.4768	8 100	1.8502	8 744	1.7255	6 566	1.6138	8 169	2.0463	8 918	2.0553
97		<i>train loss / train diff %</i>	0.7600	0.24%	0.6014	-0.18%	0.6299	0.03%	0.6890	0.16%	0.5465	-0.36%	0.5536	-0.09%

98		<i>val loss / val diff %</i>	1.1224	0.15%	1.1127	-0.07%	1.0869	0.45%	1.1119	-1.40%	1.1183	1.29%	1.1378	1.34%
99	46 000	<i>duration / val/train</i>	6 664	1.4812	8 490	1.8497	9 140	1.7488	6 856	1.6530	8 546	2.0404	9 325	2.0756
100		<i>train loss / train diff %</i>	0.7485	-1.51%	0.5996	-0.30%	0.6238	-0.97%	0.6763	-1.84%	0.5440	-0.46%	0.5461	-1.35%
101		<i>val loss / val diff %</i>	1.1087	-1.22%	1.1091	-0.32%	1.0909	0.37%	1.1179	0.54%	1.1100	-0.74%	1.1335	-0.38%
102	48 000	<i>duration / val/train</i>	6 953	1.4906	8 865	1.8810	9 547	1.7387	7 146	1.6501	8 925	2.0670	9 721	2.0818
103		<i>train loss / train diff %</i>	0.7482	-0.04%	0.5925	-1.18%	0.6199	-0.63%	0.6757	-0.09%	0.5370	-1.29%	0.5425	-0.66%
104		<i>val loss / val diff %</i>	1.1153	0.60%	1.1145	0.49%	1.0778	-1.20%	1.1150	-0.26%	1.1100	0.00%	1.1294	-0.36%
105	50 000	<i>duration / val/train</i>	7 243	1.4944	9 240	1.8726	9 966	1.7818	7 437	1.6364	9 306	2.0945	10 112	2.1144
106		<i>train loss / train diff %</i>	0.7439	-0.57%	0.5948	0.39%	0.6200	0.02%	0.6713	-0.65%	0.5367	-0.06%	0.5401	-0.44%
107		<i>val loss / val diff %</i>	1.1117	-0.32%	1.1138	-0.06%	1.1047	2.50%	1.0985	-1.48%	1.1241	1.27%	1.1420	1.12%
108	52 000	<i>duration / val/train</i>	7 533	1.5158	9 621	1.8804	10 364	1.7878	7 733	1.6758	9 690	2.1025	10 501	2.1329
109		<i>train loss / train diff %</i>	0.7377	-0.83%	0.5885	-1.06%	0.6107	-1.50%	0.6662	-0.76%	0.5307	-1.12%	0.5320	-1.50%
110		<i>val loss / val diff %</i>	1.1182	0.58%	1.1066	-0.65%	1.0918	-1.17%	1.1164	1.63%	1.1158	-0.74%	1.1347	-0.64%
111	54 000	<i>duration / val/train</i>	7 823	1.5271	9 996	1.8669	10 765	1.7811	8 032	1.6930	10 071	2.0843	10 891	2.1582
112		<i>train loss / train diff %</i>	0.7335	-0.57%	0.5905	0.34%	0.6080	-0.44%	0.6613	-0.74%	0.5325	0.34%	0.5273	-0.88%
113		<i>val loss / val diff %</i>	1.1201	0.17%	1.1024	-0.38%	1.0829	-0.82%	1.1196	0.29%	1.1099	-0.53%	1.1380	0.29%
114	56 000	<i>duration / val/train</i>	8 112	1.5242	10 375	1.9303	11 159	1.8131	8 326	1.6967	10 440	2.1353	11 281	2.2073
115		<i>train loss / train diff %</i>	0.7310	-0.34%	0.5885	-0.34%	0.6063	-0.28%	0.6598	-0.23%	0.5284	-0.77%	0.5252	-0.40%
116		<i>val loss / val diff %</i>	1.1142	-0.53%	1.1360	3.05%	1.0993	1.51%	1.1195	-0.01%	1.1283	1.66%	1.1593	1.87%
117	58 000	<i>duration / val/train</i>	8 402	1.5386	10 742	1.9318	11 564	1.8128	8 637	1.6935	10 816	2.1364	11 690	2.1894
118		<i>train loss / train diff %</i>	0.7276	-0.47%	0.5836	-0.83%	0.6052	-0.18%	0.6551	-0.71%	0.5236	-0.91%	0.5216	-0.69%
119		<i>val loss / val diff %</i>	1.1195	0.48%	1.1274	-0.76%	1.0971	-0.20%	1.1094	-0.90%	1.1186	-0.86%	1.1420	-1.49%
120	60 000	<i>duration / val/train</i>	8 692	1.5226	11 117	1.9456	11 964	1.8362	8 943	1.6781	11 190	2.1693	12 093	2.2586
121		<i>train loss / train diff %</i>	0.7290	0.19%	0.5768	-1.17%	0.5954	-1.62%	0.6542	-0.14%	0.5162	-1.41%	0.5127	-1.71%
122		<i>val loss / val diff %</i>	1.1100	-0.85%	1.1222	-0.46%	1.0933	-0.35%	1.0978	-1.05%	1.1198	0.11%	1.1580	1.40%
123	62 000	<i>duration / val/train</i>	8 981	1.5536	11 510	1.9442	12 369	1.8223	9 246	1.7012	11 566	2.1639	12 497	2.2226
124		<i>train loss / train diff %</i>	0.7200	-1.23%	0.5826	1.01%	0.6004	0.84%	0.6500	-0.64%	0.5205	0.83%	0.5135	0.16%
125		<i>val loss / val diff %</i>	1.1186	0.77%	1.1327	0.94%	1.0941	0.07%	1.1058	0.73%	1.1263	0.58%	1.1413	-1.44%
126	64 000	<i>duration / val/train</i>	9 271	1.5186	11 898	1.9744	12 791	1.8481	9 546	1.6757	11 947	2.2031	12 899	2.2598
127		<i>train loss / train diff %</i>	0.7250	0.69%	0.5816	-0.17%	0.5979	-0.42%	0.6540	0.62%	0.5189	-0.31%	0.5134	-0.02%
128		<i>val loss / val diff %</i>	1.1010	-1.57%	1.1483	1.38%	1.1050	1.00%	1.0959	-0.90%	1.1432	1.50%	1.1602	1.66%
129	66 000	<i>duration / val/train</i>	9 564	1.5568	12 271	1.9332	13 199	1.8535	9 853	1.7189	12 321	2.1898	13 302	2.2651
130		<i>train loss / train diff %</i>	0.7134	-1.60%	0.5764	-0.89%	0.5958	-0.35%	0.6413	-1.94%	0.5127	-1.19%	0.5070	-1.25%
131		<i>val loss / val diff %</i>	1.1106	0.87%	1.1143	-2.96%	1.1043	-0.06%	1.1023	0.58%	1.1227	-1.79%	1.1484	-1.02%

132	68 000	duration / val/train	9 860	1.5463	12 638	1.9678	13 593	1.8538	10 164	1.7162	12 701	2.2276	13 705	2.2519
133		train loss / train diff %	0.7164	0.42%	0.5717	-0.82%	0.5930	-0.47%	0.6435	0.34%	0.5088	-0.76%	0.5077	0.14%
134		val loss / val diff %	1.1078	-0.25%	1.1250	0.96%	1.0993	-0.45%	1.1044	0.19%	1.1334	0.95%	1.1433	-0.44%
135	70 000	duration / val/train	10 158	1.5413	13 001	1.9543	13 982	1.8677	10 481	1.7215	13 086	2.2373	14 106	2.3084
136		train loss / train diff %	0.7163	-0.01%	0.5714	-0.05%	0.5880	-0.84%	0.6449	0.22%	0.5065	-0.45%	0.5006	-1.40%
137		val loss / val diff %	1.1040	-0.34%	1.1167	-0.74%	1.0982	-0.10%	1.1102	0.53%	1.1332	-0.02%	1.1556	1.08%
138	72 000	duration / val/train	10 457	1.5670	13 365	1.9977	14 370	1.8427	10 798	1.7401	13 489	2.2580	14 509	2.3075
139		train loss / train diff %	0.7094	-0.96%	0.5684	-0.53%	0.5874	-0.10%	0.6411	-0.59%	0.5047	-0.36%	0.5011	0.10%
140		val loss / val diff %	1.1116	0.69%	1.1355	1.68%	1.0824	-1.44%	1.1156	0.49%	1.1396	0.56%	1.1563	0.06%
141	74 000	duration / val/train	10 760	1.5646	13 734	1.9338	14 758	1.8718	11 103	1.7354	13 858	2.2542	14 912	2.3465
142		train loss / train diff %	0.7106	0.17%	0.5724	0.70%	0.5857	-0.29%	0.6398	-0.20%	0.5047	0.00%	0.4944	-1.34%
143		val loss / val diff %	1.1118	0.02%	1.1069	-2.52%	1.0963	1.28%	1.1103	-0.48%	1.1377	-0.17%	1.1601	0.33%
144	76 000	duration / val/train	11 073	1.5799	14 099	1.9645	15 146	1.8938	11 396	1.7769	14 225	2.2601	15 314	2.3768
145		train loss / train diff %	0.7032	-1.04%	0.5664	-1.05%	0.5800	-0.97%	0.6310	-1.38%	0.5002	-0.89%	0.4902	-0.85%
146		val loss / val diff %	1.1110	-0.07%	1.1127	0.52%	1.0984	0.19%	1.1212	0.98%	1.1305	-0.63%	1.1651	0.43%
147	78 000	duration / val/train	11 367	1.6019	14 479	1.9899	15 534	1.8932	11 688	1.7793	14 589	2.3144	15 707	2.3530
148		train loss / train diff %	0.7019	-0.18%	0.5644	-0.35%	0.5796	-0.07%	0.6312	0.03%	0.4984	-0.36%	0.4907	0.10%
149		val loss / val diff %	1.1244	1.21%	1.1231	0.93%	1.0973	-0.10%	1.1231	0.17%	1.1535	2.03%	1.1546	-0.90%
150	80 000	duration / val/train	11 664	1.5925	14 868	2.0000	15 923	1.9074	11 979	1.7701	14 954	2.3012	16 099	2.4054
151		train loss / train diff %	0.6953	-0.94%	0.5654	0.18%	0.5808	0.21%	0.6251	-0.97%	0.4964	-0.40%	0.4899	-0.16%
152		val loss / val diff %	1.1073	-1.52%	1.1308	0.69%	1.1078	0.96%	1.1065	-1.48%	1.1423	-0.97%	1.1784	2.06%
153	82 000	duration / val/train	11 963	1.5688	15 236	2.0087	16 318	1.8947	12 269	1.7727	15 319	2.3017	16 493	2.4200
154		train loss / train diff %	0.6969	0.23%	0.5622	-0.57%	0.5791	-0.29%	0.6274	0.37%	0.4939	-0.50%	0.4871	-0.57%
155		val loss / val diff %	1.0933	-1.26%	1.1293	-0.13%	1.0972	-0.96%	1.1122	0.52%	1.1368	-0.48%	1.1788	0.03%
156	84 000	duration / val/train	12 264	1.6005	15 599	2.0231	16 706	1.8916	12 559	1.8087	15 683	2.3367	16 887	2.4424
157		train loss / train diff %	0.6934	-0.50%	0.5583	-0.69%	0.5776	-0.26%	0.6236	-0.61%	0.4910	-0.59%	0.4858	-0.27%
158		val loss / val diff %	1.1098	1.51%	1.1295	0.02%	1.0926	-0.42%	1.1279	1.41%	1.1473	0.92%	1.1865	0.65%
159	86 000	duration / val/train	12 575	1.5544	15 965	2.0237	17 094	1.9356	12 850	1.7467	16 048	2.3143	17 288	2.4219
160		train loss / train diff %	0.6952	0.26%	0.5611	0.50%	0.5763	-0.23%	0.6266	0.48%	0.4957	0.96%	0.4842	-0.33%
161		val loss / val diff %	1.0806	-2.63%	1.1355	0.53%	1.1155	2.10%	1.0945	-2.96%	1.1472	-0.01%	1.1727	-1.16%
162	88 000	duration / val/train	12 882	1.5884	16 354	2.0401	17 483	1.9226	13 140	1.7766	16 412	2.3650	17 686	2.4448
163		train loss / train diff %	0.6925	-0.39%	0.5538	-1.30%	0.5737	-0.45%	0.6212	-0.86%	0.4869	-1.78%	0.4786	-1.16%
164		val loss / val diff %	1.1000	1.80%	1.1298	-0.50%	1.1030	-1.12%	1.1036	0.83%	1.1515	0.37%	1.1701	-0.22%
165	90 000	duration / val/train	13 182	1.5662	16 744	2.0207	17 871	1.9139	13 430	1.7555	16 777	2.3448		

166		<i>train loss / train diff %</i>	0.6922	-0.04%	0.5556	0.33%	0.5711	-0.45%	0.6224	0.19%	0.4861	-0.16%	
167		<i>val loss / val diff %</i>	1.0841	-1.45%	1.1227	-0.63%	1.0930	-0.91%	1.0926	-1.00%	1.1398	-1.02%	
168	92 000	<i>duration / val/train</i>	13 482	1.5927	17 121	2.0647	18 259	1.9486	13 720	1.8067	17 142	2.4041	
169		<i>train loss / train diff %</i>	0.6867	-0.79%	0.5546	-0.18%	0.5642	-1.21%	0.6177	-0.76%	0.4848	-0.27%	
170		<i>val loss / val diff %</i>	1.0937	0.89%	1.1451	2.00%	1.0994	0.59%	1.1160	2.14%	1.1655	2.25%	
171	94 000	<i>duration / val/train</i>	13 783	1.6091	17 513	2.0436	18 647	1.9086	14 011	1.7776	17 506	2.3786	
172		<i>train loss / train diff %</i>	0.6830	-0.54%	0.5555	0.16%	0.5699	1.01%	0.6133	-0.71%	0.4850	0.04%	
173		<i>val loss / val diff %</i>	1.0990	0.48%	1.1352	-0.86%	1.0877	-1.06%	1.0902	-2.31%	1.1536	-1.02%	
174	96 000	<i>duration / val/train</i>	14 080	1.6095	17 907	2.0252	19 035	1.9298	14 301	1.7996	17 871	2.3684	
175		<i>train loss / train diff %</i>	0.6820	-0.15%	0.5526	-0.52%	0.5658	-0.72%	0.6119	-0.23%	0.4816	-0.70%	
176		<i>val loss / val diff %</i>	1.0977	-0.12%	1.1191	-1.42%	1.0919	0.39%	1.1012	1.01%	1.1406	-1.13%	
177	98 000	<i>duration / val/train</i>	14 382	1.5779	18 301	2.0558	19 423	1.9788	14 591	1.7790	18 236	2.3993	
178		<i>train loss / train diff %</i>	0.6875	0.81%	0.5504	-0.40%	0.5613	-0.80%	0.6180	1.00%	0.4796	-0.42%	
179		<i>val loss / val diff %</i>	1.0848	-1.18%	1.1315	1.11%	1.1107	1.72%	1.0994	-0.16%	1.1507	0.89%	
180	100 000	<i>duration / val/train</i>	14 677	1.6064	18 682	2.0865	19 831	1.9255	14 882	1.8152	18 600	2.4368	
181		<i>train loss / train diff %</i>	0.6812	-0.92%	0.5494	-0.18%	0.5648	0.62%	0.6125	-0.89%	0.4782	-0.29%	
182		<i>val loss / val diff %</i>	1.0943	0.88%	1.1463	1.31%	1.0875	-2.09%	1.1118	1.13%	1.1653	1.27%	
183	102 000	<i>duration / val/train</i>	14 970	1.5802	19 083	2.0664	20 235	1.9827	15 173	1.7988	18 965	2.4197	
184		<i>train loss / train diff %</i>	0.6874	0.91%	0.5514	0.36%	0.5592	-0.99%	0.6168	0.70%	0.4789	0.15%	
185		<i>val loss / val diff %</i>	1.0862	-0.74%	1.1394	-0.60%	1.1087	1.95%	1.1095	-0.21%	1.1588	-0.56%	
186	104 000	<i>duration / val/train</i>	15 259	1.6004	19 462	2.1012	20 638	1.9386	15 478	1.8014	19 329	2.4493	
187		<i>train loss / train diff %</i>	0.6804	-1.02%	0.5464	-0.91%	0.5639	0.84%	0.6127	-0.66%	0.4765	-0.50%	
188		<i>val loss / val diff %</i>	1.0889	0.25%	1.1481	0.76%	1.0932	-1.40%	1.1037	-0.52%	1.1671	0.72%	
189	106 000	<i>duration / val/train</i>	15 549	1.6021	19 858	2.0863	21 033	2.0079	15 777	1.8155	19 694	2.4304	
190		<i>train loss / train diff %</i>	0.6800	-0.06%	0.5480	0.29%	0.5543	-1.70%	0.6120	-0.11%	0.4744	-0.44%	
191		<i>val loss / val diff %</i>	1.0894	0.05%	1.1433	-0.42%	1.1130	1.81%	1.1111	0.67%	1.1530	-1.21%	
192	108 000	<i>duration / val/train</i>	15 839	1.6082	20 236	2.0711	21 433	2.0118	16 082	1.8156	20 059	2.4455	
193		<i>train loss / train diff %</i>	0.6746	-0.79%	0.5475	-0.09%	0.5530	-0.23%	0.6056	-1.05%	0.4730	-0.30%	
194		<i>val loss / val diff %</i>	1.0849	-0.41%	1.1339	-0.82%	1.1125	-0.04%	1.0995	-1.04%	1.1567	0.32%	
195	110 000	<i>duration / val/train</i>	16 128	1.6369	20 634	2.1001	21 826	2.0209	16 382	1.8468	20 424	2.4594	
196		<i>train loss / train diff %</i>	0.6733	-0.19%	0.5423	-0.95%	0.5505	-0.45%	0.6057	0.02%	0.4697	-0.70%	
197		<i>val loss / val diff %</i>	1.1021	1.59%	1.1389	0.44%	1.1125	0.00%	1.1186	1.74%	1.1552	-0.13%	
198	112 000	<i>duration / val/train</i>	16 418	1.6048	21 002	2.1021	22 218	2.0155	16 686	1.8597	20 788	2.4828	
199		<i>train loss / train diff %</i>	0.6711	-0.33%	0.5414	-0.17%	0.5560	1.00%	0.6030	-0.45%	0.4685	-0.26%	

200		<i>val loss / val diff %</i>	1.0770	-2.28%	1.1381	-0.07%	1.1206	0.73%	1.1214	0.25%	1.1632	0.69%	
201	114 000	<i>duration / val/train</i>	16 708	1.6104	21 367	2.1380	22 609	1.9975	16 991	1.8395	21 153	2.5078	
202		<i>train loss / train diff %</i>	0.6779	1.01%	0.5404	-0.18%	0.5514	-0.83%	0.6069	0.65%	0.4665	-0.43%	
203		<i>val loss / val diff %</i>	1.0917	1.36%	1.1554	1.52%	1.1014	-1.71%	1.1164	-0.45%	1.1699	0.58%	
204	116 000	<i>duration / val/train</i>	16 997	1.6112	21 740	2.1108	22 997	2.0135	17 290	1.8409	21 518	2.5091	
205		<i>train loss / train diff %</i>	0.6693	-1.27%	0.5371	-0.61%	0.5494	-0.36%	0.6002	-1.10%	0.4655	-0.21%	
206		<i>val loss / val diff %</i>	1.0784	-1.22%	1.1337	-1.88%	1.1062	0.44%	1.1049	-1.03%	1.1680	-0.16%	
207	118 000	<i>duration / val/train</i>	17 287	1.6047	22 121	2.1645	23 385	2.0110	17 593	1.8623	21 882	2.5704	
208		<i>train loss / train diff %</i>	0.6691	-0.03%	0.5355	-0.30%	0.5529	0.64%	0.5984	-0.30%	0.4597	-1.25%	
209		<i>val loss / val diff %</i>	1.0737	-0.44%	1.1591	2.24%	1.1119	0.52%	1.1144	0.86%	1.1816	1.16%	
210	120 000	<i>duration / val/train</i>	17 577	1.6380	22 497	2.1025	23 773	2.0115	17 894	1.8478	22 247	2.4665	
211		<i>train loss / train diff %</i>	0.6691	0.00%	0.5395	0.75%	0.5558	0.52%	0.6019	0.58%	0.4669	1.57%	
212		<i>val loss / val diff %</i>	1.0960	2.08%	1.1343	-2.14%	1.1180	0.55%	1.1122	-0.20%	1.1516	-2.54%	
213	122 000	<i>duration / val/train</i>	17 867	1.6272	22 879	2.0945	24 161	2.0378	18 196	1.8675	22 633	2.4986	
214		<i>train loss / train diff %</i>	0.6647	-0.66%	0.5388	-0.13%	0.5527	-0.56%	0.5955	-1.06%	0.4645	-0.51%	
215		<i>val loss / val diff %</i>	1.0816	-1.31%	1.1285	-0.51%	1.1263	0.74%	1.1121	-0.01%	1.1606	0.78%	
216	124 000	<i>duration / val/train</i>	18 165	1.6335	23 256	2.1163	24 550	2.0251	18 501	1.8671	23 062	2.5430	
217		<i>train loss / train diff %</i>	0.6650	0.05%	0.5339	-0.91%	0.5466	-1.10%	0.5952	-0.05%	0.4589	-1.21%	
218		<i>val loss / val diff %</i>	1.0863	0.43%	1.1299	0.12%	1.1069	-1.72%	1.1113	-0.07%	1.1670	0.55%	
219	126 000	<i>duration / val/train</i>	18 474	1.6361	23 625	2.1160	24 938	2.0258	18 807	1.8512			
220		<i>train loss / train diff %</i>	0.6686	0.54%	0.5344	0.09%	0.5502	0.66%	0.5994	0.71%			
221		<i>val loss / val diff %</i>	1.0939	0.70%	1.1308	0.08%	1.1146	0.70%	1.1096	-0.15%			
222	128 000	<i>duration / val/train</i>	18 788	1.6364			25 326	2.0292	19 109	1.8819			
223		<i>train loss / train diff %</i>	0.6626	-0.90%			0.5439	-1.15%	0.5935	-0.98%			
224		<i>val loss / val diff %</i>	1.0843	-0.88%			1.1037	-0.98%	1.1169	0.66%			
225	130 000	<i>duration / val/train</i>	19 087	1.6449			25 726	2.0171	19 408	1.8780			
226		<i>train loss / train diff %</i>	0.6561	-0.98%			0.5448	0.17%	0.5886	-0.83%			
227		<i>val loss / val diff %</i>	1.0792	-0.47%			1.0989	-0.43%	1.1054	-1.03%			
228	132 000	<i>duration / val/train</i>	19 391	1.6201			26 127	2.0801	19 712	1.8595			
229		<i>train loss / train diff %</i>	0.6631	1.07%			0.5416	-0.59%	0.5942	0.95%			
230		<i>val loss / val diff %</i>	1.0743	-0.45%			1.1266	2.52%	1.1049	-0.05%			
231	134 000	<i>duration / val/train</i>	19 697	1.6458			26 530	2.0615	20 016	1.8827			
232		<i>train loss / train diff %</i>	0.6572	-0.89%			0.5417	0.02%	0.5898	-0.74%			
233		<i>val loss / val diff %</i>	1.0816	0.68%			1.1167	-0.88%	1.1104	0.50%			

234	136 000	<i>duration / val/train</i>	20 000	1.6294		26 933	2.0446	20 316	1.8760		
235		<i>train loss / train diff %</i>	0.6613	0.62%		0.5425	0.15%	0.5935	0.63%		
236		<i>val loss / val diff %</i>	1.0775	-0.38%		1.1092	-0.67%	1.1134	0.27%		
237	138 000	<i>duration / val/train</i>	20 301	1.6476		27 328	2.0863	20 618	1.8908		
238		<i>train loss / train diff %</i>	0.6583	-0.45%		0.5402	-0.42%	0.5904	-0.52%		
239		<i>val loss / val diff %</i>	1.0846	0.66%		1.1270	1.60%	1.1163	0.26%		
240	140 000	<i>duration / val/train</i>	20 601	1.6430		27 722	2.1065	20 921	1.9115		
241		<i>train loss / train diff %</i>	0.6574	-0.14%		0.5372	-0.56%	0.5896	-0.14%		
242		<i>val loss / val diff %</i>	1.0801	-0.41%		1.1316	0.41%	1.1270	0.96%		
243	142 000	<i>duration / val/train</i>	20 906	1.6576		28 126	2.0966	21 225	1.9017		
244		<i>train loss / train diff %</i>	0.6548	-0.40%		0.5340	-0.60%	0.5867	-0.49%		
245		<i>val loss / val diff %</i>	1.0854	0.49%		1.1196	-1.06%	1.1157	-1.00%		
246	144 000	<i>duration / val/train</i>	21 212	1.6564		28 522	2.1216	21 527	1.8900		
247		<i>train loss / train diff %</i>	0.6527	-0.32%		0.5344	0.07%	0.5852	-0.26%		
248		<i>val loss / val diff %</i>	1.0811	-0.40%		1.1338	1.27%	1.1060	-0.87%		
249	146 000	<i>duration / val/train</i>	21 510	1.6457		28 924	2.0760	21 838	1.8945		
250		<i>train loss / train diff %</i>	0.6605	1.20%		0.5379	0.65%	0.5912	1.03%		
251		<i>val loss / val diff %</i>	1.0870	0.55%		1.1167	-1.51%	1.1200	1.27%		
252	148 000	<i>duration / val/train</i>	21 802	1.6488		29 328	2.0721	22 145	1.9052		
253		<i>train loss / train diff %</i>	0.6557	-0.73%		0.5338	-0.76%	0.5862	-0.85%		
254		<i>val loss / val diff %</i>	1.0811	-0.54%		1.1061	-0.95%	1.1168	-0.29%		
255	150 000	<i>duration / val/train</i>	22 102	1.6495		29 732	2.1009	22 446	1.9009		
256		<i>train loss / train diff %</i>	0.6483	-1.13%		0.5350	0.22%	0.5813	-0.84%		
257		<i>val loss / val diff %</i>	1.0694	-1.08%		1.1240	1.62%	1.1050	-1.06%		
258	152 000	<i>duration / val/train</i>	22 413	1.6620		30 130	2.0813	22 744	1.9136		
259		<i>train loss / train diff %</i>	0.6533	0.77%		0.5360	0.19%	0.5856	0.74%		
260		<i>val loss / val diff %</i>	1.0858	1.53%		1.1156	-0.75%	1.1206	1.41%		
261	154 000	<i>duration / val/train</i>	22 721	1.6821		30 521	2.1074	23 042	1.8982		
262		<i>train loss / train diff %</i>	0.6481	-0.80%		0.5325	-0.65%	0.5797	-1.01%		
263		<i>val loss / val diff %</i>	1.0902	0.41%		1.1222	0.59%	1.1004	-1.80%		
264	156 000	<i>duration / val/train</i>	23 021	1.6393		30 927	2.1357				
265		<i>train loss / train diff %</i>	0.6468	-0.20%		0.5291	-0.64%				
266		<i>val loss / val diff %</i>	1.0603	-2.74%		1.1300	0.70%				
267	158 000	<i>duration / val/train</i>	23 322	1.6974		31 336	2.1122				

268		<i>train loss / train diff %</i>	0.6457	-0.17%		0.5312	0.40%			
269		<i>val loss / val diff %</i>	1.0960	3.37%		1.1220	-0.71%			
270	160 000	<i>duration / val/train</i>	23 627	1.6570		31 737	2.0898			
271		<i>train loss / train diff %</i>	0.6499	0.65%		0.5334	0.41%			
272		<i>val loss / val diff %</i>	1.0769	-1.74%		1.1147	-0.65%			
273	162 000	<i>duration / val/train</i>	23 925	1.6707						
274		<i>train loss / train diff %</i>	0.6468	-0.48%						
275		<i>val loss / val diff %</i>	1.0806	0.34%						
276	164 000	<i>duration / val/train</i>	24 228	1.6581						
277		<i>train loss / train diff %</i>	0.6491	0.36%						
278		<i>val loss / val diff %</i>	1.0763	-0.40%						
279	166 000	<i>duration / val/train</i>	24 530	1.6423						
280		<i>train loss / train diff %</i>	0.6463	-0.43%						
281		<i>val loss / val diff %</i>	1.0614	-1.38%						
282	168 000	<i>duration / val/train</i>	24 843	1.6580						
283		<i>train loss / train diff %</i>	0.6421	-0.65%						
284		<i>val loss / val diff %</i>	1.0646	0.30%						
285	170 000	<i>duration / val/train</i>	25 140	1.6751						
286		<i>train loss / train diff %</i>	0.6441	0.31%						
287		<i>val loss / val diff %</i>	1.0789	1.34%						
288	172 000	<i>duration / val/train</i>	25 446	1.6877						
289		<i>train loss / train diff %</i>	0.6400	-0.64%						
290		<i>val loss / val diff %</i>	1.0801	0.11%						
291	174 000	<i>duration / val/train</i>	25 748	1.6835						
292		<i>train loss / train diff %</i>	0.6401	0.02%						
293		<i>val loss / val diff %</i>	1.0776	-0.23%						
294	176 000	<i>duration / val/train</i>	26 046	1.6805						
295		<i>train loss / train diff %</i>	0.6431	0.47%						
296		<i>val loss / val diff %</i>	1.0807	0.29%						
297	178 000	<i>duration / val/train</i>	26 344	1.6697						
298		<i>train loss / train diff %</i>	0.6363	-1.06%						
299		<i>val loss / val diff %</i>	1.0624	-1.69%						
300	180 000	<i>duration / val/train</i>	26 650	1.6510						
301		<i>train loss / train diff %</i>	0.6421	0.91%						

302		<i>val loss / val diff %</i>	1.0601	-0.22%					
303	182 000	<i>duration / val/train</i>	26 942	1.6832					
304		<i>train loss / train diff %</i>	0.6429	0.12%					
305		<i>val loss / val diff %</i>	1.0821	2.08%					
306	184 000	<i>duration / val/train</i>	27 233	1.6818					
307		<i>train loss / train diff %</i>	0.6420	-0.14%					
308		<i>val loss / val diff %</i>	1.0797	-0.22%					
309	186 000	<i>duration / val/train</i>	27 523	1.6771					
310		<i>train loss / train diff %</i>	0.6373	-0.73%					
311		<i>val loss / val diff %</i>	1.0688	-1.01%					
312	188 000	<i>duration / val/train</i>	27 813	1.6908					
313		<i>train loss / train diff %</i>	0.6358	-0.24%					
314		<i>val loss / val diff %</i>	1.0750	0.58%					
315	190 000	<i>duration / val/train</i>	28 107	1.6956					
316		<i>train loss / train diff %</i>	0.6331	-0.42%					
317		<i>val loss / val diff %</i>	1.0735	-0.14%					
318	192 000	<i>duration / val/train</i>	28 406	1.6657					
319		<i>train loss / train diff %</i>	0.6362	0.49%					
320		<i>val loss / val diff %</i>	1.0597	-1.29%					
321	194 000	<i>duration / val/train</i>	28 698	1.6542					
322		<i>train loss / train diff %</i>	0.6379	0.27%					
323		<i>val loss / val diff %</i>	1.0552	-0.42%					
324	196 000	<i>duration / val/train</i>	28 990	1.6729					
325		<i>train loss / train diff %</i>	0.6362	-0.27%					
326		<i>val loss / val diff %</i>	1.0643	0.86%					
327	198 000	<i>duration / val/train</i>	29 279	1.6923					
328		<i>train loss / train diff %</i>	0.6321	-0.64%					
329		<i>val loss / val diff %</i>	1.0697	0.51%					
330	200 000	<i>duration / val/train</i>	29 569	1.6623					
331		<i>train loss / train diff %</i>	0.6358	0.59%					
332		<i>val loss / val diff %</i>	1.0569	-1.20%					
333	202 000	<i>duration / val/train</i>	29 861	1.6770					
334		<i>train loss / train diff %</i>	0.6343	-0.24%					
335		<i>val loss / val diff %</i>	1.0637	0.64%					

336	204 000	<i>duration / val/train</i>	30 153	1.6795					
337		<i>train loss / train diff %</i>	0.6321	-0.35%					
338		<i>val loss / val diff %</i>	1.0616	-0.20%					
339	206 000	<i>duration / val/train</i>	30 443	1.6769					
340		<i>train loss / train diff %</i>	0.6350	0.46%					
341		<i>val loss / val diff %</i>	1.0648	0.30%					
342	208 000	<i>duration / val/train</i>	30 733	1.7000					
343		<i>train loss / train diff %</i>	0.6271	-1.24%					
344		<i>val loss / val diff %</i>	1.0661	0.12%					
345	210 000	<i>duration / val/train</i>	31 022	1.7110					
346		<i>train loss / train diff %</i>	0.6302	0.49%					
347		<i>val loss / val diff %</i>	1.0783	1.14%					
348	212 000	<i>duration / val/train</i>	31 312	1.6835					
349		<i>train loss / train diff %</i>	0.6353	0.81%					
350		<i>val loss / val diff %</i>	1.0695	-0.82%					
351	214 000	<i>duration / val/train</i>	31 602	1.6879					
352		<i>train loss / train diff %</i>	0.6316	-0.58%					
353		<i>val loss / val diff %</i>	1.0661	-0.32%					
354	216 000	<i>duration / val/train</i>	31 891	1.7139					
355		<i>train loss / train diff %</i>	0.6232	-1.33%					
356		<i>val loss / val diff %</i>	1.0681	0.19%					
357	218 000	<i>duration / val/train</i>	32 181	1.6745					
358		<i>train loss / train diff %</i>	0.6305	1.17%					
359		<i>val loss / val diff %</i>	1.0558	-1.15%					
360	220 000	<i>duration / val/train</i>	32 470	1.6704					
361		<i>train loss / train diff %</i>	0.6302	-0.05%					
362		<i>val loss / val diff %</i>	1.0527	-0.29%					
363	222 000	<i>duration / val/train</i>	32 760	1.7324					
364		<i>train loss / train diff %</i>	0.6263	-0.62%					
365		<i>val loss / val diff %</i>	1.0850	3.07%					
366	224 000	<i>duration / val/train</i>	33 050	1.7124					
367		<i>train loss / train diff %</i>	0.6227	-0.57%					
368		<i>val loss / val diff %</i>	1.0663	-1.72%					
369	226 000	<i>duration / val/train</i>	33 339	1.7227					

370		<i>train loss / train diff %</i>	0.6257	0.48%					
371		<i>val loss / val diff %</i>	1.0779	1.09%					
372	228 000	<i>duration / val/train</i>	33 629	1.7408					
373		<i>train loss / train diff %</i>	0.6185	-1.15%					
374		<i>val loss / val diff %</i>	1.0767	-0.11%					
375	230 000	<i>duration / val/train</i>	33 918	1.7246					
376		<i>train loss / train diff %</i>	0.6267	1.33%					
377		<i>val loss / val diff %</i>	1.0808	0.38%					
378	232 000	<i>duration / val/train</i>	34 208	1.7221					
379		<i>train loss / train diff %</i>	0.6262	-0.08%					
380		<i>val loss / val diff %</i>	1.0784	-0.22%					
381	234 000	<i>duration / val/train</i>	34 498	1.7128					
382		<i>train loss / train diff %</i>	0.6229	-0.53%					
383		<i>val loss / val diff %</i>	1.0669	-1.07%					
384	236 000	<i>duration / val/train</i>	34 787	1.7050					
385		<i>train loss / train diff %</i>	0.6180	-0.79%					
386		<i>val loss / val diff %</i>	1.0537	-1.24%					
387	238 000	<i>duration / val/train</i>	35 077	1.7182					
388		<i>train loss / train diff %</i>	0.6289	1.76%					
389		<i>val loss / val diff %</i>	1.0806	2.55%					

Table 7. Results of training models at Step 2.

7.2. Step-by-step model training. Step 3.

The models are modified in the same way as in Step 2, but in addition to the model size, the training dataset is also modified at each step. The model modification plan for Step 3 is shown in Table 8.

No	Stage/Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1	<i>model size, M</i>	2.69	4.91	6.94
2	<i>n_layer</i>	12	10	8
3	<i>n_embd</i>	128	192	256
4	<i>n_head</i>	4	6	8
5	<i>block_size</i>	2 048	2 048	2 048
6	<i>dropout</i>	0.10	0.10	0.15
7	<i>batch_size</i>	4	4	4
8	Start stage 1	2/32/1/128/0.00/64	2/64/2/128/0.00/64	2/64/2/128/0.00/64
9	Stage 2	4/64/2/256/0.05/64	4/96/3/256/0.05/64	3/96/3/256/0.05/64
10	Stage 3	6/64/2/512/0.05/64	6/128/4/512/0.10/64	4/128/4/512/0.05/64
11	Stage 4	9/96/3/1024/0.05/32	8/160/5/1536/0.10/8	6/192/6/1536/0.10/8
12	Final Stage 5	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4

Table 8. Step 3 model change plan.

The results of training the models in Step 3 are shown in Table 9.

No	iterations	Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1		model size, M	2.69	4.91	6.94	2.69	4.91	6.94
2		n_layer	12	10	8	12	10	8
3		n_embd	128	192	256	128	192	256
4		n_head	4	6	8	4	6	8
5		block_size	2 048	2 048	2 048	2 048	2 048	2 048
6		dropout	0.10	0.10	0.15	0.10	0.10	0.15
7		batch_size	4	4	4	4	4	4
8		zero_new_weights	True	True	True	False	False	False
9	Stage 1	model	2/32/1/128/0.00/64/80k	2/64/2/128/0.00/64/80k	2/64/2/128/0.00/64/80k	2/32/1/128/0.00/64/80k	2/64/2/128/0.00/64/80k	2/64/2/128/0.00/64/80k
10		iterations	80 000	80 000	80 000	80 000	80 000	80 000
11		duration / val/train	434 1.2828	437 1.5386	420 1.5386	432 1.2828	437 1.5386	445 1.5386
12		train loss	1.5441	1.2235	1.2235	1.5441	1.2235	1.2235
13		val loss	1.9808	1.8825	1.8825	1.9808	1.8825	1.8825
14	Stage 2	model	4/64/2/256/0.05/64/115k	4/96/3/256/0.05/64/100k	3/96/3/256/0.05/64/175k	4/64/2/256/0.05/64/110k	4/96/3/256/0.05/64/80k	3/96/3/256/0.05/64/125k
15		iterations	115 000	100 000	175 000	110 000	80 000	125 000
16		duration / val/train	1 101 1.4515	1 490 1.6929	2 003 1.5811	1 070 1.4682	1 170 1.7866	1 468 1.6230
17		train loss	1.1590 -24.94%	0.9222 -24.63%	1.0081 -17.61%	1.1096 -28.14%	0.9001 -26.43%	0.9615 -21.41%
18		val loss	1.6823 -15.07%	1.5612 -17.07%	1.5939 -15.33%	1.6291 -17.76%	1.6081 -14.58%	1.5605 -17.10%
19	Stage 3	model	6/64/2/512/0.05/64/75k	6/128/4/512/0.10/64/25k	4/128/4/512/0.05/64/40k	6/64/2/512/0.05/64/35k	6/128/4/512/0.10/64/20k	4/128/4/512/0.05/64/20k
20		iterations	75 000	25 000	40 000	35 000	20 000	20 000
21		duration / val/train	3 407 1.5426	2 299 1.8021	2 497 1.8216	1 574 1.5140	1 768 1.9385	1 241 1.9331
22		train loss	0.9655 -16.70%	0.7934 -13.97%	0.8021 -20.43%	0.9766 -11.99%	0.7575 -15.84%	0.7727 -19.64%
23		val loss	1.4894 -11.47%	1.4298 -8.42%	1.4611 -8.33%	1.4786 -9.24%	1.4684 -8.69%	1.4937 -4.28%
24	Stage 4	model	9/96/3/1024/0.05/32/75k	8/160/5/1536/0.10/8/95k	6/192/6/1536/0.10/8/110k	9/96/3/1024/0.05/32/95k	8/160/5/1536/0.10/8/95k	6/192/6/1536/0.10/8/85k
25		iterations	75 000	95 000	110 000	95 000	95 000	85 000
26		duration / val/train	13 334 1.6797	14 050 1.9600	14 644 1.9162	16 583 1.9072	13 995 2.2210	11 269 2.2054

27		train loss	0.7496	-22.36%	0.6270	-20.97%	0.6300	-21.46%	0.6758	-	0.5612	-	0.5706	-26.16%
28		val loss	1.2591	-15.46%	1.2289	-14.05%	1.2072	-17.38%	1.2889	30.80%	1.2464	25.91%	1.2584	-15.75%
29		Total time	18 276		18 276		19 564		19 659	12.83%	17 370	15.12%	14 423	
30	0	duration / val/train	0	1.0404	0	1.0457	0	1.0937	0	1.0332	0	1.0077	0	0.9912
31		train loss	3.8201		3.4260		3.4082		3.8306		3.5825		3.6263	
32		val loss	3.9744		3.5825		3.7276		3.9579		3.6102		3.5945	
33	2 000	duration / val/train	290	1.2888	364	1.5045	404	1.3729	291	1.3171	382	1.5411	390	1.3729
34		train loss / train diff %	1.0042	-73.71%	0.7739	-77.41%	0.8222	-75.88%	0.9463	-	0.7497	-	0.8924	-75.39%
35		val loss / val diff %	1.2942	-67.44%	1.1643	-67.50%	1.1288	-69.72%	1.2464	75.30%	1.1554	79.07%	1.2252	-65.91%
36	4 000	duration / val/train	580	1.3563	728	1.5624	799	1.4259	581	1.3785	761	1.6482	779	1.4929
37		train loss / train diff %	0.9062	-9.76%	0.7340	-5.16%	0.7791	-5.24%	0.8454	-	0.6936	-7.48%	0.7723	-13.46%
38		val loss / val diff %	1.2291	-5.03%	1.1468	-1.50%	1.1109	-1.59%	1.1654	10.66%	1.1432	-1.06%	1.1530	-5.89%
39	6 000	duration / val/train	869	1.3587	1 098	1.5856	1 206	1.4639	876	1.4168	1 140	1.6497	1 168	1.5341
40		train loss / train diff %	0.8803	-2.86%	0.7116	-3.05%	0.7487	-3.90%	0.8133	-3.80%	0.6746	-2.74%	0.7319	-5.23%
41		val loss / val diff %	1.1961	-2.68%	1.1283	-1.61%	1.0960	-1.34%	1.1523	-1.12%	1.1129	-2.65%	1.1228	-2.62%
42	8 000	duration / val/train	1 159	1.3728	1 464	1.5821	1 608	1.4964	1 177	1.4171	1 521	1.7290	1 569	1.5530
43		train loss / train diff %	0.8604	-2.26%	0.7037	-1.11%	0.7308	-2.39%	0.7965	-2.07%	0.6549	-2.92%	0.7110	-2.86%
44		val loss / val diff %	1.1812	-1.25%	1.1133	-1.33%	1.0936	-0.22%	1.1287	-2.05%	1.1323	1.74%	1.1042	-1.66%
45	10 000	duration / val/train	1 449	1.3826	1 828	1.6069	2 007	1.5087	1 486	1.4559	1 899	1.7202	1 975	1.6069
46		train loss / train diff %	0.8421	-2.13%	0.6878	-2.26%	0.7163	-1.98%	0.7791	-2.18%	0.6430	-1.82%	0.6912	-2.78%
47		val loss / val diff %	1.1643	-1.43%	1.1052	-0.73%	1.0807	-1.18%	1.1343	0.50%	1.1061	-2.31%	1.1107	0.59%
48	12 000	duration / val/train	1 738	1.3878	2 192	1.6550	2 417	1.5457	1 787	1.4563	2 268	1.7333	2 376	1.6374
49		train loss / train diff %	0.8353	-0.81%	0.6756	-1.77%	0.7050	-1.58%	0.7648	-1.84%	0.6379	-0.79%	0.6821	-1.32%
50		val loss / val diff %	1.1592	-0.44%	1.1181	1.17%	1.0897	0.83%	1.1138	-1.81%	1.1057	-0.04%	1.1169	0.56%
51	14 000	duration / val/train	2 028	1.4067	2 568	1.6832	2 820	1.5722	2 104	1.4803	2 650	1.7658	2 776	1.6816
52		train loss / train diff %	0.8180	-2.07%	0.6674	-1.21%	0.6935	-1.63%	0.7558	-1.18%	0.6278	-1.58%	0.6614	-3.03%
53		val loss / val diff %	1.1507	-0.73%	1.1234	0.47%	1.0903	0.06%	1.1188	0.45%	1.1086	0.26%	1.1122	-0.42%
54	16 000	duration / val/train	2 318	1.4022	2 945	1.6905	3 225	1.5902	2 429	1.4747	3 028	1.8330	3 178	1.7150
55		train loss / train diff %	0.8110	-0.86%	0.6598	-1.14%	0.6849	-1.24%	0.7415	-1.89%	0.6162	-1.85%	0.6466	-2.24%
56		val loss / val diff %	1.1372	-1.17%	1.1154	-0.71%	1.0891	-0.11%	1.0935	-2.26%	1.1295	1.89%	1.1089	-0.30%
57	18 000	duration / val/train	2 607	1.4276	3 322	1.7189	3 626	1.5967	2 753	1.4857	3 404	1.7972	3 592	1.7185

58		<i>train loss / train diff %</i>	0.7987	-1.52%	0.6538	-0.91%	0.6794	-0.80%	0.7348	-0.90%	0.6140	-0.36%	0.6391	-1.16%
59		<i>val loss / val diff %</i>	1.1402	0.26%	1.1238	0.75%	1.0848	-0.39%	1.0917	-0.16%	1.1035	-2.30%	1.0983	-0.96%
60	20 000	<i>duration / val/train</i>	2 897	1.4484	3 713	1.7536	4 023	1.6205	3 075	1.5103	3 772	1.8714	4 001	1.7383
61		<i>train loss / train diff %</i>	0.7955	-0.40%	0.6403	-2.06%	0.6711	-1.22%	0.7257	-1.24%	0.5987	-2.49%	0.6294	-1.52%
62		<i>val loss / val diff %</i>	1.1522	1.05%	1.1228	-0.09%	1.0875	0.25%	1.0960	0.39%	1.1204	1.53%	1.0941	-0.38%
63	22 000	<i>duration / val/train</i>	3 187	1.4640	4 095	1.7239	4 419	1.6136	3 388	1.5226	4 137	1.8745	4 403	1.7933
64		<i>train loss / train diff %</i>	0.7865	-1.13%	0.6433	0.47%	0.6675	-0.54%	0.7185	-0.99%	0.5966	-0.35%	0.6164	-2.07%
65		<i>val loss / val diff %</i>	1.1514	-0.07%	1.1090	-1.23%	1.0771	-0.96%	1.0940	-0.18%	1.1183	-0.19%	1.1054	1.03%
66	24 000	<i>duration / val/train</i>	3 477	1.4680	4 492	1.7568	4 811	1.6400	3 690	1.5261	4 502	1.8812	4 802	1.7894
67		<i>train loss / train diff %</i>	0.7735	-1.65%	0.6292	-2.19%	0.6598	-1.15%	0.7117	-0.95%	0.5859	-1.79%	0.6068	-1.56%
68		<i>val loss / val diff %</i>	1.1355	-1.38%	1.1054	-0.32%	1.0821	0.46%	1.0861	-0.72%	1.1022	-1.44%	1.0858	-1.77%
69	26 000	<i>duration / val/train</i>	3 766	1.4895	4 881	1.7620	5 199	1.6779	3 988	1.5722	4 867	1.9210	5 203	1.8099
70		<i>train loss / train diff %</i>	0.7642	-1.20%	0.6295	0.05%	0.6560	-0.58%	0.6999	-1.66%	0.5783	-1.30%	0.5980	-1.45%
71		<i>val loss / val diff %</i>	1.1383	0.25%	1.1092	0.34%	1.1007	1.72%	1.1004	1.32%	1.1109	0.79%	1.0823	-0.32%
72	28 000	<i>duration / val/train</i>	4 056	1.4571	5 256	1.7760	5 587	1.6738	4 282	1.5816	5 232	1.9284	5 621	1.8330
73		<i>train loss / train diff %</i>	0.7694	0.68%	0.6282	-0.21%	0.6490	-1.07%	0.6951	-0.69%	0.5744	-0.67%	0.5905	-1.25%
74		<i>val loss / val diff %</i>	1.1211	-1.51%	1.1157	0.59%	1.0863	-1.31%	1.0994	-0.09%	1.1077	-0.29%	1.0824	0.01%
75	30 000	<i>duration / val/train</i>	4 346	1.4348	5 634	1.8023	5 975	1.7099	4 578	1.5389	5 597	1.9781	6 020	1.8490
76		<i>train loss / train diff %</i>	0.7696	0.03%	0.6119	-2.59%	0.6428	-0.96%	0.6977	0.37%	0.5659	-1.48%	0.5859	-0.78%
77		<i>val loss / val diff %</i>	1.1042	-1.51%	1.1028	-1.16%	1.0991	1.18%	1.0737	-2.34%	1.1194	1.06%	1.0833	0.08%
78	32 000	<i>duration / val/train</i>	4 636	1.4667	6 012	1.8257	6 363	1.7225	4 878	1.5587	5 962	2.0152	6 419	1.8756
79		<i>train loss / train diff %</i>	0.7553	-1.86%	0.6139	0.33%	0.6356	-1.12%	0.6965	-0.17%	0.5595	-1.13%	0.5834	-0.43%
80		<i>val loss / val diff %</i>	1.1078	0.33%	1.1208	1.63%	1.0948	-0.39%	1.0856	1.11%	1.1275	0.72%	1.0942	1.01%
81	34 000	<i>duration / val/train</i>	4 925	1.4904	6 389	1.8208	6 751	1.7341	5 177	1.5759	6 327	2.0582	6 824	1.9212
82		<i>train loss / train diff %</i>	0.7537	-0.21%	0.6072	-1.09%	0.6366	0.16%	0.6833	-1.90%	0.5569	-0.46%	0.5750	-1.44%
83		<i>val loss / val diff %</i>	1.1233	1.40%	1.1056	-1.36%	1.1039	0.83%	1.0768	-0.81%	1.1462	1.66%	1.1047	0.96%
84	36 000	<i>duration / val/train</i>	5 215	1.4764	6 755	1.8426	7 139	1.7234	5 470	1.5843	6 692	2.0996	7 221	1.9672
85		<i>train loss / train diff %</i>	0.7515	-0.29%	0.6049	-0.38%	0.6324	-0.66%	0.6795	-0.56%	0.5524	-0.81%	0.5669	-1.41%
86		<i>val loss / val diff %</i>	1.1095	-1.23%	1.1146	0.81%	1.0899	-1.27%	1.0765	-0.03%	1.1598	1.19%	1.1152	0.95%
87	38 000	<i>duration / val/train</i>	5 505	1.4917	7 145	1.8652	7 527	1.7379	5 766	1.5910	7 057	2.1177	7 615	1.9533
88		<i>train loss / train diff %</i>	0.7444	-0.94%	0.6059	0.17%	0.6268	-0.89%	0.6760	-0.52%	0.5439	-1.54%	0.5656	-0.23%
89		<i>val loss / val diff %</i>	1.1104	0.08%	1.1301	1.39%	1.0893	-0.06%	1.0755	-0.09%	1.1518	-0.69%	1.1048	-0.93%
90	40 000	<i>duration / val/train</i>	5 794	1.5024	7 542	1.8670	7 916	1.7473	6 063	1.6047	7 421	2.1089	8 012	2.0269
91		<i>train loss / train diff %</i>	0.7385	-0.79%	0.5999	-0.99%	0.6273	0.08%	0.6712	-0.71%	0.5444	0.09%	0.5506	-2.65%

92		<i>val loss / val diff %</i>	1.1095	-0.08%	1.1200	-0.89%	1.0961	0.62%	1.0771	0.15%	1.1481	-0.32%	1.1160	1.01%
93	42 000	<i>duration / val/train</i>	6 084	1.5101	7 923	1.8863	8 304	1.7286	6 356	1.6384	7 786	2.1322	8 421	2.0547
94		<i>train loss / train diff %</i>	0.7377	-0.11%	0.5918	-1.35%	0.6234	-0.62%	0.6674	-0.57%	0.5363	-1.49%	0.5466	-0.73%
95		<i>val loss / val diff %</i>	1.1140	0.41%	1.1163	-0.33%	1.0776	-1.69%	1.0935	1.52%	1.1435	-0.40%	1.1231	0.64%
96	44 000	<i>duration / val/train</i>	6 374	1.4928	8 299	1.8874	8 692	1.7659	6 657	1.6423	8 151	2.1700	8 825	2.0430
97		<i>train loss / train diff %</i>	0.7341	-0.49%	0.5924	0.10%	0.6117	-1.88%	0.6657	-0.25%	0.5393	0.56%	0.5436	-0.55%
98		<i>val loss / val diff %</i>	1.0959	-1.62%	1.1181	0.16%	1.0802	0.24%	1.0933	-0.02%	1.1703	2.34%	1.1106	-1.11%
99	46 000	<i>duration / val/train</i>	6 664	1.5089	8 675	1.8763	9 087	1.7821	6 973	1.6208	8 516	2.1764	9 224	2.0712
100		<i>train loss / train diff %</i>	0.7268	-0.99%	0.5943	0.32%	0.6109	-0.13%	0.6592	-0.98%	0.5351	-0.78%	0.5422	-0.26%
101		<i>val loss / val diff %</i>	1.0967	0.07%	1.1151	-0.27%	1.0887	0.79%	1.0684	-2.28%	1.1646	-0.49%	1.1230	1.12%
102	48 000	<i>duration / val/train</i>	6 953	1.5357	9 049	1.9375	9 485	1.8106	7 286	1.6554	8 881	2.2014	9 627	2.1060
103		<i>train loss / train diff %</i>	0.7246	-0.30%	0.5822	-2.04%	0.6087	-0.36%	0.6590	-0.03%	0.5302	-0.92%	0.5302	-2.21%
104		<i>val loss / val diff %</i>	1.1128	1.47%	1.1280	1.16%	1.1021	1.23%	1.0909	2.11%	1.1672	0.22%	1.1166	-0.57%
105	50 000	<i>duration / val/train</i>	7 243	1.5041	9 431	1.9478	9 887	1.8005	7 585	1.6291	9 246	2.2464	10 024	2.0921
106		<i>train loss / train diff %</i>	0.7262	0.22%	0.5806	-0.27%	0.6081	-0.10%	0.6565	-0.38%	0.5216	-1.62%	0.5311	0.17%
107		<i>val loss / val diff %</i>	1.0923	-1.84%	1.1309	0.26%	1.0949	-0.65%	1.0695	-1.96%	1.1717	0.39%	1.1111	-0.49%
108	52 000	<i>duration / val/train</i>	7 533	1.5144	9 804	1.9068	10 299	1.8105	7 882	1.6342	9 611	2.2250	10 416	2.0990
109		<i>train loss / train diff %</i>	0.7222	-0.55%	0.5859	0.91%	0.6068	-0.21%	0.6589	0.37%	0.5266	0.96%	0.5324	0.24%
110		<i>val loss / val diff %</i>	1.0937	0.13%	1.1172	-1.21%	1.0986	0.34%	1.0768	0.68%	1.1717	0.00%	1.1175	0.58%
111	54 000	<i>duration / val/train</i>	7 823	1.5404	10 175	1.9227	10 709	1.7943	8 182	1.6789	9 975	2.2304	10 805	2.1786
112		<i>train loss / train diff %</i>	0.7197	-0.35%	0.5833	-0.44%	0.6062	-0.10%	0.6440	-2.26%	0.5247	-0.36%	0.5242	-1.54%
113		<i>val loss / val diff %</i>	1.1086	1.36%	1.1215	0.38%	1.0877	-0.99%	1.0812	0.41%	1.1703	-0.12%	1.1420	2.19%
114	56 000	<i>duration / val/train</i>	8 113	1.5460	10 542	1.9306	11 109	1.8237	8 491	1.6829	10 340	2.2647	11 195	2.1879
115		<i>train loss / train diff %</i>	0.7190	-0.10%	0.5751	-1.41%	0.5995	-1.11%	0.6486	0.71%	0.5194	-1.01%	0.5210	-0.61%
116		<i>val loss / val diff %</i>	1.1116	0.27%	1.1103	-1.00%	1.0933	0.51%	1.0915	0.95%	1.1763	0.51%	1.1399	-0.18%
117	58 000	<i>duration / val/train</i>	8 410	1.5533	10 907	1.9528	11 508	1.8247	8 791	1.6716	10 705	2.2839	11 584	2.1998
118		<i>train loss / train diff %</i>	0.7117	-1.02%	0.5760	0.16%	0.5995	0.00%	0.6479	-0.11%	0.5167	-0.52%	0.5140	-1.34%
119		<i>val loss / val diff %</i>	1.1055	-0.55%	1.1248	1.31%	1.0939	0.05%	1.0830	-0.78%	1.1801	0.32%	1.1307	-0.81%
120	60 000	<i>duration / val/train</i>	8 712	1.5421	11 271	1.9532	11 909	1.8157	9 084	1.7119	11 070	2.2973	11 974	2.2336
121		<i>train loss / train diff %</i>	0.7120	0.04%	0.5752	-0.14%	0.5973	-0.37%	0.6439	-0.62%	0.5139	-0.54%	0.5132	-0.16%
122		<i>val loss / val diff %</i>	1.0980	-0.68%	1.1235	-0.12%	1.0845	-0.86%	1.1023	1.78%	1.1806	0.04%	1.1463	1.38%
123	62 000	<i>duration / val/train</i>	9 013	1.5659	11 635	1.9622	12 311	1.8766	9 375	1.6871	11 435	2.2812	12 363	2.2252
124		<i>train loss / train diff %</i>	0.7056	-0.90%	0.5692	-1.04%	0.5891	-1.37%	0.6418	-0.33%	0.5107	-0.62%	0.5088	-0.86%
125		<i>val loss / val diff %</i>	1.1049	0.63%	1.1169	-0.59%	1.1055	1.94%	1.0828	-1.77%	1.1650	-1.32%	1.1322	-1.23%

126	64 000	<i>duration / val/train</i>	9 313	1.5625	11 999	1.9596	12 725	1.8174	9 665	1.6859	11 800	2.2927	12 753	2.2672
127		<i>train loss / train diff %</i>	0.7059	0.04%	0.5719	0.47%	0.5975	1.43%	0.6351	-1.04%	0.5105	-0.04%	0.5015	-1.43%
128		<i>val loss / val diff %</i>	1.1030	-0.17%	1.1207	0.34%	1.0859	-1.77%	1.0707	-1.12%	1.1704	0.46%	1.1370	0.42%
129	66 000	<i>duration / val/train</i>	9 607	1.5512	12 363	1.9830	13 117	1.8550	9 955	1.7109	12 165	2.3499	13 149	2.2453
130		<i>train loss / train diff %</i>	0.7018	-0.58%	0.5699	-0.35%	0.5841	-2.24%	0.6347	-0.06%	0.5056	-0.96%	0.5022	0.14%
131		<i>val loss / val diff %</i>	1.0886	-1.31%	1.1301	0.84%	1.0835	-0.22%	1.0859	1.42%	1.1881	1.51%	1.1276	-0.83%
132	68 000	<i>duration / val/train</i>	9 906	1.5785	12 726	2.0148	13 506	1.8742	10 246	1.7172	12 530	2.3502	13 540	2.2795
133		<i>train loss / train diff %</i>	0.7003	-0.21%	0.5619	-1.40%	0.5844	0.05%	0.6298	-0.77%	0.5014	-0.83%	0.5009	-0.26%
134		<i>val loss / val diff %</i>	1.1054	1.54%	1.1321	0.18%	1.0953	1.09%	1.0815	-0.41%	1.1784	-0.82%	1.1418	1.26%
135	70 000	<i>duration / val/train</i>	10 206	1.5709	13 104	2.0493	13 894	1.9431	10 536	1.7184	12 894	2.3619	13 942	2.2739
136		<i>train loss / train diff %</i>	0.6963	-0.57%	0.5619	0.00%	0.5764	-1.37%	0.6296	-0.03%	0.5029	0.30%	0.4954	-1.10%
137		<i>val loss / val diff %</i>	1.0938	-1.05%	1.1515	1.71%	1.1200	2.26%	1.0819	0.04%	1.1878	0.80%	1.1265	-1.34%
138	72 000	<i>duration / val/train</i>	10 499	1.5615	13 481	2.0106	14 283	1.8743	10 826	1.7195	13 259	2.4027	14 339	2.3267
139		<i>train loss / train diff %</i>	0.6995	0.46%	0.5639	0.36%	0.5814	0.87%	0.6260	-0.57%	0.4976	-1.05%	0.4958	0.08%
140		<i>val loss / val diff %</i>	1.0923	-0.14%	1.1338	-1.54%	1.0897	-2.71%	1.0764	-0.51%	1.1956	0.66%	1.1536	2.41%
141	74 000	<i>duration / val/train</i>	10 791	1.5946	13 849	1.9971	14 671	1.9356	11 116	1.7184	13 624	2.3977	14 740	2.3342
142		<i>train loss / train diff %</i>	0.6976	-0.27%	0.5588	-0.90%	0.5778	-0.62%	0.6286	0.42%	0.4943	-0.66%	0.4937	-0.42%
143		<i>val loss / val diff %</i>	1.1124	1.84%	1.1160	-1.57%	1.1184	2.63%	1.0802	0.35%	1.1852	-0.87%	1.1524	-0.10%
144	76 000	<i>duration / val/train</i>	11 081	1.5878	14 216	2.0094	15 059	1.9193	11 407	1.7316	13 989	2.3404	15 143	2.3918
145		<i>train loss / train diff %</i>	0.6951	-0.36%	0.5614	0.47%	0.5784	0.10%	0.6225	-0.97%	0.5017	1.50%	0.4877	-1.22%
146		<i>val loss / val diff %</i>	1.1037	-0.78%	1.1281	1.08%	1.1101	-0.74%	1.0779	-0.21%	1.1742	-0.93%	1.1665	1.22%
147	78 000	<i>duration / val/train</i>	11 371	1.6013	14 584	1.9830	15 447	1.9089	11 697	1.7455	14 354	2.4288	15 536	2.3763
148		<i>train loss / train diff %</i>	0.6908	-0.62%	0.5632	0.32%	0.5776	-0.14%	0.6232	0.11%	0.4897	-2.39%	0.4858	-0.39%
149		<i>val loss / val diff %</i>	1.1062	0.23%	1.1168	-1.00%	1.1026	-0.68%	1.0878	0.92%	1.1894	1.29%	1.1544	-1.04%
150	80 000	<i>duration / val/train</i>	11 661	1.5999	14 949	2.0481	15 835	1.9376	11 987	1.7569	14 718	2.3971	15 926	2.4019
151		<i>train loss / train diff %</i>	0.6894	-0.20%	0.5525	-1.90%	0.5692	-1.45%	0.6175	-0.91%	0.4878	-0.39%	0.4792	-1.36%
152		<i>val loss / val diff %</i>	1.1030	-0.29%	1.1316	1.33%	1.1029	0.03%	1.0849	-0.27%	1.1693	-1.69%	1.1510	-0.29%
153	82 000	<i>duration / val/train</i>	11 950	1.5989	15 313	2.0497	16 223	1.9502	12 277	1.7574	15 083	2.4387	16 320	2.4034
154		<i>train loss / train diff %</i>	0.6843	-0.74%	0.5498	-0.49%	0.5740	0.84%	0.6151	-0.39%	0.4915	0.76%	0.4836	0.92%
155		<i>val loss / val diff %</i>	1.0941	-0.81%	1.1269	-0.42%	1.1194	1.50%	1.0810	-0.36%	1.1986	2.51%	1.1623	0.98%
156	84 000	<i>duration / val/train</i>	12 240	1.5677	15 677	2.0768	16 612	1.9387	12 568	1.7408	15 448	2.4970	16 748	2.4083
157		<i>train loss / train diff %</i>	0.6886	0.63%	0.5506	0.15%	0.5711	-0.51%	0.6147	-0.07%	0.4877	-0.77%	0.4791	-0.93%
158		<i>val loss / val diff %</i>	1.0795	-1.33%	1.1435	1.47%	1.1072	-1.09%	1.0701	-1.01%	1.2178	1.60%	1.1538	-0.73%
159	86 000	<i>duration / val/train</i>	12 530	1.5979	16 041	2.0387	17 000	1.9161	12 858	1.7558	15 813	2.5162		

160		<i>train loss / train diff %</i>	0.6842	-0.64%	0.5554	0.87%	0.5732	0.37%	0.6187	0.65%	0.4857	-0.41%	
161		<i>val loss / val diff %</i>	1.0933	1.28%	1.1323	-0.98%	1.0983	-0.80%	1.0863	1.51%	1.2221	0.35%	
162	88 000	<i>duration / val/train</i>	12 820	1.5960	16 405	2.1059	17 388	1.9804	13 148	1.7409	16 178	2.4676	
163		<i>train loss / train diff %</i>	0.6913	1.04%	0.5466	-1.58%	0.5609	-2.15%	0.6153	-0.55%	0.4833	-0.49%	
164		<i>val loss / val diff %</i>	1.1033	0.91%	1.1511	1.66%	1.1108	1.14%	1.0712	-1.39%	1.1926	-2.41%	
165	90 000	<i>duration / val/train</i>	13 110	1.5672	16 769	2.0466	17 776	1.9932	13 438	1.7176	16 543	2.4936	
166		<i>train loss / train diff %</i>	0.6828	-1.23%	0.5499	0.60%	0.5608	-0.02%	0.6183	0.49%	0.4810	-0.48%	
167		<i>val loss / val diff %</i>	1.0701	-3.01%	1.1254	-2.23%	1.1178	0.63%	1.0620	-0.86%	1.1994	0.57%	
168	92 000	<i>duration / val/train</i>	13 405	1.6090	17 133	2.0952	18 164	1.9527	13 729	1.7667	16 907	2.4750	
169		<i>train loss / train diff %</i>	0.6852	0.35%	0.5482	-0.31%	0.5668	1.07%	0.6141	-0.68%	0.4855	0.94%	
170		<i>val loss / val diff %</i>	1.1025	3.03%	1.1486	2.06%	1.1068	-0.98%	1.0849	2.16%	1.2016	0.18%	
171	94 000	<i>duration / val/train</i>	13 705	2.3844	17 497	2.0648	18 552	1.9823	14 020	1.7631	17 272	2.5189	
172		<i>train loss / train diff %</i>	0.6754	-1.43%	0.5447	-0.64%	0.5592	-1.34%	0.6154	0.21%	0.4777	-1.61%	
173		<i>val loss / val diff %</i>	1.0992	-0.30%	1.1247	-2.08%	1.1085	0.15%	1.0850	0.01%	1.2033	0.14%	
174	96 000	<i>duration / val/train</i>	14 007	1.6189	17 861	2.0606	18 941	1.9984	14 326	1.7892	17 637	2.5166	
175		<i>train loss / train diff %</i>	0.6769	0.22%	0.5475	0.51%	0.5593	0.02%	0.6063	-1.48%	0.4795	0.38%	
176		<i>val loss / val diff %</i>	1.0958	-31.95%	1.1282	0.31%	1.1177	0.83%	1.0848	-0.02%	1.2067	0.28%	
177	98 000	<i>duration / val/train</i>	14 313	1.6193	18 224	2.1036	19 329	1.9935	14 627	1.7906	18 002	2.5536	
178		<i>train loss / train diff %</i>	0.6737	-0.47%	0.5427	-0.88%	0.5542	-0.91%	0.6059	-0.07%	0.4763	-0.67%	
179		<i>val loss / val diff %</i>	1.0909	-0.45%	1.1416	1.19%	1.1048	-1.15%	1.0849	0.01%	1.2163	0.80%	
180	100 000	<i>duration / val/train</i>	14 610	1.5858	18 588	2.0833	19 717	2.0115	14 943	1.8033	18 367	2.5161	
181		<i>train loss / train diff %</i>	0.6815	1.16%	0.5429	0.04%	0.5568	0.47%	0.6054	-0.08%	0.4739	-0.50%	
182		<i>val loss / val diff %</i>	1.0807	-0.94%	1.1310	-0.93%	1.1200	1.38%	1.0917	0.63%	1.1924	-1.96%	
183	102 000	<i>duration / val/train</i>	14 907	1.6478	18 952	2.0871	20 105	2.0146	15 247	1.7831	18 731	2.5171	
184		<i>train loss / train diff %</i>	0.6710	-1.54%	0.5442	0.24%	0.5550	-0.32%	0.6096	0.69%	0.4715	-0.51%	
185		<i>val loss / val diff %</i>	1.1057	2.31%	1.1358	0.42%	1.1181	-0.17%	1.0870	-0.43%	1.1868	-0.47%	
186	104 000	<i>duration / val/train</i>	15 204	1.6467	19 316	2.1007	20 494	1.9905	15 548	1.8138	19 096	2.5742	
187		<i>train loss / train diff %</i>	0.6696	-0.21%	0.5370	-1.32%	0.5555	0.09%	0.6016	-1.31%	0.4699	-0.34%	
188		<i>val loss / val diff %</i>	1.1026	-0.28%	1.1281	-0.68%	1.1057	-1.11%	1.0912	0.39%	1.2096	1.92%	
189	106 000	<i>duration / val/train</i>	15 509	1.6281	19 680	2.1137	20 886	2.0187	15 853	1.8129	19 461	2.5830	
190		<i>train loss / train diff %</i>	0.6723	0.40%	0.5376	0.11%	0.5516	-0.70%	0.5998	-0.30%	0.4695	-0.09%	
191		<i>val loss / val diff %</i>	1.0946	-0.73%	1.1363	0.73%	1.1135	0.71%	1.0874	-0.35%	1.2127	0.26%	
192	108 000	<i>duration / val/train</i>	15 806	1.6592	20 044	2.1130	21 278	2.0132	16 146	1.7666	19 826	2.6091	
193		<i>train loss / train diff %</i>	0.6676	-0.70%	0.5418	0.78%	0.5516	0.00%	0.6028	0.50%	0.4640	-1.17%	

194		<i>val loss / val diff %</i>	1.1077	1.20%	1.1448	0.75%	1.1105	-0.27%	1.0649	-2.07%	1.2106	-0.17%	
195	110 000	<i>duration / val/train</i>	16 098	1.6419	20 408	2.0926	21 669	2.0389	16 438	1.8252	20 191	2.5854	
196		<i>train loss / train diff %</i>	0.6644	-0.48%	0.5357	-1.13%	0.5454	-1.12%	0.5985	-0.71%	0.4698	1.25%	
197		<i>val loss / val diff %</i>	1.0909	-1.52%	1.1210	-2.08%	1.1120	0.14%	1.0924	2.58%	1.2146	0.33%	
198	112 000	<i>duration / val/train</i>	16 388	1.6406	20 772	2.1246	22 061	1.9843	16 729	1.8174	20 556	2.5934	
199		<i>train loss / train diff %</i>	0.6745	1.52%	0.5339	-0.34%	0.5547	1.71%	0.5947	-0.63%	0.4690	-0.17%	
200		<i>val loss / val diff %</i>	1.1066	1.44%	1.1343	1.19%	1.1007	-1.02%	1.0808	-1.06%	1.2163	0.14%	
201	114 000	<i>duration / val/train</i>	16 678	1.6682	21 136	2.0969	22 453	2.0074	17 031	1.8260	20 921	2.6226	
202		<i>train loss / train diff %</i>	0.6675	-1.04%	0.5375	0.67%	0.5514	-0.59%	0.6018	1.19%	0.4637	-1.13%	
203		<i>val loss / val diff %</i>	1.1135	0.62%	1.1271	-0.63%	1.1069	0.56%	1.0989	1.67%	1.2161	-0.02%	
204	116 000	<i>duration / val/train</i>	16 968	1.6559	21 500	2.1208	22 844	2.0161	17 332	1.8473	21 305	2.6579	
205		<i>train loss / train diff %</i>	0.6602	-1.09%	0.5348	-0.50%	0.5542	0.51%	0.5959	-0.98%	0.4619	-0.39%	
206		<i>val loss / val diff %</i>	1.0932	-1.82%	1.1342	0.63%	1.1173	0.94%	1.1008	0.17%	1.2277	0.95%	
207	118 000	<i>duration / val/train</i>	17 258	1.6310	21 864	2.1378	23 235	2.0344	17 631	1.8501	21 685	2.6133	
208		<i>train loss / train diff %</i>	0.6637	0.53%	0.5328	-0.37%	0.5463	-1.43%	0.5903	-0.94%	0.4647	0.61%	
209		<i>val loss / val diff %</i>	1.0825	-0.98%	1.1390	0.42%	1.1114	-0.53%	1.0921	-0.79%	1.2144	-1.08%	
210	120 000	<i>duration / val/train</i>	17 547	1.6505	22 228	2.1280	23 627	2.0462	17 948	1.8518	22 065	2.6518	
211		<i>train loss / train diff %</i>	0.6587	-0.75%	0.5330	0.04%	0.5457	-0.11%	0.5957	0.91%	0.4641	-0.13%	
212		<i>val loss / val diff %</i>	1.0872	0.43%	1.1342	-0.42%	1.1166	0.47%	1.1031	1.01%	1.2307	1.34%	
213	122 000	<i>duration / val/train</i>	17 837	1.6690	22 592	2.1780	24 018	2.0167	18 258	1.8204			
214		<i>train loss / train diff %</i>	0.6635	0.73%	0.5291	-0.73%	0.5444	-0.24%	0.5880	-1.29%			
215		<i>val loss / val diff %</i>	1.1074	1.86%	1.1524	1.60%	1.0979	-1.67%	1.0704	-2.96%			
216	124 000	<i>duration / val/train</i>	18 127	1.6479	22 956	2.1543	24 410	2.0208	18 562	1.8106			
217		<i>train loss / train diff %</i>	0.6617	-0.27%	0.5295	0.08%	0.5425	-0.35%	0.5957	1.31%			
218		<i>val loss / val diff %</i>	1.0904	-1.54%	1.1407	-1.02%	1.0963	-0.15%	1.0786	0.77%			
219	126 000	<i>duration / val/train</i>	18 417	1.6603	23 320	2.1717	24 801	2.0538	18 863	1.8327			
220		<i>train loss / train diff %</i>	0.6594	-0.35%	0.5271	-0.45%	0.5448	0.42%	0.5937	-0.34%			
221		<i>val loss / val diff %</i>	1.0948	0.40%	1.1447	0.35%	1.1189	2.06%	1.0881	0.88%			
222	128 000	<i>duration / val/train</i>	18 706	1.6636	23 684	2.1830	25 193	2.0381	19 166	1.8426			
223		<i>train loss / train diff %</i>	0.6552	-0.64%	0.5278	0.13%	0.5402	-0.84%	0.5908	-0.49%			
224		<i>val loss / val diff %</i>	1.0900	-0.44%	1.1522	0.66%	1.1010	-1.60%	1.0886	0.05%			
225	130 000	<i>duration / val/train</i>	18 996	1.6704	24 048	2.1605	25 585	2.0283	19 460	1.8347			
226		<i>train loss / train diff %</i>	0.6584	0.49%	0.5282	0.08%	0.5435	0.61%	0.5888	-0.34%			
227		<i>val loss / val diff %</i>	1.0998	0.90%	1.1412	-0.95%	1.1024	0.13%	1.0803	-0.76%			

228	132 000	<i>duration / val/train</i>	19 286	1.6489	24 412	2.2144	25 977	2.0740	19 761	1.8305		
229		<i>train loss / train diff %</i>	0.6660	1.15%	0.5239	-0.81%	0.5407	-0.52%	0.5866	-0.37%		
230		<i>val loss / val diff %</i>	1.0982	-0.15%	1.1601	1.66%	1.1214	1.72%	1.0738	-0.60%		
231	134 000	<i>duration / val/train</i>	19 576	1.6809	24 776	2.1416	26 368	2.0373	20 057	1.8398		
232		<i>train loss / train diff %</i>	0.6577	-1.25%	0.5268	0.55%	0.5411	0.07%	0.5931	1.11%		
233		<i>val loss / val diff %</i>	1.1055	0.66%	1.1282	-2.75%	1.1024	-1.69%	1.0912	1.62%		
234	136 000	<i>duration / val/train</i>	19 865	1.6607	25 139	2.1987	26 772	2.0897	20 352	1.8479		
235		<i>train loss / train diff %</i>	0.6555	-0.33%	0.5239	-0.55%	0.5364	-0.87%	0.5892	-0.66%		
236		<i>val loss / val diff %</i>	1.0886	-1.53%	1.1519	2.10%	1.1209	1.68%	1.0888	-0.22%		
237	138 000	<i>duration / val/train</i>	20 155	1.6484	25 503	2.1802	27 175	2.0920	20 647	1.8551		
238		<i>train loss / train diff %</i>	0.6573	0.27%	0.5293	1.03%	0.5378	0.26%	0.5852	-0.68%		
239		<i>val loss / val diff %</i>	1.0835	-0.47%	1.1540	0.18%	1.1251	0.37%	1.0856	-0.29%		
240	140 000	<i>duration / val/train</i>	20 445	1.6634	25 867	2.1777	27 572	2.0813	20 942	1.8741		
241		<i>train loss / train diff %</i>	0.6512	-0.93%	0.5229	-1.21%	0.5365	-0.24%	0.5869	0.29%		
242		<i>val loss / val diff %</i>	1.0832	-0.03%	1.1387	-1.33%	1.1166	-0.76%	1.0999	1.32%		
243	142 000	<i>duration / val/train</i>	20 734	1.6684	26 231	2.1893	27 977	2.0789	21 239	1.8699		
244		<i>train loss / train diff %</i>	0.6511	-0.02%	0.5235	0.11%	0.5350	-0.28%	0.5813	-0.95%		
245		<i>val loss / val diff %</i>	1.0863	0.29%	1.1461	0.65%	1.1122	-0.39%	1.0870	-1.17%		
246	144 000	<i>duration / val/train</i>	21 024	1.6995	26 595	2.1811	28 369	2.0884	21 538	1.8654		
247		<i>train loss / train diff %</i>	0.6486	-0.38%	0.5213	-0.42%	0.5285	-1.21%	0.5808	-0.09%		
248		<i>val loss / val diff %</i>	1.1023	1.47%	1.1370	-0.79%	1.1037	-0.76%	1.0834	-0.33%		
249	146 000	<i>duration / val/train</i>	21 314	1.6793	26 959	2.1654	28 763	2.0982				
250		<i>train loss / train diff %</i>	0.6508	0.34%	0.5241	0.54%	0.5335	0.95%				
251		<i>val loss / val diff %</i>	1.0929	-0.85%	1.1349	-0.18%	1.1194	1.42%				
252	148 000	<i>duration / val/train</i>	21 604	1.6790	27 323	2.1926						
253		<i>train loss / train diff %</i>	0.6514	0.09%	0.5203	-0.73%						
254		<i>val loss / val diff %</i>	1.0937	0.07%	1.1408	0.52%						
255	150 000	<i>duration / val/train</i>	21 893	1.6910	27 687	2.2547						
256		<i>train loss / train diff %</i>	0.6520	0.09%	0.5147	-1.08%						
257		<i>val loss / val diff %</i>	1.1025	0.80%	1.1605	1.73%						
258	152 000	<i>duration / val/train</i>	22 194	1.6813	28 051	2.1938						
259		<i>train loss / train diff %</i>	0.6492	-0.43%	0.5196	0.95%						
260		<i>val loss / val diff %</i>	1.0915	-1.00%	1.1399	-1.78%						
261	154 000	<i>duration / val/train</i>	22 501	1.7055	28 415	2.2402						

262		<i>train loss / train diff %</i>	0.6452	-0.62%	0.5178	-0.35%				
263		<i>val loss / val diff %</i>	1.1004	0.82%	1.1600	1.76%				
264	156 000	<i>duration / val/train</i>	22 800	1.6791	28 781	2.1813				
265		<i>train loss / train diff %</i>	0.6475	0.36%	0.5246	1.31%				
266		<i>val loss / val diff %</i>	1.0872	-1.20%	1.1443	-1.35%				
267	158 000	<i>duration / val/train</i>	23 099	1.7080	29 160	2.2285				
268		<i>train loss / train diff %</i>	0.6427	-0.74%	0.5151	-1.81%				
269		<i>val loss / val diff %</i>	1.0977	0.97%	1.1479	0.31%				
270	160 000	<i>duration / val/train</i>	23 401	1.7125	29 540	2.1977				
271		<i>train loss / train diff %</i>	0.6431	0.06%	0.5204	1.03%				
272		<i>val loss / val diff %</i>	1.1013	0.33%	1.1437	-0.37%				
273	162 000	<i>duration / val/train</i>	23 703	1.7052						
274		<i>train loss / train diff %</i>	0.6468	0.58%						
275		<i>val loss / val diff %</i>	1.1029	0.15%						
276	164 000	<i>duration / val/train</i>	24 010	1.7132						
277		<i>train loss / train diff %</i>	0.6416	-0.80%						
278		<i>val loss / val diff %</i>	1.0992	-0.34%						
279	166 000	<i>duration / val/train</i>	24 317	1.6727						
280		<i>train loss / train diff %</i>	0.6446	0.47%						
281		<i>val loss / val diff %</i>	1.0782	-1.91%						
282	168 000	<i>duration / val/train</i>	24 624	1.6912						
283		<i>train loss / train diff %</i>	0.6471	0.39%						
284		<i>val loss / val diff %</i>	1.0944	1.50%						
285	170 000	<i>duration / val/train</i>	24 920	1.6971						
286		<i>train loss / train diff %</i>	0.6442	-0.45%						
287		<i>val loss / val diff %</i>	1.0933	-0.10%						
288	172 000	<i>duration / val/train</i>	25 213	1.7121						
289		<i>train loss / train diff %</i>	0.6409	-0.51%						
290		<i>val loss / val diff %</i>	1.0973	0.37%						
291	174 000	<i>duration / val/train</i>	25 526	1.7066						
292		<i>train loss / train diff %</i>	0.6397	-0.19%						
293		<i>val loss / val diff %</i>	1.0917	-0.51%						
294	176 000	<i>duration / val/train</i>	25 831	1.6958						
295		<i>train loss / train diff %</i>	0.6348	-0.77%						

296		<i>val loss / val diff %</i>	1.0765	-1.39%					
297	178 000	<i>duration / val/train</i>	26 131	1.6730					
298		<i>train loss / train diff %</i>	0.6409	0.96%					
299		<i>val loss / val diff %</i>	1.0722	-0.40%					
300	180 000	<i>duration / val/train</i>	26 428	1.6766					
301		<i>train loss / train diff %</i>	0.6431	0.34%					
302		<i>val loss / val diff %</i>	1.0782	0.56%					
303	182 000	<i>duration / val/train</i>	26 735	1.7091					
304		<i>train loss / train diff %</i>	0.6417	-0.22%					
305		<i>val loss / val diff %</i>	1.0967	1.72%					
306	184 000	<i>duration / val/train</i>	27 039	1.7175					
307		<i>train loss / train diff %</i>	0.6365	-0.81%					
308		<i>val loss / val diff %</i>	1.0932	-0.32%					
309	186 000	<i>duration / val/train</i>	27 348	1.6818					
310		<i>train loss / train diff %</i>	0.6415	0.79%					
311		<i>val loss / val diff %</i>	1.0789	-1.31%					
312	188 000	<i>duration / val/train</i>	27 660	1.7161					
313		<i>train loss / train diff %</i>	0.6404	-0.17%					
314		<i>val loss / val diff %</i>	1.0990	1.86%					
315	190 000	<i>duration / val/train</i>	27 965	1.7015					
316		<i>train loss / train diff %</i>	0.6348	-0.87%					
317		<i>val loss / val diff %</i>	1.0801	-1.72%					
318	192 000	<i>duration / val/train</i>	28 271	1.7015					
319		<i>train loss / train diff %</i>	0.6391	0.68%					
320		<i>val loss / val diff %</i>	1.0874	0.68%					
321	194 000	<i>duration / val/train</i>	28 574	1.7561					
322		<i>train loss / train diff %</i>	0.6293	-1.53%					
323		<i>val loss / val diff %</i>	1.1051	1.63%					
324	196 000	<i>duration / val/train</i>	28 885	1.7472					
325		<i>train loss / train diff %</i>	0.6348	0.87%					
326		<i>val loss / val diff %</i>	1.1091	0.36%					
327	198 000	<i>duration / val/train</i>	29 190	1.7156					
328		<i>train loss / train diff %</i>	0.6367	0.30%					
329		<i>val loss / val diff %</i>	1.0923	-1.51%					

330	200 000	<i>duration / val/train</i>	29 494	1.7301					
331		<i>train loss / train diff %</i>	0.6379	0.19%					
332		<i>val loss / val diff %</i>	1.1036	1.03%					

Table 9. Results of training models at Step 3.

7.3. Step-by-step model training. Step 4.

Models are trained on the full dataset as in Step 2, but their sizes change less between stages. All models start with the minimum size (1/32/1). At each stage, one transformer layer is added. The remaining model parameters are also increased as minimally as possible. The model modification plan for Step 4 is shown in Table 10.

<i>Nº</i>	<i>Stage/Params</i>	<i>12/128/4/2048/0.10/4</i>	<i>10/192/6/2048/0.10/4</i>	<i>8/256/8/2048/0.15/4</i>
1	<i>model size, M</i>	2.69	4.91	6.94
2	<i>n_layer</i>	12	10	8
3	<i>n_embd</i>	128	192	256
4	<i>n_head</i>	4	6	8
5	<i>block_size</i>	2 048	2 048	2 048
6	<i>dropout</i>	0.10	0.10	0.15
7	<i>batch_size</i>	4	4	4
8	Start stage 1	1/32/1/32/0.00/64	1/32/1/32/0.00/64	1/32/1/32/0.00/64
9	Stage 2	2/32/1/64/0.00/64	2/64/2/64/0.00/64	2/64/2/64/0.00/64
10	Stage 3	3/64/2/128/0.00/64	3/96/3/128/0.05/64	3/96/3/128/0.05/64
11	Stage 4	4/64/2/256/0.00/64	4/96/3/256/0.05/64	4/128/4/256/0.05/64
12	Stage 5	5/64/2/512/0.05/64	5/128/4/512/0.05/64	5/160/5/512/0.05/32
13	Stage 6	6/96/3/512/0.05/64	6/128/4/1024/0.05/32	6/192/6/1024/0.10/16
14	Stage 7	7/96/3/1024/0.05/32	7/160/5/1536/0.10/8	7/224/7/1536/0.10/8
15	Stage 8	8/96/3/1024/0.05/32	8/160/5/1536/0.10/8	8/256/8/2048/0.15/4
16	Stage 9	9/96/3/1536/0.05/16	9/192/6/2048/0.10/4	
17	Stage 10	10/128/4/1536/0.10/8	10/192/6/2048/0.10/4	
18	Stage 11	11/128/4/2048/0.10/4		
19	Stage 12	12/128/4/2048/0.10/4		

Table 10. Step 4 model change plan.

The results of training the models in Step 4 are shown in Table 11.

No	iterations	Params	12/128/4/2048/0.10/ 4	10/192/6/2048/0.10/ 4	8/256/8/2048/0.15/4	12/128/4/2048/0.10/ 4	10/192/6/2048/0.10/ 4	8/256/8/2048/0.15/4
1		model size, M	2.69	4.91	6.94	2.69	4.91	6.94
2		n_layer	12	10	8	12	10	8
3		n_embd	128	192	256	128	192	256
4		n_head	4	6	8	4	6	8
5		block_size	2 048	2 048	2 048	2 048	2 048	2 048
6		dropout	0.10	0.10	0.15	0.10	0.10	0.15
7		batch_size	4	4	4	4	4	4
8		zero_new_weights	True	True	True	False	False	False
9	Stage 1	model	1/32/1/32/0.00/64/2 00k	1/32/1/32/0.00/64/2 00k	1/32/1/32/0.00/64/2 00k	1/32/1/32/0.00/64/2 00k	1/32/1/32/0.00/64/2 00k	1/32/1/32/0.00/64/2 00k
10		iterations	200 000	200 000	200 000	200 000	200 000	200 000
11		duration / val/train	642 1.2061	629 1.2061	634 1.2061	637 1.2061	634 1.2061	655 1.2061
12		train loss	1.6868	1.6868	1.6868	1.6868	1.6868	1.6868
13		val loss	2.0345	2.0345	2.0345	2.0345	2.0345	2.0345
14	Stage 2	model	2/32/1/64/0.00/64/1 05k	2/64/2/64/0.00/64/6 5k	2/64/2/64/0.00/64/6 5k	2/32/1/64/0.00/64/1 05k	2/64/2/64/0.00/64/6 5k	2/64/2/64/0.00/64/6 5k
15		iterations	105 000	65 000	65 000	105 000	65 000	65 000
16		duration / val/train	528 1.2782	345 1.3267	341 1.3267	533 1.2753	347 1.3518	348 1.3518
17		train loss	1.4653 -13.13%	1.2954 -23.20%	1.2954 -23.20%	1.4616 -13.35%	1.2531 -25.71%	1.2531 -25.71%
18		val loss	1.8730 -7.94%	1.7186 -15.53%	1.7186 -15.53%	1.8640 -8.38%	1.6940 -16.74%	1.6940 -16.74%
19	Stage 3	model	3/64/2/128/0.00/64/ 45k	3/96/3/128/0.05/64/ 85k	3/96/3/128/0.05/64/ 85k	3/64/2/128/0.00/64/ 45k	3/96/3/128/0.05/64/ 105k	3/96/3/128/0.05/64/ 105k
20		iterations	45 000	85 000	85 000	45 000	105 000	105 000
21		duration / val/train	311 1.3815	618 1.4246	621 1.4246	314 1.4290	767 1.4713	767 1.4713
22		train loss	1.2090 -17.49%	1.1249 -13.16%	1.1249 -13.16%	1.1429 -21.80%	1.0329 -17.57%	1.0329 -17.57%
23		val loss	1.6702 -10.83%	1.6025 -6.76%	1.6025 -6.76%	1.6332 -12.38%	1.5197 -10.29%	1.5197 -10.29%
24	Stage 4	model	4/64/2/256/0.00/64/ 95k	4/96/3/256/0.05/64/ 65k	4/128/4/256/0.05/64/ /65k	4/64/2/256/0.00/64/ 135k	4/96/3/256/0.05/64/ 85k	4/128/4/256/0.05/64/ /65k
25		iterations	95 000	65 000	65 000	135 000	85 000	65 000
26		duration	832 1.4350	954 1.4819	1 216 1.5529	1 215 1.5585	1 249 1.5536	1 237 1.6625
27		train loss	1.1284 -6.67%	0.9587 -14.77%	0.8998 -20.01%	0.9810 -14.17%	0.9015 -12.72%	0.8217 -20.45%
28		val loss	1.6192 -3.05%	1.4207 -11.34%	1.3973 -12.80%	1.5289 -6.39%	1.4006 -7.84%	1.3661 -10.11%
29	Stage 5	model	5/64/2/512/0.05/64/ 80k	5/128/4/512/0.05/64/ /60k	5/160/5/512/0.10/32/ /55k	5/64/2/512/0.05/64/ 115k	5/128/4/512/0.05/64/ /75k	5/160/5/512/0.10/32/ /85k

30		<i>iterations</i>	80 000		60 000		55 000		115 000		75 000		85 000	
31		<i>duration / val/train</i>	3 053	1.3759	4 400	1.6217	2 697	1.5570	4 459	1.4539	5 514	1.8069	4 218	1.7939
32		<i>train loss</i>	1.0154	-10.01%	0.7936	-17.22%	0.8099	-9.99%	0.9486	-3.30%	0.7157	-20.61%	0.7056	-14.13%
33		<i>val loss</i>	1.3971	-13.72%	1.2870	-9.41%	1.2610	-9.75%	1.3792	-9.79%	1.2932	-7.67%	1.2658	-7.34%
34	Stage 6	<i>model</i>	6/96/3/512/0.05/64/55k		6/128/4/1024/0.05/32/80k		6/192/6/1024/0.10/16/70k		6/96/3/512/0.05/64/85k		6/128/4/1024/0.05/32/45k		6/192/6/1024/0.10/16/90k	
35		<i>iterations</i>	55 000		80 000		70 000		85 000		45 000		90 000	
36		<i>duration</i>	3 871	1.4747	12 427	1.6899	8 328	1.7177	5 760	1.7207	7 003	1.8452	10 956	2.0767
37		<i>train loss</i>	0.8703	-14.29%	0.7059	-11.05%	0.6801	-16.03%	0.7634	-19.52%	0.6603	-7.74%	0.5661	-19.77%
38		<i>val loss</i>	1.2834	-8.14%	1.1929	-7.31%	1.1682	-7.36%	1.3136	-4.76%	1.2184	-5.78%	1.1756	-7.13%
39	Stage 7	<i>model</i>	7/96/3/1024/0.05/32/65k		7/160/5/1536/0.10/8/85k		7/224/7/1536/0.15/8/50k		7/96/3/1024/0.05/32/75k		7/160/5/1536/0.10/8/60k		7/224/7/1536/0.15/8/50k	
40		<i>iterations</i>	65 000		85 000		50 000		75 000		60 000		50 000	
41		<i>duration / val/train</i>	8 940	1.4993	11 303	1.6438	8 832	1.6914	10 205	1.6847	7 973	1.7765	9 123	1.9995
42		<i>train loss</i>	0.7871	-9.56%	0.6822	-3.36%	0.6488	-4.60%	0.7007	-8.21%	0.6269	-5.06%	0.5492	-2.99%
43		<i>val loss</i>	1.1801	-8.05%	1.1214	-5.99%	1.0974	-6.06%	1.1805	-10.13%	1.1137	-8.59%	1.0981	-6.59%
44	Stage 8	<i>model</i>	8/96/3/1024/0.05/32/65k		8/160/5/1536/0.10/8/40k				8/96/3/1024/0.05/32/70k		8/160/5/1536/0.10/8/40k			
45		<i>iterations</i>	65 000		40 000				70 000		40 000			
46		<i>duration / val/train</i>	10 067	1.5520	6 039	1.7005			10 872	1.7509	5 829	1.8735		
47		<i>train loss</i>	0.7482	-4.94%	0.6547	-4.03%			0.6704	-4.32%	0.5944	-5.18%		
48		<i>val loss</i>	1.1612	-1.60%	1.1133	-0.72%			1.1738	-0.57%	1.1136	-0.01%		
49	Stage 9	<i>model</i>	9/96/3/1536/0.05/16/60k		9/192/6/2048/0.10/4/40k				9/96/3/1536/0.05/16/50k		9/192/6/2048/0.10/4/40k			
50		<i>steps</i>	60 000		40 000				50 000		40 000			
51		<i>duration / val/train</i>	11 544	1.5611	6 838	1.7433			9 592	1.6825	6 802	1.9974		
52		<i>train loss</i>	0.7168	-4.20%	0.6155	-5.99%			0.6668	-0.54%	0.5467	-8.02%		
53		<i>val loss</i>	1.1190	-3.63%	1.0730	-3.62%			1.1219	-4.42%	1.0920	-1.94%		
54	Stage 10	<i>model</i>	10/128/4/1536/0.10/16/0k						10/128/4/1536/0.10/16/0k					
55		<i>iterations</i>												
56		<i>duration / val/train</i>												
57		<i>train loss</i>												
58		<i>val loss</i>												
59	Stage 11	<i>model</i>	11/128/4/2048/0.10/8/0k						11/128/4/2048/0.10/8/0k					

60		<i>iterations</i>							
61		<i>duration / val/train</i>							
62		<i>train loss</i>							
63		<i>val loss</i>							
64		<i>Total time</i>	39 788	43 553	22 669	43 587	36 118	27 304	
65	0	<i>duration / val/train</i>		0 1.0958	0 1.1111		0 0.9571	0 1.1130	
66		<i>train loss</i>		3.0790	3.7049		3.2246	3.4269	
67		<i>val loss</i>		3.3741	4.1166		3.0862	3.8143	
68	2 000	<i>duration / val/train</i>		378 1.6645	388 1.4824		365 1.7323	403 1.5102	
69		<i>train loss / train diff %</i>		0.6606 -78.54%	0.7351 -80.16%		0.6346 -80.32%	0.7757 -77.36%	
70		<i>val loss / val diff %</i>		1.0996 -67.41%	1.0897 -73.53%		1.0993 -64.38%	1.1715 -69.29%	
71	4 000	<i>duration / val/train</i>		759 1.6828	781 1.5263		730 1.8268	807 1.5743	
72		<i>train loss / train diff %</i>		0.6394 -3.21%	0.6987 -4.95%		0.5923 -6.67%	0.6960 -10.27%	
73		<i>val loss / val diff %</i>		1.0760 -2.15%	1.0664 -2.14%		1.0820 -1.57%	1.0957 -6.47%	
74	6 000	<i>duration / val/train</i>		1 140 1.7291	1 181 1.5953		1 095 1.8590	1 214 1.6236	
75		<i>train loss / train diff %</i>		0.6316 -1.22%	0.6740 -3.54%		0.5772 -2.55%	0.6632 -4.71%	
76		<i>val loss / val diff %</i>		1.0921 1.50%	1.0752 0.83%		1.0730 -0.83%	1.0768 -1.72%	
77	8 000	<i>duration / val/train</i>		1 517 1.7351	1 576 1.5799		1 459 1.8841	1 617 1.6642	
78		<i>train loss / train diff %</i>		0.6252 -1.01%	0.6703 -0.55%		0.5677 -1.65%	0.6506 -1.90%	
79		<i>val loss / val diff %</i>		1.0848 -0.67%	1.0590 -1.51%		1.0696 -0.32%	1.0827 0.55%	
80	10 000	<i>duration / val/train</i>		1 888 1.7604	1 969 1.6212		1 825 1.9351	2 014 1.7036	
81		<i>train loss / train diff %</i>		0.6186 -1.06%	0.6584 -1.78%		0.5566 -1.96%	0.6248 -3.97%	
82		<i>val loss / val diff %</i>		1.0890 0.39%	1.0674 0.79%		1.0771 0.70%	1.0644 -1.69%	
83	12 000	<i>duration / val/train</i>		2 260 1.7947	2 362 1.6110		2 200 1.9770	2 411 1.7472	
84		<i>train loss / train diff %</i>		0.6158 -0.45%	0.6583 -0.02%		0.5522 -0.79%	0.6052 -3.14%	
85		<i>val loss / val diff %</i>		1.1052 1.49%	1.0605 -0.65%		1.0917 1.36%	1.0574 -0.66%	
86	14 000	<i>duration / val/train</i>		2 628 1.8108	2 754 1.6543		2 579 2.0136	2 805 1.8170	
87		<i>train loss / train diff %</i>		0.6116 -0.68%	0.6477 -1.61%		0.5443 -1.43%	0.5799 -4.18%	
88		<i>val loss / val diff %</i>		1.1075 0.21%	1.0715 1.04%		1.0960 0.39%	1.0537 -0.35%	
89	16 000	<i>duration / val/train</i>		2 992 1.8304	3 143 1.6481		2 960 2.0662	3 204 1.8531	
90		<i>train loss / train diff %</i>		0.6104 -0.20%	0.6494 0.26%		0.5422 -0.39%	0.5772 -0.47%	
91		<i>val loss / val diff %</i>		1.1173 0.88%	1.0703 -0.11%		1.1203 2.22%	1.0696 1.51%	
92	18 000	<i>duration / val/train</i>		3 356 1.8200	3 530 1.6791		3 339 2.0566	3 605 1.9103	
93		<i>train loss / train diff %</i>		0.6018 -1.41%	0.6391 -1.59%		0.5340 -1.51%	0.5665 -1.85%	

94		val loss / val diff %		1.0953	-1.97%	1.0731	0.26%		1.0982	-1.97%	1.0822	1.18%
95	20 000	duration / val/train		3 720	1.8877	3 918	1.6896		3 733	2.1447	4 001	1.9426
96		train loss / train diff %		0.5957	-1.01%	0.6349	-0.66%		0.5272	-1.27%	0.5556	-1.92%
97		val loss / val diff %		1.1245	2.67%	1.0727	-0.04%		1.1307	2.96%	1.0793	-0.27%
98	22 000	duration / val/train		4 084	1.8689	4 306	1.7057		4 127	2.0894	4 398	1.9802
99		train loss / train diff %		0.6010	0.89%	0.6290	-0.93%		0.5301	0.55%	0.5496	-1.08%
100		val loss / val diff %		1.1232	-0.12%	1.0729	0.02%		1.1076	-2.04%	1.0883	0.83%
101	24 000	duration / val/train		4 448	1.8602	4 694	1.7108		4 497	2.1243	4 800	1.9856
102		train loss / train diff %		0.5960	-0.83%	0.6238	-0.83%		0.5237	-1.21%	0.5431	-1.18%
103		val loss / val diff %		1.1087	-1.29%	1.0672	-0.53%		1.1125	0.44%	1.0784	-0.91%
104	26 000	duration / val/train		4 812	1.8583	5 082	1.7075		4 863	2.1319	5 198	1.9749
105		train loss / train diff %		0.5943	-0.29%	0.6228	-0.16%		0.5223	-0.27%	0.5421	-0.18%
106		val loss / val diff %		1.1044	-0.39%	1.0634	-0.36%		1.1135	0.09%	1.0706	-0.72%
107	28 000	duration / val/train		5 176	1.8820	5 470	1.7427		5 232	2.1629	5 590	2.0395
108		train loss / train diff %		0.5906	-0.62%	0.6160	-1.09%		0.5169	-1.03%	0.5319	-1.88%
109		val loss / val diff %		1.1115	0.64%	1.0735	0.95%		1.1180	0.40%	1.0848	1.33%
110	30 000	duration / val/train		5 540	1.8941	5 858	1.7276		5 600	2.1815	5 980	2.0355
111		train loss / train diff %		0.5885	-0.36%	0.6159	-0.02%		0.5152	-0.33%	0.5303	-0.30%
112		val loss / val diff %		1.1147	0.29%	1.0640	-0.88%		1.1239	0.53%	1.0794	-0.50%
113	32 000	duration / val/train		5 904	1.9194	6 252	1.7491		5 973	2.2167	6 372	2.0551
114		train loss / train diff %		0.5894	0.15%	0.6122	-0.60%		0.5112	-0.78%	0.5262	-0.77%
115		val loss / val diff %		1.1313	1.49%	1.0708	0.64%		1.1332	0.83%	1.0814	0.19%
116	34 000	duration / val/train		6 268	1.8962	6 655	1.7533		6 341	2.1769	6 768	2.0917
117		train loss / train diff %		0.5858	-0.61%	0.6121	-0.02%		0.5111	-0.02%	0.5243	-0.36%
118		val loss / val diff %		1.1108	-1.81%	1.0732	0.22%		1.1126	-1.82%	1.0967	1.41%
119	36 000	duration / val/train		6 632	1.8926	7 048	1.7785		6 706	2.2085	7 170	2.1342
120		train loss / train diff %		0.5839	-0.32%	0.6051	-1.14%		0.5078	-0.65%	0.5164	-1.51%
121		val loss / val diff %		1.1051	-0.51%	1.0762	0.28%		1.1215	0.80%	1.1021	0.49%
122	38 000	duration / val/train		6 995	1.9368	7 454	1.7800		7 071	2.2314	7 571	2.1382
123		train loss / train diff %		0.5791	-0.82%	0.6092	0.68%		0.5022	-1.10%	0.5166	0.04%
124		val loss / val diff %		1.1216	1.49%	1.0844	0.76%		1.1206	-0.08%	1.1046	0.23%
125	40 000	duration / val/train		7 359	1.9301	7 854	1.7953		7 436	2.2508	7 967	2.1197
126		train loss / train diff %		0.5764	-0.47%	0.6024	-1.12%		0.4984	-0.76%	0.5106	-1.16%
127		val loss / val diff %		1.1125	-0.81%	1.0815	-0.27%		1.1218	0.11%	1.0823	-2.02%

128	42 000	duration / val/train		7 723	1.9016	8 246	1.7905		7 800	2.2438	8 366	2.1727
129		train loss / train diff %		0.5771	0.12%	0.6000	-0.40%		0.4996	0.24%	0.5055	-1.00%
130		val loss / val diff %		1.0974	-1.36%	1.0743	-0.67%		1.1210	-0.07%	1.0983	1.48%
131	44 000	duration / val/train		8 087	1.9804	8 642	1.7929		8 165	2.3049	8 763	2.1960
132		train loss / train diff %		0.5720	-0.88%	0.5929	-1.18%		0.4926	-1.40%	0.4999	-1.11%
133		val loss / val diff %		1.1328	3.23%	1.0630	-1.05%		1.1354	1.28%	1.0978	-0.05%
134	46 000	duration / val/train		8 451	1.9572	9 051	1.8233		8 530	2.2959	9 160	2.2260
135		train loss / train diff %		0.5766	0.80%	0.5886	-0.73%		0.4961	0.71%	0.4929	-1.40%
136		val loss / val diff %		1.1285	-0.38%	1.0732	0.96%		1.1390	0.32%	1.0972	-0.05%
137	48 000	duration / val/train		8 815	1.9959	9 447	1.8509		8 895	2.3216	9 567	2.2788
138		train loss / train diff %		0.5636	-2.25%	0.5887	0.02%		0.4851	-2.22%	0.4918	-0.22%
139		val loss / val diff %		1.1249	-0.32%	1.0896	1.53%		1.1262	-1.12%	1.1207	2.14%
140	50 000	duration / val/train		9 179	1.9207	9 853	1.8409		9 260	2.3227	9 964	2.2381
141		train loss / train diff %		0.5713	1.37%	0.5897	0.17%		0.4893	0.87%	0.4931	0.26%
142		val loss / val diff %		1.0973	-2.45%	1.0856	-0.37%		1.1365	0.91%	1.1036	-1.53%
143	52 000	duration / val/train		9 543	1.9927	10 255	1.8399		9 624	2.3658	10 358	2.2654
144		train loss / train diff %		0.5626	-1.52%	0.5877	-0.34%		0.4852	-0.84%	0.4906	-0.51%
145		val loss / val diff %		1.1211	2.17%	1.0813	-0.40%		1.1479	1.00%	1.1114	0.71%
146	54 000	duration / val/train		9 907	1.9889	10 651	1.8771		9 995	2.3336	10 751	2.2918
147		train loss / train diff %		0.5694	1.21%	0.5786	-1.55%		0.4868	0.33%	0.4815	-1.85%
148		val loss / val diff %		1.1325	1.02%	1.0861	0.44%		1.1360	-1.04%	1.1035	-0.71%
149	56 000	duration / val/train		10 271	1.9945	11 059	1.8215		10 375	2.3888	11 145	2.2862
150		train loss / train diff %		0.5650	-0.77%	0.5870	1.45%		0.4822	-0.94%	0.4882	1.39%
151		val loss / val diff %		1.1269	-0.49%	1.0692	-1.56%		1.1519	1.40%	1.1161	1.14%
152	58 000	duration / val/train		10 635	2.0138	11 464	1.8630		10 748	2.3956	11 543	2.3317
153		train loss / train diff %		0.5645	-0.09%	0.5797	-1.24%		0.4813	-0.19%	0.4812	-1.43%
154		val loss / val diff %		1.1368	0.88%	1.0800	1.01%		1.1530	0.10%	1.1220	0.53%
155	60 000	duration / val/train		10 999	1.9917	11 856	1.8796		11 128	2.4021	11 940	2.3294
156		train loss / train diff %		0.5641	-0.07%	0.5812	0.26%		0.4810	-0.06%	0.4806	-0.12%
157		val loss / val diff %		1.1235	-1.17%	1.0924	1.15%		1.1554	0.21%	1.1195	-0.22%
158	62 000	duration / val/train		11 363	1.9989	12 244	1.8572		11 509	2.4232	12 355	2.3789
159		train loss / train diff %		0.5613	-0.50%	0.5790	-0.38%		0.4776	-0.71%	0.4777	-0.60%
160		val loss / val diff %		1.1220	-0.13%	1.0753	-1.57%		1.1573	0.16%	1.1364	1.51%
161	64 000	duration / val/train		11 727	2.0255	12 632	1.8966		11 883	2.4414	12 748	2.3623

162		<i>train loss / train diff %</i>		0.5573	-0.71%	0.5753	-0.64%		0.4740	-0.75%	0.4750	-0.57%
163		<i>val loss / val diff %</i>		1.1288	0.61%	1.0911	1.47%		1.1572	-0.01%	1.1221	-1.26%
164	66 000	<i>duration / val/train</i>		12 090	2.0669	13 019	1.8535		12 264	2.4785	13 140	2.3476
165		<i>train loss / train diff %</i>		0.5603	0.54%	0.5797	0.76%		0.4750	0.21%	0.4764	0.29%
166		<i>val loss / val diff %</i>		1.1581	2.60%	1.0745	-1.52%		1.1773	1.74%	1.1184	-0.33%
167	68 000	<i>duration / val/train</i>		12 454	2.0504	13 407	1.8786		12 671	2.4740	13 552	2.3957
168		<i>train loss / train diff %</i>		0.5500	-1.84%	0.5756	-0.71%		0.4681	-1.45%	0.4728	-0.76%
169		<i>val loss / val diff %</i>		1.1277	-2.62%	1.0813	0.63%		1.1581	-1.63%	1.1327	1.28%
170	70 000	<i>duration / val/train</i>		12 818	2.0360	13 795	1.9305		13 042	2.5028	13 955	2.4283
171		<i>train loss / train diff %</i>		0.5551	0.93%	0.5723	-0.57%		0.4710	0.62%	0.4670	-1.23%
172		<i>val loss / val diff %</i>		1.1302	0.22%	1.1048	2.17%		1.1788	1.79%	1.1340	0.11%
173	72 000	<i>duration / val/train</i>		13 186	2.0317	14 183	1.9169		13 409	2.4988	14 361	2.4556
174		<i>train loss / train diff %</i>		0.5517	-0.61%	0.5695	-0.49%		0.4681	-0.62%	0.4673	0.06%
175		<i>val loss / val diff %</i>		1.1209	-0.82%	1.0917	-1.19%		1.1697	-0.77%	1.1475	1.19%
176	74 000	<i>duration / val/train</i>		13 560	2.0228	14 571	1.8872		13 774	2.4868	14 769	2.4534
177		<i>train loss / train diff %</i>		0.5575	1.05%	0.5664	-0.54%		0.4717	0.77%	0.4605	-1.46%
178		<i>val loss / val diff %</i>		1.1277	0.61%	1.0689	-2.09%		1.1730	0.28%	1.1298	-1.54%
179	76 000	<i>duration / val/train</i>		13 938	2.0444	14 959	1.9202		14 151	2.4772	15 168	2.4482
180		<i>train loss / train diff %</i>		0.5517	-1.04%	0.5666	0.04%		0.4654	-1.34%	0.4610	0.11%
181		<i>val loss / val diff %</i>		1.1279	0.02%	1.0880	1.79%		1.1529	-1.71%	1.1286	-0.11%
182	78 000	<i>duration / val/train</i>		14 316	2.0907	15 347	1.8941		14 546	2.5581	15 559	2.4420
183		<i>train loss / train diff %</i>		0.5490	-0.49%	0.5713	0.83%		0.4630	-0.52%	0.4645	0.76%
184		<i>val loss / val diff %</i>		1.1478	1.76%	1.0821	-0.54%		1.1844	2.73%	1.1343	0.51%
185	80 000	<i>duration / val/train</i>		14 687	2.0665	15 735	1.9367		14 921	2.5271	15 948	2.5190
186		<i>train loss / train diff %</i>		0.5475	-0.27%	0.5626	-1.52%		0.4627	-0.06%	0.4541	-2.24%
187		<i>val loss / val diff %</i>		1.1314	-1.43%	1.0896	0.69%		1.1693	-1.27%	1.1439	0.85%
188	82 000	<i>duration / val/train</i>		15 060	2.0701	16 123	1.9518		15 288	2.5595	16 337	2.5827
189		<i>train loss / train diff %</i>		0.5451	-0.44%	0.5585	-0.73%		0.4586	-0.89%	0.4517	-0.53%
190		<i>val loss / val diff %</i>		1.1284	-0.27%	1.0901	0.05%		1.1738	0.38%	1.1666	1.98%
191	84 000	<i>duration / val/train</i>		15 427	2.0921	16 511	1.9518		15 653	2.5602	16 727	2.5346
192		<i>train loss / train diff %</i>		0.5483	0.59%	0.5596	0.20%		0.4611	0.55%	0.4521	0.09%
193		<i>val loss / val diff %</i>		1.1471	1.66%	1.0922	0.19%		1.1805	0.57%	1.1459	-1.77%
194	86 000	<i>duration / val/train</i>		15 791	2.0839	16 898	1.9446		16 018	2.5646	17 116	2.5567
195		<i>train loss / train diff %</i>		0.5492	0.16%	0.5650	0.96%		0.4614	0.07%	0.4554	0.73%

196		val loss / val diff %		1.1445	-0.23%	1.0987	0.60%		1.1833	0.24%	1.1643	1.61%
197	88 000	duration / val/train		16 155	2.1267	17 286	1.9413		16 383	2.5522	17 505	2.5809
198		train loss / train diff %		0.5477	-0.27%	0.5587	-1.12%		0.4616	0.04%	0.4486	-1.49%
199		val loss / val diff %		1.1648	1.77%	1.0846	-1.28%		1.1781	-0.44%	1.1578	-0.56%
200	90 000	duration / val/train		16 519	2.1186	17 674	1.9398		16 747	2.6177	17 894	2.5373
201		train loss / train diff %		0.5381	-1.75%	0.5627	0.72%		0.4504	-2.43%	0.4530	0.98%
202		val loss / val diff %		1.1400	-2.13%	1.0915	0.64%		1.1790	0.08%	1.1494	-0.73%
203	92 000	duration / val/train		16 883	2.0975	18 062	1.9062		17 112	2.5830	18 283	2.5297
204		train loss / train diff %		0.5455	1.38%	0.5630	0.05%		0.4556	1.15%	0.4531	0.02%
205		val loss / val diff %		1.1442	0.37%	1.0732	-1.68%		1.1768	-0.19%	1.1462	-0.28%
206	94 000	duration / val/train		17 262	2.0864	18 450	1.9460		17 477	2.6181		
207		train loss / train diff %		0.5394	-1.12%	0.5537	-1.65%		0.4501	-1.21%		
208		val loss / val diff %		1.1254	-1.64%	1.0775	0.40%		1.1784	0.14%		
209	96 000	duration / val/train		17 648	2.1139	18 838	1.9228		17 842	2.6482		
210		train loss / train diff %		0.5393	-0.02%	0.5573	0.65%		0.4494	-0.16%		
211		val loss / val diff %		1.1400	1.30%	1.0716	-0.55%		1.1901	0.99%		
212	98 000	duration / val/train		18 036	2.1244	19 227	1.9592		18 206	2.6461		
213		train loss / train diff %		0.5402	0.17%	0.5541	-0.57%		0.4504	0.22%		
214		val loss / val diff %		1.1476	0.67%	1.0856	1.31%		1.1918	0.14%		
215	100 000	duration / val/train		18 418	2.1361	19 628	1.9871		18 571	2.6565		
216		train loss / train diff %		0.5364	-0.70%	0.5500	-0.74%		0.4486	-0.40%		
217		val loss / val diff %		1.1458	-0.16%	1.0929	0.67%		1.1917	-0.01%		
218	102 000	duration / val/train		18 796	2.1112	20 022	1.9658		18 936	2.6381		
219		train loss / train diff %		0.5414	0.93%	0.5521	0.38%		0.4510	0.53%		
220		val loss / val diff %		1.1430	-0.24%	1.0853	-0.70%		1.1898	-0.16%		
221	104 000	duration / val/train		19 184	2.1337	20 421	1.9806		19 300	2.6445		
222		train loss / train diff %		0.5348	-1.22%	0.5517	-0.07%		0.4447	-1.40%		
223		val loss / val diff %		1.1411	-0.17%	1.0927	0.68%		1.1760	-1.16%		
224	106 000	duration / val/train		19 568	2.1323	20 829	1.9640		19 665	2.6281		
225		train loss / train diff %		0.5343	-0.09%	0.5583	1.20%		0.4482	0.79%		
226		val loss / val diff %		1.1393	-0.16%	1.0965	0.35%		1.1779	0.16%		
227	108 000	duration / val/train		19 941	2.1402	21 223	1.9489		20 030	2.6429		
228		train loss / train diff %		0.5372	0.54%	0.5503	-1.43%		0.4483	0.02%		
229		val loss / val diff %		1.1497	0.91%	1.0725	-2.19%		1.1848	0.59%		

230	110 000	<i>duration / val/train</i>		20 326	2.1481	21 614	1.9678		20 395	2.6782	
231		<i>train loss / train diff %</i>		0.5347	-0.47%	0.5551	0.87%		0.4434	-1.09%	
232		<i>val loss / val diff %</i>		1.1486	-0.10%	1.0923	1.85%		1.1875	0.23%	
233	112 000	<i>duration / val/train</i>				22 003	1.9771		20 759	2.7017	
234		<i>train loss / train diff %</i>				0.5495	-1.01%		0.4425	-0.20%	
235		<i>val loss / val diff %</i>				1.0864	-0.54%		1.1955	0.67%	
236	114 000	<i>duration / val/train</i>				22 395	1.9758		21 124	2.7025	
237		<i>train loss / train diff %</i>				0.5492	-0.05%		0.4450	0.56%	
238		<i>val loss / val diff %</i>				1.0851	-0.12%		1.2026	0.59%	
239	116 000	<i>duration / val/train</i>				22 782	2.0143		21 489	2.7141	
240		<i>train loss / train diff %</i>				0.5441	-0.93%		0.4445	-0.11%	
241		<i>val loss / val diff %</i>				1.0960	1.00%		1.2064	0.32%	
242	118 000	<i>duration / val/train</i>				23 170	1.9916		21 854	2.7009	
243		<i>train loss / train diff %</i>				0.5470	0.53%		0.4433	-0.27%	
244		<i>val loss / val diff %</i>				1.0894	-0.60%		1.1973	-0.75%	
245	120 000	<i>duration / val/train</i>				23 558	1.9954		22 218	2.7363	
246		<i>train loss / train diff %</i>				0.5467	-0.05%		0.4399	-0.77%	
247		<i>val loss / val diff %</i>				1.0909	0.14%		1.2037	0.53%	
248	122 000	<i>duration / val/train</i>				23 946	2.0007		22 583	2.7208	
249		<i>train loss / train diff %</i>				0.5448	-0.35%		0.4416	0.39%	
250		<i>val loss / val diff %</i>				1.0900	-0.08%		1.2015	-0.18%	
251	124 000	<i>duration / val/train</i>				24 334	2.0390		22 948	2.7741	
252		<i>train loss / train diff %</i>				0.5408	-0.73%		0.4383	-0.75%	
253		<i>val loss / val diff %</i>				1.1027	1.17%		1.2159	1.20%	
254	126 000	<i>duration / val/train</i>				24 722	2.0163		23 313	2.8014	
255		<i>train loss / train diff %</i>				0.5415	0.13%		0.4365	-0.41%	
256		<i>val loss / val diff %</i>				1.0918	-0.99%		1.2228	0.57%	
257	128 000	<i>duration / val/train</i>				25 110	2.0307		23 677	2.8079	
258		<i>train loss / train diff %</i>				0.5403	-0.22%		0.4341	-0.55%	
259		<i>val loss / val diff %</i>				1.0972	0.49%		1.2189	-0.32%	
260	130 000	<i>duration / val/train</i>				25 503	2.0314		24 042	2.8170	
261		<i>train loss / train diff %</i>				0.5374	-0.54%		0.4311	-0.69%	
262		<i>val loss / val diff %</i>				1.0917	-0.50%		1.2144	-0.37%	
263	132 000	<i>duration / val/train</i>				25 902	1.9946		24 407	2.7689	

264		<i>train loss / train diff %</i>			0.5392	0.33%		0.4332	0.49%	
265		<i>val loss / val diff %</i>			1.0755	-1.48%		1.1995	-1.23%	
266	134 000	<i>duration / val/train</i>			26 306	2.0417		24 772	2.8269	
267		<i>train loss / train diff %</i>			0.5375	-0.32%		0.4304	-0.65%	
268		<i>val loss / val diff %</i>			1.0974	2.04%		1.2167	1.43%	
269	136 000	<i>duration / val/train</i>			26 715	2.0138		25 136	2.7739	
270		<i>train loss / train diff %</i>			0.5380	0.09%		0.4343	0.91%	
271		<i>val loss / val diff %</i>			1.0834	-1.28%		1.2047	-0.99%	
272	138 000	<i>duration / val/train</i>			27 130	2.0020		25 501	2.8010	
273		<i>train loss / train diff %</i>			0.5379	-0.02%		0.4317	-0.60%	
274		<i>val loss / val diff %</i>			1.0769	-0.60%		1.2092	0.37%	
275	140 000	<i>duration / val/train</i>			27 530	2.0332		25 866	2.8149	
276		<i>train loss / train diff %</i>			0.5386	0.13%		0.4365	1.11%	
277		<i>val loss / val diff %</i>			1.0951	1.69%		1.2287	1.61%	
278	142 000	<i>duration / val/train</i>			27 927	2.0584		26 231	2.7873	
279		<i>train loss / train diff %</i>			0.5310	-1.41%		0.4297	-1.56%	
280		<i>val loss / val diff %</i>			1.0930	-0.19%		1.1977	-2.52%	
281	144 000	<i>duration / val/train</i>			28 325	2.0532		26 595	2.8218	
282		<i>train loss / train diff %</i>			0.5339	0.55%		0.4299	0.05%	
283		<i>val loss / val diff %</i>			1.0962	0.29%		1.2131	1.29%	
284	146 000	<i>duration / val/train</i>			28 720	2.0579		26 960	2.8594	
285		<i>train loss / train diff %</i>			0.5340	0.02%		0.4274	-0.58%	
286		<i>val loss / val diff %</i>			1.0989	0.25%		1.2221	0.74%	
287	148 000	<i>duration / val/train</i>			29 110	2.0387		27 325	2.8350	
288		<i>train loss / train diff %</i>			0.5350	0.19%		0.4327	1.24%	
289		<i>val loss / val diff %</i>			1.0907	-0.75%		1.2267	0.38%	
290	150 000	<i>duration / val/train</i>			29 498	2.0573		27 689	2.8331	
291		<i>train loss / train diff %</i>			0.5356	0.11%		0.4266	-1.41%	
292		<i>val loss / val diff %</i>			1.1019	1.03%		1.2086	-1.48%	
293	152 000	<i>duration / val/train</i>			29 889	2.0536		28 054	2.8476	
294		<i>train loss / train diff %</i>			0.5339	-0.32%		0.4225	-0.96%	
295		<i>val loss / val diff %</i>			1.0964	-0.50%		1.2031	-0.46%	
296	154 000	<i>duration / val/train</i>			30 279	2.0670		28 419	2.8780	
297		<i>train loss / train diff %</i>			0.5311	-0.52%		0.4236	0.26%	

298		<i>val loss / val diff %</i>			1.0978	0.13%		1.2191	1.33%	
299	156 000	<i>duration / val/train</i>			30 669	2.0730		28 784	2.8903	
300		<i>train loss / train diff %</i>			0.5327	0.30%		0.4267	0.73%	
301		<i>val loss / val diff %</i>			1.1043	0.59%		1.2333	1.16%	
302	158 000	<i>duration / val/train</i>			31 059	2.0975		29 148	2.8930	
303		<i>train loss / train diff %</i>			0.5272	-1.03%		0.4253	-0.33%	
304		<i>val loss / val diff %</i>			1.1058	0.14%		1.2304	-0.24%	

Table 11. Results of training models at Step 4.

7.4. Analysis of the results of training models in Steps 2- 4

The selected results for each model are shown in Table 12.

No	Checkpoints		Params	12/128/4/ 2048/0.10/ 4	10/192/6/ 2048/0.10 /4	8/256/8/20 48/0.15/4
1			<i>model size, M</i>	2.69	4.91	6.94
2			<i>n_layer</i>	12	10	8
3			<i>n_embd</i>	128	192	256
4			<i>n_head</i>	4	6	8
5			<i>block_size</i>	2 048	2 048	2 048
6			<i>dropout</i>	0.10	0.10	0.15
7			<i>batch_size</i>	4	4	4
8	val loss plateau	Step 1	<i>iterations</i>	96 000	38 000	32 000
9			<i>duration</i>	13 934	7 151	6 441
10			<i>train loss</i>	0.6692	0.6842	0.6914
11			<i>val loss</i>	1.1060	1.0988	1.0867
12		Step 2 zero	<i>iterations</i>	38 000	10 000	14 000
13			<i>duration</i>	13 664	12 342	14 252
14			<i>train loss</i>	0.7626	0.6965	0.7124
15			<i>val loss</i>	1.1131	1.0940	1.0907
16			<i>duration diff %</i>	98.1%	172.6%	221.3%
17		Step 2 normal	<i>iterations</i>	22 000	4 000	12 000
18			<i>duration</i>	11 974	11 226	11 623
19			<i>train loss</i>	0.7393	0.7058	0.6860
20			<i>val loss</i>	1.1101	1.0869	1.1060
21			<i>duration diff %</i>	85.9%	157.0%	180.5%
22		Step 3 zero	<i>iterations</i>	30 000	8 000	8 000
23			<i>duration</i>	22 622	19 740	21 172
24			<i>train loss</i>	0.7696	0.7037	0.7308
25			<i>val loss</i>	1.1042	1.1133	1.0936
26			<i>duration diff %</i>	162.4%	276.0%	328.7%
27		Step 3 normal	<i>iterations</i>	12 000	10 000	18 000
28			<i>duration</i>	21 446	19 269	18 015
29			<i>train loss</i>	0.7648	0.6430	0.6391
30			<i>val loss</i>	1.1138	1.1061	1.0983
31			<i>duration diff %</i>	153.9%	269.5%	279.7%
32		Step 4 zero	<i>iterations</i>		6 000	2 000
33			<i>duration</i>		44 693	23 057
34			<i>train loss</i>		0.6316	0.7351
35			<i>val loss</i>		1.0921	1.0897
36			<i>duration diff %</i>		625.0%	358.0%
37		Step 4 normal	<i>iterations</i>		2 000	6 000

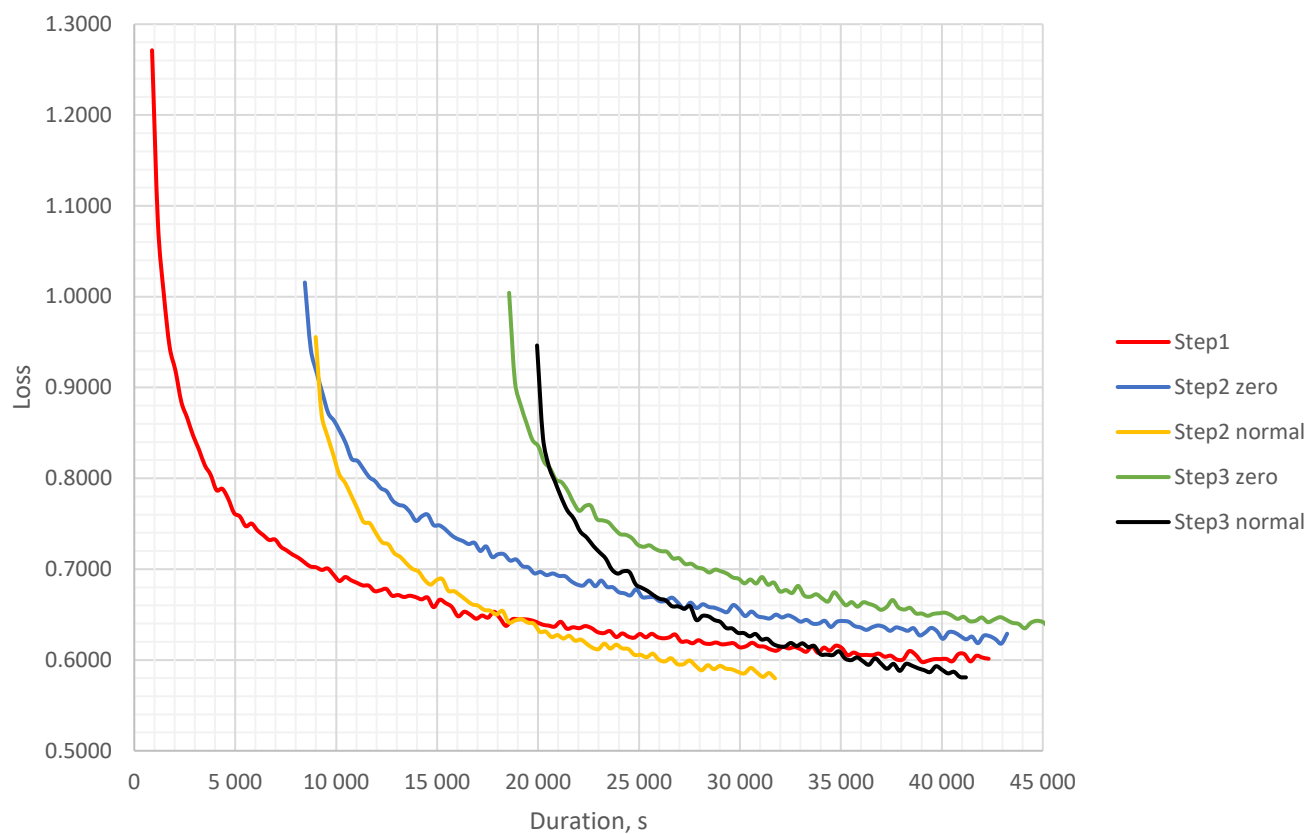
38			<i>duration</i>		36 483	28 518
39			<i>train loss</i>		0.6346	0.6632
40			<i>val loss</i>		1.0993	1.0768
41			<i>duration diff %</i>		510.2%	442.8%
42	min val loss	Step 1	<i>iterations</i>	232 000	76 000	56 000
43			<i>duration</i>	34 159	14 321	11 109
44			<i>train loss</i>	0.6136	0.6078	0.6195
45			<i>val loss</i>	1.0808	1.0877	1.0731
46		Step 2 zero	<i>iterations</i>	220 000	4 000	30 000
47			<i>duration</i>	40 629	11 250	17 365
48			<i>train loss</i>	0.6302	0.7300	0.6506
49			<i>val loss</i>	1.0527	1.0813	1.0658
50			<i>val loss diff %</i>	97.4%	99.4%	99.3%
51		Step 2 normal	<i>iterations</i>	94 000	14 000	30 000
52			<i>duration</i>	22 706	13 063	15 279
53			<i>train loss</i>	0.6133	0.6372	0.5870
54			<i>val loss</i>	1.0902	1.0671	1.0862
55			<i>val loss diff %</i>	100.9%	98.1%	101.2%
56		Step 3 zero	<i>iterations</i>	90 000	30 000	22 000
57			<i>duration</i>	31 386	23 910	23 983
58			<i>train loss</i>	0.6828	0.6119	0.6675
59			<i>val loss</i>	1.0701	1.1028	1.0771
60			<i>val loss diff %</i>	99.0%	101.4%	100.4%
61		Step 3 normal	<i>iterations</i>	90 000	24 000	28 000
62			<i>duration</i>	33 097	21 872	20 044
63			<i>train loss</i>	0.6183	0.5859	0.5905
64			<i>val loss</i>	1.0620	1.1022	1.0824
65			<i>val loss diff %</i>	98.3%	101.3%	100.9%
66		Step 4 zero	<i>iterations</i>		4 000	80 000
67			<i>duration</i>		44 312	24 245
68			<i>train loss</i>		0.6394	0.6703
69			<i>val loss</i>		1.0760	1.0590
70			<i>val loss diff %</i>		98.9%	98.7%
71		Step 4 normal	<i>iterations</i>		8 000	14 000
72			<i>duration</i>		37 577	30 109
73			<i>train loss</i>		0.5677	0.5799
74			<i>val loss</i>		1.0696	1.0537
75			<i>val loss diff %</i>		98.3%	98.2%
76	min train loss	Step 1	<i>iterations</i>	252 000	180 000	196 000
77			<i>duration</i>	37 172	33 606	39 193
78			<i>train loss</i>	0.6035	0.5224	0.5035
79			<i>val loss</i>	1.1132	1.1467	1.1503

80		Step 2 zero	<i>iterations</i>	216 000	124 000	142 000
81			<i>duration</i>	40 050	33 778	39 541
82			<i>train loss</i>	0.6232	0.5339	0.5340
83			<i>val loss</i>	1.0681	1.1299	1.1196
84			<i>duration diff %</i>	107.7%	100.5%	100.9%
85			<i>train loss diff %</i>	103.3%	102.2%	106.1%
86		Step 2 normal	<i>iterations</i>	150 000	106 000	78 000
87			<i>duration</i>	31 141	30 178	24 952
88			<i>train loss</i>	0.5813	0.4744	0.4907
89			<i>val loss</i>	1.1050	1.1530	1.1546
90			<i>duration diff %</i>	83.8%	89.8%	63.7%
91			<i>train loss diff %</i>	96.3%	90.8%	97.5%
92		Step 3 zero	<i>iterations</i>	176 000	140 000	144 000
93			<i>duration</i>	44 107	44 143	47 933
94			<i>train loss</i>	0.6348	0.5229	0.5285
95			<i>val loss</i>	1.0765	1.1387	1.1037
96			<i>duration diff %</i>	118.7%	131.4%	122.3%
97			<i>train loss diff %</i>	105.2%	100.1%	105.0%
98		Step 3 normal	<i>iterations</i>	112 000	50 000	80 000
99			<i>duration</i>	40 898	26 616	30 349
100			<i>train loss</i>	0.5813	0.5216	0.4792
101			<i>val loss</i>	1.0870	1.1717	1.1510
102			<i>duration diff %</i>	110.0%	79.2%	77.4%
103			<i>train loss diff %</i>	96.3%	99.8%	95.2%
104		Step 4 zero	<i>iterations</i>		104 000	142 000
105			<i>duration</i>		62 737	50 596
106			<i>train loss</i>		0.5348	0.5310
107			<i>val loss</i>		1.1411	1.0930
108			<i>duration diff %</i>		186.7%	129.1%
109			<i>train loss diff %</i>		102.4%	105.5%
110		Step 4 normal	<i>iterations</i>		64 000	80 000
111			<i>duration</i>		48 001	43 252
112			<i>train loss</i>		0.4740	0.4541
113			<i>val loss</i>		1.1572	1.1439
114			<i>duration diff %</i>		142.8%	110.4%
115			<i>train loss diff %</i>		90.7%	90.2%

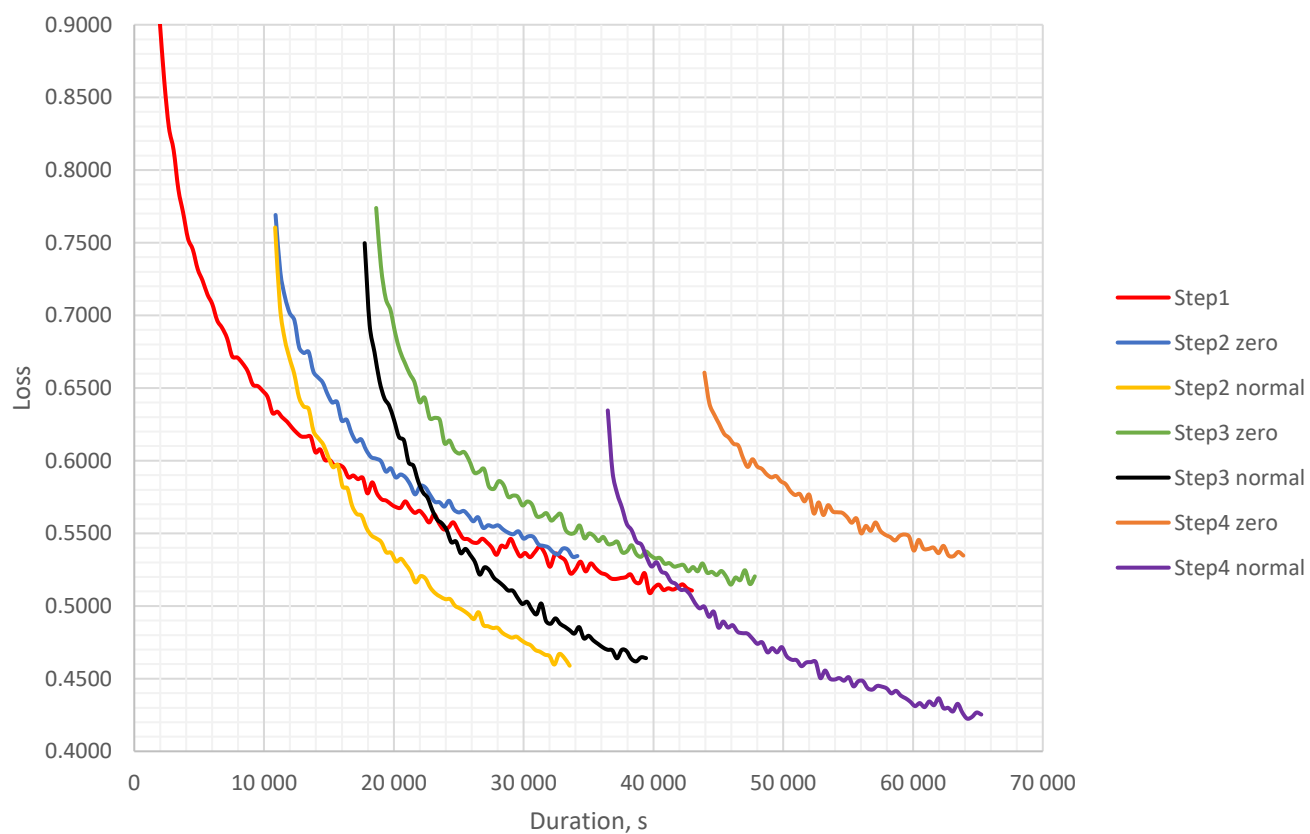
Table 12. Model checkpoints at Steps 2–4

Figure 2 shows the graphs of the change in train loss depending on the duration of model training in Steps 2-4. Figure 3 shows the graphs of the change in val loss depending on the duration of model training in Steps 2-4.

12/128/4/2048/0.10/4 Step 4 train



10/192/6/2048/0.10/4 Step 4 train



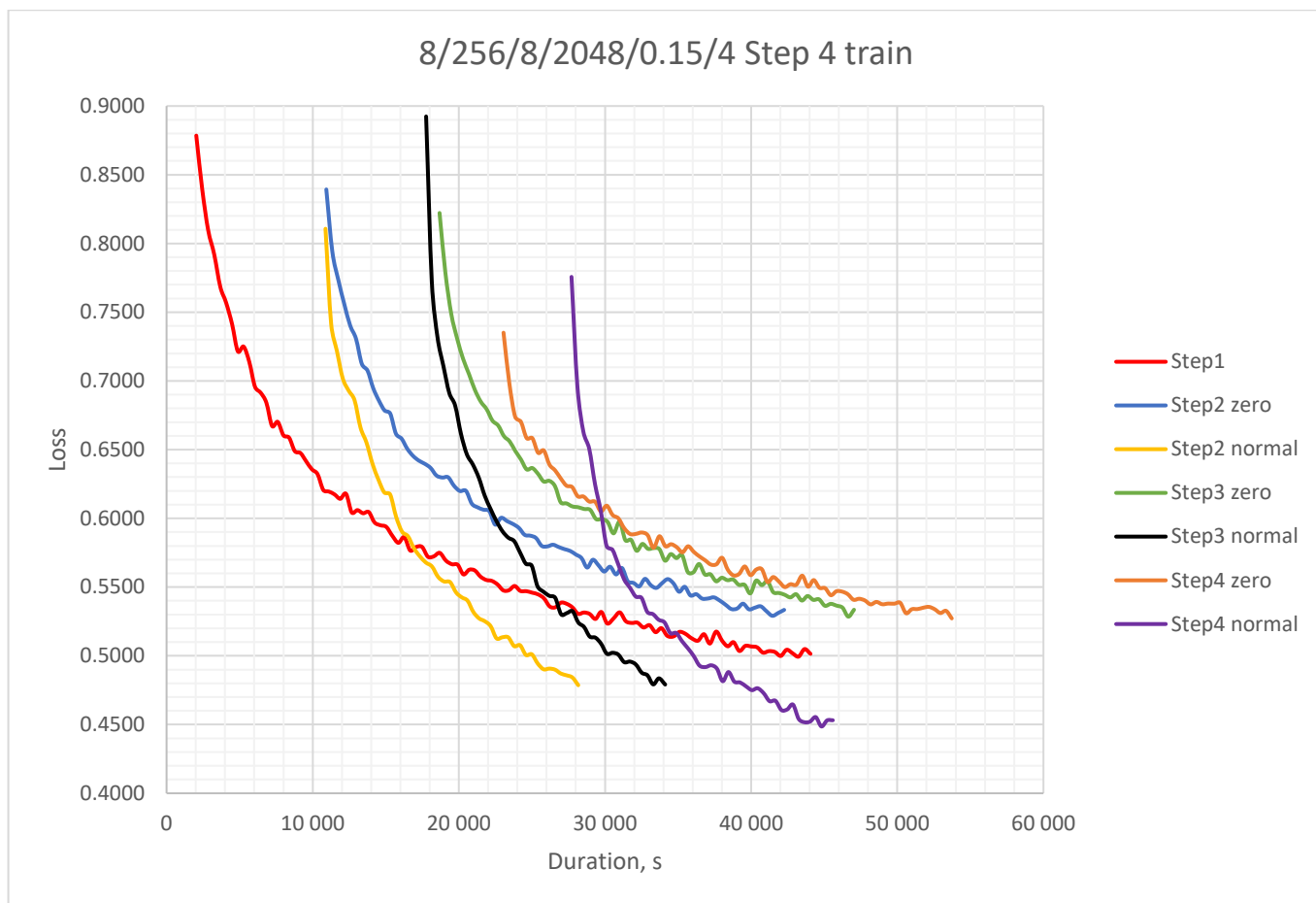
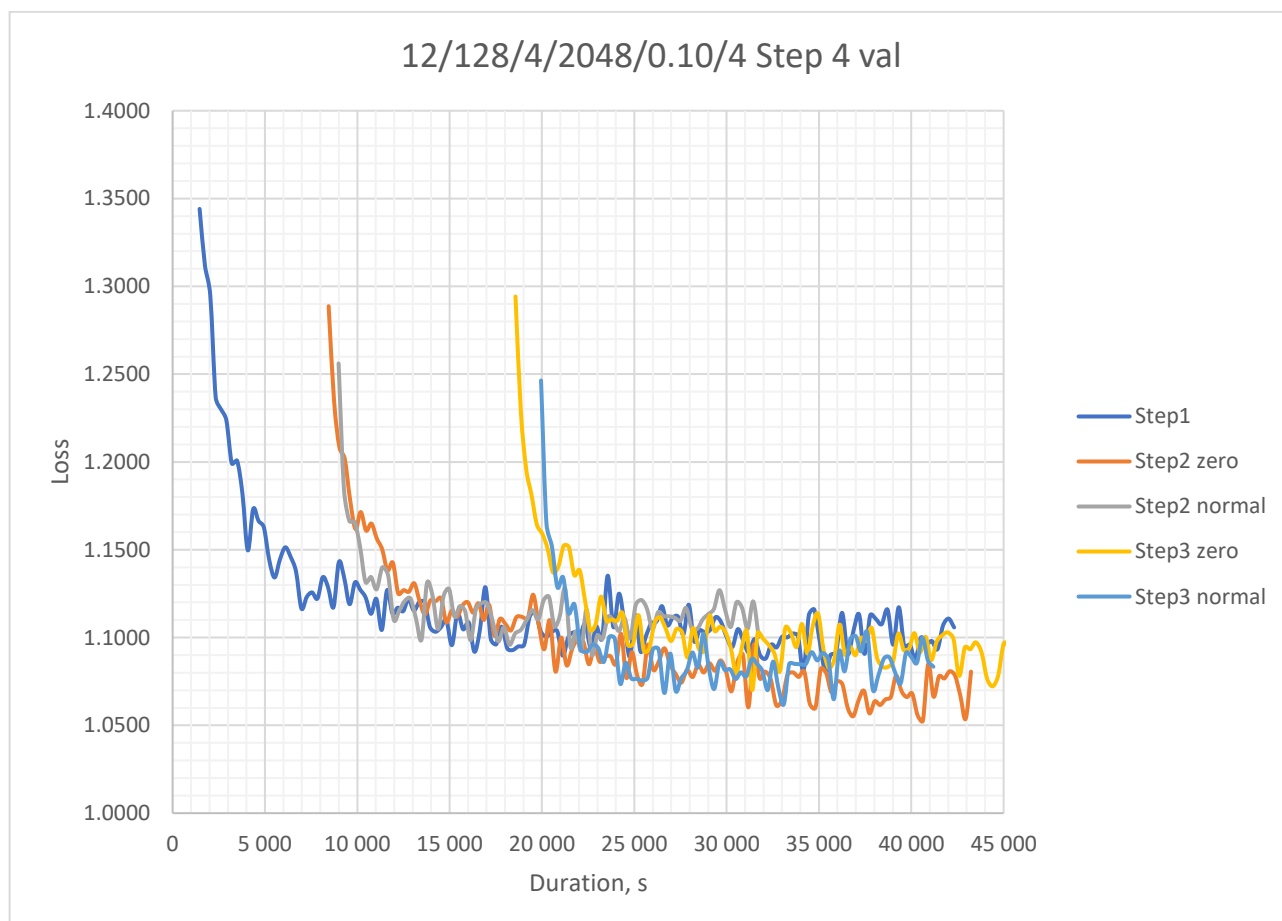


Figure 2. Train loss graphs depending on the training duration of models at Steps 2-4



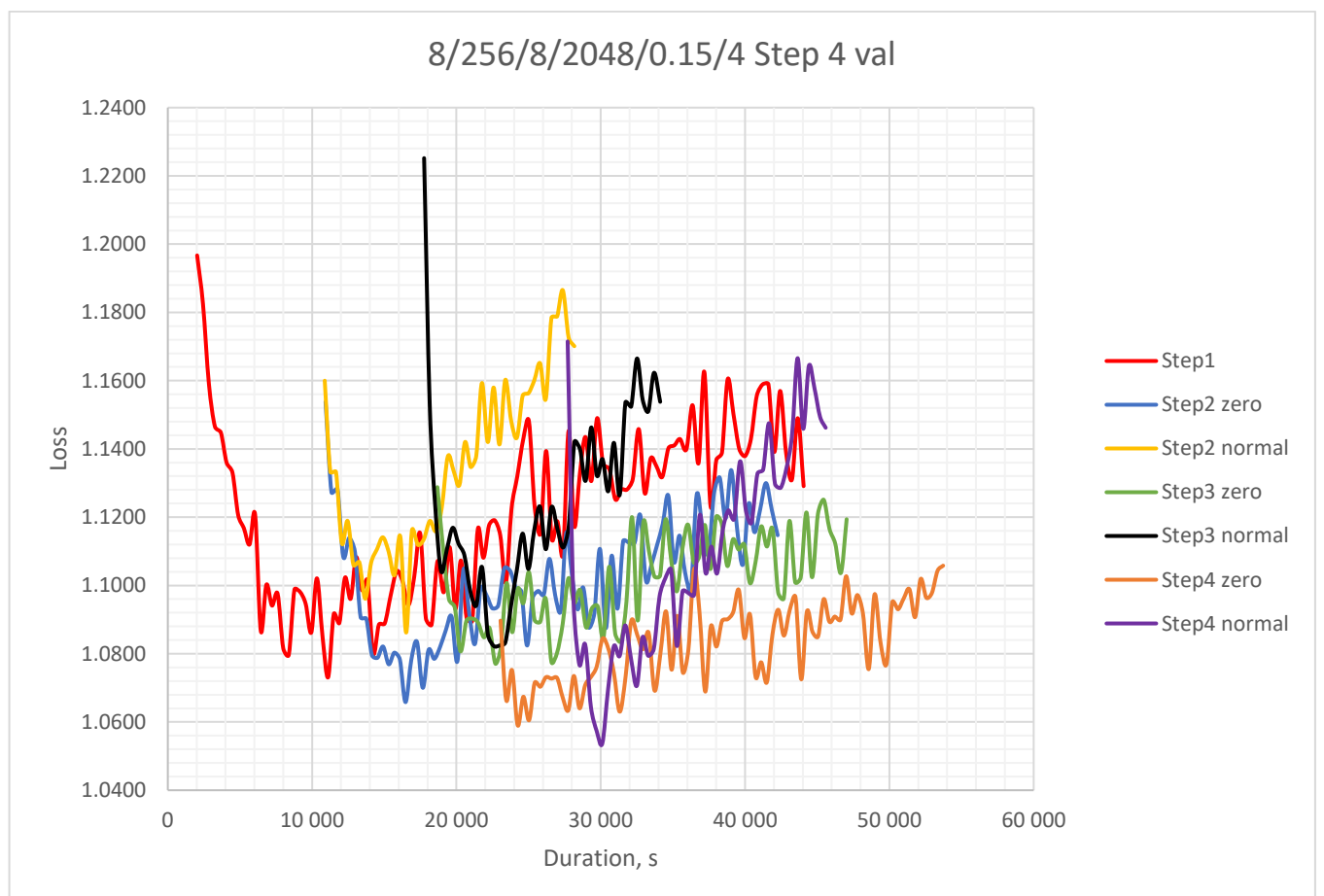
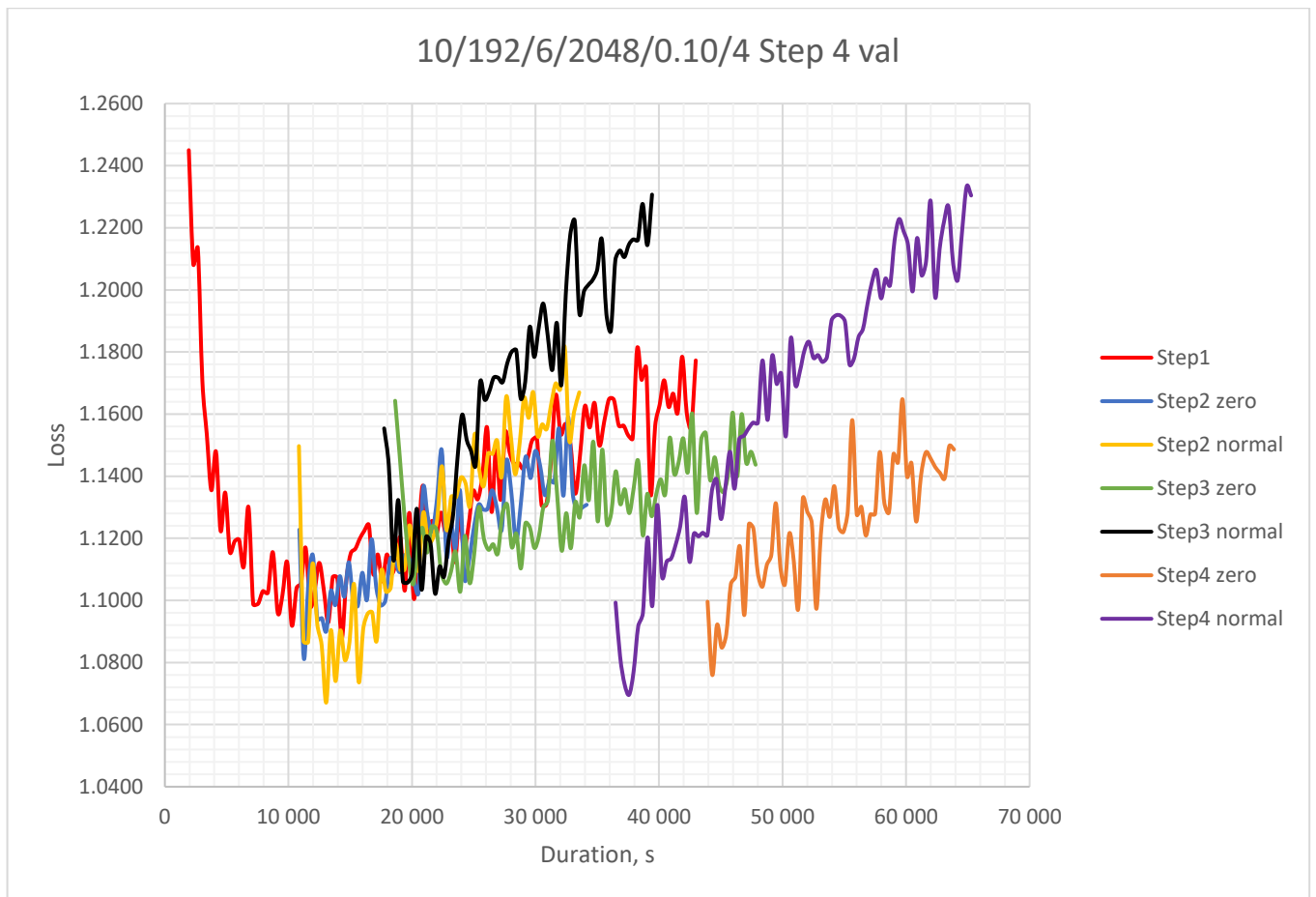


Figure 3. Graphs of val loss depending on the duration of model training at Steps 2-4

The results of steps 2–4 show that growing models by filling new weights with zeros yields worse results than filling new weights with random values. Therefore, as models grow, new weights will be filled with random values.

The idea of training in stages on partial datasets has not yet been confirmed. Step 3 not only fails to produce better results, but actually performs worse than Step 2, which trains the model in stages but on the full dataset. Perhaps the datasets need to be divided differently than in human learning, from simple to complex. However, we will not be testing this division in this study.

Staged training doesn't accelerate plateauing, which is explained by the repeated weight recalculations as the model grows. Staged training can reduce val loss, but the reduction is so small that we won't focus on it. However, the reduction in train loss or the time required to reach the same train loss is significant and can reach 10% or more.

More stages in Step 4 increases the time required to train the model without providing better results.

In the next steps, we'll try to optimize the model's training time. We'll repeat steps 2–4, but we'll interrupt training at intermediate stages not when the train loss plateaus, but when the decline in train loss is no longer sufficiently significant. As soon as the train loss has decreased by less than 1% over the previous 5,000 epochs, training for that stage will be stopped. This may be sufficient for good final results, as the model will continually retrain at the next stage, reducing the training time.

7.5. Step-by-step model training. Step 5.

Models are trained in the same way as in Step 2, but training at intermediate stages is terminated when the change in train loss becomes less than 1% over 5000 epochs. The results of model training in Step 5 are shown in Table 13.

No	iterations	Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1		model size, M	2.69	4.91	6.94
2		n_layer	12	10	8
3		n_embd	128	192	256
4		n_head	4	6	8
5		block_size	2 048	2 048	2 048
6		dropout	0.10	0.10	0.15
7		batch_size	4	4	4
8	Stage 1	model	2/32/1/128/0.00/64/40k	2/64/2/128/0.00/64/50k	2/64/2/128/0.00/64/50k
9		iterations	40 000	50 000	50 000
10		duration / val/train	216 1.2228	267 1.3309	278 1.3309
11		train loss	1.6816	1.3640	1.3640
12		val loss	2.0563	1.8154	1.8154
13	Stage 2	model	4/64/2/256/0.05/64/45k	4/96/3/256/0.05/64/55k	3/96/3/256/0.05/64/45k
14		iterations	45 000	55 000	45 000
15		duration / val/train	435 1.3782	807 1.4910	517 1.4562
16		train loss	1.1784 -29.92%	0.9753 -28.50%	1.0947 -19.74%
17		val loss	1.6241 -21.02%	1.4542 -19.90%	1.5941 -12.19%
18	Stage 3	model	6/64/2/512/0.05/64/30k	6/128/4/512/0.05/64/45k	4/128/4/512/0.05/64/35k
19		iterations	30 000	45 000	35 000
20		duration / val/train	1 430 1.3979	4 169 1.8206	2 080 1.6071
21		train loss	1.0019 -14.98%	0.7342 -24.72%	0.8277 -24.39%
22		val loss	1.4006 -13.76%	1.3367 -8.08%	1.3302 -16.55%

23	Stage 4	model	9/96/3/1024/0.05/32 /45k		8/160/5/1536/0.10/8/50 k		6/192/6/1536/0.10/8/ 65k	
24		iterations	45 000		50 000		65 000	
25		duration / val/train	8 006	1.7418	7 400	1.7193	8 714	1.7914
26		train loss	0.7343	-26.71%	0.6501	-11.45%	0.6370	-23.04%
27		val loss	1.2790	-8.68%	1.1177	-16.38%	1.1411	-14.22%
28		Total time	10 087		12 643		11 589	
29	0	duration / val/train	0	1.0746	0	1.0215	0	1.0784
30		train loss	3.6044		3.8950		3.2241	
31		val loss	3.8732		3.9787		3.4770	
32	2 000	duration / val/train	290	1.3305	374	1.4893	404	1.4333
33		train loss / train diff %	0.9473	-73.72%	0.7592	-80.51%	0.7912	-75.46%
34		val loss / val diff %	1.2604	-67.46%	1.1307	-71.58%	1.1340	-67.39%
35	4 000	duration / val/train	581	1.3786	762	1.5211	820	1.5264
36		train loss / train diff %	0.8614	-9.07%	0.7018	-7.56%	0.7209	-8.89%
37		val loss / val diff %	1.1875	-5.78%	1.0675	-5.59%	1.1004	-2.96%
38	6 000	duration / val/train	871	1.3929	1 151	1.5914	1 241	1.5627
39		train loss / train diff %	0.8382	-2.69%	0.6806	-3.02%	0.6990	-3.04%
40		val loss / val diff %	1.1675	-1.68%	1.0831	1.46%	1.0923	-0.74%
41	8 000	duration / val/train	1 161	1.4284	1 519	1.6187	1 654	1.6048
42		train loss / train diff %	0.8138	-2.91%	0.6677	-1.90%	0.6819	-2.45%
43		val loss / val diff %	1.1624	-0.44%	1.0808	-0.21%	1.0943	0.18%
44	10 000	duration / val/train	1 451	1.4503	1 894	1.6355	2 063	1.6112
45		train loss / train diff %	0.7944	-2.38%	0.6590	-1.30%	0.6726	-1.36%
46		val loss / val diff %	1.1521	-0.89%	1.0778	-0.28%	1.0837	-0.97%
47	12 000	duration / val/train	1 742	1.4593	2 278	1.6661	2 471	1.6262
48		train loss / train diff %	0.7855	-1.12%	0.6407	-2.78%	0.6659	-1.00%
49		val loss / val diff %	1.1463	-0.50%	1.0675	-0.96%	1.0829	-0.07%
50	14 000	duration / val/train	2 032	1.4823	2 646	1.6802	2 875	1.6752
51		train loss / train diff %	0.7723	-1.68%	0.6361	-0.72%	0.6503	-2.34%
52		val loss / val diff %	1.1448	-0.13%	1.0688	0.12%	1.0894	0.60%
53	16 000	duration / val/train	2 322	1.5154	3 012	1.7047	3 268	1.6569
54		train loss / train diff %	0.7571	-1.97%	0.6362	0.02%	0.6450	-0.82%
55		val loss / val diff %	1.1473	0.22%	1.0845	1.47%	1.0687	-1.90%
56	18 000	duration / val/train	2 612	1.5503	3 378	1.7245	3 658	1.7058
57		train loss / train diff %	0.7440	-1.73%	0.6215	-2.31%	0.6309	-2.19%
58		val loss / val diff %	1.1534	0.53%	1.0718	-1.17%	1.0762	0.70%
59	20 000	duration / val/train	2 902	1.5170	3 754	1.7443	4 048	1.7362
60		train loss / train diff %	0.7432	-0.11%	0.6176	-0.63%	0.6225	-1.33%
61		val loss / val diff %	1.1274	-2.25%	1.0773	0.51%	1.0808	0.43%
62	22 000	duration / val/train	3 193	1.5186	4 137	1.7514	4 437	1.7885
63		train loss / train diff %	0.7333	-1.33%	0.6122	-0.87%	0.6117	-1.73%
64		val loss / val diff %	1.1136	-1.22%	1.0722	-0.47%	1.0940	1.22%
65	24 000	duration / val/train	3 483	1.5580	4 520	1.8121	4 827	1.8003
66		train loss / train diff %	0.7218	-1.57%	0.6034	-1.44%	0.6073	-0.72%
67		val loss / val diff %	1.1246	0.99%	1.0934	1.98%	1.0933	-0.06%
68	26 000	duration / val/train	3 773	1.5642	4 900	1.8071	5 216	1.8502
69		train loss / train diff %	0.7203	-0.21%	0.5961	-1.21%	0.5926	-2.42%
70		val loss / val diff %	1.1267	0.19%	1.0772	-1.48%	1.0964	0.28%

71	28 000	duration / val/train	4 063	1.5861	5 277	1.7764	5 606	1.8566
72		train loss / train diff %	0.7103	-1.39%	0.5993	0.54%	0.5830	-1.62%
73		val loss / val diff %	1.1266	-0.01%	1.0646	-1.17%	1.0824	-1.28%
74	30 000	duration / val/train	4 353	1.5810	5 647	1.8519	5 997	1.8547
75		train loss / train diff %	0.7058	-0.63%	0.5835	-2.64%	0.5781	-0.84%
76		val loss / val diff %	1.1159	-0.95%	1.0806	1.50%	1.0722	-0.94%
77	32 000	duration / val/train	4644	1.5907	6 016	1.8588	6 398	1.9085
78		train loss / train diff %	0.6997	-0.86%	0.5823	-0.21%	0.5703	-1.35%
79		val loss / val diff %	1.1130	-0.26%	1.0824	0.17%	1.0884	1.51%
80	34 000	duration / val/train	4 934	1.6148	6 392	1.9037	6 803	1.9265
81		train loss / train diff %	0.6941	-0.80%	0.5732	-1.56%	0.5647	-0.98%
82		val loss / val diff %	1.1208	0.70%	1.0912	0.81%	1.0879	-0.05%
83	36 000	duration / val/train	5 224	1.6183	6 776	1.8901	7 210	1.9485
84		train loss / train diff %	0.6892	-0.71%	0.5671	-1.06%	0.5608	-0.69%
85		val loss / val diff %	1.1153	-0.49%	1.0719	-1.77%	1.0927	0.44%
86	38 000	duration / val/train	5 514	1.6220	7 154	1.9204	7 608	1.9638
87		train loss / train diff %	0.6818	-1.07%	0.5654	-0.30%	0.5573	-0.62%
88		val loss / val diff %	1.1059	-0.84%	1.0858	1.30%	1.0944	0.16%
89	40 000	duration / val/train	5 804	1.6482	7 545	1.9702	8 006	1.9811
90		train loss / train diff %	0.6765	-0.78%	0.5570	-1.49%	0.5489	-1.51%
91		val loss / val diff %	1.1150	0.82%	1.0974	1.07%	1.0874	-0.64%
92	42 000	duration / val/train	6 095	1.6474	7 940	1.9940	8 404	2.0117
93		train loss / train diff %	0.6784	0.28%	0.5522	-0.86%	0.5482	-0.13%
94		val loss / val diff %	1.1176	0.23%	1.1011	0.34%	1.1028	1.42%
95	44 000	duration / val/train	6 385	1.6440	8 328	1.9929	8 797	2.0314
96		train loss / train diff %	0.6794	0.15%	0.5491	-0.56%	0.5476	-0.11%
97		val loss / val diff %	1.1169	-0.06%	1.0943	-0.62%	1.1124	0.87%
98	46 000	duration / val/train	6 675	1.6679	8 695	1.9868	9 187	2.0443
99		train loss / train diff %	0.6697	-1.43%	0.5463	-0.51%	0.5412	-1.17%
100		val loss / val diff %	1.1170	0.01%	1.0854	-0.81%	1.1064	-0.54%
101	48 000	duration / val/train	6 965	1.6594	9 059	2.0464	9 576	2.0663
102		train loss / train diff %	0.6702	0.07%	0.5384	-1.45%	0.5371	-0.76%
103		val loss / val diff %	1.1121	-0.44%	1.1018	1.51%	1.1098	0.31%
104	50 000	duration / val/train	7 255	1.6718	9 424	2.0237	9 972	2.0921
105		train loss / train diff %	0.6645	-0.85%	0.5396	0.22%	0.5340	-0.58%
106		val loss / val diff %	1.1109	-0.11%	1.0920	-0.89%	1.1172	0.67%
107	52 000	duration / val/train	7 546	1.7010	9 789	2.0714	10 371	2.1066
108		train loss / train diff %	0.6591	-0.81%	0.5337	-1.09%	0.5264	-1.42%
109		val loss / val diff %	1.1211	0.92%	1.1055	1.24%	1.1089	-0.74%
110	54 000	duration / val/train	7 836	1.7184	10 154	2.0675	10 772	2.1409
111		train loss / train diff %	0.6569	-0.33%	0.5336	-0.02%	0.5216	-0.91%
112		val loss / val diff %	1.1288	0.69%	1.1032	-0.21%	1.1167	0.70%
113	56 000	duration / val/train	8 126	1.7018	10 522	2.1244	11 175	2.1593
114		train loss / train diff %	0.6557	-0.18%	0.5304	-0.60%	0.5203	-0.25%
115		val loss / val diff %	1.1159	-1.14%	1.1268	2.14%	1.1235	0.61%
116	58 000	duration / val/train	8 416	1.7210	10 899	2.1113	11 568	2.1399
117		train loss / train diff %	0.6508	-0.75%	0.5256	-0.90%	0.5169	-0.65%
118		val loss / val diff %	1.1200	0.37%	1.1097	-1.52%	1.1061	-1.55%
119	60 000	duration / val/train	8 706	1.6900	11 267	2.1755	11 957	2.2101

120		<i>train loss / train diff %</i>	0.6499	-0.14%	0.5169	-1.66%	0.5087	-1.59%
121		<i>val loss / val diff %</i>	1.0983	-1.94%	1.1245	1.33%	1.1243	1.65%
122	62 000	<i>duration / val/train</i>	8 997	1.7308	11 633	2.1541	12 346	2.1967
123		<i>train loss / train diff %</i>	0.6460	-0.60%	0.5219	0.97%	0.5083	-0.08%
124		<i>val loss / val diff %</i>	1.1181	1.80%	1.1242	-0.03%	1.1166	-0.68%
125	64 000	<i>duration / val/train</i>	9 287	1.7075	12 006	2.1575	12 736	2.1953
126		<i>train loss / train diff %</i>	0.6500	0.62%	0.5207	-0.23%	0.5094	0.22%
127		<i>val loss / val diff %</i>	1.1099	-0.73%	1.1234	-0.07%	1.1183	0.15%
128	66 000	<i>duration / val/train</i>	9 577	1.7499	12 375	2.1573	13 125	2.2407
129		<i>train loss / train diff %</i>	0.6358	-2.18%	0.5144	-1.21%	0.5044	-0.98%
130		<i>val loss / val diff %</i>	1.1126	0.24%	1.1097	-1.22%	1.1302	1.06%
131	68 000	<i>duration / val/train</i>	9 867	1.7363	12 742	2.1848	13 518	2.2254
132		<i>train loss / train diff %</i>	0.6408	0.79%	0.5109	-0.68%	0.5023	-0.42%
133		<i>val loss / val diff %</i>	1.1126	0.00%	1.1162	0.59%	1.1178	-1.10%
134	70 000	<i>duration / val/train</i>	10 158	1.7358	13 106	2.2210	13 909	2.2778
135		<i>train loss / train diff %</i>	0.6401	-0.11%	0.5085	-0.47%	0.4971	-1.04%
136		<i>val loss / val diff %</i>	1.1111	-0.13%	1.1294	1.18%	1.1323	1.30%
137	72 000	<i>duration / val/train</i>	10 448	1.7603	13 471	2.2688	14 299	2.2945
138		<i>train loss / train diff %</i>	0.6345	-0.87%	0.5067	-0.35%	0.4948	-0.46%
139		<i>val loss / val diff %</i>	1.1169	0.52%	1.1496	1.79%	1.1353	0.26%
140	74 000	<i>duration / val/train</i>	10 738	1.7606	13 836	2.2113	14 688	2.3122
141		<i>train loss / train diff %</i>	0.6348	0.05%	0.5069	0.04%	0.4898	-1.01%
142		<i>val loss / val diff %</i>	1.1176	0.06%	1.1209	-2.50%	1.1325	-0.25%
143	76 000	<i>duration / val/train</i>	11 028	1.7554	14 201	2.2514	15 083	2.3416
144		<i>train loss / train diff %</i>	0.6285	-0.99%	0.5023	-0.91%	0.4857	-0.84%
145		<i>val loss / val diff %</i>	1.1033	-1.28%	1.1309	0.89%	1.1373	0.42%
146	78 000	<i>duration / val/train</i>	11 319	1.7910	14 572	2.2772	15 490	2.3470
147		<i>train loss / train diff %</i>	0.6272	-0.21%	0.5003	-0.40%	0.4859	0.04%
148		<i>val loss / val diff %</i>	1.1233	1.81%	1.1393	0.74%	1.1404	0.27%
149	80 000	<i>duration / val/train</i>	11 609	1.8029	14 940	2.3091	15 894	2.3270
150		<i>train loss / train diff %</i>	0.6211	-0.97%	0.4986	-0.34%	0.4862	0.06%
151		<i>val loss / val diff %</i>	1.1198	-0.31%	1.1513	1.05%	1.1314	-0.79%
152	82 000	<i>duration / val/train</i>	11 899	1.7658	15 305	2.2886	16 313	2.3944
153		<i>train loss / train diff %</i>	0.6237	0.42%	0.4945	-0.82%	0.4825	-0.76%
154		<i>val loss / val diff %</i>	1.1013	-1.65%	1.1317	-1.70%	1.1553	2.11%
155	84 000	<i>duration / val/train</i>	12 191	1.7969	15 670	2.3333	16 706	2.4093
156		<i>train loss / train diff %</i>	0.6189	-0.77%	0.4918	-0.55%	0.4806	-0.39%
157		<i>val loss / val diff %</i>	1.1121	0.98%	1.1475	1.40%	1.1579	0.23%
158	86 000	<i>duration / val/train</i>	12 481	1.7832	16 035	2.3364	17 097	2.3765
159		<i>train loss / train diff %</i>	0.6227	0.61%	0.4961	0.87%	0.4802	-0.08%
160		<i>val loss / val diff %</i>	1.1104	-0.15%	1.1591	1.01%	1.1412	-1.44%
161	88 000	<i>duration / val/train</i>	12 771	1.7833	16 400	2.3424	17 502	2.3817
162		<i>train loss / train diff %</i>	0.6169	-0.93%	0.4883	-1.57%	0.4752	-1.04%
163		<i>val loss / val diff %</i>	1.1001	-0.93%	1.1438	-1.32%	1.1318	-0.82%
164	90 000	<i>duration / val/train</i>	13 062	1.7692	16 764	2.3648	17 895	2.4204
165		<i>train loss / train diff %</i>	0.6188	0.31%	0.4863	-0.41%	0.4729	-0.48%
166		<i>val loss / val diff %</i>	1.0948	-0.48%	1.1500	0.54%	1.1446	1.13%
167	92 000	<i>duration / val/train</i>	13 352	1.8183	17 129	2.3995	18 285	2.4678
168		<i>train loss / train diff %</i>	0.6147	-0.66%	0.4868	0.10%	0.4667	-1.31%

169		val loss / val diff %	1.1177	2.09%	1.1681	1.57%	1.1517	0.62%
170	94 000	duration / val/train	13 642	1.7856	17 494	2.3859	18 675	2.4475
171		train loss / train diff %	0.6114	-0.54%	0.4849	-0.39%	0.4720	1.14%
172		val loss / val diff %	1.0917	-2.33%	1.1569	-0.96%	1.1552	0.30%
173	96 000	duration / val/train	13 932	1.8182	17 859	2.4151	19 064	2.4756
174		train loss / train diff %	0.6083	-0.51%	0.4828	-0.43%	0.4689	-0.66%
175		val loss / val diff %	1.1060	1.31%	1.1660	0.79%	1.1608	0.48%
176	98 000	duration / val/train	14 223	1.7761	18 228	2.4149	19 453	2.5145
177		train loss / train diff %	0.6160	1.27%	0.4816	-0.25%	0.4626	-1.34%
178		val loss / val diff %	1.0941	-1.08%	1.1630	-0.26%	1.1632	0.21%
179	100 000	duration / val/train	14 513	1.8272	18 610	2.4598	19 843	2.4370
180		train loss / train diff %	0.6101	-0.96%	0.4787	-0.60%	0.4671	0.97%
181		val loss / val diff %	1.1148	1.89%	1.1775	1.25%	1.1383	-2.14%
182	102 000	duration / val/train	14 806	1.7934	18 999	2.4532	20 240	2.5228
183		train loss / train diff %	0.6152	0.84%	0.4810	0.48%	0.4596	-1.61%
184		val loss / val diff %	1.1033	-1.03%	1.1800	0.21%	1.1595	1.86%
185	104 000	duration / val/train	15 104	1.8218	19 381	2.4846	20 650	2.5009
186		train loss / train diff %	0.6079	-1.19%	0.4761	-1.02%	0.4624	0.61%
187		val loss / val diff %	1.1075	0.38%	1.1829	0.25%	1.1564	-0.27%
188	106 000	duration / val/train	15 403	1.8465	19 757	2.4428	21 062	2.5831
189		train loss / train diff %	0.6099	0.33%	0.4767	0.13%	0.4541	-1.79%
190		val loss / val diff %	1.1262	1.69%	1.1645	-1.56%	1.1730	1.44%
191	108 000	duration / val/train	15 704	1.8326	20 138	2.4387		
192		train loss / train diff %	0.6027	-1.18%	0.4757	-0.21%		
193		val loss / val diff %	1.1045	-1.93%	1.1601	-0.38%		
194	110 000	duration / val/train	16 007	1.8562	20 521	2.4820		
195		train loss / train diff %	0.6024	-0.05%	0.4687	-1.47%		
196		val loss / val diff %	1.1182	1.24%	1.1633	0.28%		
197	112 000	duration / val/train	16 304	1.8556	20 897	2.4906		
198		train loss / train diff %	0.6011	-0.22%	0.4696	0.19%		
199		val loss / val diff %	1.1154	-0.25%	1.1696	0.54%		
200	114 000	duration / val/train	16 602	1.8483	21 270	2.5157		
201		train loss / train diff %	0.6050	0.65%	0.4664	-0.68%		
202		val loss / val diff %	1.1182	0.25%	1.1733	0.32%		
203	116 000	duration / val/train	16 905	1.8511	21 649	2.5062		
204		train loss / train diff %	0.5965	-1.40%	0.4674	0.21%		
205		val loss / val diff %	1.1042	-1.25%	1.1714	-0.16%		
206	118 000	duration / val/train	17 198	1.8627				
207		train loss / train diff %	0.5972	0.12%				
208		val loss / val diff %	1.1124	0.74%				
209	120 000	duration / val/train	17 501	1.8726				
210		train loss / train diff %	0.6006	0.57%				
211		val loss / val diff %	1.1247	1.11%				
212	122 000	duration / val/train	17 816	1.8634				
213		train loss / train diff %	0.5923	-1.38%				
214		val loss / val diff %	1.1037	-1.87%				
215	124 000	duration / val/train	18 133	1.8753				
216		train loss / train diff %	0.5917	-0.10%				
217		val loss / val diff %	1.1096	0.53%				

218	126 000	<i>duration / val/train</i>	18 453	1.8738		
219		<i>train loss / train diff %</i>	0.5965	0.81%		
220		<i>val loss / val diff %</i>	1.1177	0.73%		
221	128 000	<i>duration / val/train</i>	18 756	1.8808		
222		<i>train loss / train diff %</i>	0.5914	-0.85%		
223		<i>val loss / val diff %</i>	1.1123	-0.48%		
224	130 000	<i>duration / val/train</i>	19 057	1.9097		
224		<i>train loss / train diff %</i>	0.5867	-0.79%		
226		<i>val loss / val diff %</i>	1.1204	0.73%		
227	132 000	<i>duration / val/train</i>	19 360	1.8616		
228		<i>train loss / train diff %</i>	0.5924	0.97%		
229		<i>val loss / val diff %</i>	1.1028	-1.57%		
230	134 000	<i>duration / val/train</i>	19 661	1.8853		
231		<i>train loss / train diff %</i>	0.5874	-0.84%		
232		<i>val loss / val diff %</i>	1.1074	0.42%		
233	136 000	<i>duration / val/train</i>	19 961	1.8862		
234		<i>train loss / train diff %</i>	0.5924	0.85%		
235		<i>val loss / val diff %</i>	1.1174	0.90%		
236	138 000	<i>duration / val/train</i>	20 259	1.9070		
237		<i>train loss / train diff %</i>	0.5881	-0.73%		
238		<i>val loss / val diff %</i>	1.1215	0.37%		
239	140 000	<i>duration / val/train</i>	20 554	1.9042		
240		<i>train loss / train diff %</i>	0.5874	-0.12%		
241		<i>val loss / val diff %</i>	1.1185	-0.27%		
242	142 000	<i>duration / val/train</i>	20 852	1.9186		
243		<i>train loss / train diff %</i>	0.5851	-0.39%		
244		<i>val loss / val diff %</i>	1.1226	0.37%		
245	144 000	<i>duration / val/train</i>	21 154	1.9079		
246		<i>train loss / train diff %</i>	0.5831	-0.34%		
247		<i>val loss / val diff %</i>	1.1125	-0.90%		
248	146 000	<i>duration / val/train</i>	21 461	1.8971		
249		<i>train loss / train diff %</i>	0.5906	1.29%		
250		<i>val loss / val diff %</i>	1.1204	0.71%		
251	148 000	<i>duration / val/train</i>	21 759	1.9290		
252		<i>train loss / train diff %</i>	0.5842	-1.08%		
253		<i>val loss / val diff %</i>	1.1269	0.58%		
254	150 000	<i>duration / val/train</i>	22 055	1.9298		
255		<i>train loss / train diff %</i>	0.5798	-0.75%		
256		<i>val loss / val diff %</i>	1.1189	-0.71%		

Table 13. Results of training models at Step 5.

7.6. Step-by-step model training. Step 6.

Models are trained in the same way as in Step 3, but training at intermediate stages is terminated when the change in train loss becomes less than 1% over 5000 epochs. The results of model training in Step 6 are shown in Table 14.

<i>No</i>	<i>iterations</i>	<i>Params</i>	<i>12/128/4/2048/0.10/4</i>	<i>10/192/6/2048/0.10/4</i>	<i>8/256/8/2048/0.15/4</i>
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1		<i>model size, M</i>	2.69		4.91		6.94
2		<i>n_layer</i>	12		10		8
3		<i>n_embd</i>	128		192		256
4		<i>n_head</i>	4		6		8
5		<i>block_size</i>	2 048		2 048		2 048
6		<i>dropout</i>	0.10		0.10		0.15
7		<i>batch_size</i>	4		4		4
8	Stage 1	<i>model</i>	2/32/1/128/0.00/64/45k		2/64/2/128/0.00/64/65k		2/64/2/128/0.00/64/65k
9		<i>iterations</i>	45 000		65 000		65 000
10		<i>duration / val/train</i>	236	1.2753	348	1.5100	356 1.5100
11		<i>train loss</i>	1.5947		1.2541		1.2541
12		<i>val loss</i>	2.0337		1.8937		1.8937
13	Stage 2	<i>model</i>	4/64/2/256/0.05/64/40k		4/96/3/256/0.05/64/40k		3/96/3/256/0.05/64/45k
14		<i>iterations</i>	40 000		40 000		45 000
15		<i>duration / val/train</i>	378	1.3978	585	1.7057	505 1.5115
16		<i>train loss</i>	1.1837	-25.77%	0.9696	-22.69%	1.0432 -16.82%
17		<i>val loss</i>	1.6546	-18.64%	1.6538	-12.67%	1.5768 -16.73%
18	Stage 3	<i>model</i>	6/64/2/512/0.05/64/35k		6/128/4/512/0.10/64/55k		4/128/4/512/0.05/64/60k
19		<i>iterations</i>	35 000		55 000		60 000
20		<i>duration / val/train</i>	1 548	1.5407	4 837	2.3066	3 547 2.1253
21		<i>train loss</i>	0.9777	-17.40%	0.6687	-31.03%	0.7087 -32.06%
22		<i>val loss</i>	1.5063	-8.96%	1.5424	-6.74%	1.5062 -4.48%
23	Stage 4	<i>model</i>	9/96/3/1024/0.05/32/45k		8/160/5/1536/0.10/8/60k		6/192/6/1536/0.10/8/65k
24		<i>iterations</i>	45 000		60 000		65 000
25		<i>duration / val/train</i>	7 914	1.8138	8 717	2.0860	8 587 2.1623
26		<i>train loss</i>	0.7207	-26.29%	0.5977	-10.62%	0.5902 -16.72%
27		<i>val loss</i>	1.3072	-13.22%	1.2468	-19.16%	1.2762 -15.27%
28		<i>Total time</i>	10 076		14 487		12 995
29	0	<i>duration / val/train</i>	0	1.0429	0	1.0282	0 0.9917
30		<i>train loss</i>	3.9123		3.7911		3.6187
31		<i>val loss</i>	4.0803		3.8981		3.5885
32	2 000	<i>duration / val/train</i>	291	1.2744	370	1.4890	400 1.4185
33		<i>train loss / train diff %</i>	1.0848	-72.27%	0.7751	-79.55%	0.8266 -77.16%
34		<i>val loss / val diff %</i>	1.3825	-66.12%	1.1541	-70.39%	1.1725 -67.33%
35	4 000	<i>duration / val/train</i>	581	1.3195	742	1.5827	803 1.5268
36		<i>train loss / train diff %</i>	0.9314	-14.14%	0.7182	-7.34%	0.7561 -8.53%
37		<i>val loss / val diff %</i>	1.2290	-11.10%	1.1367	-1.51%	1.1544 -1.54%
38	6 000	<i>duration / val/train</i>	872	1.3142	1 121	1.6067	1 205 1.5467
39		<i>train loss / train diff %</i>	0.8855	-4.93%	0.6913	-3.75%	0.7278 -3.74%
40		<i>val loss / val diff %</i>	1.1637	-5.31%	1.1107	-2.29%	1.1257 -2.49%
41	8 000	<i>duration / val/train</i>	1 162	1.3785	1 495	1.6579	1 600 1.5676
42		<i>train loss / train diff %</i>	0.8594	-2.95%	0.6806	-1.55%	0.7103 -2.40%
43		<i>val loss / val diff %</i>	1.1847	1.80%	1.1284	1.59%	1.1135 -1.08%
44	10 000	<i>duration / val/train</i>	1 453	1.4180	1 867	1.6700	2 009 1.5827
45		<i>train loss / train diff %</i>	0.8371	-2.59%	0.6654	-2.23%	0.6925 -2.51%
46		<i>val loss / val diff %</i>	1.1870	0.19%	1.1112	-1.52%	1.0960 -1.57%

47	12 000	duration / val/train	1 743	1.4154	2 245	1.6801	2 413	1.6142
48		train loss / train diff %	0.8125	-2.94%	0.6568	-1.29%	0.6799	-1.82%
49		val loss / val diff %	1.1500	-3.12%	1.1035	-0.69%	1.0975	0.14%
50	14 000	duration / val/train	2 036	1.4360	2 614	1.7437	2 809	1.6662
51		train loss / train diff %	0.7981	-1.77%	0.6433	-2.06%	0.6623	-2.59%
52		val loss / val diff %	1.1461	-0.34%	1.1217	1.65%	1.1035	0.55%
53	16 000	duration / val/train	2 342	1.4487	2 989	1.7952	3 204	1.6566
54		train loss / train diff %	0.7871	-1.38%	0.6318	-1.79%	0.6518	-1.59%
55		val loss / val diff %	1.1403	-0.51%	1.1342	1.11%	1.0798	-2.15%
56	18 000	duration / val/train	2 642	1.4817	3 371	1.8087	3 609	1.7292
57		train loss / train diff %	0.7774	-1.23%	0.6253	-1.03%	0.6374	-2.21%
58		val loss / val diff %	1.1519	1.02%	1.1310	-0.28%	1.1022	2.07%
59	20 000	duration / val/train	2 946	1.4887	3 739	1.7959	4 011	1.7807
60		train loss / train diff %	0.7632	-1.83%	0.6159	-1.50%	0.6248	-1.98%
61		val loss / val diff %	1.1362	-1.36%	1.1061	-2.20%	1.1126	0.94%
62	22 000	duration / val/train	3 245	1.4933	4 105	1.8297	4 407	1.8065
63		train loss / train diff %	0.7588	-0.58%	0.6094	-1.06%	0.6149	-1.58%
64		val loss / val diff %	1.1331	-0.27%	1.1150	0.80%	1.1108	-0.16%
65	24 000	duration / val/train	3 550	1.5037	4 477	1.8919	4 801	1.8272
66		train loss / train diff %	0.7455	-1.75%	0.5969	-2.05%	0.6081	-1.11%
67		val loss / val diff %	1.1210	-1.07%	1.1293	1.28%	1.1111	0.03%
68	26 000	duration / val/train	3 860	1.5348	4 855	1.8607	5 204	1.8464
69		train loss / train diff %	0.7345	-1.48%	0.5982	0.22%	0.5997	-1.38%
70		val loss / val diff %	1.1273	0.56%	1.1131	-1.43%	1.1073	-0.34%
71	28 000	duration / val/train	4 177	1.5422	5 234	1.9192	5 605	1.8632
72		train loss / train diff %	0.7303	-0.57%	0.5853	-2.16%	0.5885	-1.87%
73		val loss / val diff %	1.1263	-0.09%	1.1233	0.92%	1.0965	-0.98%
74	30 000	duration / val/train	4 484	1.5454	5 602	1.9842	6 006	1.9100
75		train loss / train diff %	0.7246	-0.78%	0.5808	-0.77%	0.5845	-0.68%
76		val loss / val diff %	1.1198	-0.58%	1.1524	2.59%	1.1164	1.81%
77	32 000	duration / val/train	4 788	1.5592	5 967	1.9452	6 403	1.9102
78		train loss / train diff %	0.7178	-0.94%	0.5799	-0.15%	0.5769	-1.30%
79		val loss / val diff %	1.1192	-0.05%	1.1280	-2.12%	1.1020	-1.29%
80	34 000	duration / val/train	5 096	1.5856	6 332	1.9929	6 796	1.9402
81		train loss / train diff %	0.7078	-1.39%	0.5661	-2.38%	0.5707	-1.07%
82		val loss / val diff %	1.1223	0.28%	1.1282	0.02%	1.1073	0.48%
83	36 000	duration / val/train	5 397	1.5813	6 697	2.0138	7 186	1.9657
84		train loss / train diff %	0.7036	-0.59%	0.5666	0.09%	0.5648	-1.03%
85		val loss / val diff %	1.1126	-0.86%	1.1410	1.13%	1.1102	0.26%
86	38 000	duration / val/train	5 695	1.5938	7 062	2.0411	7 576	2.0118
87		train loss / train diff %	0.7043	0.10%	0.5573	-1.64%	0.5576	-1.27%
88		val loss / val diff %	1.1225	0.89%	1.1375	-0.31%	1.1218	1.04%
89	40 000	duration / val/train	5 989	1.6087	7 427	2.0484	7 978	2.0060
90		train loss / train diff %	0.7003	-0.57%	0.5558	-0.27%	0.5502	-1.33%
91		val loss / val diff %	1.1266	0.37%	1.1385	0.09%	1.1037	-1.61%
92	42 000	duration / val/train	6 282	1.6135	7 792	2.0332	8 375	2.0213
93		train loss / train diff %	0.6968	-0.50%	0.5545	-0.23%	0.5481	-0.38%
94		val loss / val diff %	1.1243	-0.20%	1.1274	-0.97%	1.1079	0.38%
95	44 000	duration / val/train	6 575	1.6338	8 161	2.0781	8 774	2.0307

96		<i>train loss / train diff %</i>	0.6901	-0.96%	0.5468	-1.39%	0.5446	-0.64%
97		<i>val loss / val diff %</i>	1.1275	0.28%	1.1363	0.79%	1.1059	-0.18%
98	46 000	<i>duration / val/train</i>	6 880	1.6285	8 539	2.1087	9 176	2.1076
99		<i>train loss / train diff %</i>	0.6846	-0.80%	0.5401	-1.23%	0.5399	-0.86%
100		<i>val loss / val diff %</i>	1.1149	-1.12%	1.1389	0.23%	1.1379	2.89%
101	48 000	<i>duration / val/train</i>	7 173	1.6359	8 915	2.1150	9 583	2.1261
102		<i>train loss / train diff %</i>	0.6847	0.01%	0.5410	0.17%	0.5378	-0.39%
103		<i>val loss / val diff %</i>	1.1201	0.47%	1.1442	0.47%	1.1434	0.48%
104	50 000	<i>duration / val/train</i>	7 465	1.6365	9 295	2.1366	9 985	2.1383
105		<i>train loss / train diff %</i>	0.6767	-1.17%	0.5403	-0.13%	0.5299	-1.47%
106		<i>val loss / val diff %</i>	1.1074	-1.13%	1.1544	0.89%	1.1331	-0.90%
107	52 000	<i>duration / val/train</i>	7 755	1.6310	9 668	2.1927	10 387	2.1754
108		<i>train loss / train diff %</i>	0.6773	0.09%	0.5277	-2.33%	0.5244	-1.04%
109		<i>val loss / val diff %</i>	1.1047	-0.24%	1.1571	0.23%	1.1408	0.68%
110	54 000	<i>duration / val/train</i>	8 045	1.6600	10 053	2.2063	10 789	2.1574
111		<i>train loss / train diff %</i>	0.6691	-1.21%	0.5239	-0.72%	0.5224	-0.38%
112		<i>val loss / val diff %</i>	1.1107	0.54%	1.1559	-0.10%	1.1270	-1.21%
113	56 000	<i>duration / val/train</i>	8 341	1.6381	10 429	2.1855	11 195	2.2167
114		<i>train loss / train diff %</i>	0.6678	-0.19%	0.5300	1.16%	0.5165	-1.13%
115		<i>val loss / val diff %</i>	1.0939	-1.51%	1.1583	0.21%	1.1449	1.59%
116	58 000	<i>duration / val/train</i>	8 643	1.6970	10 814	2.1858	11 607	2.1918
117		<i>train loss / train diff %</i>	0.6585	-1.39%	0.5247	-1.00%	0.5183	0.35%
118		<i>val loss / val diff %</i>	1.1175	2.16%	1.1469	-0.98%	1.1360	-0.78%
119	60 000	<i>duration / val/train</i>	8 943	1.6862	11 194	2.2673	12 010	2.2344
120		<i>train loss / train diff %</i>	0.6600	0.23%	0.5182	-1.24%	0.5103	-1.54%
121		<i>val loss / val diff %</i>	1.1129	-0.41%	1.1749	2.44%	1.1402	0.37%
122	62 000	<i>duration / val/train</i>	9 239	1.6686	11 571	2.2680	12 413	2.2486
123		<i>train loss / train diff %</i>	0.6627	0.41%	0.5165	-0.33%	0.5128	0.49%
124		<i>val loss / val diff %</i>	1.1058	-0.64%	1.1714	-0.30%	1.1531	1.13%
125	64 000	<i>duration / val/train</i>	9 532	1.6720	11 950	2.2643	12 807	2.2523
126		<i>train loss / train diff %</i>	0.6494	-2.01%	0.5157	-0.15%	0.5085	-0.84%
127		<i>val loss / val diff %</i>	1.0858	-1.81%	1.1677	-0.32%	1.1453	-0.68%
128	66 000	<i>duration / val/train</i>	9 827	1.6808	12 326	2.3189	13 196	2.2459
129		<i>train loss / train diff %</i>	0.6504	0.15%	0.5095	-1.20%	0.5055	-0.59%
130		<i>val loss / val diff %</i>	1.0932	0.68%	1.1815	1.18%	1.1353	-0.87%
131	68 000	<i>duration / val/train</i>	10 131	1.6894	12 694	2.2731	13 586	2.2855
132		<i>train loss / train diff %</i>	0.6548	0.68%	0.5101	0.12%	0.5033	-0.44%
133		<i>val loss / val diff %</i>	1.1062	1.19%	1.1595	-1.86%	1.1503	1.32%
134	70 000	<i>duration / val/train</i>	10 424	1.7096	13 059	2.3205	13 975	2.3188
135		<i>train loss / train diff %</i>	0.6487	-0.93%	0.5079	-0.43%	0.4938	-1.89%
136		<i>val loss / val diff %</i>	1.1090	0.25%	1.1786	1.65%	1.1450	-0.46%
137	72 000	<i>duration / val/train</i>	10 731	1.6803	13 424	2.3786	14 365	2.3031
138		<i>train loss / train diff %</i>	0.6499	0.18%	0.5010	-1.36%	0.5002	1.30%
139		<i>val loss / val diff %</i>	1.0920	-1.53%	1.1917	1.11%	1.1520	0.61%
140	74 000	<i>duration / val/train</i>	11 048	1.7060	13 789	2.3232	14 754	2.4029
141		<i>train loss / train diff %</i>	0.6418	-1.25%	0.4990	-0.40%	0.4882	-2.40%
142		<i>val loss / val diff %</i>	1.0949	0.27%	1.1593	-2.72%	1.1731	1.83%
143	76 000	<i>duration / val/train</i>	11 345	1.7363	14 154	2.3437	15 144	2.3785
144		<i>train loss / train diff %</i>	0.6401	-0.26%	0.5019	0.58%	0.4874	-0.16%

145		val loss / val diff %	1.1114	1.51%	1.1763	1.47%	1.1593	-1.18%
146	78 000	duration / val/train	11 653	1.6867	14 519	2.3722	15 534	2.4643
147		train loss / train diff %	0.6445	0.69%	0.4952	-1.33%	0.4816	-1.19%
148		val loss / val diff %	1.0871	-2.19%	1.1747	-0.14%	1.1868	2.37%
149	80 000	duration / val/train	11 951	1.6986	14 884	2.4043	15 923	2.3962
150		train loss / train diff %	0.6454	0.14%	0.4981	0.59%	0.4843	0.56%
151		val loss / val diff %	1.0963	0.85%	1.1976	1.95%	1.1605	-2.22%
152	82 000	duration / val/train	12 244	1.7137			16 313	2.3896
153		train loss / train diff %	0.6419	-0.54%			0.4823	-0.41%
154		val loss / val diff %	1.1000	0.34%			1.1525	-0.69%
155	84 000	duration / val/train	12 538	1.7190			16 702	2.4262
156		train loss / train diff %	0.6357	-0.97%			0.4812	-0.23%
157		val loss / val diff %	1.0928	-0.65%			1.1675	1.30%
158	86 000	duration / val/train	12 831	1.7306			17 092	2.4418
159		train loss / train diff %	0.6318	-0.61%			0.4794	-0.37%
160		val loss / val diff %	1.0934	0.05%			1.1706	0.27%
161	88 000	duration / val/train	13 122	1.7317			17 482	2.4988
162		train loss / train diff %	0.6336	0.28%			0.4725	-1.44%
163		val loss / val diff %	1.0972	0.35%			1.1807	0.86%
164	90 000	duration / val/train	13 412	1.7625			17 871	2.5015
165		train loss / train diff %	0.6317	-0.30%			0.4736	0.23%
166		val loss / val diff %	1.1134	1.48%			1.1847	0.34%
167	92 000	duration / val/train	13 730	1.7372			18 261	2.4943
168		train loss / train diff %	0.6313	-0.06%			0.4700	-0.76%
169		val loss / val diff %	1.0967	-1.50%			1.1723	-1.05%
170	94 000	duration / val/train	14 023	1.7483			18 650	2.5258
171		train loss / train diff %	0.6310	-0.05%			0.4690	-0.21%
172		val loss / val diff %	1.1032	0.59%			1.1846	1.05%
173	96 000	duration / val/train	14 316	1.7784			19 040	2.5391
174		train loss / train diff %	0.6218	-1.46%			0.4626	-1.36%
175		val loss / val diff %	1.1058	0.24%			1.1746	-0.84%
176	98 000	duration / val/train	14 616	1.7840			19 430	2.5559
177		train loss / train diff %	0.6226	0.13%			0.4591	-0.76%
178		val loss / val diff %	1.1107	0.44%			1.1734	-0.10%
179	100 000	duration / val/train	14 918	1.7692			19 819	2.5566
180		train loss / train diff %	0.6204	-0.35%			0.4644	1.15%
181		val loss / val diff %	1.0976	-1.18%			1.1873	1.18%
182	102 000	duration / val/train	15 225	1.7809			20 209	2.6257
183		train loss / train diff %	0.6216	0.19%			0.4550	-2.02%
184		val loss / val diff %	1.1070	0.86%			1.1947	0.62%
185	104 000	duration / val/train	15 524	1.7650			20 598	2.5656
186		train loss / train diff %	0.6204	-0.19%			0.4597	1.03%
187		val loss / val diff %	1.0950	-1.08%			1.1794	-1.28%
188	106 000	duration / val/train	15 818	1.7937			20 988	2.6117
189		train loss / train diff %	0.6209	0.08%			0.4533	-1.39%
190		val loss / val diff %	1.1137	1.71%			1.1839	0.38%
191	108 000	duration / val/train	16 111	1.7815			21 378	2.6345
192		train loss / train diff %	0.6216	0.11%			0.4534	0.02%
193		val loss / val diff %	1.1074	-0.57%			1.1945	0.90%

194	110 000	<i>duration / val/train</i>	16 401	1.7588		21 767	2.6257
195		<i>train loss / train diff %</i>	0.6219	0.05%		0.4513	-0.46%
196		<i>val loss / val diff %</i>	1.0938	-1.23%		1.1850	-0.80%
197	112 000	<i>duration / val/train</i>	16 692	1.7957		22 157	2.6761
198		<i>train loss / train diff %</i>	0.6123	-1.54%		0.4504	-0.20%
199		<i>val loss / val diff %</i>	1.0995	0.52%		1.2053	1.71%
200	114 000	<i>duration / val/train</i>	16 982	1.7974		22 546	2.6618
201		<i>train loss / train diff %</i>	0.6134	0.18%		0.4474	-0.67%
202		<i>val loss / val diff %</i>	1.1025	0.27%		1.1909	-1.19%
203	116 000	<i>duration / val/train</i>	17 273	1.7906		22 936	2.6763
204		<i>train loss / train diff %</i>	0.6150	0.26%		0.4477	0.07%
205		<i>val loss / val diff %</i>	1.1012	-0.12%		1.1982	0.61%
206	118 000	<i>duration / val/train</i>	17 563	1.7786		23 326	2.7029
207		<i>train loss / train diff %</i>	0.6143	-0.11%		0.4429	-1.07%
208		<i>val loss / val diff %</i>	1.0926	-0.78%		1.1971	-0.09%
209	120 000	<i>duration / val/train</i>	17 854	1.8292		23 715	2.7180
210		<i>train loss / train diff %</i>	0.6111	-0.52%		0.4479	1.13%
211		<i>val loss / val diff %</i>	1.1178	2.31%		1.2174	1.70%
212	122 000	<i>duration / val/train</i>	18 144	1.8000		24 105	2.7166
213		<i>train loss / train diff %</i>	0.6129	0.29%		0.4488	0.20%
214		<i>val loss / val diff %</i>	1.1032	-1.31%		1.2192	0.15%
215	124 000	<i>duration / val/train</i>	18 435	1.7915		24 494	2.7023
216		<i>train loss / train diff %</i>	0.6116	-0.21%		0.4468	-0.45%
217		<i>val loss / val diff %</i>	1.0957	-0.68%		1.2074	-0.97%
218	126 000	<i>duration / val/train</i>	18 725	1.8020		24 884	2.7031
219		<i>train loss / train diff %</i>	0.6106	-0.16%		0.4425	-0.96%
220		<i>val loss / val diff %</i>	1.1003	0.42%		1.1961	-0.94%
221	128 000	<i>duration / val/train</i>	19 017	1.8309		25 274	2.7579
222		<i>train loss / train diff %</i>	0.6079	-0.44%		0.4420	-0.11%
223		<i>val loss / val diff %</i>	1.1130	1.15%		1.2190	1.91%
224	130 000	<i>duration / val/train</i>	19 316	1.8384			
225		<i>train loss / train diff %</i>	0.6053	-0.43%			
226		<i>val loss / val diff %</i>	1.1128	-0.02%			
227	132 000	<i>duration / val/train</i>	19 614	1.8330			
228		<i>train loss / train diff %</i>	0.6071	0.30%			
229		<i>val loss / val diff %</i>	1.1128	0.00%			
230	134 000	<i>duration / val/train</i>	19 907	1.8358			
231		<i>train loss / train diff %</i>	0.6046	-0.41%			
232		<i>val loss / val diff %</i>	1.1099	-0.26%			
233	136 000	<i>duration / val/train</i>	20 197	1.8199			
234		<i>train loss / train diff %</i>	0.6035	-0.18%			
235		<i>val loss / val diff %</i>	1.0983	-1.05%			
236	138 000	<i>duration / val/train</i>	20 488	1.8619			
237		<i>train loss / train diff %</i>	0.5987	-0.80%			
238		<i>val loss / val diff %</i>	1.1147	1.49%			
239	140 000	<i>duration / val/train</i>	20 788	1.8392			
240		<i>train loss / train diff %</i>	0.6057	1.17%			
241		<i>val loss / val diff %</i>	1.1140	-0.06%			
242	142 000	<i>duration / val/train</i>	21 097	1.8237			

243		<i>train loss / train diff %</i>	0.6040	-0.28%		
244		<i>val loss / val diff %</i>	1.1015	-1.12%		
245	144 000	<i>duration / val/train</i>	21 407	1.8679		
246		<i>train loss / train diff %</i>	0.5981	-0.98%		
247		<i>val loss / val diff %</i>	1.1172	1.43%		
248	146 000	<i>duration / val/train</i>	21 721	1.8535		
249		<i>train loss / train diff %</i>	0.5973	-0.13%		
250		<i>val loss / val diff %</i>	1.1071	-0.90%		
251	148 000	<i>duration / val/train</i>	22 019	1.8420		
252		<i>train loss / train diff %</i>	0.5999	0.44%		
253		<i>val loss / val diff %</i>	1.1050	-0.19%		
254	150 000	<i>duration / val/train</i>	22 328	1.8658		
255		<i>train loss / train diff %</i>	0.5963	-0.60%		
256		<i>val loss / val diff %</i>	1.1126	0.69%		
257	152 000	<i>duration / val/train</i>	22 645	1.8304		
258		<i>train loss / train diff %</i>	0.6057	1.58%		
259		<i>val loss / val diff %</i>	1.1087	-0.35%		
260	154 000	<i>duration / val/train</i>	22 956	1.8378		
261		<i>train loss / train diff %</i>	0.5975	-1.35%		
262		<i>val loss / val diff %</i>	1.0981	-0.96%		
263	156 000	<i>duration / val/train</i>	23 263	1.8508		
264		<i>train loss / train diff %</i>	0.5992	0.28%		
265		<i>val loss / val diff %</i>	1.1090	0.99%		
266	158 000	<i>duration / val/train</i>	23 574	1.8707		
267		<i>train loss / train diff %</i>	0.5907	-1.42%		
268		<i>val loss / val diff %</i>	1.1050	-0.36%		
269	160 000	<i>duration / val/train</i>	23 888	1.8803		
270		<i>train loss / train diff %</i>	0.5938	0.52%		
271		<i>val loss / val diff %</i>	1.1165	1.04%		
272	162 000	<i>duration / val/train</i>	24 187	1.8676		
273		<i>train loss / train diff %</i>	0.5975	0.62%		
274		<i>val loss / val diff %</i>	1.1159	-0.05%		
275	164 000	<i>duration / val/train</i>	24 480	1.8639		
276		<i>train loss / train diff %</i>	0.5930	-0.75%		
277		<i>val loss / val diff %</i>	1.1053	-0.95%		

Table 14. Results of training models at Step 6.

7.7. Step-by-step model training. Step 7.

Models are trained in the same way as in Step 4, but training at intermediate stages is terminated when the change in train loss becomes less than 1% over 5000 epochs. The results of model training in Step 7 are shown in Table 15.

No	iterations	Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1		<i>model size, M</i>	2.69	4.91	6.94
2		<i>n_layer</i>	12	10	8
3		<i>n_embd</i>	128	192	256
4		<i>n_head</i>	4	6	8

5		<i>block_size</i>	2 048	2 048	2 048
6		<i>dropout</i>	0.10	0.10	0.15
7		<i>batch_size</i>	4	4	4
8	Stage 1	<i>model</i>	1/32/1/32/0.00/64/35 k	1/32/1/32/0.00/64/35 k	1/32/1/32/0.00/64/35 k
9		<i>iterations</i>	35 000	35 000	35 000
10		<i>duration / val/train</i>	112 1.1874	111 1.1874	111 1.1874
11		<i>train loss</i>	1.8340	1.8340	1.8340
12		<i>val loss</i>	2.1776	2.1776	2.1776
13	Stage 2	<i>model</i>	2/32/1/64/0.00/64/30 k	2/64/2/64/0.00/64/35 k	2/64/2/64/0.00/64/35 k
14		<i>iterations</i>	30 000	35 000	35 000
15		<i>duration / val/train</i>	153 1.2550	184 1.3452	182 1.3452
16		<i>train loss</i>	1.5615 -14.86%	1.3000 -29.12%	1.3000 -29.12%
17		<i>val loss</i>	1.9597 -10.01%	1.7487 -19.70%	1.7487 -19.70%
18	Stage 3	<i>model</i>	3/64/2/128/0.00/64/3 5k	3/96/3/128/0.05/64/4 0k	3/96/3/128/0.05/64/4 0k
19		<i>iterations</i>	35 000	40 000	40 000
20		<i>duration / val/train</i>	244 1.4254	293 1.4238	293 1.4238
21		<i>train loss</i>	1.1685 -25.17%	1.0974 -15.58%	1.0974 -15.58%
22		<i>val loss</i>	1.6656 -15.01%	1.5625 -10.65%	1.5625 -10.65%
23	Stage 4	<i>model</i>	4/64/2/256/0.00/64/5 0k	4/96/3/256/0.05/64/3 0k	4/128/4/256/0.05/64/ 40k
24		<i>iterations</i>	50 000	30 000	40 000
25		<i>duration</i>	454 1.4538	438 1.5048	777 1.6145
26		<i>train loss</i>	1.0909 -6.64%	0.9504 -13.40%	0.8518 -22.38%
27		<i>val loss</i>	1.5859 -4.79%	1.4302 -8.47%	1.3752 -11.99%
28	Stage 5	<i>model</i>	5/64/2/512/0.05/64/4 5k	5/128/4/512/0.05/64/ 35k	5/160/5/512/0.10/32/ 35k
29		<i>iterations</i>	45 000	35 000	35 000
30		<i>duration / val/train</i>	1 739 1.4191	2 623 1.7005	1 777 1.7139
31		<i>train loss</i>	1.0054 -7.84%	0.7503 -21.05%	0.7507 -11.87%
32		<i>val loss</i>	1.4268 -10.03%	1.2759 -10.79%	1.2866 -6.44%
33	Stage 6	<i>model</i>	6/96/3/512/0.05/64/4 0k	6/128/4/1024/0.05/32 /35k	6/192/6/1024/0.10/16 /45k
34		<i>iterations</i>	40 000	35 000	45 000
35		<i>duration</i>	2 752 1.5937	5 439 1.7753	5 531 1.9412
36		<i>train loss</i>	0.8103 -19.41%	0.6733 -10.26%	0.6107 -18.65%
37		<i>val loss</i>	1.2914 -9.49%	1.1953 -6.32%	1.1855 -7.86%
38	Stage 7	<i>model</i>	7/96/3/1024/0.05/32/ 25k	7/160/5/1536/0.10/8/ 40k	7/224/7/1536/0.15/8/ 45k
39		<i>iterations</i>	25 000	40 000	45 000
40		<i>duration / val/train</i>	3 402 1.6065	5 100 1.7061	8 165 1.9705
41		<i>train loss</i>	0.7579 -6.47%	0.6542 -2.84%	0.5698 -6.70%
42		<i>val loss</i>	1.2176 -5.71%	1.1161 -6.63%	1.1228 -5.29%
43	Stage 8	<i>model</i>	8/96/3/1024/0.05/32/ 20k	8/160/5/1536/0.10/8/35k	
44		<i>iterations</i>	20 000	35 000	
45		<i>duration / val/train</i>	3 107 1.6291	5 208 1.8444	
46		<i>train loss</i>	0.7317 -3.46%	0.6081 -7.05%	
47		<i>val loss</i>	1.1920 -2.10%	1.1216 0.49%	

48	Stage 9	model	9/96/3/1536/0.05/16/30k		9/192/6/2048/0.10/4/35k			
49		steps	30 000		35 000			
50		duration / val/train	5 734	1.6384	5 817	1.8795		
51		train loss	0.7035	-3.85%	0.5694	-6.36%		
52		val loss	1.1526	-3.31%	1.0702	-4.58%		
53	Stage 10	model	10/128/4/1536/0.10/8/50k					
54		iterations	50 000					
55		duration / val/train	7 030	1.6682				
56		train loss	0.6646	-5.53%				
57		val loss	1.1087	-3.81%				
58	Stage 11	model	11/128/4/2048/0.10/4/30k					
59		iterations	30 000					
60		duration / val/train	3 995	1.6121				
61		train loss	0.6618	-0.42%				
62		val loss	1.0669	-3.77%				
63		Total time	28 722		25 213		16 836	
64	0	duration / val/train	0	1.0647	0	1.0105	0	1.1022
65		train loss	3.5631		3.4444		3.7110	
66		val loss	3.7937		3.4807		4.0901	
67	2 000	duration / val/train	291	1.4634	382	1.7257	390	1.5043
68		train loss / train diff %	0.8411	-76.39%	0.6391	-81.45%	0.7591	-79.54%
69		val loss / val diff %	1.2309	-67.55%	1.1029	-68.31%	1.1419	-72.08%
70	4 000	duration / val/train	581	1.5359	761	1.8067	779	1.5978
71		train loss / train diff %	0.7372	-12.35%	0.6057	-5.23%	0.6852	-9.74%
72		val loss / val diff %	1.1323	-8.01%	1.0943	-0.78%	1.0948	-4.12%
73	6 000	duration / val/train	871	1.5332	1 138	1.8060	1 169	1.6505
74		train loss / train diff %	0.7110	-3.55%	0.5907	-2.48%	0.6567	-4.16%
75		val loss / val diff %	1.0901	-3.73%	1.0668	-2.51%	1.0839	-1.00%
76	8 000	duration / val/train	1 161	1.5693	1 516	1.8791	1 558	1.6808
77		train loss / train diff %	0.6954	-2.19%	0.5772	-2.29%	0.6435	-2.01%
78		val loss / val diff %	1.0913	0.11%	1.0846	1.67%	1.0816	-0.21%
79	10 000	duration / val/train	1 452	1.5772	1 891	1.8893	1 948	2.7045
80		train loss / train diff %	0.6837	-1.68%	0.5753	-0.33%	0.6295	-2.18%
81		val loss / val diff %	1.0783	-1.19%	1.0869	0.21%	1.7025	57.41%
82	12 000	duration / val/train	1 742	1.5641	2 264	1.9287	2 337	1.7663
83		train loss / train diff %	0.6814	-0.34%	0.5642	-1.93%	0.5978	-5.04%
84		val loss / val diff %	1.0658	-1.16%	1.0882	0.12%	1.0559	-37.98%
85	14 000	duration / val/train	2 032	1.5876	2 642	1.9452	2 727	1.8312
86		train loss / train diff %	0.6782	-0.47%	0.5570	-1.28%	0.5859	-1.99%
87		val loss / val diff %	1.0767	1.02%	1.0835	-0.43%	1.0729	1.61%
88	16 000	duration / val/train	2 322	1.5960	3 029	1.9344	3 116	1.8831
89		train loss / train diff %	0.6737	-0.66%	0.5535	-0.63%	0.5733	-2.15%
90		val loss / val diff %	1.0752	-0.14%	1.0707	-1.18%	1.0796	0.62%
91	18 000	duration / val/train	2 613	1.6218	3 410	2.0117	3 506	1.9028
92		train loss / train diff %	0.6652	-1.26%	0.5476	-1.07%	0.5669	-1.12%
93		val loss / val diff %	1.0788	0.33%	1.1016	2.89%	1.0787	-0.08%
94	20 000	duration / val/train	2 903	1.6193	3 799	2.0050	3 895	1.9443
95		train loss / train diff %	0.6601	-0.77%	0.5454	-0.40%	0.5584	-1.50%

96		<i>val loss / val diff %</i>	1.0689	-0.92%	1.0935	-0.74%	1.0857	0.65%
97	22 000	<i>duration / val/train</i>	3 193	1.6254	4 167	1.9948	4 285	1.9672
98		<i>train loss / train diff %</i>	0.6543	-0.88%	0.5430	-0.44%	0.5494	-1.61%
99		<i>val loss / val diff %</i>	1.0635	-0.51%	1.0832	-0.94%	1.0808	-0.45%
100	24 000	<i>duration / val/train</i>	3 483	1.6359	4 534	2.0683	4 674	1.9670
101		<i>train loss / train diff %</i>	0.6565	0.34%	0.5345	-1.57%	0.5514	0.36%
102		<i>val loss / val diff %</i>	1.0740	0.99%	1.1055	2.06%	1.0846	0.35%
103	26 000	<i>duration / val/train</i>	3 774	1.6868	4 912	2.0882	5 064	2.0072
104		<i>train loss / train diff %</i>	0.6548	-0.26%	0.5295	-0.94%	0.5431	-1.51%
105		<i>val loss / val diff %</i>	1.1045	2.84%	1.1057	0.02%	1.0901	0.51%
106	28 000	<i>duration / val/train</i>	4 064	1.6521	5 281	2.0901	5 453	2.0462
107		<i>train loss / train diff %</i>	0.6432	-1.77%	0.5359	1.21%	0.5347	-1.55%
108		<i>val loss / val diff %</i>	1.0626	-3.79%	1.1201	1.30%	1.0941	0.37%
109	30 000	<i>duration / val/train</i>	4 354	1.6934	5 647	2.1430	5 843	2.0561
110		<i>train loss / train diff %</i>	0.6351	-1.26%	0.5224	-2.52%	0.5382	0.65%
111		<i>val loss / val diff %</i>	1.0755	1.21%	1.1195	-0.05%	1.1066	1.14%
112	32 000	<i>duration / val/train</i>	4 644	1.6772	6 012	2.1799	6 232	2.0840
113		<i>train loss / train diff %</i>	0.6395	0.69%	0.5196	-0.54%	0.5288	-1.75%
114		<i>val loss / val diff %</i>	1.0726	-0.27%	1.1327	1.18%	1.1020	-0.42%
115	34 000	<i>duration / val/train</i>	4 935	1.7107	6 377	2.1495	6 622	2.1002
116		<i>train loss / train diff %</i>	0.6326	-1.08%	0.5205	0.17%	0.5251	-0.70%
117		<i>val loss / val diff %</i>	1.0822	0.90%	1.1188	-1.23%	1.1028	0.07%
118	36 000	<i>duration / val/train</i>	5 225	1.6964	6 742	2.1680	7 011	2.0795
119		<i>train loss / train diff %</i>	0.6360	0.54%	0.5144	-1.17%	0.5235	-0.30%
120		<i>val loss / val diff %</i>	1.0789	-0.30%	1.1152	-0.32%	1.0886	-1.29%
121	38 000	<i>duration / val/train</i>	5 515	1.6848	7 107	2.1568	7 401	2.1227
122		<i>train loss / train diff %</i>	0.6279	-1.27%	0.5167	0.45%	0.5175	-1.15%
123		<i>val loss / val diff %</i>	1.0579	-1.95%	1.1144	-0.07%	1.0985	0.91%
124	40 000	<i>duration / val/train</i>	5 805	1.6971	7 472	2.2199	7 790	2.1767
125		<i>train loss / train diff %</i>	0.6319	0.64%	0.5066	-1.95%	0.5100	-1.45%
126		<i>val loss / val diff %</i>	1.0724	1.37%	1.1246	0.92%	1.1101	1.06%
127	42 000	<i>duration / val/train</i>	6 096	1.7266	7 837	2.2104	8 180	2.1703
128		<i>train loss / train diff %</i>	0.6214	-1.66%	0.5066	0.00%	0.5098	-0.04%
129		<i>val loss / val diff %</i>	1.0729	0.05%	1.1198	-0.43%	1.1064	-0.33%
130	44 000	<i>duration / val/train</i>	6 386	1.7258	8 202	2.2144	8 569	2.1852
131		<i>train loss / train diff %</i>	0.6236	0.35%	0.4995	-1.40%	0.5043	-1.08%
132		<i>val loss / val diff %</i>	1.0762	0.31%	1.1061	-1.22%	1.1020	-0.40%
133	46 000	<i>duration / val/train</i>	6 676	1.7404	8 567	2.2740	8 959	2.2195
134		<i>train loss / train diff %</i>	0.6230	-0.10%	0.5014	0.38%	0.4984	-1.17%
135		<i>val loss / val diff %</i>	1.0843	0.75%	1.1402	3.08%	1.1062	0.38%
136	48 000	<i>duration / val/train</i>	6 966	1.7328	8 932	2.2764	9 349	2.2049
137		<i>train loss / train diff %</i>	0.6228	-0.03%	0.4971	-0.86%	0.5027	0.86%
138		<i>val loss / val diff %</i>	1.0792	-0.47%	1.1316	-0.75%	1.1084	0.20%
139	50 000	<i>duration / val/train</i>	7 257	1.7269	9 297	2.3063	9 738	2.2325
140		<i>train loss / train diff %</i>	0.6207	-0.34%	0.4975	0.08%	0.4993	-0.68%
141		<i>val loss / val diff %</i>	1.0719	-0.68%	1.1474	1.40%	1.1147	0.57%
142	52 000	<i>duration / val/train</i>	7 548	1.7369	9 661	2.2807	10 128	2.2366
143		<i>train loss / train diff %</i>	0.6162	-0.72%	0.5001	0.52%	0.4946	-0.94%
144		<i>val loss / val diff %</i>	1.0703	-0.15%	1.1406	-0.59%	1.1062	-0.76%

145	54 000	<i>duration / val/train</i>	7 849	1.7675	10 026	2.3329	10 517	2.2695
146		<i>train loss / train diff %</i>	0.6163	0.02%	0.4878	-2.46%	0.4910	-0.73%
147		<i>val loss / val diff %</i>	1.0893	1.78%	1.1380	-0.23%	1.1143	0.73%
148	56 000	<i>duration / val/train</i>	8 143	1.7328	10 391	2.3059	10 907	2.2877
149		<i>train loss / train diff %</i>	0.6167	0.06%	0.4929	1.05%	0.4904	-0.12%
150		<i>val loss / val diff %</i>	1.0686	-1.90%	1.1366	-0.12%	1.1219	0.68%
151	58 000	<i>duration / val/train</i>	8 444	1.7769	10 756	2.3489	11 296	2.3382
152		<i>train loss / train diff %</i>	0.6083	-1.36%	0.4887	-0.85%	0.4849	-1.12%
153		<i>val loss / val diff %</i>	1.0809	1.15%	1.1479	0.99%	1.1338	1.06%
154	60 000	<i>duration / val/train</i>	8 740	1.7711	11 121	2.3276	11 686	2.3033
155		<i>train loss / train diff %</i>	0.6076	-0.12%	0.4905	0.37%	0.4836	-0.27%
156		<i>val loss / val diff %</i>	1.0761	-0.44%	1.1417	-0.54%	1.1139	-1.76%
157	62 000	<i>duration / val/train</i>	9 040	1.7604	11 486	2.3684	12 075	2.3182
158		<i>train loss / train diff %</i>	0.6085	0.15%	0.4829	-1.55%	0.4831	-0.10%
159		<i>val loss / val diff %</i>	1.0712	-0.46%	1.1437	0.18%	1.1199	0.54%
160	64 000	<i>duration / val/train</i>	9 348	1.7971	11 851	2.3752	12 465	2.3877
161		<i>train loss / train diff %</i>	0.6028	-0.94%	0.4782	-0.97%	0.4774	-1.18%
162		<i>val loss / val diff %</i>	1.0833	1.13%	1.1358	-0.69%	1.1399	1.79%
163	66 000	<i>duration / val/train</i>	9 656	1.7660	12 216	2.4117	12 854	2.3955
164		<i>train loss / train diff %</i>	0.6081	0.88%	0.4795	0.27%	0.4763	-0.23%
165		<i>val loss / val diff %</i>	1.0739	-0.87%	1.1564	1.81%	1.1410	0.10%
166	68 000	<i>duration / val/train</i>	9 949	1.7826	12 581	2.4157	13 244	2.4180
167		<i>train loss / train diff %</i>	0.6085	0.07%	0.4816	0.44%	0.4727	-0.76%
168		<i>val loss / val diff %</i>	1.0847	1.01%	1.1634	0.61%	1.1430	0.18%
169	70 000	<i>duration / val/train</i>	10 239	1.8029	12 946	2.3921	13 633	2.4487
170		<i>train loss / train diff %</i>	0.6043	-0.69%	0.4777	-0.81%	0.4700	-0.57%
171		<i>val loss / val diff %</i>	1.0895	0.44%	1.1427	-1.78%	1.1509	0.69%
172	72 000	<i>duration / val/train</i>	10 539	1.7788	13 311	2.4609	14 023	2.4494
173		<i>train loss / train diff %</i>	0.6086	0.71%	0.4715	-1.30%	0.4702	0.04%
174		<i>val loss / val diff %</i>	1.0826	-0.63%	1.1603	1.54%	1.1517	0.07%
175	74 000	<i>duration / val/train</i>	10 846	1.8020	13 676	2.4532	14 412	2.4429
176		<i>train loss / train diff %</i>	0.6016	-1.15%	0.4724	0.19%	0.4665	-0.79%
177		<i>val loss / val diff %</i>	1.0841	0.14%	1.1589	-0.12%	1.1396	-1.05%
178	76 000	<i>duration / val/train</i>	11 141	1.7849	14 041	2.4541	14 802	2.4223
179		<i>train loss / train diff %</i>	0.6001	-0.25%	0.4724	0.00%	0.4682	0.36%
180		<i>val loss / val diff %</i>	1.0711	-1.20%	1.1593	0.03%	1.1341	-0.48%
181	78 000	<i>duration / val/train</i>	11 449	1.7738	14 406	2.4598	15 191	2.4530
182		<i>train loss / train diff %</i>	0.6004	0.05%	0.4696	-0.59%	0.4627	-1.17%
183		<i>val loss / val diff %</i>	1.0650	-0.57%	1.1551	-0.36%	1.1350	0.08%
184	80 000	<i>duration / val/train</i>	11 746	1.7922	14 771	2.5111	15 581	2.4901
185		<i>train loss / train diff %</i>	0.5982	-0.37%	0.4684	-0.26%	0.4544	-1.79%
186		<i>val loss / val diff %</i>	1.0721	0.67%	1.1762	1.83%	1.1315	-0.31%
187	82 000	<i>duration / val/train</i>	12 054	1.7860	15 136	2.4851	15 970	2.5049
188		<i>train loss / train diff %</i>	0.6050	1.14%	0.4692	0.17%	0.4605	1.34%
189		<i>val loss / val diff %</i>	1.0805	0.78%	1.1660	-0.87%	1.1535	1.94%
190	84 000	<i>duration / val/train</i>	12 361	1.8238	15 501	2.5094	16 360	2.4977
191		<i>train loss / train diff %</i>	0.5955	-1.57%	0.4631	-1.30%	0.4641	0.78%
192		<i>val loss / val diff %</i>	1.0861	0.52%	1.1621	-0.33%	1.1592	0.49%
193	86 000	<i>duration / val/train</i>	12 666	1.8158	15 866	2.5115	16 749	2.5218

194		<i>train loss / train diff %</i>	0.5957	0.03%	0.4637	0.13%	0.4565	-1.64%
195		<i>val loss / val diff %</i>	1.0817	-0.41%	1.1646	0.22%	1.1512	-0.69%
196	88 000	<i>duration / val/train</i>	12 959	1.8072	16 231	2.5173	17 139	2.5441
197		<i>train loss / train diff %</i>	0.5964	0.12%	0.4643	0.13%	0.4543	-0.48%
198		<i>val loss / val diff %</i>	1.0778	-0.36%	1.1688	0.36%	1.1558	0.40%
199	90 000	<i>duration / val/train</i>	13 255	1.8574	16 596	2.5084	17 528	2.5354
200		<i>train loss / train diff %</i>	0.5897	-1.12%	0.4624	-0.41%	0.4535	-0.18%
201		<i>val loss / val diff %</i>	1.0953	1.62%	1.1599	-0.76%	1.1498	-0.52%
202	92 000	<i>duration / val/train</i>	13 554	1.8185	16 960	2.5493	17 918	2.5048
203		<i>train loss / train diff %</i>	0.5944	0.80%	0.4586	-0.82%	0.4566	0.68%
204		<i>val loss / val diff %</i>	1.0809	-1.31%	1.1691	0.79%	1.1437	-0.53%
205	94 000	<i>duration / val/train</i>	13 847	1.8375	17 325	2.5435	18 307	2.5581
206		<i>train loss / train diff %</i>	0.5914	-0.50%	0.4572	-0.31%	0.4515	-1.12%
207		<i>val loss / val diff %</i>	1.0867	0.54%	1.1629	-0.53%	1.1550	0.99%
208	96 000	<i>duration / val/train</i>	14 143	1.8433	17 690	2.5294	18 697	2.5935
209		<i>train loss / train diff %</i>	0.5922	0.14%	0.4558	-0.31%	0.4482	-0.73%
210		<i>val loss / val diff %</i>	1.0916	0.45%	1.1529	-0.86%	1.1624	0.64%
211	98 000	<i>duration / val/train</i>	14 436	1.8433	18 055	2.5416	19 086	2.5787
212		<i>train loss / train diff %</i>	0.5895	-0.46%	0.4564	0.13%	0.4450	-0.71%
213		<i>val loss / val diff %</i>	1.0866	-0.46%	1.1600	0.62%	1.1475	-1.28%
214	100 000	<i>duration / val/train</i>	14 740	1.8439	18 420	2.5574	19 476	2.5878
215		<i>train loss / train diff %</i>	0.5874	-0.36%	0.4541	-0.50%	0.4432	-0.40%
216		<i>val loss / val diff %</i>	1.0831	-0.32%	1.1613	0.11%	1.1469	-0.05%
217	102 000	<i>duration / val/train</i>	15 042	1.8195	18 785	2.6190	19 874	2.5749
218		<i>train loss / train diff %</i>	0.5934	1.02%	0.4564	0.51%	0.4486	1.22%
219		<i>val loss / val diff %</i>	1.0797	-0.31%	1.1953	2.93%	1.1551	0.71%
220	104 000	<i>duration / val/train</i>	15 336	1.8314	19 150	2.5920	20 273	2.6099
221		<i>train loss / train diff %</i>	0.5888	-0.78%	0.4507	-1.25%	0.4424	-1.38%
222		<i>val loss / val diff %</i>	1.0783	-0.13%	1.1682	-2.27%	1.1546	-0.04%
223	106 000	<i>duration / val/train</i>	15 637	1.8599	19 515	2.6398	20 678	2.5787
224		<i>train loss / train diff %</i>	0.5840	-0.82%	0.4539	0.71%	0.4432	0.18%
225		<i>val loss / val diff %</i>	1.0862	0.73%	1.1982	2.57%	1.1429	-1.01%
226	108 000	<i>duration / val/train</i>	15 935	1.8925	19 880	2.6201	21 081	2.6314
227		<i>train loss / train diff %</i>	0.5813	-0.46%	0.4491	-1.06%	0.4381	-1.15%
228		<i>val loss / val diff %</i>	1.1001	1.28%	1.1767	-1.79%	1.1528	0.87%
229	110 000	<i>duration / val/train</i>	16 235	1.8485	20 245	2.5980	21 483	2.6289
230		<i>train loss / train diff %</i>	0.5868	0.95%	0.4500	0.20%	0.4408	0.62%
231		<i>val loss / val diff %</i>	1.0847	-1.40%	1.1691	-0.65%	1.1588	0.52%
232	112 000	<i>duration / val/train</i>			20 610	2.6089		
233		<i>train loss / train diff %</i>			0.4503	0.07%		
234		<i>val loss / val diff %</i>			1.1748	0.49%		
235	114 000	<i>duration / val/train</i>			20 975	2.6548		
236		<i>train loss / train diff %</i>			0.4484	-0.42%		
237		<i>val loss / val diff %</i>			1.1904	1.33%		
238	116 000	<i>duration / val/train</i>			21 340	2.6910		
239		<i>train loss / train diff %</i>			0.4431	-1.18%		
240		<i>val loss / val diff %</i>			1.1924	0.17%		
241	118 000	<i>duration / val/train</i>			21 705	2.6512		
242		<i>train loss / train diff %</i>			0.4512	1.83%		

243		<i>val loss / val diff %</i>		1.1962	0.32%	
244	120 000	<i>duration / val/train</i>		22 070	2.6926	
245		<i>train loss / train diff %</i>		0.4441	-1.57%	
246		<i>val loss / val diff %</i>		1.1958	-0.03%	
247	122 000	<i>duration / val/train</i>		22 435	2.6849	
248		<i>train loss / train diff %</i>		0.4436	-0.11%	
249		<i>val loss / val diff %</i>		1.1910	-0.40%	
250	124 000	<i>duration / val/train</i>		22 800	2.6879	
251		<i>train loss / train diff %</i>		0.4435	-0.02%	
252		<i>val loss / val diff %</i>		1.1921	0.09%	
253	126 000	<i>duration / val/train</i>		23 165	2.6995	
254		<i>train loss / train diff %</i>		0.4403	-0.72%	
255		<i>val loss / val diff %</i>		1.1886	-0.29%	
256	128 000	<i>duration / val/train</i>		23 529	2.6772	
257		<i>train loss / train diff %</i>		0.4430	0.61%	
258		<i>val loss / val diff %</i>		1.1860	-0.22%	
259	130 000	<i>duration / val/train</i>		23 894	2.7741	
260		<i>train loss / train diff %</i>		0.4356	-1.67%	
261		<i>val loss / val diff %</i>		1.2084	1.89%	
262	132 000	<i>duration / val/train</i>		24 259	2.7078	
263		<i>train loss / train diff %</i>		0.4398	0.96%	
264		<i>val loss / val diff %</i>		1.1909	-1.45%	
265	134 000	<i>duration / val/train</i>		24 624	2.7679	
266		<i>train loss / train diff %</i>		0.4343	-1.25%	
267		<i>val loss / val diff %</i>		1.2021	0.94%	
268	136 000	<i>duration / val/train</i>		24 989	2.6706	
269		<i>train loss / train diff %</i>		0.4417	1.70%	
270		<i>val loss / val diff %</i>		1.1796	-1.87%	
271	138 000	<i>duration / val/train</i>		25 354	2.6848	
272		<i>train loss / train diff %</i>		0.4419	0.05%	
273		<i>val loss / val diff %</i>		1.1864	0.58%	
274	140 000	<i>duration / val/train</i>		25 719	2.7430	
275		<i>train loss / train diff %</i>		0.4369	-1.13%	
276		<i>val loss / val diff %</i>		1.1984	1.01%	
277	142 000	<i>duration / val/train</i>		26 084	2.7866	
278		<i>train loss / train diff %</i>		0.4321	-1.10%	
279		<i>val loss / val diff %</i>		1.2041	0.48%	
280	144 000	<i>duration / val/train</i>		26 449	2.7689	
281		<i>train loss / train diff %</i>		0.4309	-0.28%	
282		<i>val loss / val diff %</i>		1.1931	-0.91%	
283	146 000	<i>duration / val/train</i>		26 814	2.7590	
284		<i>train loss / train diff %</i>		0.4348	0.91%	
285		<i>val loss / val diff %</i>		1.1996	0.54%	
286	148 000	<i>duration / val/train</i>		27 179	2.7661	
287		<i>train loss / train diff %</i>		0.4336	-0.28%	
288		<i>val loss / val diff %</i>		1.1994	-0.02%	

Table 15. Results of training models at Step 7.

7.8. Analysis of the results of training models in Steps 2- 7

The selected results for each model are shown in Table 16.

No	Checkpoints		Params	12/128/4/ 2048/0.10/ 4	10/192/6/ 2048/0.10 /4	8/256/8/20 48/0.15/4
1			<i>model size, M</i>	2.69	4.91	6.94
2			<i>n_layer</i>	12	10	8
3			<i>n_embd</i>	128	192	256
4			<i>n_head</i>	4	6	8
5			<i>block_size</i>	2 048	2 048	2 048
6			<i>dropout</i>	0.10	0.10	0.15
7			<i>batch_size</i>	4	4	4
8	val loss plateau	Step 1	<i>iterations</i>	96 000	38 000	32 000
9			<i>duration</i>	13 934	7 151	6 441
10			<i>train loss</i>	0.6692	0.6842	0.6914
11			<i>val loss</i>	1.1060	1.0988	1.0867
12		Step 2 zero	<i>iterations</i>	38 000	10 000	14 000
13			<i>duration</i>	13 664	12 342	14 252
14			<i>train loss</i>	0.7626	0.6965	0.7124
15			<i>val loss</i>	1.1131	1.0940	1.0907
16			<i>duration diff %</i>	98.1%	172.6%	221.3%
17		Step 2 normal	<i>iterations</i>	22 000	4 000	12 000
18			<i>duration</i>	11 974	11 226	11 623
19			<i>train loss</i>	0.7393	0.7058	0.6860
20			<i>val loss</i>	1.1101	1.0869	1.1060
21			<i>duration diff %</i>	85.9%	157.0%	180.5%
22		Step 3 zero	<i>iterations</i>	30 000	8 000	8 000
23			<i>duration</i>	22 622	19 740	21 172
24			<i>train loss</i>	0.7696	0.7037	0.7308
25			<i>val loss</i>	1.1042	1.1133	1.0936
26			<i>duration diff %</i>	162.4%	276.0%	328.7%
27		Step 3 normal	<i>iterations</i>	12 000	10 000	18 000
28			<i>duration</i>	21 446	19 269	18 015
29			<i>train loss</i>	0.7648	0.6430	0.6391
30			<i>val loss</i>	1.1138	1.1061	1.0983
31			<i>duration diff %</i>	153.9%	269.5%	279.7%
32		Step 4 zero	<i>iterations</i>		6 000	2 000
33			<i>duration</i>		44 693	23 057
34			<i>train loss</i>		0.6316	0.7351
35			<i>val loss</i>		1.0921	1.0897
36			<i>duration diff %</i>		625.0%	358.0%
37		Step 4 normal	<i>iterations</i>		2 000	6 000

38			<i>duration</i>		36 483	28 518
39			<i>train loss</i>		0.6346	0.6632
40			<i>val loss</i>		1.0993	1.0768
41			<i>duration diff %</i>		510.2%	442.8%
42		Step 5	<i>iterations</i>	32 000	6 000	6 000
43			<i>duration</i>	14 731	13 794	12 830
44			<i>train loss</i>	0.6997	0.6806	0.6990
45			<i>val loss</i>	1.1130	1.0831	1.0923
46			<i>duration diff %</i>	105.7%	192.9%	199.2%
47		Step 6	<i>iterations</i>	36 000	10 000	8 000
48			<i>duration</i>	15 473	16 354	14 595
49			<i>train loss</i>	0.7036	0.6654	0.7103
50			<i>val loss</i>	1.1126	1.1112	1.1135
51			<i>duration diff %</i>	111.0%	228.7%	226.6%
52		Step 7	<i>iterations</i>	6 000	2 000	4 000
53			<i>duration</i>	29 593	25 595	17 615
54			<i>train loss</i>	0.7110	0.6391	0.6852
55			<i>val loss</i>	1.0901	1.1029	1.0948
56			<i>duration diff %</i>	212.4%	357.9%	273.5%
57	min val loss	Step 1	<i>iterations</i>	232 000	76 000	56 000
58			<i>duration</i>	34 159	14 321	11 109
59			<i>train loss</i>	0.6136	0.6078	0.6195
60			<i>val loss</i>	1.0808	1.0877	1.0731
61		Step 2 zero	<i>iterations</i>	220 000	4 000	30 000
62			<i>duration</i>	40 629	11 250	17 365
63			<i>train loss</i>	0.6302	0.7300	0.6506
64			<i>val loss</i>	1.0527	1.0813	1.0658
65			<i>val loss diff %</i>	97.4%	99.4%	99.3%
66		Step 2 normal	<i>iterations</i>	94 000	14 000	30 000
67			<i>duration</i>	22 706	13 063	15 279
68			<i>train loss</i>	0.6133	0.6372	0.5870
69			<i>val loss</i>	1.0902	1.0671	1.0862
70			<i>val loss diff %</i>	100.9%	98.1%	101.2%
71		Step 3 zero	<i>iterations</i>	90 000	30 000	22 000
72			<i>duration</i>	31 386	23 910	23 983
73			<i>train loss</i>	0.6828	0.6119	0.6675
74			<i>val loss</i>	1.0701	1.1028	1.0771
75			<i>val loss diff %</i>	99.0%	101.4%	100.4%
76		Step 3 normal	<i>iterations</i>	90 000	24 000	28 000
77			<i>duration</i>	33 097	21 872	20 044
78			<i>train loss</i>	0.6183	0.5859	0.5905
79			<i>val loss</i>	1.0620	1.1022	1.0824
80			<i>val loss diff %</i>	98.3%	101.3%	100.9%
81		Step 4 zero	<i>iterations</i>		4 000	80 000

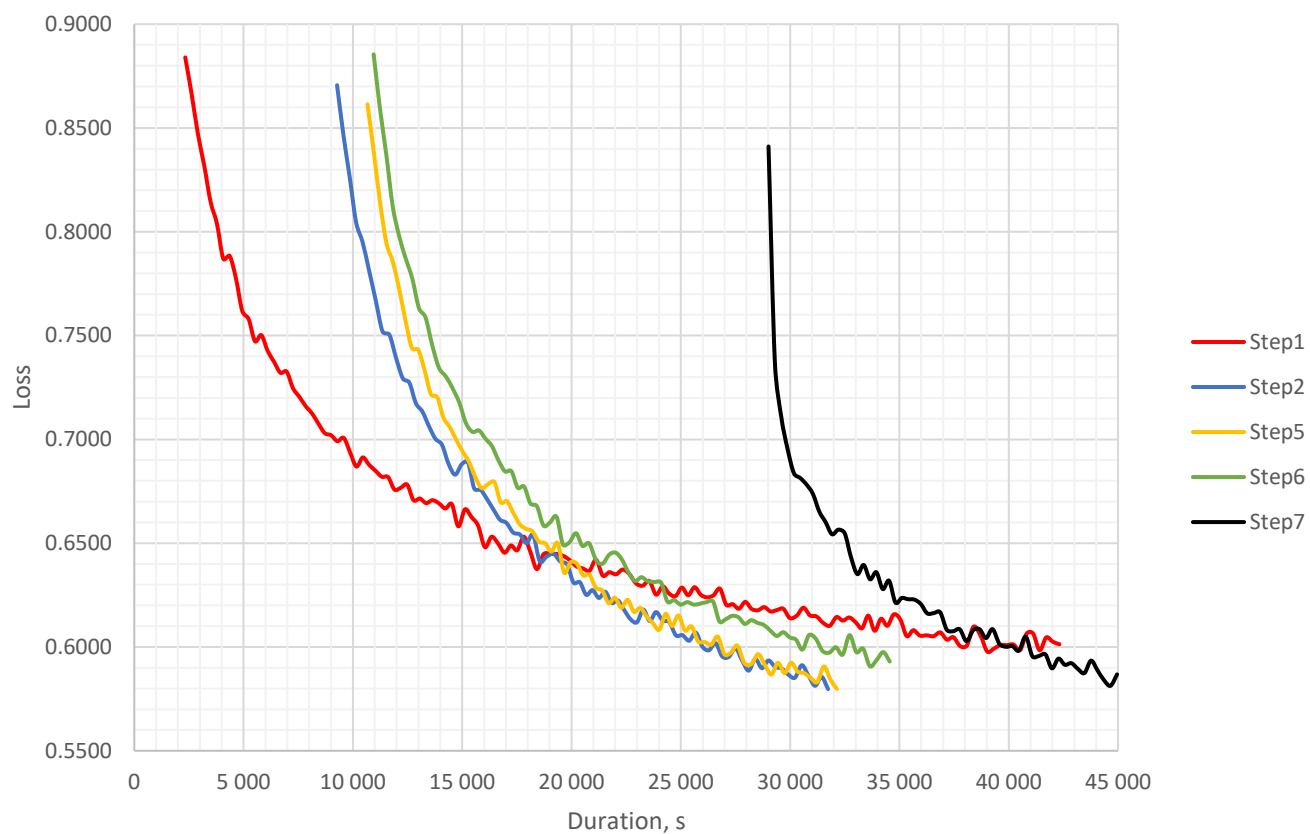
82			<i>duration</i>		44 312	24 245
83			<i>train loss</i>		0.6394	0.6703
84			<i>val loss</i>		1.0760	1.0590
85			<i>val loss diff %</i>		98.9%	98.7%
86		Step 4 normal	<i>iterations</i>		8 000	14 000
87			<i>duration</i>		37 577	30 109
88			<i>train loss</i>		0.5677	0.5799
89			<i>val loss</i>		1.0696	1.0537
90			<i>val loss diff %</i>		98.3%	98.2%
91		Step 5	<i>iterations</i>	94 000	28 000	16 000
92			<i>duration</i>	23 729	17 920	14 857
93			<i>train loss</i>	0.6114	0.5993	0.6450
94			<i>val loss</i>	1.0917	1.0646	1.0687
95			<i>val loss diff %</i>	101.0%	97.9%	99.6%
96		Step 6	<i>iterations</i>	64 000	12 000	16 000
97			<i>duration</i>	19 608	16 732	16 199
98			<i>train loss</i>	0.6494	0.6568	0.6518
99			<i>val loss</i>	1.0858	1.1035	1.0798
100			<i>val loss diff %</i>	100.5%	101.5%	100.6%
101		Step 7	<i>iterations</i>	38 000	6 000	12 000
102			<i>duration</i>	34 237	26 351	19 173
103			<i>train loss</i>	0.6279	0.5907	0.5978
104			<i>val loss</i>	1.0579	1.0668	1.0559
105			<i>val loss diff %</i>	97.9%	98.1%	98.4%
106	min train loss	Step 1	<i>iterations</i>	252 000	180 000	196 000
107			<i>duration</i>	37 172	33 606	39 193
108			<i>train loss</i>	0.6035	0.5224	0.5035
109			<i>val loss</i>	1.1132	1.1467	1.1503
110		Step 2 zero	<i>iterations</i>	216 000	124 000	142 000
111			<i>duration</i>	40 050	33 778	39 541
112			<i>train loss</i>	0.6232	0.5339	0.5340
113			<i>val loss</i>	1.0681	1.1299	1.1196
114			<i>duration diff %</i>	107.7%	100.5%	100.9%
115			<i>train loss diff %</i>	103.3%	102.2%	106.1%
116		Step 2 normal	<i>iterations</i>	150 000	106 000	78 000
117			<i>duration</i>	31 141	30 178	24 952
118			<i>train loss</i>	0.5813	0.4744	0.4907
119			<i>val loss</i>	1.1050	1.1530	1.1546
120			<i>duration diff %</i>	83.8%	89.8%	63.7%
121			<i>train loss diff %</i>	96.3%	90.8%	97.5%
122		Step 3 zero	<i>iterations</i>	176 000	140 000	144 000
123			<i>duration</i>	44 107	44 143	47 933
124			<i>train loss</i>	0.6348	0.5229	0.5285
125			<i>val loss</i>	1.0765	1.1387	1.1037

126			<i>duration diff %</i>	118.7%	131.4%	122.3%
127			<i>train loss diff %</i>	105.2%	100.1%	105.0%
128		Step 3 normal	<i>iterations</i>	112 000	50 000	80 000
129			<i>duration</i>	40 898	26 616	30 349
130			<i>train loss</i>	0.5813	0.5216	0.4792
131			<i>val loss</i>	1.0870	1.1717	1.1510
132			<i>duration diff %</i>	110.0%	79.2%	77.4%
133			<i>train loss diff %</i>	96.3%	99.8%	95.2%
134		Step 4 zero	<i>iterations</i>		104 000	142 000
135			<i>duration</i>		62 737	50 596
136			<i>train loss</i>		0.5348	0.5310
137			<i>val loss</i>		1.1411	1.0930
138			<i>duration diff %</i>		186.7%	129.1%
139			<i>train loss diff %</i>		102.4%	105.5%
140		Step 4 normal	<i>iterations</i>		64 000	80 000
141			<i>duration</i>		48 001	43 252
142			<i>train loss</i>		0.4740	0.4541
143			<i>val loss</i>		1.1572	1.1439
144			<i>duration diff %</i>		142.8%	110.4%
145			<i>train loss diff %</i>		90.7%	90.2%
146		Step 5	<i>iterations</i>	144 000	110 000	90 000
147			<i>duration</i>	31 241	33 164	29 484
148			<i>train loss</i>	0.5831	0.4687	0.4729
149			<i>val loss</i>	1.1125	1.1633	1.1446
150			<i>duration diff %</i>	84.0%	98.7%	75.2%
151			<i>train loss diff %</i>	96.6%	89.7%	93.9%
152		Step 6	<i>iterations</i>	158 000	74 000	82 000
153			<i>duration</i>	33 650	28 276	29 308
154			<i>train loss</i>	0.5907	0.4990	0.4823
155			<i>val loss</i>	1.1050	1.1593	1.1525
156			<i>duration diff %</i>	90.5%	84.1%	74.8%
157			<i>train loss diff %</i>	97.9%	95.5%	95.8%
158		Step 7	<i>iterations</i>	106 000	100 000	100 000
159			<i>duration</i>	44 359	43 633	36 312
160			<i>train loss</i>	0.5840	0.4541	0.4432
161			<i>val loss</i>	1.0862	1.1613	1.1469
162			<i>duration diff %</i>	119.3%	129.8%	92.6%
163			<i>train loss diff %</i>	96.8%	86.9%	88.0%

Table 16. Checkpoints of the models at Steps 2 - 7

Figure 4 shows graphs of changes in train loss depending on the duration of model training at Steps 5-7.

12/128/4/2048/0.10/4 Step 7



10/192/6/2048/0.10/4 Step 7

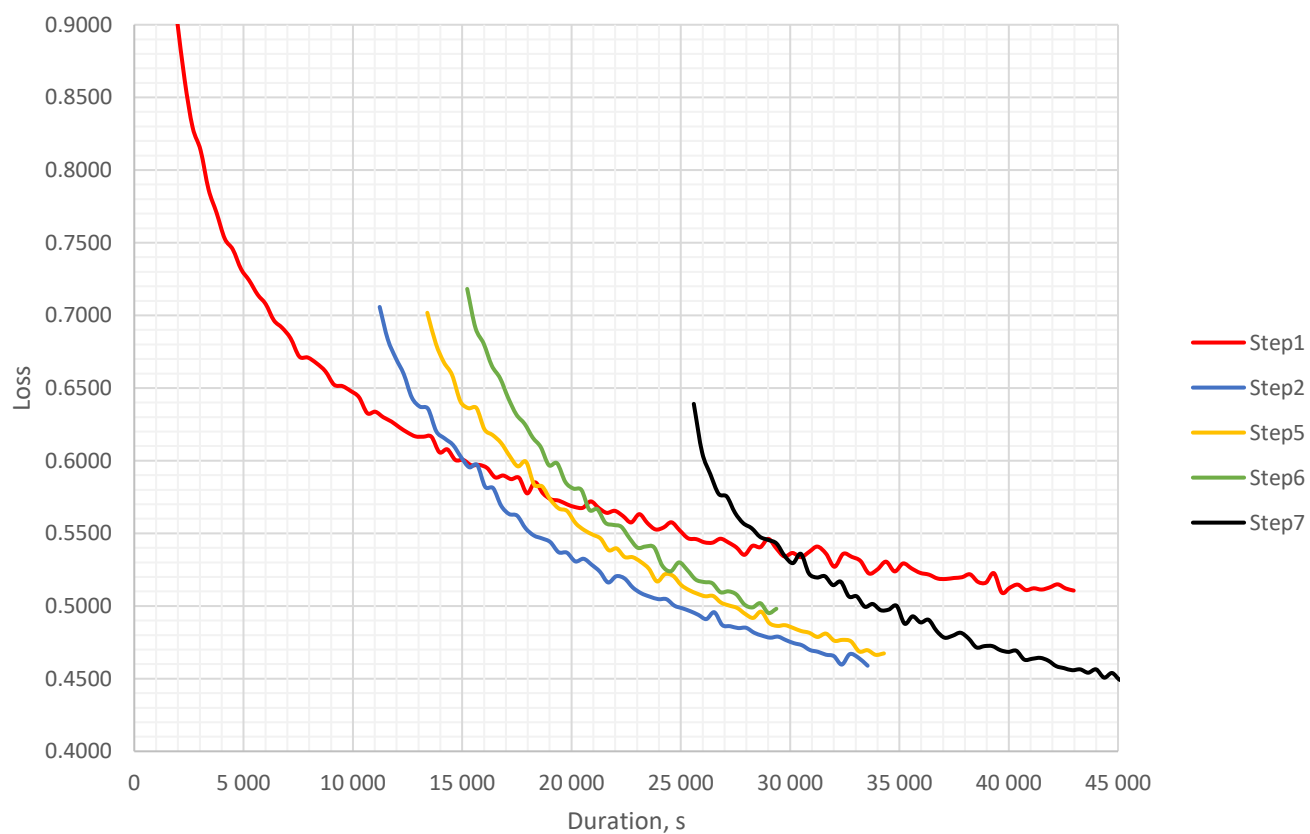




Figure 4. Train loss graphs depending on the training duration of models at Steps 5-7

A larger number of stages (Step 7) predictably leads to increased model training time (Step 5), but does not yield significant improvements in performance. There appears to be an optimal number of stages for a given dataset and model sizes, but we will not investigate this within the scope of this study.

Stepwise training on a partial dataset (Step 6) still yields worse results than stepwise training on the full dataset (Step 5). Perhaps the dataset needs to be divided differently, but we won't test this in this study.

Stopping model training at intermediate stages when a certain train loss level is reached (Steps 5-7) yields the same results as training the model until it reaches a plateau (Step 2). However, monitoring whether to stop training the model when a certain loss change is reached is easier than determining when the plateau has been reached. There is likely an optimal train loss threshold for stopping training for a given dataset and model sizes, but we will not investigate this in this study. We will continue to use a threshold of $>-1\%$ for 5000 iterations.

In the next chapter, we will take the models from Step 5 and try to improve them further by using constraints on weight changes.

8. Optimization of LLM training phases

We'll use the models from Step 5 as a basis, training on the full dataset and stopping training at intermediate stages after the train loss changes to $>-1\%$ after 5000 iterations. The plan for modifying the models in the following steps is shown in Table 17.

No	Stage/Params	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4
1	model size, M	2.69	4.91	6.94
2	n_{layer}	12	10	8

3	<i>n_embd</i>	128	192	256
4	<i>n_head</i>	4	6	8
5	<i>block_size</i>	2 048	2 048	2 048
6	<i>dropout</i>	0.10	0.10	0.15
7	<i>batch_size</i>	4	4	4
8	Start stage 1	2/32/1/128/0.00/64	2/64/2/128/0.00/64	2/64/2/128/0.00/64
9	Stage 2	4/64/2/256/0.05/64	4/96/3/256/0.05/64	3/96/3/256/0.05/64
10	Stage 3	6/64/2/512/0.05/64	6/128/4/512/0.05/64	4/128/4/512/0.05/64
11	Stage 4	9/96/3/1024/0.05/32	8/160/5/1536/0.10/8	6/192/6/1536/0.10/8
12	Final Stage 5	12/128/4/2048/0.10/4	10/192/6/2048/0.10/4	8/256/8/2048/0.15/4

Table 17. Model Change Plan Steps 8-10.

The first method for optimizing model training is warming up the new weights. After each model growth, we experience an explosion of gradients, which breaks previously learned weights. We will allow time for the new weights to warm up. After the model grows, at the beginning of each stage, the old weights (from previous stages) will not be modified for a certain number of iterations. Once the gradients are no longer so large, the model will train as before. The duration of this warm-up (the warmup parameter) is determined experimentally. A warmup value of 0 disables optimization.

The second method for optimizing model training is freezing the warm-up of new weights. This means freezing weights from previous stages. The *weight_freeze* parameter specifies the multiplier by which the gradient for old parameters is multiplied. This is done so that at each stage, the model learns mostly using the new weights. The *weight_freeze* parameter is inherited from previous stages. That is, new weights have a gradient multiplier of 1, weights from the previous stage have a multiplier equal to *weight_freeze*, weights from the stage before the previous stage have a multiplier of *weight_freeze* * *weight_freeze*, and so on. Optimal values for *weight_freeze* are determined experimentally. When *weight_freeze* is equal to 1, optimization is disabled. When *weight_freeze* is equal to 0, weights from previous stages are not changed at all.

The third optimization method will be a combination of the previous two. We'll use both warming up the new weights and freezing the old ones.

8.1. Warming up the new weights. Step 8.

The results of training the models in Step 8 are shown in Table 18.

No	iterations	Params	12/128/4/2048/0.10/4/250	10/192/6/2048/0.10/4/400	8/256/8/2048/0.15/4/250
1		<i>model size, M</i>	2.69	4.91	6.94
2		<i>n_layer</i>	12	10	8
3		<i>n_embd</i>	128	192	256
4		<i>n_head</i>	4	6	8
5		<i>block_size</i>	2 048	2 048	2 048
6		<i>dropout</i>	0.10	0.10	0.15
7		<i>batch_size</i>	4	4	4
8		<i>warmup</i>	250	400	250
9	Stage 1	<i>model</i>	2/32/1/128/0.00/64/40k	2/64/2/128/0.00/64/50k	2/64/2/128/0.00/64/50k
10		<i>iterations</i>	40 000	50 000	50 000
11		<i>duration / val/train</i>	211 1.2228	269 1.3309	278 1.3309
12		<i>train loss</i>	1.6816	1.3640	1.3640
13		<i>val loss</i>	2.0563	1.8154	1.8154

14	Stage 2	model	4/64/2/256/0.05/64/400/40k		4/96/3/256/0.05/64/400/45k		3/96/3/256/0.05/64/300/40k	
15		iterations	40 000		45 000		40 000	
16		duration / val/train	397	1.3637	687	1.4908	458	1.4015
17		train loss	1.1786	-29.91%	0.9895	-27.46%	1.0524	-22.84%
18		val loss	1.6073	-21.84%	1.4751	-18.75%	1.4749	-18.76%
19	Stage 3	model	6/64/2/512/0.05/64/300/30k		6/128/4/512/0.05/64/250/40k		4/128/4/512/0.05/64/300/40k	
20		iterations	30 000		40 000		40 000	
21		duration / val/train	1 376	1.3986	3 729	1.8237	2 397	1.6316
22		train loss	1.0006	-15.10%	0.7420	-25.01%	0.8168	-22.39%
23		val loss	1.3994	-12.93%	1.3532	-8.26%	1.3327	-9.64%
24	Stage 4	model	9/96/3/1024/0.05/32/250/40k		8/160/5/1536/0.10/8/400/55k		6/192/6/1536/0.10/8/250/50k	
25		iterations	40 000		55 000		50 000	
26		duration / val/train	7 193	1.6682	8 199	1.7685	6 553	1.7473
27		train loss	0.7345	-26.59%	0.6390	-13.88%	0.6455	-20.97%
28		val loss	1.2253	-12.44%	1.1301	-16.49%	1.1279	-15.37%
29		Total time	9 177		12 884		9 686	
30	0	duration / val/train	0	1.0565	0	1.0325	0	1.0335
31		train loss	4.6373		5.0253		4.4669	
32		val loss	4.8994		5.1884		4.6166	
33	2 000	duration / val/train	300	1.3324	388	1.5154	388	1.4391
34		train loss / train diff %	0.9379	-79.77%	0.7613	-84.85%	0.8198	-81.65%
35		val loss / val diff %	1.2497	-74.49%	1.1537	-77.76%	1.1798	-74.44%
36	4 000	duration / val/train	602	1.4049	775	1.5656	779	1.4937
37		train loss / train diff %	0.8446	-9.95%	0.6952	-8.68%	0.7363	-10.19%
38		val loss / val diff %	1.1866	-5.05%	1.0884	-5.66%	1.0998	-6.78%
39	6 000	duration / val/train	897	1.4159	1 159	1.6282	1 170	1.5425
40		train loss / train diff %	0.8221	-2.66%	0.6721	-3.32%	0.7172	-2.59%
41		val loss / val diff %	1.1640	-1.90%	1.0943	0.54%	1.1063	0.59%
42	8 000	duration / val/train	1 195	1.4150	1 529	1.6642	1 561	1.5489
43		train loss / train diff %	0.8116	-1.28%	0.6656	-0.97%	0.7071	-1.41%
44		val loss / val diff %	1.1484	-1.34%	1.1077	1.22%	1.0952	-1.00%
45	10 000	duration / val/train	1 491	1.4386	1 906	1.6720	1 952	1.5858
46		train loss / train diff %	0.7885	-2.85%	0.6531	-1.88%	0.6902	-2.39%
47		val loss / val diff %	1.1343	-1.23%	1.0920	-1.42%	1.0945	-0.06%
48	12 000	duration / val/train	1 787	1.4633	2 285	1.7350	2 343	1.6085
49		train loss / train diff %	0.7756	-1.64%	0.6386	-2.22%	0.6756	-2.12%
50		val loss / val diff %	1.1349	0.05%	1.1080	1.47%	1.0867	-0.71%
51	14 000	duration / val/train	2 091	1.4728	2 670	1.7077	2 733	1.6231
52		train loss / train diff %	0.7646	-1.42%	0.6349	-0.58%	0.6668	-1.30%
53		val loss / val diff %	1.1261	-0.78%	1.0842	-2.15%	1.0823	-0.40%
54	16 000	duration / val/train	2 396	1.5167	3 056	1.7578	3 124	1.6677
55		train loss / train diff %	0.7528	-1.54%	0.6226	-1.94%	0.6427	-3.61%
56		val loss / val diff %	1.1418	1.39%	1.0944	0.94%	1.0718	-0.97%
57	18 000	duration / val/train	2 705	1.5488	3 448	1.7571	3 515	1.7015
58		train loss / train diff %	0.7400	-1.70%	0.6234	0.13%	0.6321	-1.65%
59		val loss / val diff %	1.1461	0.38%	1.0954	0.09%	1.0755	0.35%
60	20 000	duration / val/train	3 007	1.5418	3 832	1.7726	3 906	1.7541

61		<i>train loss / train diff %</i>	0.7349	-0.69%	0.6125	-1.75%	0.6226	-1.50%
62		<i>val loss / val diff %</i>	1.1331	-1.13%	1.0857	-0.89%	1.0921	1.54%
63	22 000	<i>duration / val/train</i>	3 311	1.5544	4 216	1.8270	4 297	1.7882
64		<i>train loss / train diff %</i>	0.7228	-1.65%	0.6017	-1.76%	0.6162	-1.03%
65		<i>val loss / val diff %</i>	1.1235	-0.85%	1.0993	1.25%	1.1019	0.90%
66	24 000	<i>duration / val/train</i>	3 614	1.5617	4 596	1.8709	4 688	1.8327
67		<i>train loss / train diff %</i>	0.7148	-1.11%	0.5885	-2.19%	0.5997	-2.68%
68		<i>val loss / val diff %</i>	1.1163	-0.64%	1.1010	0.15%	1.0991	-0.25%
69	26 000	<i>duration / val/train</i>	3 929	1.5779	4 978	1.8975	5 079	1.8394
70		<i>train loss / train diff %</i>	0.7069	-1.11%	0.5885	0.00%	0.5973	-0.40%
71		<i>val loss / val diff %</i>	1.1154	-0.08%	1.1167	1.43%	1.0987	-0.04%
72	28 000	<i>duration / val/train</i>	4 246	1.5525	5 366	1.8603	5 470	1.8432
73		<i>train loss / train diff %</i>	0.7102	0.47%	0.5850	-0.59%	0.5932	-0.69%
74		<i>val loss / val diff %</i>	1.1026	-1.15%	1.0883	-2.54%	1.0934	-0.48%
75	30 000	<i>duration / val/train</i>	4 551	1.6290	5 755	1.8753	5 861	1.8665
76		<i>train loss / train diff %</i>	0.6916	-2.62%	0.5784	-1.13%	0.5819	-1.90%
77		<i>val loss / val diff %</i>	1.1266	2.18%	1.0847	-0.33%	1.0861	-0.67%
78	32 000	<i>duration / val/train</i>	4849	1.6198	6 142	1.9319	6 252	1.8965
79		<i>train loss / train diff %</i>	0.6905	-0.16%	0.5725	-1.02%	0.5761	-1.00%
80		<i>val loss / val diff %</i>	1.1185	-0.72%	1.1060	1.96%	1.0926	0.60%
81	34 000	<i>duration / val/train</i>	5 144	1.6054	6 512	1.9714	6 643	1.9298
82		<i>train loss / train diff %</i>	0.6877	-0.41%	0.5656	-1.21%	0.5673	-1.53%
83		<i>val loss / val diff %</i>	1.1040	-1.30%	1.1150	0.81%	1.0948	0.20%
84	36 000	<i>duration / val/train</i>	5 437	1.6340	6 883	1.9435	7 033	1.9485
85		<i>train loss / train diff %</i>	0.6801	-1.11%	0.5615	-0.72%	0.5646	-0.48%
86		<i>val loss / val diff %</i>	1.1113	0.66%	1.0913	-2.13%	1.1001	0.48%
87	38 000	<i>duration / val/train</i>	5 733	1.6206	7 252	1.9821	7 424	1.9814
88		<i>train loss / train diff %</i>	0.6777	-0.35%	0.5590	-0.45%	0.5599	-0.83%
89		<i>val loss / val diff %</i>	1.0983	-1.17%	1.1080	1.53%	1.1094	0.85%
90	40 000	<i>duration / val/train</i>	6 028	1.6527	7 618	1.9989	7 815	1.9802
91		<i>train loss / train diff %</i>	0.6678	-1.46%	0.5551	-0.70%	0.5508	-1.63%
92		<i>val loss / val diff %</i>	1.1037	0.49%	1.1096	0.14%	1.0907	-1.69%
93	42 000	<i>duration / val/train</i>	6 320	1.6688	7 985	2.0359	8 206	2.0364
94		<i>train loss / train diff %</i>	0.6670	-0.12%	0.5460	-1.64%	0.5463	-0.82%
95		<i>val loss / val diff %</i>	1.1131	0.85%	1.1116	0.18%	1.1125	2.00%
96	44 000	<i>duration / val/train</i>	6 613	1.6945	8 352	2.0772	8 597	2.0458
97		<i>train loss / train diff %</i>	0.6580	-1.35%	0.5438	-0.40%	0.5438	-0.46%
98		<i>val loss / val diff %</i>	1.1150	0.17%	1.1296	1.62%	1.1125	0.00%
99	46 000	<i>duration / val/train</i>	6 905	1.6788	8 718	2.0379	8 988	2.0561
100		<i>train loss / train diff %</i>	0.6582	0.03%	0.5412	-0.48%	0.5380	-1.07%
101		<i>val loss / val diff %</i>	1.1050	-0.90%	1.1029	-2.36%	1.1062	-0.57%
102	48 000	<i>duration / val/train</i>	7 198	1.6735	9 083	2.0834	9 378	2.0813
103		<i>train loss / train diff %</i>	0.6533	-0.74%	0.5334	-1.44%	0.5374	-0.11%
104		<i>val loss / val diff %</i>	1.0933	-1.06%	1.1113	0.76%	1.1185	1.11%
105	50 000	<i>duration / val/train</i>	7 490	1.6947	9 451	2.1095	9 769	2.0766
106		<i>train loss / train diff %</i>	0.6534	0.02%	0.5362	0.52%	0.5312	-1.15%
107		<i>val loss / val diff %</i>	1.1073	1.28%	1.1311	1.78%	1.1031	-1.38%
108	52 000	<i>duration / val/train</i>	7 783	1.7210	9 818	2.1484	10 160	2.0842
109		<i>train loss / train diff %</i>	0.6472	-0.95%	0.5284	-1.45%	0.5288	-0.45%

110		<i>val loss / val diff %</i>	1.1138	0.59%	1.1352	0.36%	1.1021	-0.09%
111	54 000	<i>duration / val/train</i>	8 075	1.7269	10 184	2.1343	10 550	2.1490
112		<i>train loss / train diff %</i>	0.6453	-0.29%	0.5213	-1.34%	0.5214	-1.40%
113		<i>val loss / val diff %</i>	1.1144	0.05%	1.1126	-1.99%	1.1205	1.67%
114	56 000	<i>duration / val/train</i>	8 368	1.7187	10 551	2.2090	10 941	2.1995
115		<i>train loss / train diff %</i>	0.6396	-0.88%	0.5238	0.48%	0.5129	-1.63%
116		<i>val loss / val diff %</i>	1.0993	-1.35%	1.1571	4.00%	1.1281	0.68%
117	58 000	<i>duration / val/train</i>	8 661	1.7357	10 918	2.1585	11 339	2.1572
118		<i>train loss / train diff %</i>	0.6363	-0.52%	0.5204	-0.65%	0.5164	0.68%
119		<i>val loss / val diff %</i>	1.1044	0.46%	1.1233	-2.92%	1.1140	-1.25%
120	60 000	<i>duration / val/train</i>	8 953	1.7240	11 284	2.2005	11 737	2.1912
121		<i>train loss / train diff %</i>	0.6338	-0.39%	0.5178	-0.50%	0.5115	-0.95%
122		<i>val loss / val diff %</i>	1.0927	-1.06%	1.1394	1.43%	1.1208	0.61%
123	62 000	<i>duration / val/train</i>	9 246	1.7334	11 651	2.2209	12 137	2.2063
124		<i>train loss / train diff %</i>	0.6373	0.55%	0.5138	-0.77%	0.5074	-0.80%
125		<i>val loss / val diff %</i>	1.1047	1.10%	1.1411	0.15%	1.1195	-0.12%
126	64 000	<i>duration / val/train</i>	9 538	1.7582	12 017	2.2678	12 547	2.2488
127		<i>train loss / train diff %</i>	0.6315	-0.91%	0.5100	-0.74%	0.5077	0.06%
128		<i>val loss / val diff %</i>	1.1103	0.51%	1.1566	1.36%	1.1417	1.98%
129	66 000	<i>duration / val/train</i>	9 831	1.7295	12 384	2.2062	12 943	2.2507
130		<i>train loss / train diff %</i>	0.6365	0.79%	0.5135	0.69%	0.4994	-1.63%
131		<i>val loss / val diff %</i>	1.1008	-0.86%	1.1329	-2.05%	1.1240	-1.55%
132	68 000	<i>duration / val/train</i>	10 123	1.7680	12 750	2.2539	13 347	2.2330
133		<i>train loss / train diff %</i>	0.6301	-1.01%	0.5045	-1.75%	0.5021	0.54%
134		<i>val loss / val diff %</i>	1.1140	1.20%	1.1371	0.37%	1.1212	-0.25%
135	70 000	<i>duration / val/train</i>	10 416	1.7748	13 117	2.2495	13 749	2.2751
136		<i>train loss / train diff %</i>	0.6310	0.14%	0.5047	0.04%	0.4933	-1.75%
137		<i>val loss / val diff %</i>	1.1199	0.53%	1.1353	-0.16%	1.1223	0.10%
138	72 000	<i>duration / val/train</i>	10 708	1.8103	13 484	2.2816	14 144	2.3209
139		<i>train loss / train diff %</i>	0.6209	-1.60%	0.5017	-0.59%	0.4899	-0.69%
140		<i>val loss / val diff %</i>	1.1240	0.37%	1.1447	0.83%	1.1370	1.31%
141	74 000	<i>duration / val/train</i>	11 005	1.7740	13 850	2.3040	14 534	2.3270
142		<i>train loss / train diff %</i>	0.6217	0.13%	0.4977	-0.80%	0.4863	-0.73%
143		<i>val loss / val diff %</i>	1.1029	-1.88%	1.1467	0.17%	1.1316	-0.47%
144	76 000	<i>duration / val/train</i>	11 308	1.7786	14 217	2.2949	14 925	2.3642
145		<i>train loss / train diff %</i>	0.6180	-0.60%	0.4954	-0.46%	0.4888	0.51%
146		<i>val loss / val diff %</i>	1.0992	-0.34%	1.1369	-0.85%	1.1556	2.12%
147	78 000	<i>duration / val/train</i>	11 613	1.7629	14 583	2.3543	15 316	2.3336
148		<i>train loss / train diff %</i>	0.6225	0.73%	0.4937	-0.34%	0.4853	-0.72%
149		<i>val loss / val diff %</i>	1.0974	-0.16%	1.1623	2.23%	1.1325	-2.00%
150	80 000	<i>duration / val/train</i>	11 911	1.8251	14 950	2.3620	15 707	2.4065
151		<i>train loss / train diff %</i>	0.6136	-1.43%	0.4911	-0.53%	0.4809	-0.91%
152		<i>val loss / val diff %</i>	1.1199	2.05%	1.1600	-0.20%	1.1573	2.19%
153	82 000	<i>duration / val/train</i>	12 206	1.8264	15 316	2.3586	16 098	2.3860
154		<i>train loss / train diff %</i>	0.6117	-0.31%	0.4850	-1.24%	0.4832	0.48%
155		<i>val loss / val diff %</i>	1.1172	-0.24%	1.1439	-1.39%	1.1529	-0.38%
156	84 000	<i>duration / val/train</i>	12 499	1.8197	15 686	2.3713	16 489	2.4013
157		<i>train loss / train diff %</i>	0.6141	0.39%	0.4921	1.46%	0.4807	-0.52%
158		<i>val loss / val diff %</i>	1.1175	0.03%	1.1669	2.01%	1.1543	0.12%

159	86 000	duration / val/train	12 791	1.8141	16 063	2.3949	16 888	2.4021
160		train loss / train diff %	0.6078	-1.03%	0.4905	-0.33%	0.4817	0.21%
161		val loss / val diff %	1.1026	-1.33%	1.1747	0.67%	1.1571	0.24%
162	88 000	duration / val/train	13 084	1.8245	16 448	2.3889	17 291	2.4054
163		train loss / train diff %	0.6050	-0.46%	0.4811	-1.92%	0.4716	-2.10%
164		val loss / val diff %	1.1038	0.11%	1.1493	-2.16%	1.1344	-1.96%
165	90 000	duration / val/train	13 377	1.7914	16 827	2.4240	17 696	2.4456
166		train loss / train diff %	0.6130	1.32%	0.4826	0.31%	0.4746	0.64%
167		val loss / val diff %	1.0981	-0.52%	1.1698	1.78%	1.1607	2.32%
168	92 000	duration / val/train	13 669	1.8407	17 211	2.4157	18 124	2.4559
169		train loss / train diff %	0.6059	-1.16%	0.4826	0.00%	0.4666	-1.69%
170		val loss / val diff %	1.1153	1.57%	1.1658	-0.34%	1.1459	-1.28%
171	94 000	duration / val/train	13 962	1.8326	17 583	2.4303	18 525	2.4610
172		train loss / train diff %	0.6011	-0.79%	0.4818	-0.17%	0.4668	0.04%
173		val loss / val diff %	1.1016	-1.23%	1.1709	0.44%	1.1488	0.25%
174	96 000	duration / val/train	14 254	1.8276	17 952	2.4172	18 924	2.5275
175		train loss / train diff %	0.6031	0.33%	0.4751	-1.39%	0.4639	-0.62%
176		val loss / val diff %	1.1022	0.05%	1.1484	-1.92%	1.1725	2.06%
177	98 000	duration / val/train	14 547	1.8336	18 319	2.4740	19 319	2.5048
178		train loss / train diff %	0.5991	-0.66%	0.4743	-0.17%	0.4612	-0.58%
179		val loss / val diff %	1.0985	-0.34%	1.1734	2.18%	1.1552	-1.48%
180	100 000	duration / val/train	14 839	1.8345	18 691	2.4597	19 710	2.4625
181		train loss / train diff %	0.6049	0.97%	0.4723	-0.42%	0.4618	0.13%
182		val loss / val diff %	1.1097	1.02%	1.1617	-1.00%	1.1372	-1.56%
183	102 000	duration / val/train	15 132	1.8464	19 069	2.5013	20 101	2.5042
184		train loss / train diff %	0.5997	-0.86%	0.4720	-0.06%	0.4601	-0.37%
185		val loss / val diff %	1.1073	-0.22%	1.1806	1.63%	1.1522	1.32%
186	104 000	duration / val/train	15 424	1.8234			20 491	2.5830
187		train loss / train diff %	0.5986	-0.18%			0.4554	-1.02%
188		val loss / val diff %	1.0915	-1.43%			1.1763	2.09%
189	106 000	duration / val/train	15 717	1.8550			20 891	2.6147
190		train loss / train diff %	0.5959	-0.45%			0.4524	-0.66%
191		val loss / val diff %	1.1054	1.27%			1.1829	0.56%
192	108 000	duration / val/train	16 009	1.8594			21 295	2.6191
193		train loss / train diff %	0.5947	-0.20%			0.4495	-0.64%
194		val loss / val diff %	1.1058	0.04%			1.1773	-0.47%
195	110 000	duration / val/train	16 302	1.8686				
196		train loss / train diff %	0.5999	0.87%				
197		val loss / val diff %	1.1210	1.37%				
198	112 000	duration / val/train	16 594	1.8733				
199		train loss / train diff %	0.5945	-0.90%				
200		val loss / val diff %	1.1137	-0.65%				
201	114 000	duration / val/train	16 887	1.8794				
202		train loss / train diff %	0.5902	-0.72%				
203		val loss / val diff %	1.1092	-0.40%				
204	116 000	duration / val/train	17 179	1.8690				
205		train loss / train diff %	0.5914	0.20%				
206		val loss / val diff %	1.1053	-0.35%				
207	118 000	duration / val/train	17 472	1.9017				

208		<i>train loss / train diff %</i>	0.5878	-0.61%		
209		<i>val loss / val diff %</i>	1.1178	1.13%		
210	120 000	<i>duration / val/train</i>	17 764	1.8943		
211		<i>train loss / train diff %</i>	0.5867	-0.19%		
212		<i>val loss / val diff %</i>	1.1114	-0.57%		
213	122 000	<i>duration / val/train</i>	18 057	1.9007		
214		<i>train loss / train diff %</i>	0.5842	-0.43%		
215		<i>val loss / val diff %</i>	1.1104	-0.09%		
216	124 000	<i>duration / val/train</i>	18 350	1.8731		
217		<i>train loss / train diff %</i>	0.5895	0.91%		
218		<i>val loss / val diff %</i>	1.1042	-0.56%		
219	126 000	<i>duration / val/train</i>	18 642	1.9256		
220		<i>train loss / train diff %</i>	0.5859	-0.61%		
221		<i>val loss / val diff %</i>	1.1282	2.17%		
222	128 000	<i>duration / val/train</i>	18 935	1.9105		
223		<i>train loss / train diff %</i>	0.5854	-0.09%		
224		<i>val loss / val diff %</i>	1.1184	-0.87%		
225	130 000	<i>duration / val/train</i>	19 227	1.8908		
226		<i>train loss / train diff %</i>	0.5824	-0.51%		
227		<i>val loss / val diff %</i>	1.1012	-1.54%		
228	132 000	<i>duration / val/train</i>	19 520	1.9248		
229		<i>train loss / train diff %</i>	0.5814	-0.17%		
230		<i>val loss / val diff %</i>	1.1191	1.63%		
231	134 000	<i>duration / val/train</i>	19 812	1.9021		
232		<i>train loss / train diff %</i>	0.5855	0.71%		
233		<i>val loss / val diff %</i>	1.1137	-0.48%		
234	136 000	<i>duration / val/train</i>	20 105	1.9108		
235		<i>train loss / train diff %</i>	0.5816	-0.67%		
236		<i>val loss / val diff %</i>	1.1113	-0.22%		
237	138 000	<i>duration / val/train</i>	20 397	1.9513		
238		<i>train loss / train diff %</i>	0.5768	-0.83%		
239		<i>val loss / val diff %</i>	1.1255	1.28%		
240	140 000	<i>duration / val/train</i>	20 690	1.9306		
241		<i>train loss / train diff %</i>	0.5824	0.97%		
242		<i>val loss / val diff %</i>	1.1244	-0.10%		
243	142 000	<i>duration / val/train</i>	20 982	1.9487		
244		<i>train loss / train diff %</i>	0.5756	-1.17%		
245		<i>val loss / val diff %</i>	1.1217	-0.24%		
246	144 000	<i>duration / val/train</i>	21 275	1.9079		
247		<i>train loss / train diff %</i>	0.5777	0.36%		
248		<i>val loss / val diff %</i>	1.1022	-1.74%		
249	146 000	<i>duration / val/train</i>	21 568	1.9284		
250		<i>train loss / train diff %</i>	0.5797	0.35%		
251		<i>val loss / val diff %</i>	1.1179	1.42%		
252	148 000	<i>duration / val/train</i>	21 860	1.9274		
253		<i>train loss / train diff %</i>	0.5789	-0.14%		
254		<i>val loss / val diff %</i>	1.1158	-0.19%		
255	150 000	<i>duration / val/train</i>	22 153	1.9269		
256		<i>train loss / train diff %</i>	0.5743	-0.79%		

257		<i>val loss / val diff %</i>	1.1066	-0.82%		
258	152 000	<i>duration / val/train</i>	22 445	1.9543		
259		<i>train loss / train diff %</i>	0.5758	0.26%		
260		<i>val loss / val diff %</i>	1.1253	1.69%		
261	154 000	<i>duration / val/train</i>	22 738	1.9521		
262		<i>train loss / train diff %</i>	0.5684	-1.29%		
263		<i>val loss / val diff %</i>	1.1096	-1.40%		
264	156 000	<i>duration / val/train</i>	23 030	1.9654		
265		<i>train loss / train diff %</i>	0.5721	0.65%		
266		<i>val loss / val diff %</i>	1.1244	1.33%		
267	158 000	<i>duration / val/train</i>	23 323	1.9430		
268		<i>train loss / train diff %</i>	0.5751	0.52%		
269		<i>val loss / val diff %</i>	1.1174	-0.62%		
270	160 000	<i>duration / val/train</i>	23 615	1.9704		
271		<i>train loss / train diff %</i>	0.5749	-0.03%		
272		<i>val loss / val diff %</i>	1.1328	1.38%		
273	162 000	<i>duration / val/train</i>	23 908	1.9600		
274		<i>train loss / train diff %</i>	0.5704	-0.78%		
275		<i>val loss / val diff %</i>	1.1180	-1.31%		
276	164 000	<i>duration / val/train</i>	24 200	1.9243		
277		<i>train loss / train diff %</i>	0.5757	0.93%		
278		<i>val loss / val diff %</i>	1.1078	-0.91%		
279	166 000	<i>duration / val/train</i>	24 493	1.9199		
280		<i>train loss / train diff %</i>	0.5741	-0.28%		
281		<i>val loss / val diff %</i>	1.1022	-0.51%		
282	168 000	<i>duration / val/train</i>	24 786	1.9764		
283		<i>train loss / train diff %</i>	0.5666	-1.31%		
284		<i>val loss / val diff %</i>	1.1198	1.60%		
285	170 000	<i>duration / val/train</i>	25 078	1.9654		
286		<i>train loss / train diff %</i>	0.5719	0.94%		
287		<i>val loss / val diff %</i>	1.1240	0.38%		
288	172 000	<i>duration / val/train</i>	25 371	1.9915		
289		<i>train loss / train diff %</i>	0.5619	-1.75%		
290		<i>val loss / val diff %</i>	1.1190	-0.44%		
291	174 000	<i>duration / val/train</i>	25 663	1.9563		
292		<i>train loss / train diff %</i>	0.5677	1.03%		
293		<i>val loss / val diff %</i>	1.1106	-0.75%		
294	176 000	<i>duration / val/train</i>	25 956	1.9676		
295		<i>train loss / train diff %</i>	0.5675	-0.04%		
296		<i>val loss / val diff %</i>	1.1166	0.54%		
297	178 000	<i>duration / val/train</i>	26 248	1.9846		
298		<i>train loss / train diff %</i>	0.5665	-0.18%		
299		<i>val loss / val diff %</i>	1.1243	0.69%		
300	180 000	<i>duration / val/train</i>	26 541	1.9831		
301		<i>train loss / train diff %</i>	0.5611	-0.95%		
302		<i>val loss / val diff %</i>	1.1127	-1.03%		
303	182 000	<i>duration / val/train</i>	26 833	1.9757		
304		<i>train loss / train diff %</i>	0.5674	1.12%		
305		<i>val loss / val diff %</i>	1.1210	0.75%		

306	184 000	<i>duration / val/train</i>	27 126	1.9515		
307		<i>train loss / train diff %</i>	0.5668	-0.11%		
308		<i>val loss / val diff %</i>	1.1061	-1.33%		
309	186 000	<i>duration / val/train</i>	27 419	1.9693		
310		<i>train loss / train diff %</i>	0.5637	-0.55%		
311		<i>val loss / val diff %</i>	1.1101	0.36%		
312	188 000	<i>duration / val/train</i>	27 711	1.9761		
313		<i>train loss / train diff %</i>	0.5607	-0.53%		
314		<i>val loss / val diff %</i>	1.1080	-0.19%		
315	190 000	<i>duration / val/train</i>	28 004	1.9899		
316		<i>train loss / train diff %</i>	0.5617	0.18%		
317		<i>val loss / val diff %</i>	1.1177	0.88%		
318	192 000	<i>duration / val/train</i>	28 296	1.9842		
319		<i>train loss / train diff %</i>	0.5561	-1.00%		
320		<i>val loss / val diff %</i>	1.1034	-1.28%		
321	194 000	<i>duration / val/train</i>	28 589	1.9775		
322		<i>train loss / train diff %</i>	0.5639	1.40%		
323		<i>val loss / val diff %</i>	1.1151	1.06%		
324	196 000	<i>duration / val/train</i>	28 881	1.9817		
325		<i>train loss / train diff %</i>	0.5639	0.00%		
326		<i>val loss / val diff %</i>	1.1175	0.22%		
327	198 000	<i>duration / val/train</i>	29 174	1.9840		
328		<i>train loss / train diff %</i>	0.5610	-0.51%		
329		<i>val loss / val diff %</i>	1.1130	-0.40%		
330	200 000	<i>duration / val/train</i>	29 466	1.9973		
331		<i>train loss / train diff %</i>	0.5544	-1.18%		
332		<i>val loss / val diff %</i>	1.1073	-0.51%		
333	202 000	<i>duration / val/train</i>	29 759	1.9689		
334		<i>train loss / train diff %</i>	0.5619	1.35%		
335		<i>val loss / val diff %</i>	1.1063	-0.09%		
336	204 000	<i>duration / val/train</i>	30 051	2.0063		
337		<i>train loss / train diff %</i>	0.5558	-1.09%		
338		<i>val loss / val diff %</i>	1.1151	0.80%		
339	206 000	<i>duration / val/train</i>	30 344	1.9966		
340		<i>train loss / train diff %</i>	0.5624	1.19%		
341		<i>val loss / val diff %</i>	1.1229	0.70%		
342	208 000	<i>duration / val/train</i>	30 637	2.0066		
343		<i>train loss / train diff %</i>	0.5604	-0.36%		
344		<i>val loss / val diff %</i>	1.1245	0.14%		
345	210 000	<i>duration / val/train</i>	30 929	1.9859		
346		<i>train loss / train diff %</i>	0.5587	-0.30%		
347		<i>val loss / val diff %</i>	1.1095	-1.33%		
348	212 000	<i>duration / val/train</i>	31 222	2.0237		
349		<i>train loss / train diff %</i>	0.5558	-0.52%		
350		<i>val loss / val diff %</i>	1.1248	1.38%		
351	214 000	<i>duration / val/train</i>	31 514	2.0111		
352		<i>train loss / train diff %</i>	0.5574	0.29%		
353		<i>val loss / val diff %</i>	1.1210	-0.34%		
354	216 000	<i>duration / val/train</i>	31 807	2.0013		

355		<i>train loss / train diff %</i>	0.5590	0.29%		
356		<i>val loss / val diff %</i>	1.1187	-0.21%		
357	218 000	<i>duration / val/train</i>	32 099	2.0029		
358		<i>train loss / train diff %</i>	0.5566	-0.43%		
359		<i>val loss / val diff %</i>	1.1148	-0.35%		
360	220 000	<i>duration / val/train</i>	32 392	2.0125		
361		<i>train loss / train diff %</i>	0.5527	-0.70%		
362		<i>val loss / val diff %</i>	1.1123	-0.22%		
363	222 000	<i>duration / val/train</i>	32 684	2.0453		
364		<i>train loss / train diff %</i>	0.5568	0.74%		
365		<i>val loss / val diff %</i>	1.1388	2.38%		
366	224 000	<i>duration / val/train</i>	32 977	2.0056		
367		<i>train loss / train diff %</i>	0.5533	-0.63%		
368		<i>val loss / val diff %</i>	1.1097	-2.56%		
369	226 000	<i>duration / val/train</i>	33 269	2.0056		
370		<i>train loss / train diff %</i>	0.5553	0.36%		
371		<i>val loss / val diff %</i>	1.1137	0.36%		
372	228 000	<i>duration / val/train</i>	33 562	2.0074		
373		<i>train loss / train diff %</i>	0.5553	0.00%		
374		<i>val loss / val diff %</i>	1.1147	0.09%		
375	230 000	<i>duration / val/train</i>	33 855	2.0319		
376		<i>train loss / train diff %</i>	0.5553	0.00%		
377		<i>val loss / val diff %</i>	1.1283	1.22%		
378	232 000	<i>duration / val/train</i>	34 147	2.0173		
379		<i>train loss / train diff %</i>	0.5540	-0.23%		
380		<i>val loss / val diff %</i>	1.1176	-0.95%		
381	234 000	<i>duration / val/train</i>	34 440	2.0169		
382		<i>train loss / train diff %</i>	0.5514	-0.47%		
383		<i>val loss / val diff %</i>	1.1121	-0.49%		
384	236 000	<i>duration / val/train</i>	34 732	2.0203		
385		<i>train loss / train diff %</i>	0.5512	-0.04%		
386		<i>val loss / val diff %</i>	1.1136	0.13%		
387	238 000	<i>duration / val/train</i>	35 025	2.0556		
388		<i>train loss / train diff %</i>	0.5507	-0.09%		
389		<i>val loss / val diff %</i>	1.1320	1.65%		
390	240 000	<i>duration / val/train</i>	35 317	2.0214		
391		<i>train loss / train diff %</i>	0.5509	0.04%		
392		<i>val loss / val diff %</i>	1.1136	-1.63%		
393	242 000	<i>duration / val/train</i>	35 610	2.0706		
394		<i>train loss / train diff %</i>	0.5468	-0.74%		
395		<i>val loss / val diff %</i>	1.1322	1.67%		
396	244 000	<i>duration / val/train</i>	35 902	2.0366		
397		<i>train loss / train diff %</i>	0.5497	0.53%		
398		<i>val loss / val diff %</i>	1.1195	-1.12%		
399	246 000	<i>duration / val/train</i>	36 195	2.0463		
400		<i>train loss / train diff %</i>	0.5446	-0.93%		
401		<i>val loss / val diff %</i>	1.1144	-0.46%		
402	248 000	<i>duration / val/train</i>	36 487	2.0292		
403		<i>train loss / train diff %</i>	0.5483	0.68%		

404		<i>val loss / val diff %</i>	1.1126	-0.16%		
405	250 000	<i>duration / val/train</i>	36 780	2.0348		
406		<i>train loss / train diff %</i>	0.5490	0.13%		
407		<i>val loss / val diff %</i>	1.1171	0.40%		
408	252 000	<i>duration / val/train</i>	37 072	2.0439		
409		<i>train loss / train diff %</i>	0.5530	0.73%		
410		<i>val loss / val diff %</i>	1.1303	1.18%		
411	254 000	<i>duration / val/train</i>	37 370	2.0226		
412		<i>train loss / train diff %</i>	0.5498	-0.58%		
413		<i>val loss / val diff %</i>	1.1120	-1.62%		
414	256 000	<i>duration / val/train</i>	37 673	2.0324		
415		<i>train loss / train diff %</i>	0.5486	-0.22%		
416		<i>val loss / val diff %</i>	1.1150	0.27%		
417	258 000	<i>duration / val/train</i>	37 974	2.0497		
418		<i>train loss / train diff %</i>	0.5476	-0.18%		
419		<i>val loss / val diff %</i>	1.1224	0.66%		
420	260 000	<i>duration / val/train</i>	38 272	2.0153		
421		<i>train loss / train diff %</i>	0.5504	0.51%		
422		<i>val loss / val diff %</i>	1.1092	-1.18%		
423	262 000	<i>duration / val/train</i>	38 576	2.0394		
424		<i>train loss / train diff %</i>	0.5484	-0.36%		
425		<i>val loss / val diff %</i>	1.1184	0.83%		
426	264 000	<i>duration / val/train</i>	38 878	2.0515		
427		<i>train loss / train diff %</i>	0.5472	-0.22%		
428		<i>val loss / val diff %</i>	1.1226	0.38%		
429	266 000	<i>duration / val/train</i>	39 183	2.0567		
430		<i>train loss / train diff %</i>	0.5464	-0.15%		
431		<i>val loss / val diff %</i>	1.1238	0.11%		
432	268 000	<i>duration / val/train</i>	39 487	2.0049		
433		<i>train loss / train diff %</i>	0.5517	0.97%		
434		<i>val loss / val diff %</i>	1.1061	-1.58%		
435	270 000	<i>duration / val/train</i>	39 800	2.0537		
436		<i>train loss / train diff %</i>	0.5435	-1.49%		
437		<i>val loss / val diff %</i>	1.1162	0.91%		

Table 18. Results of training models at Step 8.

8.2. Freezing old weights. Step 9.

Testing showed that at intermediate steps, the results change only when the freezing coefficients are close to zero, or at least less than 0.1. This reduces the accuracy of the neural network. Therefore, a single coefficient of 0.8 was used for all stages of this step. This will either have no effect on the results or will at least serve as an additional mechanism against overfitting.

The results of training the models in Step 9 are shown in Table 19.

<i>Nº</i>	<i>iterations</i>	<i>Params</i>	<i>12/128/4/2048/0.10/4/0</i> <i>.8</i>	<i>10/192/6/2048/0.10/4/0</i> <i>.8</i>	<i>8/256/8/2048/0.15/4</i> <i>/0.8</i>
1		<i>model size, M</i>	2.69	4.91	6.94
2		<i>n_layer</i>	12	10	8

3		<i>n_embd</i>	128		192		256
4		<i>n_head</i>	4		6		8
5		<i>block_size</i>	2 048		2 048		2 048
6		<i>dropout</i>	0.10		0.10		0.15
7		<i>batch_size</i>	4		4		4
8		<i>weight_freeze</i>	0.8		0.8		0.8
9	Stage 1	<i>model</i>	2/32/1/128/0.00/64/40k		2/64/2/128/0.00/64/50k		2/64/2/128/0.00/64/50k
10		<i>iterations</i>	40 000		50 000		50 000
11		<i>duration / val/train</i>	217	1.2228	274	1.3309	264 1.3309
12		<i>train loss</i>	1.6816		1.3640		1.3640
13		<i>val loss</i>	2.0563		1.8154		1.8154
14	Stage 2	<i>model</i>	4/64/2/256/0.05/64/0.8/45k		4/96/3/256/0.05/64/0.8/50k		3/96/3/256/0.05/64/0.8/75k
15		<i>iterations</i>	45 000		50 000		75 000
16		<i>duration / val/train</i>	441	1.3835	765	1.4781	886 1.4671
17		<i>train loss</i>	1.1791	-29.88%	0.9824	-27.98%	1.0347 -24.14%
18		<i>val loss</i>	1.6313	-20.67%	1.4521	-20.01%	1.5180 -16.38%
19	Stage 3	<i>model</i>	6/64/2/512/0.05/64/0.8/40k		6/128/4/512/0.05/64/0.8/40k		4/128/4/512/0.05/64/0.8/40k
20		<i>iterations</i>	40 000		40 000		40 000
21		<i>duration / val/train</i>	1 791	1.4178	3 693	1.7745	2 516 1.6700
22		<i>train loss</i>	0.9858	-16.39%	0.7438	-24.29%	0.8128 -21.45%
23		<i>val loss</i>	1.3977	-14.32%	1.3199	-9.10%	1.3574 -10.58%
24	Stage 4	<i>model</i>	9/96/3/1024/0.05/32/0.8/40k		8/160/5/1536/0.10/8/0.8/60k		6/192/6/1536/0.10/8/0.8/65k
25		<i>iterations</i>	40 000		60 000		65 000
26		<i>duration / val/train</i>	7 017	1.6901	8 908	1.8162	8 563 1.8297
27		<i>train loss</i>	0.7402	-24.91%	0.6372	-14.33%	0.6276 -22.79%
28		<i>val loss</i>	1.2510	-10.50%	1.1573	-12.32%	1.1483 -15.40%
29		<i>Total time</i>	9 466		13 640		12 229
30	0	<i>duration / val/train</i>	0	1.0487	0	0.9746	0 1.0217
31		<i>train loss</i>	3.8775		3.6840		3.3243
32		<i>val loss</i>	4.0663		3.5903		3.3964
33	2 000	<i>duration / val/train</i>	303	1.3219	367	1.5564	412 1.4751
34		<i>train loss / train diff %</i>	0.9725	-74.92%	0.7497	-79.65%	0.7978 -76.00%
35		<i>val loss / val diff %</i>	1.2855	-68.39%	1.1668	-67.50%	1.1768 -65.35%
36	4 000	<i>duration / val/train</i>	629	1.3879	734	1.5875	811 1.5603
37		<i>train loss / train diff %</i>	0.8765	-9.87%	0.6890	-8.10%	0.7251 -9.11%
38		<i>val loss / val diff %</i>	1.2165	-5.37%	1.0938	-6.26%	1.1314 -3.86%
39	6 000	<i>duration / val/train</i>	939	1.3909	1 100	1.6586	1 219 1.5980
40		<i>train loss / train diff %</i>	0.8470	-3.37%	0.6691	-2.89%	0.7015 -3.25%
41		<i>val loss / val diff %</i>	1.1781	-3.16%	1.1098	1.46%	1.1210 -0.92%
42	8 000	<i>duration / val/train</i>	1 243	1.4324	1 467	1.6862	1 624 1.6192
43		<i>train loss / train diff %</i>	0.8233	-2.80%	0.6564	-1.90%	0.6823 -2.74%
44		<i>val loss / val diff %</i>	1.1793	0.10%	1.1068	-0.27%	1.1048 -1.45%
45	10 000	<i>duration / val/train</i>	1 538	1.4270	1 833	1.6964	2 023 1.6506
46		<i>train loss / train diff %</i>	0.8032	-2.44%	0.6485	-1.20%	0.6723 -1.47%
47		<i>val loss / val diff %</i>	1.1462	-2.81%	1.1001	-0.61%	1.1097 0.44%
48	12 000	<i>duration / val/train</i>	1 833	1.4667	2 200	1.7280	2 429 1.6325

49		<i>train loss / train diff %</i>	0.7911	-1.51%	0.6316	-2.61%	0.6662	-0.91%
50		<i>val loss / val diff %</i>	1.1603	1.23%	1.0914	-0.79%	1.0876	-1.99%
51	14 000	<i>duration / val/train</i>	2 128	1.4680	2 567	1.7326	2 831	1.6899
52		<i>train loss / train diff %</i>	0.7797	-1.44%	0.6268	-0.76%	0.6498	-2.46%
53		<i>val loss / val diff %</i>	1.1446	-1.35%	1.0860	-0.49%	1.0981	0.97%
54	16 000	<i>duration / val/train</i>	2 420	1.5005	2 933	1.7504	3 227	1.7135
55		<i>train loss / train diff %</i>	0.7627	-2.18%	0.6251	-0.27%	0.6394	-1.60%
56		<i>val loss / val diff %</i>	1.1444	-0.02%	1.0942	0.76%	1.0956	-0.23%
57	18 000	<i>duration / val/train</i>	2 713	1.5519	3 300	1.7672	3 630	1.7566
58		<i>train loss / train diff %</i>	0.7469	-2.07%	0.6152	-1.58%	0.6256	-2.16%
59		<i>val loss / val diff %</i>	1.1591	1.28%	1.0872	-0.64%	1.0989	0.30%
60	20 000	<i>duration / val/train</i>	3 005	1.5319	3 667	1.8005	4 035	1.8007
61		<i>train loss / train diff %</i>	0.7462	-0.09%	0.6094	-0.94%	0.6135	-1.93%
62		<i>val loss / val diff %</i>	1.1431	-1.38%	1.0972	0.92%	1.1047	0.53%
63	22 000	<i>duration / val/train</i>	3 298	1.5551	4 033	1.8053	4 439	1.8209
64		<i>train loss / train diff %</i>	0.7364	-1.31%	0.6051	-0.71%	0.6064	-1.16%
65		<i>val loss / val diff %</i>	1.1452	0.18%	1.0924	-0.44%	1.1042	-0.05%
66	24 000	<i>duration / val/train</i>	3 590	1.5781	4 400	1.8455	4 833	1.8352
67		<i>train loss / train diff %</i>	0.7258	-1.44%	0.5966	-1.40%	0.6019	-0.74%
68		<i>val loss / val diff %</i>	1.1454	0.02%	1.1010	0.79%	1.1046	0.04%
69	26 000	<i>duration / val/train</i>	3 883	1.5890	4 766	1.8771	5 229	1.8793
70		<i>train loss / train diff %</i>	0.7226	-0.44%	0.5899	-1.12%	0.5880	-2.31%
71		<i>val loss / val diff %</i>	1.1482	0.24%	1.1073	0.57%	1.1050	0.04%
72	28 000	<i>duration / val/train</i>	4 175	1.5961	5 133	1.8623	5 651	1.9000
73		<i>train loss / train diff %</i>	0.7128	-1.36%	0.5899	0.00%	0.5799	-1.38%
74		<i>val loss / val diff %</i>	1.1377	-0.91%	1.0986	-0.79%	1.1018	-0.29%
75	30 000	<i>duration / val/train</i>	4 468	1.6225	5 500	1.9103	6 065	1.9107
76		<i>train loss / train diff %</i>	0.7086	-0.59%	0.5763	-2.31%	0.5745	-0.93%
77		<i>val loss / val diff %</i>	1.1497	1.05%	1.1009	0.21%	1.0977	-0.37%
78	32 000	<i>duration / val/train</i>	4760	1.6235	5 866	1.8995	6 468	1.9554
79		<i>train loss / train diff %</i>	0.7010	-1.07%	0.5754	-0.16%	0.5672	-1.27%
80		<i>val loss / val diff %</i>	1.1381	-1.01%	1.0930	-0.72%	1.1091	1.04%
81	34 000	<i>duration / val/train</i>	5 052	1.6337	6 233	1.9535	6 881	1.9419
82		<i>train loss / train diff %</i>	0.6977	-0.47%	0.5656	-1.70%	0.5626	-0.81%
83		<i>val loss / val diff %</i>	1.1398	0.15%	1.1049	1.09%	1.0925	-1.50%
84	36 000	<i>duration / val/train</i>	5 345	1.6572	6 608	1.9519	7 309	1.9652
85		<i>train loss / train diff %</i>	0.6922	-0.79%	0.5595	-1.08%	0.5578	-0.85%
86		<i>val loss / val diff %</i>	1.1471	0.64%	1.0921	-1.16%	1.0962	0.34%
87	38 000	<i>duration / val/train</i>	5 637	1.6404	6 988	1.9912	7 726	2.0020
88		<i>train loss / train diff %</i>	0.6835	-1.26%	0.5585	-0.18%	0.5565	-0.23%
89		<i>val loss / val diff %</i>	1.1212	-2.26%	1.1121	1.83%	1.1141	1.63%
90	40 000	<i>duration / val/train</i>	5 936	1.6887	7 382	2.0071	8 141	2.0269
91		<i>train loss / train diff %</i>	0.6788	-0.69%	0.5519	-1.18%	0.5500	-1.17%
92		<i>val loss / val diff %</i>	1.1463	2.24%	1.1077	-0.40%	1.1148	0.06%
93	42 000	<i>duration / val/train</i>	6 239	1.6739	7 752	2.0335	8 540	2.0595
94		<i>train loss / train diff %</i>	0.6829	0.60%	0.5469	-0.91%	0.5462	-0.69%
95		<i>val loss / val diff %</i>	1.1431	-0.28%	1.1121	0.40%	1.1249	0.91%
96	44 000	<i>duration / val/train</i>	6 547	1.6696	8 130	2.0390	8 938	2.0490
97		<i>train loss / train diff %</i>	0.6806	-0.34%	0.5437	-0.59%	0.5448	-0.26%

98		<i>val loss / val diff %</i>	1.1363	-0.59%	1.1086	-0.31%	1.1163	-0.76%
99	46 000	<i>duration / val/train</i>	6 855	1.6964	8 509	2.0481	9 344	2.0968
100		<i>train loss / train diff %</i>	0.6703	-1.51%	0.5425	-0.22%	0.5402	-0.84%
101		<i>val loss / val diff %</i>	1.1371	0.07%	1.1111	0.23%	1.1327	1.47%
102	48 000	<i>duration / val/train</i>	7 150	1.7033	8 887	2.1134	9 739	2.1216
103		<i>train loss / train diff %</i>	0.6707	0.06%	0.5343	-1.51%	0.5355	-0.87%
104		<i>val loss / val diff %</i>	1.1424	0.47%	1.1292	1.63%	1.1361	0.30%
105	50 000	<i>duration / val/train</i>	7 452	1.6865	9 273	2.0932	10 137	2.1327
106		<i>train loss / train diff %</i>	0.6645	-0.92%	0.5376	0.62%	0.5322	-0.62%
107		<i>val loss / val diff %</i>	1.1207	-1.90%	1.1253	-0.35%	1.1350	-0.10%
108	52 000	<i>duration / val/train</i>	7 769	1.7169	9 656	2.0975	10 564	2.1522
109		<i>train loss / train diff %</i>	0.6596	-0.74%	0.5303	-1.36%	0.5251	-1.33%
110		<i>val loss / val diff %</i>	1.1325	1.05%	1.1123	-1.16%	1.1301	-0.43%
111	54 000	<i>duration / val/train</i>	8 070	1.7463	10 034	2.0742	10 963	2.1576
112		<i>train loss / train diff %</i>	0.6559	-0.56%	0.5307	0.08%	0.5210	-0.78%
113		<i>val loss / val diff %</i>	1.1454	1.14%	1.1008	-1.03%	1.1241	-0.53%
114	56 000	<i>duration / val/train</i>	8 365	1.7448	10 427	2.1572	11 364	2.1793
115		<i>train loss / train diff %</i>	0.6555	-0.06%	0.5273	-0.64%	0.5197	-0.25%
116		<i>val loss / val diff %</i>	1.1437	-0.15%	1.1375	3.33%	1.1326	0.76%
117	58 000	<i>duration / val/train</i>	8 657	1.7585	10 809	2.1530	11 765	2.2044
118		<i>train loss / train diff %</i>	0.6494	-0.93%	0.5235	-0.72%	0.5157	-0.77%
119		<i>val loss / val diff %</i>	1.1420	-0.15%	1.1271	-0.91%	1.1368	0.37%
120	60 000	<i>duration / val/train</i>	8 950	1.7502	11 184	2.1736	12 166	2.2440
121		<i>train loss / train diff %</i>	0.6489	-0.08%	0.5162	-1.39%	0.5061	-1.86%
122		<i>val loss / val diff %</i>	1.1357	-0.55%	1.1220	-0.45%	1.1357	-0.10%
123	62 000	<i>duration / val/train</i>	9 243	1.7592	11 559	2.1804	12 563	2.2324
124		<i>train loss / train diff %</i>	0.6436	-0.82%	0.5205	0.83%	0.5077	0.32%
125		<i>val loss / val diff %</i>	1.1322	-0.31%	1.1349	1.15%	1.1334	-0.20%
126	64 000	<i>duration / val/train</i>	9 535	1.7518	11 928	2.2047	12 961	2.2800
127		<i>train loss / train diff %</i>	0.6475	0.61%	0.5202	-0.06%	0.5071	-0.12%
128		<i>val loss / val diff %</i>	1.1343	0.19%	1.1469	1.06%	1.1562	2.01%
129	66 000	<i>duration / val/train</i>	9 828	1.7756	12 295	2.1827	13 361	2.2781
130		<i>train loss / train diff %</i>	0.6349	-1.95%	0.5139	-1.21%	0.5031	-0.79%
131		<i>val loss / val diff %</i>	1.1273	-0.62%	1.1217	-2.20%	1.1461	-0.87%
132	68 000	<i>duration / val/train</i>	10 120	1.7674	12 661	2.2154	13 761	2.2871
133		<i>train loss / train diff %</i>	0.6390	0.65%	0.5093	-0.90%	0.5019	-0.24%
134		<i>val loss / val diff %</i>	1.1294	0.19%	1.1283	0.59%	1.1479	0.16%
135	70 000	<i>duration / val/train</i>	10 413	1.7907	13 028	2.2404	14 161	2.2967
136		<i>train loss / train diff %</i>	0.6399	0.14%	0.5078	-0.29%	0.4954	-1.30%
137		<i>val loss / val diff %</i>	1.1459	1.46%	1.1377	0.83%	1.1378	-0.88%
138	72 000	<i>duration / val/train</i>	10 705	1.8143	13 394	2.2505	14 562	2.3221
139		<i>train loss / train diff %</i>	0.6323	-1.19%	0.5065	-0.26%	0.4952	-0.04%
140		<i>val loss / val diff %</i>	1.1472	0.11%	1.1399	0.19%	1.1499	1.06%
141	74 000	<i>duration / val/train</i>	10 998	1.7946	13 786	2.2407	14 961	2.3491
142		<i>train loss / train diff %</i>	0.6324	0.02%	0.5060	-0.10%	0.4898	-1.09%
143		<i>val loss / val diff %</i>	1.1349	-1.07%	1.1338	-0.54%	1.1506	0.06%
144	76 000	<i>duration / val/train</i>	11 290	1.8107	14 169	2.2729	15 368	2.4102
145		<i>train loss / train diff %</i>	0.6256	-1.08%	0.5027	-0.65%	0.4856	-0.86%
146		<i>val loss / val diff %</i>	1.1328	-0.19%	1.1426	0.78%	1.1704	1.72%

147	78 000	duration / val/train	11 583	1.8213	14 567	2.2915	15 767	2.4042
148		train loss / train diff %	0.6249	-0.11%	0.4985	-0.84%	0.4847	-0.19%
149		val loss / val diff %	1.1381	0.47%	1.1423	-0.03%	1.1653	-0.44%
150	80 000	duration / val/train	11 875	1.8400	14 951	2.3249	16 161	2.3789
151		train loss / train diff %	0.6192	-0.91%	0.4974	-0.22%	0.4856	0.19%
152		val loss / val diff %	1.1393	0.11%	1.1564	1.23%	1.1552	-0.87%
153	82 000	duration / val/train	12 168	1.8117	15 330	2.3171	16 552	2.4162
154		train loss / train diff %	0.6207	0.24%	0.4961	-0.26%	0.4832	-0.49%
155		val loss / val diff %	1.1245	-1.30%	1.1495	-0.60%	1.1675	1.06%
156	84 000	duration / val/train	12 460	1.8439	15 714	2.3364	16 955	2.4315
157		train loss / train diff %	0.6174	-0.53%	0.4925	-0.73%	0.4816	-0.33%
158		val loss / val diff %	1.1384	1.24%	1.1507	0.10%	1.1710	0.30%
159	86 000	duration / val/train	12 753	1.7897	16 094	2.3427	17 357	2.4266
160		train loss / train diff %	0.6192	0.29%	0.4964	0.79%	0.4791	-0.52%
161		val loss / val diff %	1.1082	-2.65%	1.1629	1.06%	1.1626	-0.72%
162	88 000	duration / val/train	13 057	1.8119	16 493	2.3354	17 752	2.4313
163		train loss / train diff %	0.6157	-0.57%	0.4890	-1.49%	0.4739	-1.09%
164		val loss / val diff %	1.1156	0.67%	1.1420	-1.80%	1.1522	-0.89%
165	90 000	duration / val/train	13 354	1.7914	16 896	2.3746	18 143	2.4663
166		train loss / train diff %	0.6166	0.15%	0.4867	-0.47%	0.4737	-0.04%
167		val loss / val diff %	1.1046	-0.99%	1.1557	1.20%	1.1683	1.40%
168	92 000	duration / val/train	13 660	1.8453	17 282	2.3769	18 534	2.4873
169		train loss / train diff %	0.6117	-0.79%	0.4884	0.35%	0.4669	-1.44%
170		val loss / val diff %	1.1288	2.19%	1.1609	0.45%	1.1613	-0.60%
171	94 000	duration / val/train	13 963	1.8286	17 663	2.3549	18 924	2.5130
172		train loss / train diff %	0.6080	-0.60%	0.4880	-0.08%	0.4710	0.88%
173		val loss / val diff %	1.1118	-1.51%	1.1492	-1.01%	1.1836	1.92%
174	96 000	duration / val/train	14 258	1.8380	18 038	2.3725	19 315	2.5072
175		train loss / train diff %	0.6073	-0.12%	0.4840	-0.82%	0.4675	-0.74%
176		val loss / val diff %	1.1162	0.40%	1.1483	-0.08%	1.1721	-0.97%
177	98 000	duration / val/train	14 553	1.8395	18 408	2.3990	19 713	2.5658
178		train loss / train diff %	0.6123	0.82%	0.4820	-0.41%	0.4641	-0.73%
179		val loss / val diff %	1.1263	0.90%	1.1563	0.70%	1.1908	1.60%
180	100 000	duration / val/train	14 847	1.8562	18 780	2.3958	20 128	2.4955
181		train loss / train diff %	0.6071	-0.85%	0.4808	-0.25%	0.4672	0.67%
182		val loss / val diff %	1.1269	0.05%	1.1519	-0.38%	1.1659	-2.09%
183	102 000	duration / val/train	15 140	1.8420	19 155	2.4283	20 539	2.5768
184		train loss / train diff %	0.6122	0.84%	0.4833	0.52%	0.4598	-1.58%
185		val loss / val diff %	1.1277	0.07%	1.1736	1.88%	1.1848	1.62%
186	104 000	duration / val/train	15 432	1.8420				
187		train loss / train diff %	0.6062	-0.98%				
188		val loss / val diff %	1.1166	-0.98%				
189	106 000	duration / val/train	15 725	1.8838				
190		train loss / train diff %	0.6076	0.23%				
191		val loss / val diff %	1.1446	2.51%				
192	108 000	duration / val/train	16 017	1.8647				
193		train loss / train diff %	0.6008	-1.12%				
194		val loss / val diff %	1.1203	-2.12%				
195	110 000	duration / val/train	16 310	1.8963				

196		<i>train loss / train diff %</i>	0.6007	-0.02%		
197		<i>val loss / val diff %</i>	1.1391	1.68%		
198	112 000	<i>duration / val/train</i>	16 602	1.9005		
199		<i>train loss / train diff %</i>	0.5988	-0.32%		
200		<i>val loss / val diff %</i>	1.1380	-0.10%		
201	114 000	<i>duration / val/train</i>	16 895	1.8666		
202		<i>train loss / train diff %</i>	0.6026	0.63%		
203		<i>val loss / val diff %</i>	1.1248	-1.16%		
204	116 000	<i>duration / val/train</i>	17 189	1.8842		
205		<i>train loss / train diff %</i>	0.5950	-1.26%		
206		<i>val loss / val diff %</i>	1.1211	-0.33%		
207	118 000	<i>duration / val/train</i>	17 486	1.9194		
208		<i>train loss / train diff %</i>	0.5933	-0.29%		
209		<i>val loss / val diff %</i>	1.1388	1.58%		
210	120 000	<i>duration / val/train</i>	17 803	1.9111		
211		<i>train loss / train diff %</i>	0.5964	0.52%		
212		<i>val loss / val diff %</i>	1.1398	0.09%		
213	122 000	<i>duration / val/train</i>	18 122	1.9137		
214		<i>train loss / train diff %</i>	0.5901	-1.06%		
215		<i>val loss / val diff %</i>	1.1293	-0.92%		
216	124 000	<i>duration / val/train</i>	18 426	1.9133		
217		<i>train loss / train diff %</i>	0.5894	-0.12%		
218		<i>val loss / val diff %</i>	1.1277	-0.14%		
219	126 000	<i>duration / val/train</i>	18 730	1.8979		
220		<i>train loss / train diff %</i>	0.5966	1.22%		
221		<i>val loss / val diff %</i>	1.1323	0.41%		
222	128 000	<i>duration / val/train</i>	19 030	1.9322		
223		<i>train loss / train diff %</i>	0.5883	-1.39%		
224		<i>val loss / val diff %</i>	1.1367	0.39%		
225	130 000	<i>duration / val/train</i>	19 329	1.9383		
226		<i>train loss / train diff %</i>	0.5847	-0.61%		
227		<i>val loss / val diff %</i>	1.1333	-0.30%		
228	132 000	<i>duration / val/train</i>	19 651	1.9024		
229		<i>train loss / train diff %</i>	0.5893	0.79%		
230		<i>val loss / val diff %</i>	1.1211	-1.08%		
231	134 000	<i>duration / val/train</i>	19 949	1.9242		
232		<i>train loss / train diff %</i>	0.5845	-0.81%		
233		<i>val loss / val diff %</i>	1.1247	0.32%		
234	136 000	<i>duration / val/train</i>	20 264	1.9241		
235		<i>train loss / train diff %</i>	0.5891	0.79%		
236		<i>val loss / val diff %</i>	1.1335	0.78%		
237	138 000	<i>duration / val/train</i>	20 559	1.9206		
238		<i>train loss / train diff %</i>	0.5866	-0.42%		
239		<i>val loss / val diff %</i>	1.1266	-0.61%		
240	140 000	<i>duration / val/train</i>	20 854	1.9507		
241		<i>train loss / train diff %</i>	0.5861	-0.09%		
242		<i>val loss / val diff %</i>	1.1433	1.48%		
243	142 000	<i>duration / val/train</i>	21 147	1.9506		
244		<i>train loss / train diff %</i>	0.5833	-0.48%		

245		<i>val loss / val diff %</i>	1.1378	-0.48%		
246	144 000	<i>duration / val/train</i>	21 440	1.9274		
247		<i>train loss / train diff %</i>	0.5810	-0.39%		
248		<i>val loss / val diff %</i>	1.1198	-1.58%		
249	146 000	<i>duration / val/train</i>	21 740	1.9210		
250		<i>train loss / train diff %</i>	0.5888	1.34%		
251		<i>val loss / val diff %</i>	1.1311	1.01%		
252	148 000	<i>duration / val/train</i>	22 042	1.9532		
253		<i>train loss / train diff %</i>	0.5815	-1.24%		
254		<i>val loss / val diff %</i>	1.1358	0.42%		
255	150 000	<i>duration / val/train</i>	22 346	1.9299		
256		<i>train loss / train diff %</i>	0.5788	-0.46%		
257		<i>val loss / val diff %</i>	1.1170	-1.66%		

Table 19. Results of training models at Step 9.

8.3. Combining warming up new weights with freezing old ones. Step 10.

We'll take the warm-up duration from Step 8 and the old weights' freezing coefficient from Step 9 and apply them simultaneously. The results of training the models in Step 10 are shown in Table 20.

<i>No</i>	<i>iterations</i>	<i>Params</i>	12/128/4/2048/0.10/4/100/0.8	10/192/6/2048/0.10/4/200/0.8	8/256/8/2048/0.15/4/150/0.8
1		<i>model size, M</i>	2.69	4.91	6.94
2		<i>n_layer</i>	12	10	8
3		<i>n_embd</i>	128	192	256
4		<i>n_head</i>	4	6	8
5		<i>block_size</i>	2 048	2 048	2 048
6		<i>dropout</i>	0.10	0.10	0.15
7		<i>batch_size</i>	4	4	4
8		<i>warmup</i>	100	200	150
9		<i>weight_freeze</i>	0.8	0.8	0.8
10	<i>Stage 1</i>	<i>model</i>	2/32/1/128/0.00/64/40k	2/64/2/128/0.00/64/50k	2/64/2/128/0.00/64/50k
11		<i>iterations</i>	40 000	50 000	50 000
12		<i>duration / val/train</i>	216 1.2228	275 1.3309	276 1.3309
13		<i>train loss</i>	1.6816	1.3640	1.3640
14		<i>val loss</i>	2.0563	1.8154	1.8154
15	<i>Stage 2</i>	<i>model</i>	4/64/2/256/0.05/64/400/0.8/55k	4/96/3/256/0.05/64/400/0.8/45k	3/96/3/256/0.05/64/300/0.8/45k
16		<i>iterations</i>	55 000	45 000	45 000
17		<i>duration / val/train</i>	544 1.3800	673 1.4922	532 1.4117
18		<i>train loss</i>	1.1439 -31.98%	0.9881 -27.56%	1.0441 -23.45%
19		<i>val loss</i>	1.5786 -23.23%	1.4744 -18.78%	1.4740 -18.81%
20	<i>Stage 3</i>	<i>model</i>	6/64/2/512/0.05/64/300/0.8/40k	6/128/4/512/0.05/64/250/0.8/45k	4/128/4/512/0.05/64/300/0.8/40k
21		<i>iterations</i>	40 000	45 000	40 000
22		<i>duration / val/train</i>	1 791 1.4208	3 985 1.8354	2 456 1.6310
23		<i>train loss</i>	0.9855 -13.85%	0.7350 -25.61%	0.8166 -21.79%
24		<i>val loss</i>	1.4002 -11.30%	1.3490 -8.51%	1.3319 -9.64%

25	Stage 4	<i>model</i>	9/96/3/1024/0.05/32/25 0/0.8/40k		8/160/5/1536/0.10/8/ 400/0.8/65k		6/192/6/1536/0.10/8/25 0/0.8/65k	
26		<i>iterations</i>	40 000		65 000		65 000	
27		<i>duration / val/train</i>	7 017	1.6749	9 736	1.8248	8 571	1.8304
28		<i>train loss</i>	0.7315	-25.77%	0.6280	-14.56%	0.6228	-23.73%
29		<i>val loss</i>	1.2252	-12.50%	1.1460	-15.05%	1.1400	-14.41%
30		<i>Total time</i>	9 568		14 669		11 835	
31	0	<i>duration / val/train</i>	0	1.0742	0	1.0735	0	1.0572
32		<i>train loss</i>	5.1745		4.9285		4.4454	
33		<i>val loss</i>	5.5584		5.2908		4.6998	
34	2 000	<i>duration / val/train</i>	293	1.3306	383	1.5335	391	1.4423
35		<i>train loss / train diff %</i>	0.9222	-82.18%	0.7559	-84.66%	0.7984	-82.04%
36		<i>val loss / val diff %</i>	1.2271	-77.92%	1.1592	-78.09%	1.1515	-75.50%
37	4 000	<i>duration / val/train</i>	586	1.3870	767	1.5994	782	1.5269
38		<i>train loss / train diff %</i>	0.8455	-8.32%	0.6879	-9.00%	0.7220	-9.57%
39		<i>val loss / val diff %</i>	1.1727	-4.43%	1.1002	-5.09%	1.1024	-4.26%
40	6 000	<i>duration / val/train</i>	878	1.4037	1 150	1.6851	1 173	1.5586
41		<i>train loss / train diff %</i>	0.8228	-2.68%	0.6646	-3.39%	0.6982	-3.30%
42		<i>val loss / val diff %</i>	1.1550	-1.51%	1.1199	1.79%	1.0882	-1.29%
43	8 000	<i>duration / val/train</i>	1 171	1.4250	1 532	1.7191	1 564	1.5886
44		<i>train loss / train diff %</i>	0.8030	-2.41%	0.6505	-2.12%	0.6836	-2.09%
45		<i>val loss / val diff %</i>	1.1443	-0.93%	1.1183	-0.14%	1.0860	-0.20%
46	10 000	<i>duration / val/train</i>	1 463	1.4387	1 914	1.7226	1 955	1.6076
47		<i>train loss / train diff %</i>	0.7814	-2.69%	0.6414	-1.40%	0.6721	-1.68%
48		<i>val loss / val diff %</i>	1.1242	-1.76%	1.1049	-1.20%	1.0805	-0.51%
49	12 000	<i>duration / val/train</i>	1 756	1.4606	2 297	1.7668	2 346	1.6222
50		<i>train loss / train diff %</i>	0.7734	-1.02%	0.6238	-2.74%	0.6692	-0.43%
51		<i>val loss / val diff %</i>	1.1296	0.48%	1.1021	-0.25%	1.0856	0.47%
52	14 000	<i>duration / val/train</i>	2 049	1.4764	2 686	1.7671	2 737	1.6563
53		<i>train loss / train diff %</i>	0.7612	-1.58%	0.6205	-0.53%	0.6502	-2.84%
54		<i>val loss / val diff %</i>	1.1238	-0.51%	1.0965	-0.51%	1.0769	-0.80%
55	16 000	<i>duration / val/train</i>	2 341	1.4811	3 072	1.7865	3 128	1.6668
56		<i>train loss / train diff %</i>	0.7470	-1.87%	0.6178	-0.44%	0.6422	-1.23%
57		<i>val loss / val diff %</i>	1.1064	-1.55%	1.1037	0.66%	1.0704	-0.60%
58	18 000	<i>duration / val/train</i>	2 634	1.5422	3 444	1.8225	3 518	1.7350
59		<i>train loss / train diff %</i>	0.7305	-2.21%	0.6056	-1.97%	0.6242	-2.80%
60		<i>val loss / val diff %</i>	1.1266	1.83%	1.1037	0.00%	1.0830	1.18%
61	20 000	<i>duration / val/train</i>	2 929	1.5210	3 813	1.8399	3 909	1.7463
62		<i>train loss / train diff %</i>	0.7279	-0.36%	0.6004	-0.86%	0.6130	-1.79%
63		<i>val loss / val diff %</i>	1.1071	-1.73%	1.1047	0.09%	1.0705	-1.15%
64	22 000	<i>duration / val/train</i>	3 231	1.5311	4 180	1.8262	4 300	1.7850
65		<i>train loss / train diff %</i>	0.7195	-1.15%	0.5966	-0.63%	0.6052	-1.27%
66		<i>val loss / val diff %</i>	1.1016	-0.50%	1.0895	-1.38%	1.0803	0.92%
67	24 000	<i>duration / val/train</i>	3 527	1.5516	4 547	1.9062	4 691	1.7782
68		<i>train loss / train diff %</i>	0.7077	-1.64%	0.5882	-1.41%	0.6010	-0.69%
69		<i>val loss / val diff %</i>	1.0981	-0.32%	1.1212	2.91%	1.0687	-1.07%
70	26 000	<i>duration / val/train</i>	3 826	1.5727	4 913	1.9175	5 082	1.8686
71		<i>train loss / train diff %</i>	0.7047	-0.42%	0.5818	-1.09%	0.5853	-2.61%
72		<i>val loss / val diff %</i>	1.1083	0.93%	1.1156	-0.50%	1.0937	2.34%

73	28 000	duration / val/train	4 150	1.5904	5 280	1.8802	5 473	1.8712
74		train loss / train diff %	0.6951	-1.36%	0.5812	-0.10%	0.5775	-1.33%
75		val loss / val diff %	1.1055	-0.25%	1.0928	-2.04%	1.0806	-1.20%
76	30 000	duration / val/train	4 451	1.5962	5 647	1.9570	5 864	1.8579
77		train loss / train diff %	0.6894	-0.82%	0.5693	-2.05%	0.5730	-0.78%
78		val loss / val diff %	1.1004	-0.46%	1.1141	1.95%	1.0646	-1.48%
79	32 000	duration / val/train	4757	1.6089	6 014	1.9549	6 255	1.9146
80		train loss / train diff %	0.6834	-0.87%	0.5671	-0.39%	0.5647	-1.45%
81		val loss / val diff %	1.0995	-0.08%	1.1086	-0.49%	1.0812	1.56%
82	34 000	duration / val/train	5 062	1.6213	6 380	2.0126	6 646	1.9433
83		train loss / train diff %	0.6799	-0.51%	0.5574	-1.71%	0.5594	-0.94%
84		val loss / val diff %	1.1023	0.25%	1.1218	1.19%	1.0871	0.55%
85	36 000	duration / val/train	5 368	1.6401	6 747	2.0258	7 037	1.9483
86		train loss / train diff %	0.6735	-0.94%	0.5499	-1.35%	0.5574	-0.36%
87		val loss / val diff %	1.1046	0.21%	1.1140	-0.70%	1.0860	-0.10%
88	38 000	duration / val/train	5 668	1.6143	7 114	2.0377	7 428	1.9684
89		train loss / train diff %	0.6663	-1.07%	0.5516	0.31%	0.5563	-0.20%
90		val loss / val diff %	1.0756	-2.63%	1.1240	0.90%	1.0950	0.83%
91	40 000	duration / val/train	5 974	1.6646	7 480	2.0619	7 818	1.9868
92		train loss / train diff %	0.6615	-0.72%	0.5441	-1.36%	0.5463	-1.80%
93		val loss / val diff %	1.1011	2.37%	1.1219	-0.19%	1.0854	-0.88%
94	42 000	duration / val/train	6 271	1.6721	7 847	2.1131	8 209	2.0424
95		train loss / train diff %	0.6642	0.41%	0.5374	-1.23%	0.5445	-0.33%
96		val loss / val diff %	1.1106	0.86%	1.1356	1.22%	1.1121	2.46%
97	44 000	duration / val/train	6 575	1.6575	8 214	2.1212	8 600	2.0508
98		train loss / train diff %	0.6633	-0.14%	0.5353	-0.39%	0.5433	-0.22%
99		val loss / val diff %	1.0994	-1.01%	1.1355	-0.01%	1.1142	0.19%
100	46 000	duration / val/train	6 883	1.6639	8 580	2.1232	8 991	2.0824
101		train loss / train diff %	0.6545	-1.33%	0.5325	-0.52%	0.5375	-1.07%
102		val loss / val diff %	1.0890	-0.95%	1.1306	-0.43%	1.1193	0.46%
103	48 000	duration / val/train	7 186	1.6602	8 947	2.1836	9 394	2.1078
104		train loss / train diff %	0.6543	-0.03%	0.5256	-1.30%	0.5325	-0.93%
105		val loss / val diff %	1.0863	-0.25%	1.1477	1.51%	1.1224	0.28%
106	50 000	duration / val/train	7 486	1.6629	9 314	2.1578	9 794	2.1221
107		train loss / train diff %	0.6490	-0.81%	0.5273	0.32%	0.5299	-0.49%
108		val loss / val diff %	1.0792	-0.65%	1.1378	-0.86%	1.1245	0.19%
109	52 000	duration / val/train	7 788	1.7104	9 680	2.1694	10 197	2.1234
110		train loss / train diff %	0.6440	-0.77%	0.5206	-1.27%	0.5220	-1.49%
111		val loss / val diff %	1.1015	2.07%	1.1294	-0.74%	1.1084	-1.43%
112	54 000	duration / val/train	8 087	1.7273	10 047	2.1848	10 595	2.1564
113		train loss / train diff %	0.6402	-0.59%	0.5206	0.00%	0.5179	-0.79%
114		val loss / val diff %	1.1058	0.39%	1.1374	0.71%	1.1168	0.76%
115	56 000	duration / val/train	8 387	1.7245	10 414	2.2406	10 994	2.1749
116		train loss / train diff %	0.6403	0.02%	0.5195	-0.21%	0.5151	-0.54%
117		val loss / val diff %	1.1042	-0.14%	1.1640	2.34%	1.1203	0.31%
118	58 000	duration / val/train	8 688	1.7335	10 781	2.2169	11 399	2.1997
119		train loss / train diff %	0.6350	-0.83%	0.5131	-1.23%	0.5133	-0.35%
120		val loss / val diff %	1.1008	-0.31%	1.1375	-2.28%	1.1291	0.79%
121	60 000	duration / val/train	8 989	1.7126	11 147	2.2724	11 791	2.2247

122		<i>train loss / train diff %</i>	0.6329	-0.33%	0.5070	-1.19%	0.5052	-1.58%
123		<i>val loss / val diff %</i>	1.0839	-1.54%	1.1521	1.28%	1.1239	-0.46%
124	62 000	<i>duration / val/train</i>	9 291	1.7527	11 514	2.2844	12 182	2.2043
125		<i>train loss / train diff %</i>	0.6301	-0.44%	0.5116	0.91%	0.5065	0.26%
126		<i>val loss / val diff %</i>	1.1044	1.89%	1.1687	1.44%	1.1165	-0.66%
127	64 000	<i>duration / val/train</i>	9 586	1.7326	11 881	2.2752	12 573	2.2810
128		<i>train loss / train diff %</i>	0.6349	0.76%	0.5099	-0.33%	0.5057	-0.16%
129		<i>val loss / val diff %</i>	1.1000	-0.40%	1.1601	-0.74%	1.1535	3.31%
130	66 000	<i>duration / val/train</i>	9 879	1.7601	12 247	2.2727	12 964	2.2520
131		<i>train loss / train diff %</i>	0.6224	-1.97%	0.5035	-1.26%	0.5003	-1.07%
132		<i>val loss / val diff %</i>	1.0955	-0.41%	1.1443	-1.36%	1.1267	-2.32%
133	68 000	<i>duration / val/train</i>	10 171	1.7754	12 614	2.3176	13 355	2.2655
134		<i>train loss / train diff %</i>	0.6248	0.39%	0.4994	-0.81%	0.4979	-0.48%
135		<i>val loss / val diff %</i>	1.1093	1.26%	1.1574	1.14%	1.1280	0.12%
136	70 000	<i>duration / val/train</i>	10 466	1.7480	12 981	2.3309	13 746	2.2903
137		<i>train loss / train diff %</i>	0.6262	0.22%	0.4998	0.08%	0.4936	-0.86%
138		<i>val loss / val diff %</i>	1.0946	-1.33%	1.1650	0.66%	1.1305	0.22%
139	72 000	<i>duration / val/train</i>	10 761	1.7787	13 347	2.3677	14 137	2.2938
140		<i>train loss / train diff %</i>	0.6208	-0.86%	0.4955	-0.86%	0.4952	0.32%
141		<i>val loss / val diff %</i>	1.1042	0.88%	1.1732	0.70%	1.1359	0.48%
142	74 000	<i>duration / val/train</i>	11 061	1.7685	13 714	2.3209	14 530	2.3212
143		<i>train loss / train diff %</i>	0.6215	0.11%	0.4964	0.18%	0.4891	-1.23%
144		<i>val loss / val diff %</i>	1.0991	-0.46%	1.1521	-1.80%	1.1353	-0.05%
145	76 000	<i>duration / val/train</i>	11 365	1.7967	14 082	2.3622	14 930	2.3516
146		<i>train loss / train diff %</i>	0.6140	-1.21%	0.4906	-1.17%	0.4827	-1.31%
147		<i>val loss / val diff %</i>	1.1032	0.37%	1.1589	0.59%	1.1351	-0.02%
148	78 000	<i>duration / val/train</i>	11 678	1.7974	14 452	2.3697	15 347	2.3764
149		<i>train loss / train diff %</i>	0.6125	-0.24%	0.4896	-0.20%	0.4825	-0.04%
150		<i>val loss / val diff %</i>	1.1009	-0.21%	1.1602	0.11%	1.1466	1.01%
151	80 000	<i>duration / val/train</i>	11 996	1.8136	14 818	2.4309	15 760	2.3584
152		<i>train loss / train diff %</i>	0.6085	-0.65%	0.4887	-0.18%	0.4838	0.27%
153		<i>val loss / val diff %</i>	1.1036	0.25%	1.1880	2.40%	1.1410	-0.49%
154	82 000	<i>duration / val/train</i>	12 292	1.8141	15 185	2.4248	16 168	2.4057
155		<i>train loss / train diff %</i>	0.6112	0.44%	0.4861	-0.53%	0.4804	-0.70%
156		<i>val loss / val diff %</i>	1.1088	0.47%	1.1787	-0.78%	1.1557	1.29%
157	84 000	<i>duration / val/train</i>	12 586	1.8412	15 552	2.4405	16 579	2.3903
158		<i>train loss / train diff %</i>	0.6071	-0.67%	0.4819	-0.86%	0.4799	-0.10%
159		<i>val loss / val diff %</i>	1.1178	0.81%	1.1761	-0.22%	1.1471	-0.74%
160	86 000	<i>duration / val/train</i>	12 879	1.7889	15 918	2.4470	16 996	2.3882
161		<i>train loss / train diff %</i>	0.6098	0.44%	0.4866	0.98%	0.4791	-0.17%
162		<i>val loss / val diff %</i>	1.0909	-2.41%	1.1907	1.24%	1.1442	-0.25%
163	88 000	<i>duration / val/train</i>	13 172	1.8382	16 285	2.4780	17 406	2.4005
164		<i>train loss / train diff %</i>	0.6049	-0.80%	0.4780	-1.77%	0.4741	-1.04%
165		<i>val loss / val diff %</i>	1.1119	1.93%	1.1845	-0.52%	1.1381	-0.53%
166	90 000	<i>duration / val/train</i>	13 464	1.7859	16 652	2.4840	17 813	2.4566
167		<i>train loss / train diff %</i>	0.6062	0.21%	0.4777	-0.06%	0.4700	-0.86%
168		<i>val loss / val diff %</i>	1.0826	-2.64%	1.1866	0.18%	1.1546	1.45%
169	92 000	<i>duration / val/train</i>	13 757	1.8725	17 018	2.5003	18 217	2.4886
170		<i>train loss / train diff %</i>	0.6010	-0.86%	0.4781	0.08%	0.4644	-1.19%

171		<i>val loss / val diff %</i>	1.1254	3.95%	1.1954	0.74%	1.1557	0.10%
172	94 000	<i>duration / val/train</i>	14 053	1.8403	17 385	2.5177	18 624	2.4622
173		<i>train loss / train diff %</i>	0.6000	-0.17%	0.4763	-0.38%	0.4688	0.95%
174		<i>val loss / val diff %</i>	1.1042	-1.88%	1.1992	0.32%	1.1543	-0.12%
175	96 000	<i>duration / val/train</i>	14 350	1.8279	17 752	2.5182	19 035	2.4727
176		<i>train loss / train diff %</i>	0.5968	-0.53%	0.4732	-0.65%	0.4671	-0.36%
177		<i>val loss / val diff %</i>	1.0909	-1.20%	1.1916	-0.63%	1.1550	0.06%
178	98 000	<i>duration / val/train</i>	14 670	1.8302	18 118	2.5030	19 439	2.5492
179		<i>train loss / train diff %</i>	0.6031	1.06%	0.4712	-0.42%	0.4614	-1.22%
180		<i>val loss / val diff %</i>	1.1038	1.18%	1.1794	-1.02%	1.1762	1.84%
181	100 000	<i>duration / val/train</i>	14 983	1.8233	18 485	2.5230	19 846	2.4471
182		<i>train loss / train diff %</i>	0.5998	-0.55%	0.4700	-0.25%	0.4659	0.98%
183		<i>val loss / val diff %</i>	1.0936	-0.92%	1.1858	0.54%	1.1401	-3.07%
184	102 000	<i>duration / val/train</i>	15 291	1.8141				
185		<i>train loss / train diff %</i>	0.6035	0.62%				
186		<i>val loss / val diff %</i>	1.0948	0.11%				
187	104 000	<i>duration / val/train</i>	15 600	1.8462				
188		<i>train loss / train diff %</i>	0.5980	-0.91%				
189		<i>val loss / val diff %</i>	1.1040	0.84%				
190	106 000	<i>duration / val/train</i>	15 903	1.8426				
191		<i>train loss / train diff %</i>	0.6003	0.38%				
192		<i>val loss / val diff %</i>	1.1061	0.19%				
193	108 000	<i>duration / val/train</i>	16 198	1.8530				
194		<i>train loss / train diff %</i>	0.5924	-1.32%				
195		<i>val loss / val diff %</i>	1.0977	-0.76%				
196	110 000	<i>duration / val/train</i>	16 506	1.8981				
197		<i>train loss / train diff %</i>	0.5917	-0.12%				
198		<i>val loss / val diff %</i>	1.1231	2.31%				
199	112 000	<i>duration / val/train</i>	16 809	1.8847				
200		<i>train loss / train diff %</i>	0.5898	-0.32%				
201		<i>val loss / val diff %</i>	1.1116	-1.02%				
202	114 000	<i>duration / val/train</i>	17 112	1.8644				
203		<i>train loss / train diff %</i>	0.5943	0.76%				
204		<i>val loss / val diff %</i>	1.1080	-0.32%				
205	116 000	<i>duration / val/train</i>	17 410	1.8703				
206		<i>train loss / train diff %</i>	0.5884	-0.99%				
207		<i>val loss / val diff %</i>	1.1005	-0.68%				
208	118 000	<i>duration / val/train</i>	17 729	1.9012				
209		<i>train loss / train diff %</i>	0.5863	-0.36%				
210		<i>val loss / val diff %</i>	1.1147	1.29%				
211	120 000	<i>duration / val/train</i>	18 034	1.8907				
212		<i>train loss / train diff %</i>	0.5894	0.53%				
213		<i>val loss / val diff %</i>	1.1144	-0.03%				
214	122 000	<i>duration / val/train</i>	18 341	1.8955				
215		<i>train loss / train diff %</i>	0.5835	-1.00%				
216		<i>val loss / val diff %</i>	1.1060	-0.75%				
217	124 000	<i>duration / val/train</i>	18 643	1.9020				
218		<i>train loss / train diff %</i>	0.5816	-0.33%				
219		<i>val loss / val diff %</i>	1.1062	0.02%				

220	126 000	<i>duration / val/train</i>	18 938	1.8901		
221		<i>train loss / train diff %</i>	0.5878	1.07%		
222		<i>val loss / val diff %</i>	1.1110	0.43%		
223	128 000	<i>duration / val/train</i>	19 246	1.9093		
224		<i>train loss / train diff %</i>	0.5813	-1.11%		
225		<i>val loss / val diff %</i>	1.1099	-0.10%		
226	130 000	<i>duration / val/train</i>	19 544	1.9075		
227		<i>train loss / train diff %</i>	0.5774	-0.67%		
228		<i>val loss / val diff %</i>	1.1014	-0.77%		
229	132 000	<i>duration / val/train</i>	19 853	1.8977		
230		<i>train loss / train diff %</i>	0.5826	0.90%		
231		<i>val loss / val diff %</i>	1.1056	0.38%		
232	134 000	<i>duration / val/train</i>	20 155	1.8824		
233		<i>train loss / train diff %</i>	0.5800	-0.45%		
234		<i>val loss / val diff %</i>	1.0918	-1.25%		
235	136 000	<i>duration / val/train</i>	20 456	1.8832		
236		<i>train loss / train diff %</i>	0.5840	0.69%		
237		<i>val loss / val diff %</i>	1.0998	0.73%		
238	138 000	<i>duration / val/train</i>	20 756	1.9204		
239		<i>train loss / train diff %</i>	0.5794	-0.79%		
240		<i>val loss / val diff %</i>	1.1127	1.17%		
241	140 000	<i>duration / val/train</i>	21 069	1.9121		
242		<i>train loss / train diff %</i>	0.5789	-0.09%		
243		<i>val loss / val diff %</i>	1.1069	-0.52%		
244	142 000	<i>duration / val/train</i>	21 374	1.9416		
245		<i>train loss / train diff %</i>	0.5755	-0.59%		
246		<i>val loss / val diff %</i>	1.1174	0.95%		
247	144 000	<i>duration / val/train</i>	21 681	1.9248		
248		<i>train loss / train diff %</i>	0.5744	-0.19%		
249		<i>val loss / val diff %</i>	1.1056	-1.06%		
250	146 000	<i>duration / val/train</i>	21 976	1.9305		
251		<i>train loss / train diff %</i>	0.5810	1.15%		
252		<i>val loss / val diff %</i>	1.1216	1.45%		
253	148 000	<i>duration / val/train</i>	22 270	1.9232		
254		<i>train loss / train diff %</i>	0.5754	-0.96%		
255		<i>val loss / val diff %</i>	1.1066	-1.34%		
256	150 000	<i>duration / val/train</i>	22 562	1.9466		
257		<i>train loss / train diff %</i>	0.5732	-0.38%		
258		<i>val loss / val diff %</i>	1.1158	0.83%		

Table 20. Results of training models at Step 10.

8.4. Analysis of the results of training models at Steps 2-10

The selected results for each model are shown in Table 21.

No	Checkpoints	Params	12/128/4/ 2048/0.10/ 4	10/192/6/ 2048/0.10 /4	8/256/8/20 48/0.15/4
1		<i>model size, M</i>	2.69	4.91	6.94

2			<i>n_layer</i>	12	10	8
3			<i>n_embd</i>	128	192	256
4			<i>n_head</i>	4	6	8
5			<i>block_size</i>	2 048	2 048	2 048
6			<i>dropout</i>	0.10	0.10	0.15
7			<i>batch_size</i>	4	4	4
8	val loss plateau	Step 1	<i>iterations</i>	96 000	38 000	32 000
9			<i>duration</i>	13 934	7 151	6 441
10			<i>train loss</i>	0.6692	0.6842	0.6914
11			<i>val loss</i>	1.1060	1.0988	1.0867
12		Step 2 zero	<i>iterations</i>	38 000	10 000	14 000
13			<i>duration</i>	13 664	12 342	14 252
14			<i>train loss</i>	0.7626	0.6965	0.7124
15			<i>val loss</i>	1.1131	1.0940	1.0907
16			<i>duration diff %</i>	98.1%	172.6%	221.3%
17		Step 2 normal	<i>iterations</i>	22 000	4 000	12 000
18			<i>duration</i>	11 974	11 226	11 623
19			<i>train loss</i>	0.7393	0.7058	0.6860
20			<i>val loss</i>	1.1101	1.0869	1.1060
21			<i>duration diff %</i>	85.9%	157.0%	180.5%
22		Step 3 zero	<i>iterations</i>	30 000	8 000	8 000
23			<i>duration</i>	22 622	19 740	21 172
24			<i>train loss</i>	0.7696	0.7037	0.7308
25			<i>val loss</i>	1.1042	1.1133	1.0936
26			<i>duration diff %</i>	162.4%	276.0%	328.7%
27		Step 3 normal	<i>iterations</i>	12 000	10 000	18 000
28			<i>duration</i>	21 446	19 269	18 015
29			<i>train loss</i>	0.7648	0.6430	0.6391
30			<i>val loss</i>	1.1138	1.1061	1.0983
31			<i>duration diff %</i>	153.9%	269.5%	279.7%
32		Step 4 zero	<i>iterations</i>		6 000	2 000
33			<i>duration</i>		44 693	23 057
34			<i>train loss</i>		0.6316	0.7351
35			<i>val loss</i>		1.0921	1.0897
36			<i>duration diff %</i>		625.0%	358.0%
37		Step 4 normal	<i>iterations</i>		2 000	6 000
38			<i>duration</i>		36 483	28 518
39			<i>train loss</i>		0.6346	0.6632
40			<i>val loss</i>		1.0993	1.0768
41			<i>duration diff %</i>		510.2%	442.8%
42		Step 5	<i>iterations</i>	32 000	6 000	6 000
43			<i>duration</i>	14 731	13 794	12 830
44			<i>train loss</i>	0.6997	0.6806	0.6990

45			<i>val loss</i>	1.1130	1.0831	1.0923
46			<i>duration diff %</i>	105.7%	192.9%	199.2%
47		Step 6	<i>iterations</i>	36 000	10 000	8 000
48			<i>duration</i>	15 473	16 354	14 595
49			<i>train loss</i>	0.7036	0.6654	0.7103
50			<i>val loss</i>	1.1126	1.1112	1.1135
51			<i>duration diff %</i>	111.0%	228.7%	226.6%
52		Step 7	<i>iterations</i>	6 000	2 000	4 000
53			<i>duration</i>	29 593	25 595	17 615
54			<i>train loss</i>	0.7110	0.6391	0.6852
55			<i>val loss</i>	1.0901	1.1029	1.0948
56			<i>duration diff %</i>	212.4%	357.9%	273.5%
42		Step 8	<i>iterations</i>	28 000	4 000	8 000
43			<i>duration</i>	13 423	13 659	11 247
44			<i>train loss</i>	0.7102	0.6952	0.7071
45			<i>val loss</i>	1.1026	1.0884	1.0952
46			<i>duration diff %</i>	96.3%	191.0%	174.6%
47		Step 9	<i>iterations</i>	32 000	4 000	8 000
48			<i>duration</i>	14 226	14 374	13 853
49			<i>train loss</i>	0.7010	0.6890	0.6823
50			<i>val loss</i>	1.1381	1.0938	1.1048
51			<i>duration diff %</i>	102.1%	201.0%	215.1%
52		Step 10	<i>iterations</i>	18 000	4 000	6 000
53			<i>duration</i>	12 202	15 436	13 008
54			<i>train loss</i>	0.7470	0.6879	0.6982
55			<i>val loss</i>	1.1064	1.1002	1.0882
56			<i>duration diff %</i>	87.6%	215.9%	202.0%
57	min val loss	Step 1	<i>iterations</i>	232 000	76 000	56 000
58			<i>duration</i>	34 159	14 321	11 109
59			<i>train loss</i>	0.6136	0.6078	0.6195
60			<i>val loss</i>	1.0808	1.0877	1.0731
61		Step 2 zero	<i>iterations</i>	220 000	4 000	30 000
62			<i>duration</i>	40 629	11 250	17 365
63			<i>train loss</i>	0.6302	0.7300	0.6506
64			<i>val loss</i>	1.0527	1.0813	1.0658
65			<i>val loss diff %</i>	97.4%	99.4%	99.3%
66		Step 2 normal	<i>iterations</i>	94 000	14 000	30 000
67			<i>duration</i>	22 706	13 063	15 279
68			<i>train loss</i>	0.6133	0.6372	0.5870
69			<i>val loss</i>	1.0902	1.0671	1.0862
70			<i>val loss diff %</i>	100.9%	98.1%	101.2%
71		Step 3 zero	<i>iterations</i>	90 000	30 000	22 000
72			<i>duration</i>	31 386	23 910	23 983
73			<i>train loss</i>	0.6828	0.6119	0.6675
74			<i>val loss</i>	1.0701	1.1028	1.0771
75			<i>val loss diff %</i>	99.0%	101.4%	100.4%

76		Step 3 normal	<i>iterations</i>	90 000	24 000	28 000
77			<i>duration</i>	33 097	21 872	20 044
78			<i>train loss</i>	0.6183	0.5859	0.5905
79			<i>val loss</i>	1.0620	1.1022	1.0824
80			<i>val loss diff %</i>	98.3%	101.3%	100.9%
81		Step 4 zero	<i>iterations</i>		4 000	80 000
82			<i>duration</i>		44 312	24 245
83			<i>train loss</i>		0.6394	0.6703
84			<i>val loss</i>		1.0760	1.0590
85			<i>val loss diff %</i>		98.9%	98.7%
86		Step 4 normal	<i>iterations</i>		8 000	14 000
87			<i>duration</i>		37 577	30 109
88			<i>train loss</i>		0.5677	0.5799
89			<i>val loss</i>		1.0696	1.0537
90			<i>val loss diff %</i>		98.3%	98.2%
91		Step 5	<i>iterations</i>	94 000	28 000	16 000
92			<i>duration</i>	23 729	17 920	14 857
93			<i>train loss</i>	0.6114	0.5993	0.6450
94			<i>val loss</i>	1.0917	1.0646	1.0687
95			<i>val loss diff %</i>	101.0%	97.9%	99.6%
96		Step 6	<i>iterations</i>	64 000	12 000	16 000
97			<i>duration</i>	19 608	16 732	16 199
98			<i>train loss</i>	0.6494	0.6568	0.6518
99			<i>val loss</i>	1.0858	1.1035	1.0798
100			<i>val loss diff %</i>	100.5%	101.5%	100.6%
101		Step 7	<i>iterations</i>	38 000	6 000	12 000
102			<i>duration</i>	34 237	26 351	19 173
103			<i>train loss</i>	0.6279	0.5907	0.5978
104			<i>val loss</i>	1.0579	1.0668	1.0559
105			<i>val loss diff %</i>	97.9%	98.1%	98.4%
91		Step 8	<i>iterations</i>	104 000	14 000	16 000
92			<i>duration</i>	24 601	15 554	12 810
93			<i>train loss</i>	0.5986	0.6349	0.6427
94			<i>val loss</i>	1.0915	1.0842	1.0718
95			<i>val loss diff %</i>	101.0%	99.7%	99.9%
96		Step 9	<i>iterations</i>	90 000	14 000	12 000
97			<i>duration</i>	22 820	16 207	14 658
98			<i>train loss</i>	0.6166	0.6268	0.6662
99			<i>val loss</i>	1.1046	1.0860	1.0876
100			<i>val loss diff %</i>	102.2%	99.8%	101.4%
101		Step 10	<i>iterations</i>	38 000	22 000	30 000
102			<i>duration</i>	15 236	18 849	17 699
103			<i>train loss</i>	0.6663	0.5966	0.5730
104			<i>val loss</i>	1.0756	1.0895	1.0646
105			<i>val loss diff %</i>	99.5%	100.2%	99.2%

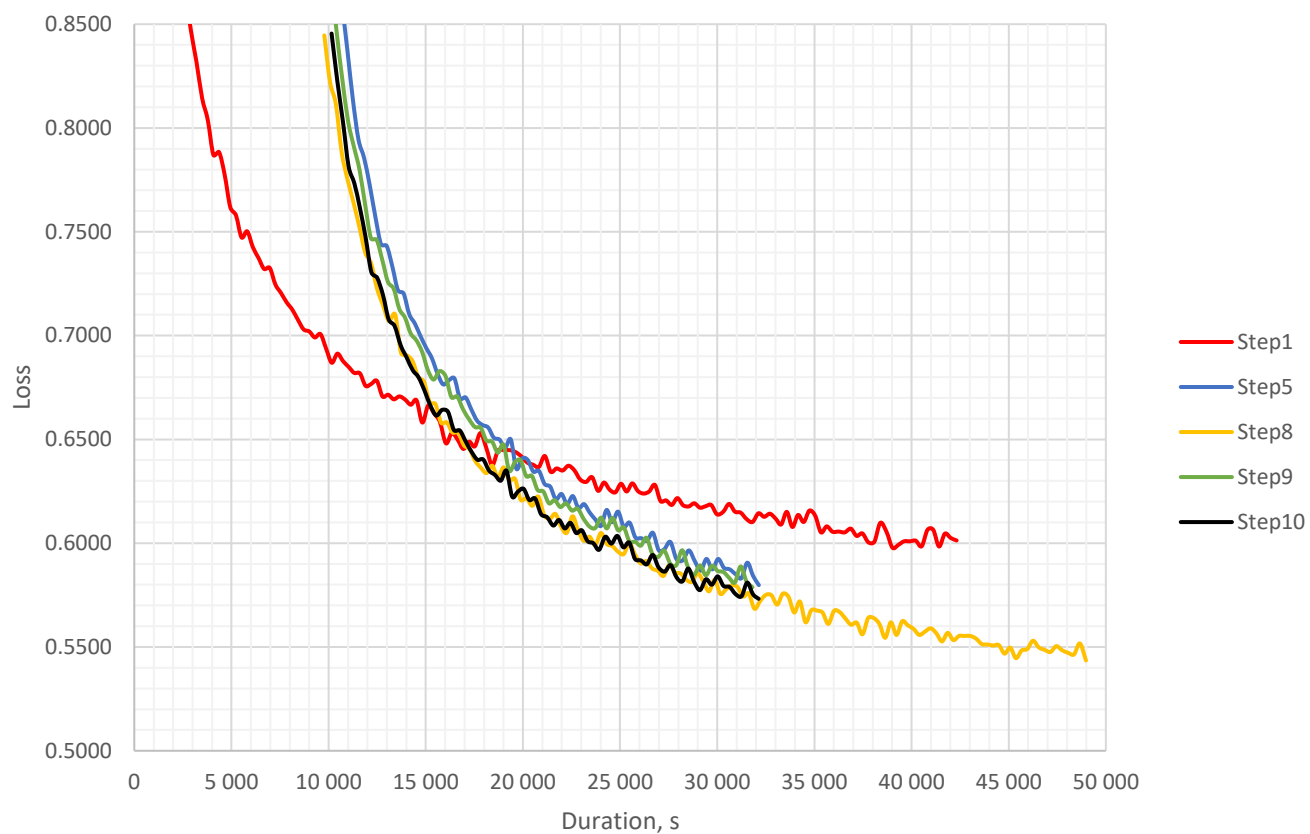
106	min train loss	Step 1	<i>iterations</i>	252 000	180 000	196 000
107			<i>duration</i>	37 172	33 606	39 193
108			<i>train loss</i>	0.6035	0.5224	0.5035
109			<i>val loss</i>	1.1132	1.1467	1.1503
110		Step 2 zero	<i>iterations</i>	216 000	124 000	142 000
111			<i>duration</i>	40 050	33 778	39 541
112			<i>train loss</i>	0.6232	0.5339	0.5340
113			<i>val loss</i>	1.0681	1.1299	1.1196
114			<i>duration diff %</i>	107.7%	100.5%	100.9%
115			<i>train loss diff %</i>	103.3%	102.2%	106.1%
116		Step 2 normal	<i>iterations</i>	150 000	106 000	78 000
117			<i>duration</i>	31 141	30 178	24 952
118			<i>train loss</i>	0.5813	0.4744	0.4907
119			<i>val loss</i>	1.1050	1.1530	1.1546
120			<i>duration diff %</i>	83.8%	89.8%	63.7%
121			<i>train loss diff %</i>	96.3%	90.8%	97.5%
122		Step 3 zero	<i>iterations</i>	176 000	140 000	144 000
123			<i>duration</i>	44 107	44 143	47 933
124			<i>train loss</i>	0.6348	0.5229	0.5285
125			<i>val loss</i>	1.0765	1.1387	1.1037
126			<i>duration diff %</i>	118.7%	131.4%	122.3%
127			<i>train loss diff %</i>	105.2%	100.1%	105.0%
128		Step 3 normal	<i>iterations</i>	112 000	50 000	80 000
129			<i>duration</i>	40 898	26 616	30 349
130			<i>train loss</i>	0.5813	0.5216	0.4792
131			<i>val loss</i>	1.0870	1.1717	1.1510
132			<i>duration diff %</i>	110.0%	79.2%	77.4%
133			<i>train loss diff %</i>	96.3%	99.8%	95.2%
134		Step 4 zero	<i>iterations</i>		104 000	142 000
135			<i>duration</i>		62 737	50 596
136			<i>train loss</i>		0.5348	0.5310
137			<i>val loss</i>		1.1411	1.0930
138			<i>duration diff %</i>		186.7%	129.1%
139			<i>train loss diff %</i>		102.4%	105.5%
140		Step 4 normal	<i>iterations</i>		64 000	80 000
141			<i>duration</i>		48 001	43 252
142			<i>train loss</i>		0.4740	0.4541
143			<i>val loss</i>		1.1572	1.1439
144			<i>duration diff %</i>		142.8%	110.4%
145			<i>train loss diff %</i>		90.7%	90.2%
146		Step 5	<i>iterations</i>	144 000	110 000	90 000
147			<i>duration</i>	31 241	33 164	29 484

148			<i>train loss</i>	0.5831	0.4687	0.4729
149			<i>val loss</i>	1.1125	1.1633	1.1446
150			<i>duration diff %</i>	84.0%	98.7%	75.2%
151			<i>train loss diff %</i>	96.6%	89.7%	93.9%
152		Step 6	<i>iterations</i>	158 000	74 000	82 000
153			<i>duration</i>	33 650	28 276	29 308
154			<i>train loss</i>	0.5907	0.4990	0.4823
155			<i>val loss</i>	1.1050	1.1593	1.1525
156			<i>duration diff %</i>	90.5%	84.1%	74.8%
157			<i>train loss diff %</i>	97.9%	95.5%	95.8%
158		Step 7	<i>iterations</i>	106 000	100 000	100 000
159			<i>duration</i>	44 359	43 633	36 312
160			<i>train loss</i>	0.5840	0.4541	0.4432
161			<i>val loss</i>	1.0862	1.1613	1.1469
162			<i>duration diff %</i>	119.3%	129.8%	92.6%
163			<i>train loss diff %</i>	96.8%	86.9%	88.0%
146		Step 8	<i>iterations</i>	150 000	100 000	100 000
147			<i>duration</i>	31 330	31 575	29 396
148			<i>train loss</i>	0.5743	0.4723	0.4618
149			<i>val loss</i>	1.1066	1.1617	1.1372
150			<i>duration diff %</i>	84.3%	94.0%	75.0%
151			<i>train loss diff %</i>	95.2%	90.4%	91.7%
152		Step 9	<i>iterations</i>	130 000	96 000	88 000
153			<i>duration</i>	28 795	31 678	29 981
154			<i>train loss</i>	0.5847	0.4840	0.4739
155			<i>val loss</i>	1.1333	1.1483	1.1522
156			<i>duration diff %</i>	77.5%	94.3%	76.5%
157			<i>train loss diff %</i>	96.9%	92.6%	94.1%
158		Step 10	<i>iterations</i>	144 000	76 000	92 000
159			<i>duration</i>	31 249	28 751	30 052
160			<i>train loss</i>	0.5744	0.4906	0.4644
161			<i>val loss</i>	1.1056	1.1589	1.1557
162			<i>duration diff %</i>	84.1%	85.6%	76.7%
163			<i>train loss diff %</i>	95.2%	93.9%	92.2%

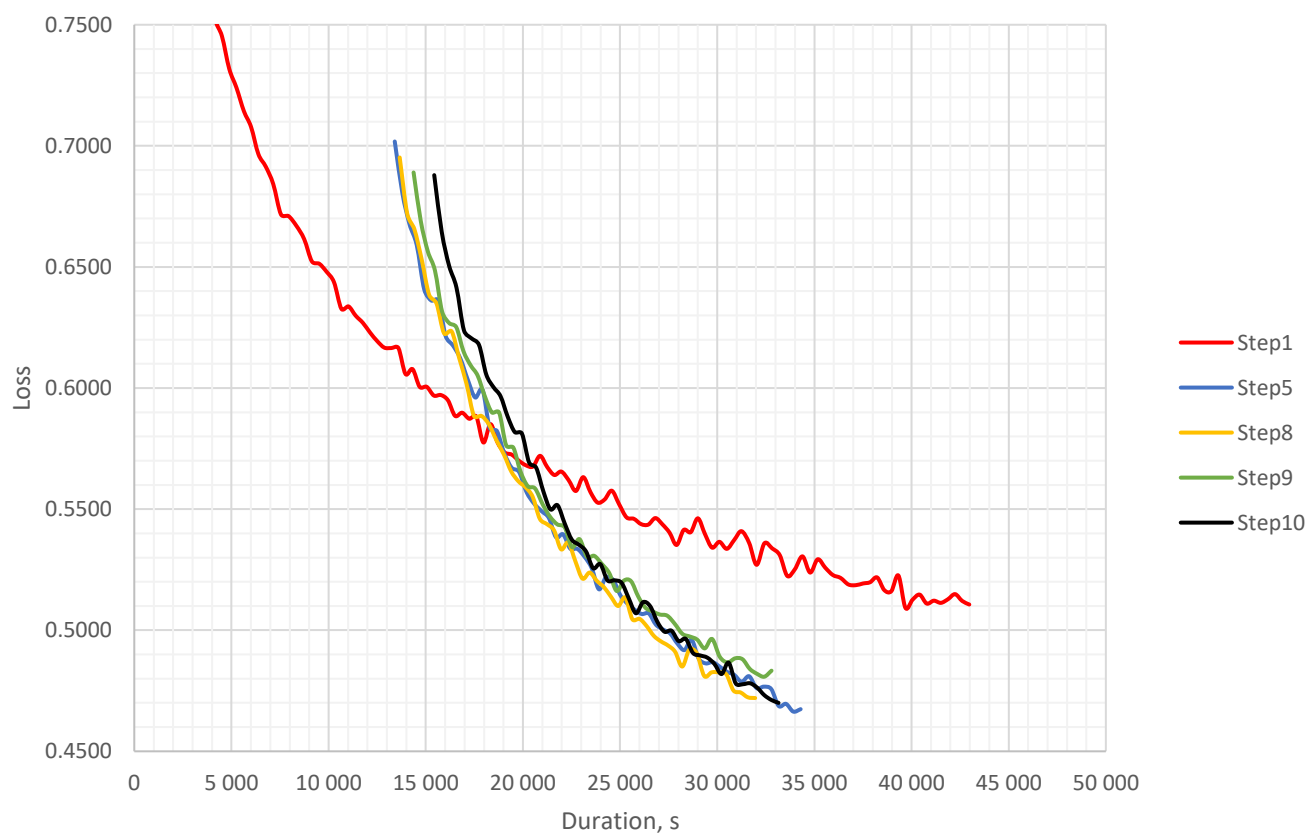
Table 21. Model checkpoints at Steps 2–10

Figure 5 shows graphs of changes in train loss depending on the duration of model training at Steps 8-10.

12/128/4/2048/0.10/4 Step 10



10/192/6/2048/0.10/4 Step 10



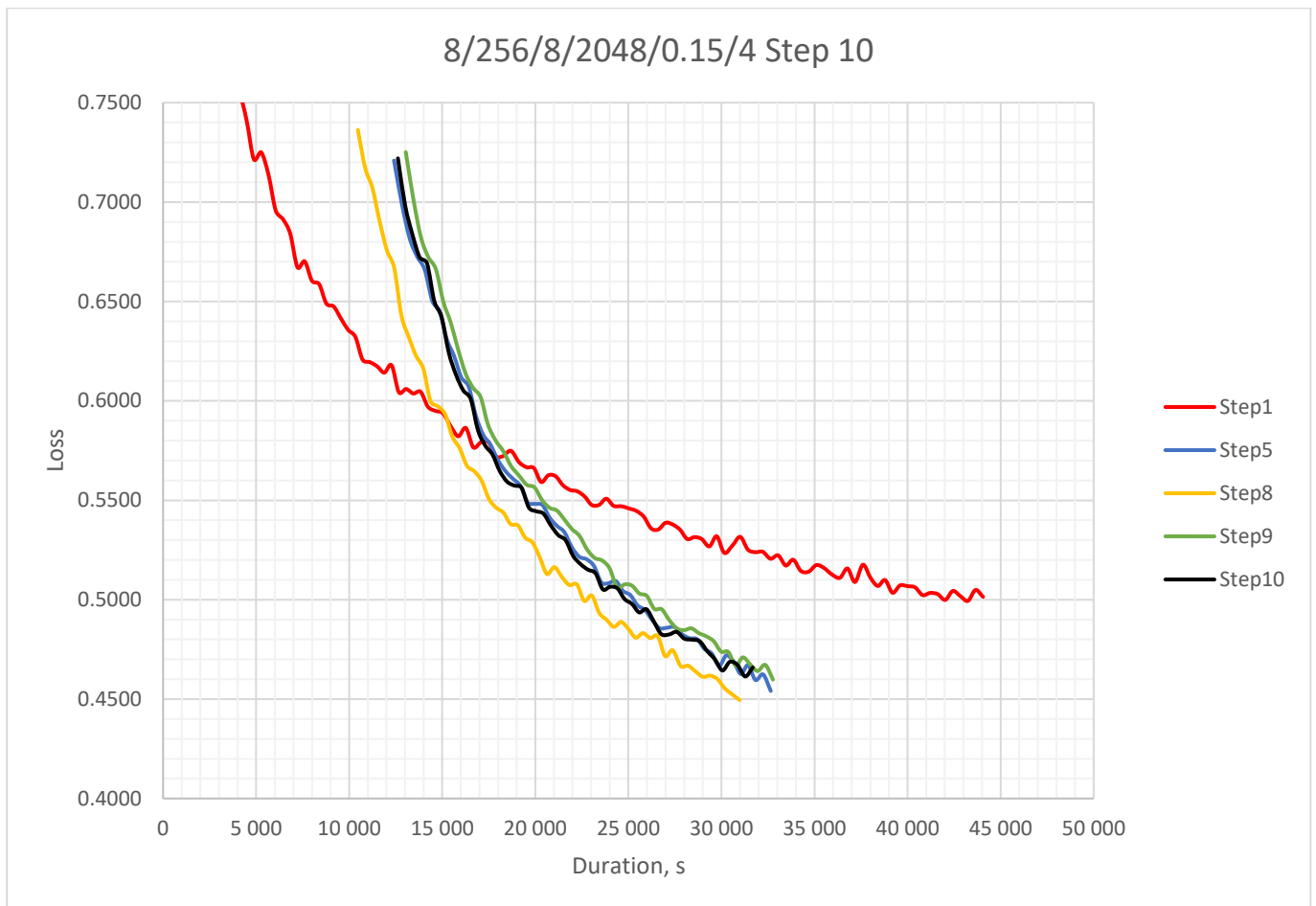


Figure 5. Train loss graphs depending on the training duration of models at Steps 8-10

Warming up new weights (Step 8) takes little time and can improve results. Freezing old weights doesn't work (Step 9) and can even worsen results. Combining warming up and freezing (Step 10) also doesn't produce any positive results.

9. Conclusions from the research results

Based on the research results, the following conclusions can be drawn:

- Splitting a dataset with non-repetitive texts into stages and training a growing model on these partial datasets doesn't work. The reason is apparently that LLMs reflect token usage statistics, and this approach doesn't work for weakly dependent and/or unrelated texts;
- However, training models with gradual growth on the full dataset works. Train loss and/or training duration can be reduced by 5-25%. This may be because a given training duration allows for more epochs to be accommodated by training smaller models;
- Filling new weights with random values gives better results than filling them with zeros;
- Models of different sizes behave approximately the same, the results differ numerically, but not qualitatively;
- Different options for stopping training at intermediate stages lead to roughly the same results, so it makes sense to choose clearer and simpler conditions. This may be because anything not trained now will be trained in subsequent stages. Increasing training time in one stage reduces it in another, and vice versa;
- Preserving the old weights while warming up the new weights at the start of each stage improves results.

In the next study, the author will select a different dataset with repeated information and again attempt to divide it into stages. The optimal number of stages and model growth parameters between stages will also be explored.

To be continued ...